AsOfGov: As-Of Correctness for Governed Queries with Typed Refusals and Certificates — Extended Technical Report with Full Proofs

Hongsheng Wang
College of Computer Science and Technology, Zhejiang University, Hangzhou, China
Loctek Ergonomic Technology Corp., China
wanghongsheng@zju.edu.cn

Xiubo Liang*
School of Software Technology, Zhejiang University, Hangzhou, China
xiubo@zju.edu.cn

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Abstract

Enterprise analytics increasingly runs through a governed semantic layer driven by a language model. Both move: calibers are re-scoped, valid-time anchors re-declared, graph versions advance. We show that answering a governed question "as of T" is a compile-time *binding* problem, not a prompting problem. On a fully public 60-question suite — nine BIRD/Spider databases, 3.83M real rows, a machine-checked non-degenerate governance layer with two committed versions per database — four frontier model families err on 60.0–76.7% of questions, answering through most of its 15 typed-refusal questions; three prompt variants (53.3–66.7%) never eliminate more than 22% of the same backbone's errors, under the frozen 40% line. Handed its database's entire governance layer verbatim, a pre-registered arm still errs on 46.7%, and its errors concentrate on exactly the decisions that need a data probe rather than a registry line: under this protocol, in-context governance yields no substantial elimination (0.361, below the frozen 0.40 line). We define a bi-temporal governed query semantics in which typed refusals are values, prove that bindability plus three refusal classes partition well-formed queries decidably, extend the compilation under a disclosure policy to MLR-COMPILE — a maximal legal rewrite or a complete structured refusal — and define point-in-time certificates checked by a verifier sharing no code with the compiler. The compiler resolves all 60 questions (33 answers, 12 rewrites, 15 typed refusals), a constructive upper bound, with the disclosure gate and the version axis exercised on gold for the first time; the verifier accepts 60 of 60 certificates, re-derives 50 of 60 window coordinates failing closed on the rest, and rejects 34 of 34 forgeries. Verification costs a median 17.7× the answering query it certifies (17.0 ms against 1.31 ms) and the certificate 10.2× the query's bytes.

How to read this report

This technical report is the extended version of the PVLDB Vol. 20 submission *AsOfGov: As-Of Correctness for Governed Queries with Typed Refusals and Certificates.* Part I (Sections 1–9) is the

*Xiubo Liang is the corresponding author.

complete submitted body, included from the same source files, so section and statement numbering coincides with the submitted PDF. Part II carries everything the body cites this report for: the full proofs and omitted constructions, the disclosure-constrained rewriting development, the per-check and per-question tables, the complete forgery enumeration, the two pre-registration protocols in full, and the three frozen theory chapters. Table 1 maps every citation of this report in the body — all 15 of them — to the appendix that discharges it.

Table 1: Every body citation of this report, and where it lands here.

#	body site and promise	discharged in
1	§1 (end): full proofs, omitted constructions, per-check tables, dissections of our own relaxations	Apps. D, C, B; Tables 9, 8; §G.1
2	§2 takeaway: proofs, notation table, omitted constructions	App. D; Table 7 (App. A); App. C
3	Example 2.1: two more worked requests — a certified OOV, a certified DISCLOSURE-BLOCKED listing	App. B
4	Lemma 2.2 (version-axis stability): proof	App. D.1
5	Proposition 2.7 (well-definedness): proof	App. D.1
6	Theorem 2.9 (degeneration): proof	App. D.1
7	Theorem 3.1 (coverage partition): proof	App. D.2
8	Proposition 3.2 (decidability and cost): proof	App. D.2
9	§4: algorithm, lattice coordinates, frontier and pruning lemmas, polynomial probe bound	App. C (Alg. 1; §C.1–C.4)
10	Theorems 4.1 and 4.2 (MLR soundness, maximality, refusal completeness): proofs	App. D.3
11	§5: per-check argument and answer/refusal tables	Tables 9, 8; App. D.4; Table 10 (App. H); App. I
12	Theorem 5.2 (verification soundness): proofs (with completeness)	App. D.4
13	Proposition 5.4 (field non-redundancy): the five constructions	App. D.4
14	§6: per-step compiler walk-through	App. E (with App. B)
15	Table 5 caption: both readings per divergence row	§F.1

Provenance and freeze discipline

The three theory chapters reproduced in Appendices N–P are frozen upstream sources of the paper (freeze manifest `theory/FREEZE_SHA256_20260731.txt`); their content is included *unchanged* — the LaTeX files are produced mechanically from the frozen Markdown (pandoc 3.9, one command recorded in `tr/build.sh`), and the SHA-256 of each source was re-computed and matched against the manifest at assembly time (2026-08-04):

frozen source	SHA-256
`C3_bitemporal_semantics.md`	`f7d454caecf644399340e2eb8e49c52ddc1a550c9339603747290960e186d179`
`C4_maximal_legal_rewrite.md`	`cb57ca611a12ef03935a6763a717048b71f05d58d97b158d434589789a31b69a`
`C5_pointintime_certificates.md`	`bc5bc160eaf57828b405c805f95c08642af9410116b6ffcbf417c954de55f34b`

The chapters are written in Chinese; the English development of Appendices A–F is their rendering for the paper, not a revision of them. The acceptance report of Appendix I and the two pre-registration protocols of Appendices J and K are likewise mechanical typesettings of the frozen originals (`pilot2/ACCEPTANCE_REPORT.md`, `pilot2/PREREG_pilot2_arms.md`, `pilot/PREREG_governance_arm.md`); the per-question table of Appendix H is generated from the frozen gold (`pilot2/domains/*/questions.json`) by a script that re-tallies the sealed suite composition and fails rather than emit stale content.

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1 INTRODUCTION

Enterprise analytics is increasingly mediated by a *governed semantic layer*: metrics, calibers — the registered conventions fixing which population a metric attributes over — and valid-time anchors are registered as machine-readable objects over the warehouse, so that "problem rate" denotes one agreed computation rather than whatever an analyst types. Large language models now sit in front of that layer, and the two interact badly along one axis neither owns: *time*. The layer is no static dictionary — calibers are re-scoped, anchors re-declared, versions advance — so a question asked "as of March" must be answered under the governance semantics in force for March, over facts whose valid time falls in March: the *as-of correctness* problem.

Observed in production, measured in public. The problem statement is industrial: on the production governance line behind this work (five governed domains), one problem-rate question evaluates to a caliber-aware 16.2% against a caliber-blind 3.8% — a 4.2× gap on the same rows. Those are *production observations*: they motivate the problem and play no other role. Every result number from here on — save the two provenance-marked enterprise measurements (these rates and the §7.6 ledger) — is computed on a fully public base — nine versioned databases of BIRD/Spider lineage [20, 35], 3,830,036 real rows, a frozen suite of 60 Chinese questions — whose complete chain ships in the artifact: the motivation is proprietary, the evidence is not.

The gap is measurable, and one-line prompting does not close it. Gold on the 60 Chinese questions is 33 values, 12 governed rewrites and 15 typed refusals (§7); every arm answers from one statement, without execution feedback, three families behind the gateway caveat of §7.1. Four frontier model families answer 60.0–76.7% incorrectly, answering straight through 11, 11, 9 and 13 of the 15 refusals; for three of the four at most a quarter of errors are failing or malformed SQL — the gap is semantic, not syntactic. Three prompt variants of the reference backbone err on 53.3–66.7%, the best eliminating 22% of its errors against the pre-registered 40% line (§7.2). An arm handing that backbone its database's *entire* governance layer verbatim still errs on 46.7% (elimination 36%): *under this protocol* — one statement, no execution feedback — in-context governance does not substantially close the gap — answering through 6 of the 7 probe-decidable refusals and all 3 policy-on-file questions, correct on 4 of the 5 registry-decidable (§7.3).

A running example. Figure 1, from the public `card_games` database, recurs through §2–§3 and is scored in §7: `ruling_intensity` relates a month's card rulings to the same month's newly released printings. In February 2017 the numerator leg is populated, the same-window denominator empty, the anchor's coverage still contains February: no point-in-time correct value exists, and the governed response is a typed refusal carrying a re-executable witness. Six of the eight LLM arms answer it anyway — the arm holding the entire governance layer among them; the two refusals cannot resolve the metric's definition; the all-history base reports ≈ 3.8% (Figure 1, bottom row).

Why this is hard. Three obstacles organise the technical part. *(1) What is the correct answer at a declared time point?* The semantics must span the valid time of facts and the version axis of the governance graph, and be *total*: refusal is a value it assigns, not an exception raised. *(2) Is refusal the last word?* A coarser, masked or narrowed query is often faithful and legal, so choosing among rewrites needs a fidelity order and a guarantee that refusal follows an empty legal set, not a search that gave up. *(3) Why believe the compiler?* A component that both answers and refuses is a single point of trust, so the decision must travel with an artifact a third party re-executes without importing the compiler.

This paper. We present AsOfGov, which treats as-of correctness as a *compile-time binding* problem: a question is bound to a point on the ⟨valid-time anchor, governance-graph version⟩ plane, and binding either succeeds or fails in one of three decidable ways.

- **A bi-temporal governed query semantics** (§2): graph versions, valid-time anchors with declared coverage modes, and a total point-in-time semantics whose failure values $\bot_{\mathsf{OOV}}$, $\bot_{\mathsf{AM}}$, $\bot_{\mathsf{MC}}$ are typed refusals; its degeneration theorem exhibits single-version read consistency (defined in §2) as the one-dimensional *shape* of the two-axis binding, at construction strength.

- **Binding compilation and its decidability** (§3–§4): a coverage theorem partitioning well-formed single-period queries into bindable and three disjoint unbindable classes, each decidable at a stated cost, the guard order itself part of the semantics. Under a disclosure policy it extends to a bounded search, MLR-COMPILE, proved legal, maximal under deterministic selection, complete on refusal.

- **Point-in-time certificates and independent verification** (§5): certificate contents — anchor assignment, version pin, routing path, disclosure decision, witnesses, probe records — and a checking relation a verifier decides from the governance graph, the data and the request context, with checking completeness, refusal-reason soundness under the frozen guard order, answer soundness on a template query class, and per-field non-redundancy.

Evidence (§6–§7): a compiler and a verifier sharing no project-internal module, enforced by a CI import assertion; nine arms on 60 questions over nine clusters, bootstrap intervals and coverage beside every rate. The compiler matches the frozen gold on all 60 (33 answers, 12 rewrites, 15 refusals with reason class), a *constructive upper bound*; the base exercises the disclosure layer (3 policy blocks, 5 roll-ups, 2 masks) and the version axis (two committed versions per database, four value-flipping pairs, one certified off-diagonal pin). The verifier accepts 60/60 certificates while re-deriving 50 of the 60 window coordinates from registered as-of conventions — run strictly, it fails closed on the 10 the question alone states — and rejects 34/34 forgeries. Verifying a decision costs a median 17.7× the answering query in time and 10.2× in bytes (§7.7). The first cross-check of the two implementations rejected 25 of 51 certificates over 15 root causes (§7.6).

The governed store. Everything below is stated over one object: a *versioned governance graph* G_v on an ordinary warehouse instance D, registering metrics, caliber routes, valid-time anchors, temporal bindings and disclosure policies on an append-only version axis (§2); the nine public databases ship these tables (847 registry rows), authored by us under three pre-registered non-degeneracy gates (§7.1). Full proofs, omitted constructions and per-check tables are in the extended version [33], shipped with the artifact.

2 BI-TEMPORAL GOVERNED QUERY SEMANTICS

Takeaway. *As-of correctness is a property of a two-axis binding — the valid-time anchor axis in the data, the version axis of the governance layer — and a structured refusal is a first-class value of it, not an error path.* Proofs are in [33]; Table 7 collects the notation of §2–§5.

Example 2.1 (Running example: CARD_GAMES)**.** The governed `card_games` warehouse registers `ruling_intensity` as a rulings leg over a new-printings leg, the denominator attributed along the caliber route `cards⋈sets`, both legs bound by `same_valid_time_window` over the day-granular anchors of Figure 1. Four requests against that one registration carry the paper, worked out with every mass in Figure 2.

Anchors, versions, queries, and the answer semantics. Time and windows. $\mathbb{T}$ is a discrete, effectively presented time domain with decidable order. A finite granularity family $\mathbb{G}$ (day...year) is ordered by refinement $\sqsubseteq$ and *assumed* a meet-semilattice, $g \sqcap g'$ being the commonly

Table 2: Notation of §2–§5, in order of definition.

$\mathbb{T}$; $\mathbb{G}, \sqsubseteq, \sqcap$; $\mathcal{W}, \mathcal{W}_g, \top$	time; granularities (refine, meet); windows
$a = (o_a, \mathrm{ty}_a, \mathrm{vt}_a, g_a, \mathrm{cm}_a)$	anchor: object, type, vt_a, granule, mode
$V_v(a)$; $\mathrm{Cov}_v(a)$	marking set; coverage domain (hull/strict)
$G_v = (N_v, E_v, \lambda_v; M_v, R_v, A_v, \beta_v)$	graph at v: nodes, edges, labels; registries
M_v; a_m	metric set (atomic/ratio); atomic-metric anchor
$R_v(m)=\rho=(\kappa, o_{\mathrm{via}}, K_\rho, \mathrm{align}_\rho)$	caliber route: key, via-object, joins, alignment
$\beta_v(m) = (a^*_{\mathrm{n}}, a^*_{\mathrm{d}}, \mathrm{rule})$	temporal binding: leg anchors, rule
$\mathcal{G} = (\{G_v\}_{v \in \mathcal{V}}, \preceq, t)$; $\mathrm{ver}(T)$	version sequence, commits; in-effect version
$q = (\sigma, T)$; $\sigma = (m, s, \vec{\omega}, \bar{a})$	query: intent (metric, scope, $\vec{\omega}$, $\bar{a}$), instant
$R(q)$; ω_r; $\uparrow$; RefA_v	roles; window map; unresolvable; absent names
$\alpha_{q,v}$; P_{rule}; svw; g^{cmp}	candidate asgn.; rule pred.; ref. rule; granule
$\mathrm{Pop}^r_{v,T}(q)$; $\mu^r_{v,T}(q)$; $\models_\rho$	leg population; leg mass; scope match along ρ
$\mathsf{Bindable}$, (b1)–(b4); $[\![q]\!]_{\mathcal{G},T}$; $\mathbb{Q}$	bindability, its guards; semantics; value domain
$\perp_{\mathsf{OOV}}, \perp_{\mathsf{AM}}, \perp_{\mathsf{MC}}$; DB	typed refusals; disclosure-blocked class
$[\![\sigma]\!]^1_{G_v,D}$; SVRC; (A1)–(A6); $n, \lvert W \rvert, \lvert [\![o]\!] \rvert$	1-axis semantics; consistency; assumptions; costs
$\mathcal{D}_v$; $L(q, T, \mathcal{G}, \mathrm{ctx}, D)$; $X(q, T)$	policies; legal set; candidate space
$b = \langle \alpha, w, v, g \rangle$; $\mathrm{cl}(b, \mathrm{ctx})$; $\preceq_D, \approx_D$	binding point; its closure; fidelity order, kernel
$\prec_{\mathrm{det}}$; CT; $U_{\min}$	deterministic key; reason table; legal grains
$B = \langle \alpha, \nu, \rho, \delta \rangle$; $C = (q, T, B, \mathrm{out})$	binding quadruple; certificate
ν; $\delta = (\mathrm{dec}, \Pi, h_{\mathrm{ctx}}, g, \mu^*)$; w	version pin; disclosure decision; refusal witness
Chk; $\mathsf{ACCEPT}/\mathsf{REJECT}$; $\mathcal{Q}_{\mathrm{tmpl}}$	checking relation; verdicts; template class

expressible granularity — an assumption, not a fact: weeks and months refine each other in neither direction. The window algebra $\mathcal{W}$ holds half-open intervals and finite date sets under intersection and containment; $\mathcal{W}_g$ collects g-aligned unions plus $\top$. Request windows are *capped* to $\bigcup_g \mathcal{W}_g$, so decidability in §3 is relative to that cap; lifting it to arbitrary definable windows is open.

Valid-time anchors. Under version v, an anchor is $a = (o_a, \mathrm{ty}_a, \mathrm{vt}_a, g_a, \mathrm{cm}_a)$: a semantic object, an anchor type, a valid-time assignment $\mathrm{vt}_a : [\![o_a]\!]_v \to \mathcal{W}$, a marking granularity $g_a \in \mathbb{G} \cup \{\mathsf{itv}\}$, and a coverage mode $\mathrm{cm}_a \in \{\mathrm{hull}, \mathsf{strict\_member}\}$. A *snapshot anchor* carries an effective date column, $\mathrm{vt}_a(r) = \mathrm{gr}_{g_a}(r.\mathrm{eff})$; an *interval anchor* ($g_a = \mathsf{itv}$) carries $(\mathrm{vf}, \mathrm{vtc})$ and sets $\mathrm{vt}_a(r) = [\,r.\mathrm{vf}, r.\mathrm{vtc})$ under point-window semantics. The *marking set* is $V_v(a) = \bigcup_{r \in [\![o_a]\!]_v} \mathrm{vt}_a(r)$; the *coverage domain* $\mathrm{Cov}_v(a)$ is $\mathrm{hull}(V_v(a))$ under hull and $V_v(a)$ under $\mathsf{strict\_member}$. Codomain $\mathcal{W}$ admits $\emptyset$: zero-length dimension rows ($\mathrm{vf} = \mathrm{vtc}$) meet every window emptily. Coverage mode is *per-anchor and versioned*, not global, and every refusal witness replays it under the declared mode.

Graph versions and the version axis. $G_v = (N_v, E_v, \lambda_v; M_v, R_v, A_v, \beta_v)$ (Table 7): layer-labelled nodes, typed edges including `numerator_of`$\langle m \rangle$ and `denominator_of`$\langle m \rangle$; a metric set M_v whose members are *atomic* (o_m, f_m) or *ratio* $((o^{\mathrm{n}}_m, f^{\mathrm{n}}_m), (o^{\mathrm{d}}_m, f^{\mathrm{d}}_m))$; a partial routing $R_v(m) = \rho = (\kappa, o_{\mathrm{via}}, K_\rho, \mathrm{align}_\rho)$ fixing the *scoped* denominator population; an anchor set A_v, carrying the atomic-metric anchor assignment $m \mapsto a_m$; and a partial *temporal binding* $\beta_v(m) = (a^*_{\mathrm{n}}, a^*_{\mathrm{d}}, \mathrm{rule})$, with "$m$ ratio $\Rightarrow \beta_v(m)$ defined, each prescribed anchor on its own leg, and a *distinct*-anchor row declaring a realisation-sensitive clause-(iv) member" a well-formedness invariant — the second clause is what sends *mixed* coverage to AM(iv) and closes the forward-cited Theorem 3.1 of §3.
A *version sequence* $\mathcal{G} = (\{G_v\}_{v \in \mathcal{V}}, \preceq, t)$ has $\mathcal{V}$ finite non-empty under a total commit order and a computable strictly monotone commit map $t : \mathcal{V} \to \mathbb{T}$; writing $t_i = t(v_i)$, the *in-effect version* is $\mathrm{ver}(T) = v_i$ for $T \in [t_i, t_{i+1})$ ($t_{n+1} {=} {+\infty}$), undefined for $T < t_1$. The axis is *append-only*: committed triples are never deleted or renumbered, a rollback being a re-commit under a fresh number.

Caliber — the route ρ fixing over which population a metric's legs are computed — *is part of the value, not of execution.* On the shipped `financial` warehouse the same (m, s, T) — the 1997 penalty-transaction rate — evaluates to 0.143% under the registered route (denominator: all transactions) and to 2.174% under a control route whose denominator is the penalised-account base itself, a 15.2× gap on the same anchors and window. Hence the bracket carries G_v as a subscript and ρ must be a certificate field.

Lemma 2.2 (version-axis stability)**.** *Let $\mathcal{G}'$ extend $\mathcal{G}$ by appending v_{n+1} with $t_{n+1} > t_n$. Then* $\mathrm{ver}_{\mathcal{G}'}(T) = \mathrm{ver}_{\mathcal{G}}(T)$ *and* $[\![q]\!]_{\mathcal{G}',T} = [\![q]\!]_{\mathcal{G},T}$ *for every* $T < t_{n+1}$ *and every* q *declared at* T*.*

Proof in [33]. Point-in-time answers, refusals included, are durable assertions about (T, v). *In our seeds*, four question pairs flip value under the version change alone (§7).

Definition 2.3 (bi-temporal query and candidate assignment)**.** A query is $q = (\sigma, T)$ with declared instant T and structured intent $\sigma = (m, s, \vec{\omega}, \bar{a})$.
(1) Intent. m is a metric reference and s a scope selector; $\vec{\omega}$ assigns each role $r \in R(q) \subseteq \{\mathrm{num}, \mathrm{den}, \mathrm{atom}\}$ a computable *window derivation* $\omega_r : \mathbb{T} \to \mathcal{W} \uplus \{\uparrow\}$ with an explicit unresolvable sentinel; $\bar{a}$ is a partial *anchor override* from roles to anchor *references by name*, not presumed to resolve in A_v.
(2) Candidate assignment. For $v = \mathrm{ver}(T)$ and $m \in M_v$, with RefA_v the override names failing to resolve in A_v, the candidate assignment is the *function* $\alpha_{q,v} : R(q) \rightharpoonup (A_v \uplus \mathrm{RefA}_v) \times \mathcal{W}$, $\alpha_{q,v}(r) = (\bar{a}(r) \text{ else prescribed},\ \omega_r(T))$, the prescribed anchors being the two legs of $\beta_v(m)$ (ratio) or a_m (atomic).
(3) Unresolved overrides. An override resolving into RefA_v keeps the role out of the OOV coverage disjunct and classifies the query $\perp_{\mathsf{AM}}$, witness "declared name absent from A_v", for ratio and atomic metrics alike (once $\neg\mathsf{OOV}$; Definition 2.8(2), a forward reference).

That $\alpha_{q,v}$ is a function — no search — is deliberate: the existential quantifier over assignments is §4's. The definitions sit on the *diagonal* $T_{\mathrm{vt}} = T_{\mathrm{ver}} = T$; our suite runs the two-parameter bracket (T_{ver} declaring, T_{vt} as-of), carried statement by statement in [33]; admissibility stays a disclosure question.

Definition 2.4 (binding rule; `same_valid_time_window`)**.** Each rule denotes a decidable $P_{\mathrm{rule}}(\alpha; G_v, D)$. The reference rule svw holds on $\alpha = \{\mathrm{num} \mapsto (a_{\mathrm{n}}, W_{\mathrm{n}}), \mathrm{den} \mapsto (a_{\mathrm{d}}, W_{\mathrm{d}})\}$ when all four clauses hold:
(i) anchor fidelity: $a_{\mathrm{n}} = a^*_{\mathrm{n}}, a_{\mathrm{d}} = a^*_{\mathrm{d}}$;
(ii) same window: $W_{\mathrm{n}} = W_{\mathrm{d}} =: W$;
(iii) window–granularity expressibility: $W \in \mathcal{W}_{g^{\mathrm{cmp}}}$ for the declared comparison granularity, by default $g_{a^*_{\mathrm{n}}} \sqcap g_{a^*_{\mathrm{d}}}$ over the legs whose g_a lies in $\mathbb{G}$ (an interval-anchored leg imposes no granule constraint; where neither leg has one, g^{cmp} must be declared);
(iv) anchor-pair admissibility: a member of a finite decidable check library fixed by the binding row's anchor pair — trivially true, or a symmetric-difference audit on the marking domains or on the window-restricted realisations.

Clause (iv) reaches into governance content: the check library is versioned data, not part of the abstraction (its teeth: Example 3.3).

Definition 2.5 (bindability)**.** With $v = \mathrm{ver}(T)$, $\rho = R_v(m)$ and $\alpha_{q,v}(r) = (a_r, W_r)$, the leg population and mass are $\mathrm{Pop}^r_{v,T}(q) = \{x \in [\![o^r_m]\!]_v : x \models_\rho s,\ \mathrm{vt}_{a_r}(x) \cap W_r \neq \emptyset\}$ and $\mu^r_{v,T}(q) = \sum_{x \in \mathrm{Pop}^r_{v,T}(q)} f^r_m(x)$, where $x \models_\rho s$ means x matches the scope after transfer along the route's join keys and attribution alignment. $\mathsf{Bindable}(q, \mathcal{G}, D)$ holds when:
(b1) *both axes in effect* — $\mathrm{ver}(T)$ defined, each $\omega_r(T)$ defined, and per role $a_r \in A_v$ with $W_r \cap \mathrm{Cov}_v(a_r) \neq \emptyset$;
(b2) *metric, route and binding registered* — $m \in M_v$, and for a ratio $R_v(m)$ defined and recomputable and $\beta_v(m)$ defined;
(b3) *rule satisfied* — $P_{\mathrm{rule}}(\alpha_{q,v}; G_v, D)$ whenever $\beta_v(m)$ is defined;
(b4) *positive denominator mass* — $\mu^{\mathrm{den}}_{v,T}(q) > 0$ for a ratio.

The legs are asymmetric: a zero numerator mass is a lawful 0 where a zero denominator mass destroys the ratio.

Definition 2.6 (point-in-time answer semantics)**.** The denotation is the *total* function $[\![q]\!]_{\mathcal{G},T}$: queries $\to \mathbb{Q} \cup \{\bot_{\mathsf{OOV}}, \bot_{\mathsf{AM}}, \bot_{\mathsf{MC}}\}$ given by $\mu^{\mathrm{num}}_{v,T}(q)/\mu^{\mathrm{den}}_{v,T}(q)$ if $\mathsf{Bindable}$ and m is a ratio, $\mu^{\mathrm{atom}}_{v,T}(q)$ if $\mathsf{Bindable}$ and m is atomic, and otherwise $\bot_c$ for the first failing guard c. Numerator and denominator are bound separately to the valid-time window of T through their own anchors, and jointly to the version $\mathrm{ver}(T)$ in effect at T: this is the two-dimensional binding. Delta and multi-period intents expand by role $\times$ period and resolve lexicographically (Corollary 3.4, a forward reference), keeping $[\![\cdot]\!]$ single-valued.

Proposition 2.7 (well-definedness)**.** *When* $\mathsf{Bindable}(q, \mathcal{G}, D)$ *holds,* $[\![q]\!]_{\mathcal{G},T}$ *is determined by* $(q, \mathcal{G}, D)$ *and independent of evaluation order.*

Proof in [33]. *Ratio-of-sums invariant:* each leg is aggregated over the window before dividing, since for $W = \biguplus_j W_j$ $\sum_j \mu^{\mathrm{n}}_j / \sum_j \mu^{\mathrm{d}}_j$ and the average of per-part ratios differ; the template's `sum(rate)` ban and the verifier's ratio-shape check are its two projections.

Definition 2.8 (the three refusal classes, as a guard chain)**.** Each predicate is defined under the negation of its predecessors, which makes them pairwise disjoint and keeps every object they mention nameable. **(1) OOV (out-of-validity)**: $\mathrm{ver}(T)$ undefined, or some $\omega_r(T) = \uparrow$, or coverage vacuity over the single-period roles — *pair-level* where $\beta_v(m)$'s rule pairs two legs (both realisations empty; mixed is Definition 2.4(iv)'s audit jurisdiction), per-role $W_r \cap \mathrm{Cov}_v(a_r) = \emptyset$ for an unpaired (atom) role — the reading frozen before any model call (deviation ledger `PORT_REPORT` §4(2), in the artifact). A disjunct naming a non-nameable object counts as false, so a missing object cannot masquerade as temporal failure. **(2) AM (anchor-mismatch), guarded by $\neg\mathsf{OOV}$**: $\beta_v(m)$ defined and $\neg P_{\mathrm{rule}}(\alpha_{q,v}; G_v, D)$, witnessed by whichever clause of Definition 2.4 fails; or some declared override resolves into RefA_v — witnessed by the name absent from A_v, the AM(i) form — for ratio and atomic metrics alike. **(3) MC (missing-caliber), guarded by $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$**: *(i)* $m \notin M_v$, or m a ratio with $R_v(m)$ undefined or not recomputable or $\beta_v(m)$ undefined; or *(ii)* m a ratio with $\mu^{\mathrm{den}}_{v,T}(q) \le 0$.

The OOV/MC(ii) boundary is the paper's most consequential adjudication: OOV carries *coverage vacuity* — the window missing the coverage domain altogether — whereas zero denominator mass *inside* that domain is a caliber failure (Figure 2 exhibits both on one scope); widening MC(ii) to $\mu_{\mathrm{den}} \le 0$ also covers refund-reversal populations — non-empty, non-positive mass — a regime the §4 guarantees exclude (Theorem 4.2). The scope selector s is a predicate, not a governed object — a scope matching nothing is lawful (0 atomic, MC(ii) ratio) — a membership guard is future work.

Degeneration: the one-dimensional shadow. Fix (G_v, D). The *single-version* semantics $[\![\sigma]\!]^1_{G_v,D}$ drops anchor–window filtering but retains the anchor-validity restriction $\mathrm{vt}_{a_r}(x) \neq \emptyset$; an evaluation satisfies *single-version read consistency* (SVRC) iff a single committed version v supplies every governance object it reads — metric record, route, anchors, binding row — and the one snapshot D every fact — a definition complete here (restated in [33]).

Theorem 2.9 (degeneration)**.** *Let $\mathcal{Q}^{\downarrow}$ be the queries satisfying **(H1)** $T \ge t_n$, hence* $\mathrm{ver}(T) = v_n$*; **(H2)** $\omega_r(T) = \top$ for every role; **(H3)** $\bar{a} = \emptyset$. Then for every $q \in \mathcal{Q}^{\downarrow}$:* (1) *the evaluation satisfies SVRC, witnessed by v_n;* (2) *whenever $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q}$ it equals $[\![\sigma]\!]^1_{G_{v_n},D}$;* (3) *each failure class collapses to a non-temporal residue —* AM *to a failure of clause (iv) alone,* OOV *to $V_{v_n}(a) = \emptyset$ for every prescribed anchor, and* MC *to a missing route or a non-positive domain-wide scoped denominator mass.*

Proof in [33]. Every $\mathsf{Bindable}$ evaluation satisfies SVRC, witnessed by $\mathrm{ver}(T)$; under (H1)–(H3) the temporal content collapses entirely to it. *Honest strength:* under the snapshot assumption clause (1) holds *by construction*; re-proving a concurrency guarantee under version advance needs a per-datum read-version function a single-snapshot D cannot express — the shape of a substrate guarantee at construction strength, not its subsumption.

3 BINDING COMPILATION

Standing assumptions for this section and the next: **(A1)** each extension $[\![o]\!]_v \subseteq D$ is finite and enumerable with bounded per-fact predicate cost; **(A2)** valid-time assignments and window operations are computable and $\mathbb{G}$ is a finite meet-semilattice with day as common refinement; **(A3)** $\mathcal{V}$ is a non-empty finite total order, the commit map is computable and strictly monotone, and the version axis is append-only; **(A4)** every rule in the library denotes a decidable predicate; **(A5)** each window derivation is a total function into $\mathcal{W} \uplus \{\uparrow\}$, so "$\omega_r(T)$ unresolvable" is one evaluation and not a halting question; **(A6)** $\mathbb{T}$ is discrete, effectively presented, with decidable order — continuous time domains, where endpoint comparison is only semi-decidable, are out of scope.

Theorem 3.1 (coverage partition, single period)**.** *For every well-formed single-period query q — σ syntactically well formed, $T \in \mathbb{T}$, and $R(q) \subseteq \{\mathrm{num}, \mathrm{den}\}$ or $R(q) = \{\mathrm{atom}\}$ — under* (A1)–(A6) *the four cases **(a)** $\mathsf{Bindable}$, **(b)** OOV, **(c)** $\neg\mathsf{OOV} \wedge \mathsf{AM}$, **(d)** $\neg\mathsf{OOV} \wedge \neg\mathsf{AM} \wedge \mathsf{MC}$ are pairwise disjoint and jointly exhaustive, and $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q} \iff$ (a), $= \bot_{\mathsf{OOV}} \iff$ (b), $= \bot_{\mathsf{AM}} \iff$ (c), $= \bot_{\mathsf{MC}} \iff$ (d).*

Proof in [33]: disjointness follows from the guard chain and obligations (b1)–(b4) of Definition 2.5; exhaustiveness uses Proposition 3.2, whose proof does not use this theorem, to make each guard two-valued on its domain, and the vacuity convention of Definition 2.8 to keep a missing object out of the temporal door. Figure 2 walks one request into each of the four outcomes on a single store.

Decidability, cost, multiple periods, and guard order.

Proposition 3.2 (decidability and cost)**.** *Under* (A1)–(A6)*, OOV, and AM and MC on their guard domains, are decidable by constructive procedures at the following costs ($n = |\mathcal{V}|$, $|W|$ the window description length, $|[\![o]\!]|$ extension cardinality). OOV**:** locate $\mathrm{ver}(T)$ by binary search over $t_1 < \cdots < t_n$; evaluate $\omega_r(T)$ through its total presentation, sentinel included; compute coverage by the declared mode — one scan taking $\min/\max$ for hull, a sort-and-merge into an interval union for $\mathsf{strict\_member}$, zero-length rows skipped — and intersect with the normal form of W_r; cost $O(\log n) + \sum_r O(|[\![o_{a_r}]\!]| \log |[\![o_{a_r}]\!]| + |W_r|)$. AM**:** clause (i) is an identifier comparison, (ii) a normal-form equality, (iii) a granule-boundary alignment test on each maximal interval of W, and (iv) a call into the declared check library — constant for the trivial variant, one scan over both anchor objects for the symmetric-difference audit, windowed for the realisation variant; cost $O(|W| \log |W|) + O(|[\![o_{a^*_{\mathrm{n}}}]\!]| + |[\![o_{a^*_{\mathrm{d}}}]\!]|)$. MC**:** mode (i) is a key lookup in M_v, R_v, mode (ii) one scan over $[\![o^{\mathrm{d}}_m]\!]_v$ accumulating f^{d}_m; cost $O(|[\![o^{\mathrm{d}}_m]\!]|)$.*

Proof in [33]; (A5) is where it would break: from partial computability alone, deciding $\omega_r(T) = \uparrow$ is a halting question. Each procedure also emits, as a by-product, the failing clause or the probe with its asserted result — the witness content §5 requires: decision procedure and witness generator are one object seen twice.

Example 3.3 (clause (iv) with teeth)**.** `EF2-Q6` asks the rating-weighted daily win rate of `european_football_2` as of 2013-12-01: the numerator leg binds to the match-day anchor under hull coverage, the denominator leg to the rating-snapshot anchor, a date-set anchor under strict_member whose weekly marking dates surround the day (2013-11-29, 2013-12-06) without containing it. The discriminant is clause (iv), the window-realisation audit this pair prescribes: on the single-day window the match anchor realises 33 rows, the snapshot anchor none, differing on exactly one date; the certificate carries 2013-12-01 with both counts, recounted by the verifier from D. A check on the rule *literal* alone would accept (§4): the same seed binds `win_rate` under that literal with both legs on one anchor.

Corollary 3.4 (per-period classification)**.** *A delta or multi-period query expands by role × period into a finite family* $\{q^{(j)}\}_j$ *of single-period queries in the declared period order. Theorem 3.1 applies to each* $q^{(j)}$*, and the value of the whole query is lexicographic: if every* $q^{(j)}$ *falls in case (a),* $[\![q]\!]_{\mathcal{G},T}$ *is the declared arithmetic combination of the per-period values; otherwise it is* $[\![q^{(j_0)}]\!]_{\mathcal{G},T}$ *for the first failing period* j_0*. Two side conditions make this well posed: guard predicates and* P_{svw} *are evaluated on the single-period restriction* $\alpha_{q,v}|_{R(q^{(j)})}$*, and the existential quantifier over roles inside* OOV *does not range across periods.*

The second condition is forced by an instance: the delta over $[\,2017\text{-}02,\ 2021\text{-}06\,]$ on Figure 2's store (b,d) has periods $\perp_{\mathsf{MC}}$ and OOV; a cross-period quantifier would report $\perp_{\mathsf{OOV}}$, the lexicographic rule $\perp_{\mathsf{MC}}$.

Guard order is not cosmetic. The three raw conditions can hold at once; the chain resolves the class to the first. Example 3.3 is the suite's live collision: on the single-day window the snapshot leg's strict_member coverage is empty, so a leg-at-a-time OOV reading fires before the audit and returns $\perp_{\mathsf{OOV}}$ — same witness replay, wrong reason class. Definition 2.8(1)'s pair-level clause rules that out — the realisation is mixed — and gold is $\perp_{\mathsf{AM}}$(iv). Adjudications of this kind, left unregistered, are what §7.6 measures. We report no permutation ablation over the six guard orders.

4 DISCLOSURE-CONSTRAINED REWRITING

Under a disclosure policy, binding compilation extends from a decision to a bounded search over binding points, with refusal a proof that the legal set is empty rather than an abstention. The search runs *after* the guard order OOV → AM → MC has been discharged on a slice and DB follows all three, so a legally trimmed answer never becomes an answer to a different question at a different time point.

The policy model, MLR, and the compiler. G_v declares a finite grain lattice whose roll-up edges are registered and never inferred from data, a finite mask lattice, and a finite versioned set $\mathcal{D}_v$ of *disclosure policies*, each pairing a protected attribute set with a role branch, a minimum mask, a grain floor and a small-cell threshold; a candidate is *admissible* when every policy whose protected closure it touches is discharged by one of those branches. The *legal set* $L(q,T,\mathcal{G},\mathrm{ctx},D)$ collects the closures of the *binding points* $b=\langle\alpha,w,v,g\rangle$ of the *structural* candidate space $X(q,T)$ of [33] — one window candidate per slice, no data-driven shrinking — faithful to the question's rigid commitments, bindable (Definition 2.5) and admissible; one masking a rigidly requested attribute away entirely is refused *disclosure-blocked* (DB), not answered with an empty shell. **MLR** (*maximal legal rewrite*) asks for either ANSWER from a $\preceq_D$-maximal element of $L/\approx_D$ under the *fidelity order* $\preceq_D$ — which compares only same-anchor, same-version points, $b\preceq_D b'$ when b' trims no more of the window at no coarser a grain, $\approx_D$ its symmetric kernel — or REFUSE with a per-slice reason table CT over $\{\mathsf{MC},\mathsf{AM},\mathsf{OOV},\mathsf{DB}\}$ evidencing $L=\emptyset$; MLR-COMPILE solves it slice by slice under a declared total order; the algorithm, the lattices' coordinates, the frontier and pruning lemmas and a polynomial probe bound are in [33]. One guard must be named because both statements below rest on it: the rule test evaluates the *full* predicate of Definition 2.4, clauses (i)–(iv), not the rule literal — the binding row of Example 3.3 carries the same `same_valid_time_window` literal as its trivially compatible siblings, and a literal-only check answers the request its audit refuses. *Scope:* four of the nine domains register policies (18 instance rows), the other five carrying the `ungoverned-disclosure` annotation of §5.1. The suite binds the layer our development seeds only formalised: 3 questions refuse DISCLOSURE-BLOCKED on a replayed small-cell probe, 5 answer only after a granularity roll-up — one under a threshold the version change raises — and 2 only under a mask closure (§7.4); 10 of the 34 forgeries attack this layer.

Theorem 4.1 (soundness, maximality, determinism)**.** *Assume* (A1)–(A6) *of §3 together with: the declared lattices are finite and the grain lattice's roll-up edges are registered; $\mathcal{D}_v$ is finite and every policy is coarsening-compatible; $\prec_{\text{det}}$ is a total key over versioned components of G_v; and* (A-emit)*: the code emitter is point-in-time correct on bindable closures — the syntax-to-semantics bridge assumed as A4.1 in [33] and recorded open there (gap G2), an obligation of the same kind as Theorem 5.2(b)'s declared scope.* (a) *If* MLR-COMPILE *returns* ANSWER *then* $\text{cl}(\hat{b}, \text{ctx}) \in L(q, T, \mathcal{G}, \text{ctx}, D)$ *and the emitted plan is point-in-time correct in the sense of Definition 2.6.* (b) *Every element of the frontier is $\preceq_D$-maximal in $L/\approx_D$, and the returned $\hat{b}$ is one of them.* (c) *The returned representative is a function of $\langle q, T, \mathcal{G}, \mathcal{D}, \text{ctx}, D\rangle$, since all three keys of $\prec_{\text{det}}$ are versioned components of G_v; re-declaring $\prec_{\text{det}}$ changes the representative, not the maximal set and not the* ANSWER/REFUSE *verdict.*

Theorem 4.2 (refusal completeness)**.** *Under the same assumptions, and with the denominator measure non-negative,* MLR-COMPILE *returns* REFUSE *iff $L(q, T, \mathcal{G}, \text{ctx}, D) = \emptyset$. On refusal,* CT *carries a per-slice emptiness witness whose reasons decompose into the three binding-layer classes of §2 plus the disclosure-layer class* DB*; when $\mathcal{D}_v = \emptyset$,* DB *is unreachable and the statement reduces to binding completeness.*

Both proofs are in [33]; refusal completeness is precisely where the full rule predicate is required. **Boundaries.** Three stay open: window bargaining needs a joint window–grain maximality theory (both theorems above are relative to X); we return one maximal element per slice frontier and do not bound the cost of enumerating all of them; and the rule language is confined to window-equality rules, so lag bindings need an extended predicate. Theorem 4.2's stated premise, like the probe bound, assumes a non-negative denominator measure — with refund-reversal columns the mass is not monotone along window containment and the pruning argument fails.

5 POINT-IN-TIME CERTIFICATES

Compilation and verification read the same (G_v, D) snapshot, and two commit axioms — typed commit identifiers unique and fresh, version numbers never reused — keep a certificate valid *at its coordinate* (T, v).

5.1 What a Certificate Contains

Binding quadruple and certificate. $B = \langle \alpha, \nu, \rho, \delta\rangle$ where *(1)* $\alpha : R(q) \rightharpoonup A(G_v) \times \mathcal{W}$ is the *anchor assignment*, sending each metric role to the anchor and window resolved for it (delta intents expand by role × period); *(2)* $\nu = (\text{domain}, v, c\,[, \text{offdiag}])$ is the *graph version pin*, c a true typed-commit identifier and $\text{offdiag} = (p, \text{ver}(T))$ the marker a certificate must carry whenever the pinned version departs from the one in effect at T — its absence commits to a diagonal read; *(3)* $\rho = \langle e_1, \ldots, e_k\rangle$ is the *caliber routing path* ($\langle\rangle$ for atomic metrics); *(4)* $\delta = (\text{dec}, \Pi, h_{\text{ctx}}, g, \mu^*)$ is the *disclosure decision*, with $\mathrm{dec} \in \{\mathsf{ANSWER}, \mathsf{REWRITE}(q'), \mathsf{REFUSE}(r)\}$, the set Π of policies in force, a hash of the request context, the delivered grain and the least mask closure; a policy-free domain sets $\Pi = \emptyset$ and must carry an `ungoverned-disclosure` annotation. A *certificate* is $C = (q, T, B, \mathrm{out})$: for $\mathrm{dec} \in \{\mathsf{ANSWER}, \mathsf{REWRITE}(q')\}$, out is the emitted query text, a $\mathsf{REWRITE}$ additionally carrying a *cut trace* — retained and trimmed slices with their reason table CT, and the declared keys of $\prec_{\mathrm{det}}$; for $\mathrm{dec} = \mathsf{REFUSE}(r)$, $r \in \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}, \mathsf{DB}\}$, out is a witness w. On a multi-slice question (r, w) are taken *slice-major*: the earliest failing slice under $\prec_{\mathrm{det}}$, then the guard order $\mathsf{OOV} \to \mathsf{AM} \to \mathsf{MC}$ — pinned, since with one slice failing MC and another AM the two readings report different classes.

Refusal witnesses. Each refusal class carries a witness shaped to replay as one governance-table lookup or one single-scan probe with an assertion about its result — refusals are *falsifiable*: issued for a reason that does not hold, no witness survives replay. MC carries a routing lookup (m, κ) showing no recomputable route under the caliber key, or a denominator probe (m, p, W) whose observed value is recorded verbatim and lies in the widened empty set (empty, null, $\mu_{\mathrm{den}} \leq 0$); AM the failing clause $c \in \{\text{i–iv}\}$ of Definition 2.4 with its payload — an anchor pair, an offending window pair, a misaligned boundary point replayed against $\mathcal{W}_{g^{\mathrm{cmp}}}$, or a discriminant date in the symmetric difference of the marking sets; OOV a triple (a, W_T, π_V) whose coverage profile asserts $W_T \cap \mathrm{Cov}_v(a) = \emptyset$ under the declared mode, at the level Definition 2.8(1) adjudicates — both legs where $\beta_v(m)$ pairs them; and DB the blocked slices, triggering policies and support-probe transcript with its snapshot version, jointly asserting an empty legal-grain frontier, $U_{\min} = \emptyset$. Replay *modes* are fixed by machine-readable declaration, never by free text. Well-formedness adds version closure, role coverage, routing continuity and decision–output matching, plus three widened variants so that even a metric with no assignable anchor, an unresolvable as-of or an undefined $\mathrm{ver}(T)$ admits a certificate.

The independent verifier. $\mathsf{Chk}(C, G_v, D, \mathrm{ctx})$ is the conjunction of the ten checks below, with G_v resolved through $C.\nu$ and ctx matched against $\delta.h_{\mathrm{ctx}}$; a *verifier* is any procedure deciding Chk from those four arguments alone whose implementation shares no project-internal module, template or code path with the compiler; no coordinate the certificate asserts may be inherited. **V0** therefore re-checks the anchor assignment role by role through an independently written window arithmetic *and* re-derives each role window from the registered as-of convention over (G_v, D), exempting only the machine-readably declared deviations it enumerates — an override or a cut-trace-backed $\mathsf{REWRITE}$, each still replayed by V3/V6c (§7.5). Then: **V1** the version pin, its unique typed commit and its diagonality; **V2** anchor existence and window typing; **V3** the binding looked up *by identifier* — (domain, metric) is not a functional key — with its rule re-executed on α's windows and matched three-valued; **V4** routing-edge existence, join keys and path continuity; **V5** the policies that should be in force, with grain, mask, role-branch and small-cell clauses replayed and the `ungoverned-disclosure` annotation checked, since absence is not passing; **V6a** that the emitted text touches only certified objects, that every time predicate *denotes* a subset of the certified window, and that the ratio shape aggregates each leg before dividing; **V6b(0)** the negation of every guard preceding r in the frozen order — an MC refusal must replay $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$; **V6b(1)** the witness under its declared replay modes, scoped to the first failing slice, other CT rows travelling as annotations; and **V6c**, for every answer and each retained slice of a rewrite, the negation of *all* refusal guards. Per-check argument and answer/refusal tables: [33].

V6b(0) and V6c are the checks a definition-body verifier lacks: without V6b(0) a refusal certificate replays correctly with the wrong class (§3); without V6c an "answered where the semantics denotes $\perp$" certificate passes every remaining check — the dominant baseline failure mode turned into a certificate. *Non-circularity:* the trusted base is the declarative Chk specification with G_v and D (§6), clause-(iii) and clause-(iv) replay variants included.

5.2 Guarantees, and What They Do Not Cover

Theorem 5.1 (verification completeness)**.** *Under the snapshot and commit assumptions, (A1)–(A6), and a compiler each of whose certificate fields is derived from facts or probe results over* (G_v, D, ctx) *using the frozen guard order and the replay modes* G_v *determines: every well-formed certificate the compiler emits, degenerate variants included, satisfies* $\mathsf{Chk}(C, G_v, D, \mathrm{ctx}) = \mathsf{ACCEPT}$*.*

Theorem 5.2 (verification soundness)**.** *Suppose* $\mathsf{Chk}(C, G_v, D, \mathrm{ctx}) = \mathsf{ACCEPT}$*.* (a) *If* C *is a refusal certificate, its reason holds at the coordinate* (T, v, D) *under the guard semantics of §2 for* $r \in \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}\}$ *and under the disclosure semantics of §4 for* $r = \mathsf{DB}$*.* (b) *If* C *is an answer certificate whose emitted text belongs to* $\mathcal{Q}_{\mathrm{tmpl}}(C)$ *(Definition 5.3 below) then, under* (A-tmpl)*, that text is point-in-time correct at* (T, v)*; for a rewrite, correctness is relative to the trimmed request.*

Proofs in [33]: (a) is a case analysis on r over V6b(0) plus V6b(1), turning on the fact that replaying r's definition body alone establishes only the weaker conjunct of Theorem 3.1; (b) layers $\mathsf{Bindable}$ at the coordinate, from V0 and V6c, over a leg-population agreement forced by V2–V4 with V6a on $\mathcal{Q}_{\mathrm{tmpl}}$.

Definition 5.3 (template class $\mathcal{Q}_{\mathrm{tmpl}}$)**.** Relative to an answer certificate C: $Q \to L \mid (L)/(L)$, the ratio form iff B certifies both legs; $L \to$ `SELECT` $f(e)$ `FROM` J `WHERE` P, with f one unnested aggregate, J exactly the leg's certified anchor and routing-path tables on the registered keys, e and P over certified columns only, and P carrying one time predicate denoting a subset of the leg's certified window. Membership is decidable — each clause a parse-tree test — and V6a's containment check decides exactly these clauses (object set, window denotation, leg-before-divide shape).

Declared scope of Theorem 5.2(b). Semantic equivalence is undecidable for arbitrary SQL with arithmetic and aggregation, so (b) is asserted on Definition 5.3's class only and rests on *(A-tmpl)*: on $\mathcal{Q}_{\mathrm{tmpl}}$, V6a's syntactic containment implies that the emitted text denotes the certified plan — the "syntax implies semantics" lemma, still open in [33] and an obligation of the same kind as Theorem 4.1's *(A-emit)*; a delta template subtracting two rates already sits outside the class. Rewrite maximality is a declared *non-goal*: re-checking it would mean re-implementing §4's search inside a verifier forbidden its code, so Chk verifies only that q' is legal and its cut trace consistent. The statement is written on the diagonal; the suite's one off-diagonal certificate (§7.4) reads under the two-parameter bracket of [33].

Proposition 5.4 (field non-redundancy)**.** *For each* $x \in \{\nu, \alpha, \rho, \delta, w\}$*, let* Chk_{-x} *weaken* Chk *by dropping the checks that consume* x*. Then there is an instance that* Chk_{-x} *accepts although point-in-time correctness fails.*

The five constructions are the proof [33]; §7.5 exhibits the deletions empirically, the version-swap on both committed versions. The proposition is per-field non-redundancy under Chk, not minimality.

6 SYSTEM

AsOfGov is two programs deliberately kept apart: a *binding compiler* (≈3.8k lines of Python: a shared core plus one adapter per evidence base) turning a question and a declared time point into a decision plus a certificate, and an *independent verifier* (≈5.1k lines) deciding the checking relation of §5 without running, importing or consulting the compiler — that separation is the system contribution.

Compiler. Given (σ, T) it pins ver(T) with its typed commit, reads G_v at that version, derives α, evaluates the guards in the frozen order $\mathsf{OOV} \to \mathsf{AM} \to \mathsf{MC}$ leaving each probe transcript in the certificate, applies the disclosure gate, and emits SQL over the certified objects, a narrowed rewrite with its cut trace, or a typed refusal with a witness (walk-through in [33]). *Registration over hard-coding*: a metric's anchor, caliber route and coverage-domain reading must be readable from G_v, not a dictionary inside the compiler (§7.6 measures the cost of violating it). *No post-hoc narration*: the certificate is emitted field by field from the decision that produced the plan.

Verifier and the anti-certification-loop rule. The verifier takes the certificate envelope, the structured question, the governance tables at the pinned version, the data and the request context, and returns accept or reject with the failing check. The question is a shared input to both sides — what defeats a self-signed override — but read for its *intent* fields only: metric, the two declared instants, scope, and, on the 10 questions of §7.5, the presented window. Every other window coordinate is *re-derived* from the registered as-of convention rather than consumed. What the verifier never takes is compiler internals: no templates, no shared module, no call into the compilation entry point; every fact the compiler looked up is *re-queried* from (G_v, D), so certificate fields are assertions awaiting verification. Even neutral-looking helpers are re-implemented: a shared wrong month expansion cancels.

The rule is enforced mechanically. A CI check parses every verifier file and asserts that (a) each import resolves to the Python standard library, the database driver, or a module inside the verifier package; (b) no verifier file imports any compiler location, including the pilot's per-domain compilers; (c) the two sides' project-internal import roots intersect in the empty set; and (d) no dynamic-import or path-manipulation escape hatch points at a compiler directory. On the released artifact the intersection is empty. Under this discipline "the compiler is buggy but the certificate verifies" requires the same defect in two independent implementations *and* in the declarative checking relation; the audit surface shrinks from a compiler to a one-page specification. A certificate travels as a JSON envelope: body (§5.1), probe transcripts, machine-readable deviation list.

7 EVALUATION

Takeaway. *The as-of gap is large (E1), survives one-line prompting (E2) and survives the entire governance layer in context, its refusal-side errors landing where a decision needs a data probe rather than a registry line (E3). The compiler closes it as a constructive upper bound with disclosure and version axes exercised (E4); the independent checker accepts all 60 certificates and rejects all 34 forgeries (E5); its first cross-check's ledger stays the most useful thing we measured (E6).*

7.1 Setup

Benchmark. 60 questions over *nine public databases* (eight BIRD [20], one Spider [35]; Table 3): 3,830,036 real rows — `financial.trans` alone holds 1,056,320, zero synthetic — plus 20,094 authored rows declared row-by-row (20,076 of them `world_1`'s synthesised 84-month history). Questions and instruction blocks are authored in Chinese over the databases' English schemas. Over each database we register a governance layer as data: 847 seed rows across ten registry tables, two committed graph versions per database, and 18 disclosure-policy instances in four domains, the other five declared ungoverned. Gold: 33 answers, 12 rewrites (5 roll-ups, 5 coverage trims, 2 masks) and 15 typed refusals covering all four classes and subtypes (Table 3), independently re-derived before freezing. *Non-degeneracy is designed and machine-checked*: registries carry no SQL fragments, snapshot pointers, coverage literals or verdict tokens (0 hits over 90 seed files;

Table 3: The suite: real rows as shipped (authored rows declared separately), questions, the compiler's answer/rewrite/refusal split, and each cluster's gold refusal classes.

Database (source)	rows	Qs	A/RW/RF	Refusal classes
`financial` (B)	1,079,680	8	5/1/2	OOV, AM(ii)
`card_games` (B)	803,445	7	4/1/2	MC(i), MC(ii)
`codebase_comm.` (B)	740,646	7	3/3/1	DB
`formula_1` (B)	514,287	8	5/1/2	OOV, MC(ii)
`debit_card_sp.` (B)	423,050	7	4/1/2	OOV, AM(iii)
`european_football` (B)	222,796	6	3/1/2	OOV, AM(iv)
`california_schools` (B)	29,941	6	4/1/1	AM(i)
`thrombosis_pred.` (B)	15,952	6	2/2/2	DB × 2
`world_1` (S)	239	5	3/1/1	AM(ii)
total (Σ real rows)	3,830,036	60	33/12/15	OOV4 AM5 MC3 DB3

ND-2); for eight refusal/rewrite questions a *witness pair* (ND-1) materialises two warehouses — byte-identical seeds, one edited data row — that flip the gold label, so those labels are underdetermined by everything a no-execution system is given; 57.1% of seed rows are distractors (ND-3); and a string-level audit keeps every gold SQL, window, refusal class, and every gold value of $\geq$8 characters out of every prompt (23 shorter gold literals fall under the audit's length guard — 22 numeric plus the two-character string `UK` — disclosed in the pre-registration).

Systems. Nine arms (Table 4): four *plain baselines*, frontier models of four families given the full schema (registry DDL and row counts, not rows), the question, and the right to refuse that gold exercises; three *prompt variants* on the reference backbone (`claude-opus-4-6`) — an as-of reminder, a registry-join instruction, a worked example (115, 226 and 187 characters); a *governance-informed* arm (§7.3); and the binding compiler, whose row transcribes the frozen certificate acceptance of §7.5 — zero model calls (criterion B$'$: compiler–gold agreement 60/60, displayed, not re-run).

Protocol, scoring, statistics. One statement per question, first response final: temperature unset (provider default), one sample per question, no execution feedback, no repair round, retry only on an empty completion (3 attempts, then scored as error), 512-token output cap. Scoring was frozen with the suite: numeric answers within 0.5% relative error of gold executed on the same warehouse, four report-shaped questions by row multiset, two by string; a refusal question is correct when the system refuses — *LLM arms owe no refusal class*, the compiler is charged with it. Errors are typed six ways (Figure 3); every rate below sits beside its answered/refused shares on the stated denominator (60 questions; 45 value+rewrite; 15 refusals). Questions come in nine database clusters, so we report cluster bootstrap 95% intervals (B=2000, frozen seed). Arms, prompt byte budgets, scoring dispatch, per-arm point predictions with 80% intervals and the paper-rewrite rule for each outcome were pre-registered and hash-frozen before the first call; every raw response is cached. *Deviations touching scored calls, in full:* `trivial_v2`'s first run returned 4 empty responses, over the pre-registered $\geq$3 threshold; the *whole arm* was voided and rerun once, both runs ship (run 1: 27 correct/33 errors; run 2, the scored one: 25/35 — same direction), and the pre-registration carries a dated post-hoc addendum, tamper evidence intact. A crashed `baseline_claude` session — the reference arm — was resumed under the pre-registered resume clause (append-only cache, zero re-queries; the 21 questions already written are the frozen enumeration prefix, not a selected subset). Five other empty responses (`baseline_claude` 1, `trivial_claude` 2, `trivial_v3` 2) count as errors for their arms; the governance-informed arm had none. Finally, the three non-Claude models ride a gateway a probe showed prepends ~4k tokens of third-party system prompt; the four Claude-backbone arms share one channel, so every contrast we lean on is channel-matched, cross-family readings carrying this caveat.

Table 4: Error over all 60 questions with cluster bootstrap 95% intervals over nine databases (B=2000); answered/refused shares; correct refusals of 15; elim, the share of the reference backbone's 36 errors the arm answers correctly (pre-registered C′ line 0.40); errors by gold form (v/33, rw/12, rf/15; split mandatory, §7.7). The compiler's 0% is a *constructive upper bound* (§7.4); its row is a deterministic acceptance transcript, not a sample, so it carries no interval. The pre-registered gov. arm (§7.3) is the strongest comparator. *Scoring is asymmetric in the baselines' favour*: ok credits an LLM any refusal, the compiler only when the refusal *class* matches gold; rewrites scored by value, no declaration asked.

	System	E/60	Err.	95% CI	elim	Ans.	Ref.	ok/15	v/33	rw/12	rf/15
plain	`claude-opus-4-6`	36	60.0%	[44.1, 74.6]	—	0.83	0.15	4	18	7	11
plain	`qwen3-coder-next`	43	71.7%	[61.0, 83.3]	0.111	0.82	0.18	4	23	9	11
plain	`deepseek-3.2`	44	73.3%	[56.7, 88.5]	0.083	0.58	0.42	6	27	8	9
plain	`minimax-m2.5`	46	76.7%	[62.7, 89.8]	0.056	0.97	0.03	2	25	8	13
prompt	v1 one-line reminder	32	53.3%	[40.7, 64.5]	0.222	0.82	0.15	6	16	7	9
prompt	v2 registry-join hint	35	58.3%	[48.3, 69.4]	0.167	0.93	0.07	3	15	8	12
prompt	v3 worked example	40	66.7%	[52.7, 80.3]	0.056	0.95	0.02	1	19	7	14
	full governance layer	28	46.7%	[28.6, 63.9]	0.361	0.90	0.10	5	11	7	10
	binding compiler	**0**	**0.0%**	—	1.000	0.75	0.25	15	0	0	0

7.2 E1–E2: The Gap, and That One-Line Prompting Does Not Close It

The four plain baselines err on 60.0%, 71.7%, 73.3% and 76.7% of the 60 Chinese questions (Table 4): every family clears the pre-registered floor (criterion A′, min ≥ 0.30; [33] PREREG §5) by 2×, and every cluster interval excludes zero while too wide to rank families. Figure 3 shows a two-sided shape: on the refusal side the baselines answer through 11, 11, 9 and 13 of the 15 refusals, refusing correctly on 4, 4, 6 and 2 — no class demanded; over the eight LLM arms the refusal classes recover as OOV 4 of 32 chances, DB 2 of 24, AM 15 of 40, MC 10 of 24. On the value side the errors are wrong values and failed executions, not missing SQL (`deepseek-3.2` instead over-refuses 19 of 45; `minimax-m2.5` fails to execute on 24), and they stack with resolution depth: `codebase_community` — the deepest chain (version-scoped aliases, a two-hop caliber route, carrying 8 of the 18 policy rows) — is among the hardest clusters for every arm, the baselines erring on $6, 6, 7, 7$ of 7.

Prompting does not close it. The three variants score 53.3%, 58.3% and 66.7% against the backbone's 60.0%: the best removes 8 of the reference's 36 errors (elim = 0.222), the worst removes 2 while adding 6, and all three sit under the pre-registered C′ line for *prompt* variants (max $= 0.222 < 0.40$). The refusal side barely moves (correct refusals $4 \to 6, 3, 1$; answering through $11 \to 9, 12, 14$); the instruction-heavy variants add wrong values and executions where the backbone was right.

7.3 E3: The Governance-Informed Arm

If the failure were an information gap, the governance layer would close it. One pre-registered arm on the reference backbone prepends, to the variant-2 instruction, its database's *entire* governance layer — all ten registries, all rows, both committed versions, serialised deterministically and packaged per *database* so no row can be selected using the question (16,176–28,279-character prompts; no truncation fired); packaging, rewrite rules and predictions were frozen before any call. It bounds what a governed metric layer's *content* [28] buys in context, not that line's deterministic resolution path.

Result. The arm errs on 46.7% (28/60), the best LLM arm. It eliminates 13 of the reference's 36 errors and introduces 5 new ones (net +8); against `trivial_v2`, which isolates the instruction, 11 fixed against 4 broken — content, not instruction, carries the gain, instruction alone moving one question (36 → 35), inside the ±4 spread of our variants (32, 35, 40), against content's 35 → 28. The pre-registered three-case rule puts the arm in case *(iii)*: $\mathrm{elim}_{\mathrm{gov}} = 13/36 = 0.361 < 0.40$, *no substantial elimination* (two short of the line; the variant-2 instruction it is anchored on is not the best-scoring of the three, 35 errors against 32, and that two-question margin sits inside the ±4 spread just quoted; the pre-registration specified no interval for elim, its frozen 80% prediction interval $[0.35, 0.75]$ straddling it; its denominator is a single measurement too, 36 against a frozen point of 25, and does not itself enter the rule — at $13/35 = 0.371$ without the reference arm's empty completion the case is unchanged) — the claim frozen before the run: **under this protocol** (single turn, 512-token cap, database-level governance content plus instruction), in-context governance does not substantially eliminate the as-of gap; we do not extrapolate.

Where the residue sits is the finding. *(a)* On the seven refusal questions whose gold label is *provably* not registry-derivable — §7.1's witness pairs materialise both labels under identical content, so only a data probe decides (OOV emptiness, MC(ii) denominator mass, AM(iv) realisation difference) — the arm errs on **6 of 7**, every one by answering. *(b)* Where a registry line *can* decide it errs on only **1 of 5** — though the two sets are not length-matched (means 24.6k vs 20.3k characters), so this contrast inherits the dilution/depth entanglement below. *(c)* On the three `DISCLOSURE-BLOCKED` questions it holds the policy rows that block the answer — and answers all three: **a policy on file is not a policy enforced**. *(d)* On the eight version-flip answers (§7.4) it errs on 3 — two by resolving the wrong version–window pair, one a failed execution. *(e)* The cluster gradient repeats it: `codebase_community` stays at 6/7 errors while shallow-chain clusters nearly repair (`financial` 4 → 1 of 8, `debit` 3 → 1, `world_1` 2 → 1), while `card_games` and `european_football_2` each regress 2 → 4 — the third- and fourth-longest packages. Correct refusals move 4 → 5 of 15, over-refusal is 1 of 45 — `trivial_v2`'s level, down from the plain backbone's 5 — and on three refusal questions the plain backbone was right where the informed arm is wrong. *(f)* The probe ceiling is measured: crediting the arm with all six probe-decidable residuals — four sit among the reference's 36 errors, two its own regressions — leaves 22/60, elimination at most $17/36 = 0.472$, past the 0.40 line; an execution loop recovers failed executions by construction, so further crediting all nine execution-error residuals (disjoint from the six; seven among the 36, two regressions) leaves 13/60, at most $24/36 = 0.667$: the verdict is a fact about probe-less, feedback-less passes, and the four policy/registry residuals are not probe-decidable at all. The content buys what a registry can state and almost nothing a probe must decide.

Three alternative explanations, checked. *Context dilution*: package length is constant within a database (spread ≤80 characters) and monotone in chain depth, so this design cannot separate dilution from depth — the question-level point-biserial is $r = 0.400$, but at the cluster unit it is $r = 0.774$ ($t = 3.24$, df = 7), and `codebase` (28.3k) errs 6/7 against `financial` (23.5k) at 1/8: we do not claim dilution excluded. *Transport faults*: the arm had zero empty responses — excluded. *Imperative compliance* (obeying a cross-window imperative instead of judging it): both AM(ii) imperatives are correctly refused — excluded as the driver.

Non-degeneracy, adversarially confirmed (ND-4). This arm is the designed attack on our own benchmark: on the retired public track, whose registries carried expressions and coverage literals, the same protocol scored 0/20 — a lookup, not a finding. Handed strictly more content, it errs on 46.7% and errs *where* the design predicts.

Prediction accounting, in full. The frozen predictions were optimistic: all 8 observed error counts exceed their point predictions and 7 of 8 exceed the 80% interval's upper end (this arm most sharply: predicted 14 [8–22], observed 28). Direction and arm ordering are unaffected; the probe prediction (≥3 of 7) lands at the interval's top with 6, correct refusals miss *below* their interval

(9/15 [6–12] predicted, 5 observed) and over-refusals sit at its floor (∼4/45 [1–9], 1), and the elimination miss (predicted 0.55, observed 0.361) moved the arm from predicted case (i) to declared (iii).

7.4 E4: The Certified Compiler

The compiler decides all 60 in agreement with the frozen gold — values, refusal classes *and subtypes*, rewrite kinds — emitting 33 answers, 12 rewrites and 15 refusals (Table 3); the refusals coincide with the 15 the specification declares unanswerable — 12 by the binding semantics of §2, 3 by the disclosure layer of §4 — so the refused share 0.25 is not caution, and re-emission is byte-stable. The 0% is a *constructive upper bound*: the registered binding rules are implementable and cover these 60 questions — not that a learned system generalises. The compiler consumes the structured intent σ of Definition 2.3; the LLM arms read the same declarations from the question's natural language (parsing language into σ is out of scope).

The disclosure layer, exercised. This suite makes §4 earn its place on gold: three `DISCLOSURE-BLOCKED` refusals (a contributor roster, two patient-level listings) where the masked, k-gated ladder has no legal point; five granularity roll-ups, one a three-level climb failing below the top exemption, one whose k moves $10 \to 20$ *between versions* (the same question answers under $v1$, rewrites under $v2$); and two mask degradations whose transform compiles into the emitted SQL — each closed by the independent verifier (§7.5), not by our own replay. Ablating the rewrite stage is arithmetic rather than a run: refusing wherever the request is not directly bindable turns the 12 rewrites into refusals, the refused share $0.25 \to 0.45$, the error rate $0\% \to 12/60 = 20.0\%$, every gold refusal class unchanged — the binding layer, not the search, carries the classes.

The version axis, exercised. Four question pairs pin one question at two declared times so the answer *flips* with the in-effect version; the sharpest is real history — Formula 1's 2009 vs. 2010 points schemes value the same 2009 season at 100.0 vs. 252.0 — alongside a problem-loan caliber change ($0.1368 \to 0.0855$), a default-caliber switch and a population re-basing. One further question reads *off-diagonal*, pinning $v1$ under $\mathrm{ver}(T) = v2$: legal exactly because the certificate carries the explicit commitment marker, whose deletion is one of the forgeries below.

7.5 E5: Certificates and Forgeries

The independent verifier accepts 60/60 certificates; the discriminating experiment is forgery: 34 forged certificates over *11 unmodified compiler certificates as bases* (all 11 accept). Coverage is family-level — at least one landing per check face. The verifier rejects 34/34, each on the check its construction pre-registered: 6 value/window lies (a whole-year window shift with coherently moved probes; an empty-denominator refusal flipped to an answer over a lied count, caught by *re-executing* the probe; the off-diagonal marker deleted; a version swap); 5 *field deletions* — $\nu, \alpha, \rho, \delta$, refusal witness — confirming each field is consumed by a non-vacuous check, the converse half of Proposition 5.4; 9 semantic lies (a dangling version pin, AM(iv) realisation-audit tampers, an out-of-closure read, a self-granted anchor override, among others); and 10 disclosure forgeries, new with the exercised layer: answering at the banned grain with roll-up traces stripped, over-coarsening past the minimal legal level (both directions), a support-minimum transcript with SQL replaced by `SELECT 999`, the policy set emptied or cited off the registry, masks dropped or version-downgraded; and 4 refusal-class substitutions — AM(iv) re-issued as OOV on one genuinely vacuous leg, and a self-granted clause-(iv) replay variant. *One boundary case is accepted and we say so*: deleting *part* of a refusal's blocking-policy list (leaving it non-empty and registered) passes — beyond non-emptiness that list is not load-bearing, decision and class pinned by checks re-deriving the policy set; the hardening is a set-equality audit.

Table 5: Every disagreement the two implementations had on their first cross-check. *Provenance*: this ledger was recorded during the two sides' integration on the production (enterprise) track that motivated this work, before the public suite of §7.1 existed; it is specification-level content — readings, checks, guard order — and carries no production data value, so it is reported as evidence about the *design*, not about the released benchmark. *Cls*: S spec-reading, D defect (subscript: which side), R registration gap; *Right*: the side the frozen text vindicated; *Check*: the check that fired († crashed). The 25 first-round rejections were V0×9, V6a×9, V3×3, V6b×2, V6c×1, V4/V6c×1, many-to-many onto these 15. *Value gold?*: would the integration suite's value-level gold — 51 questions, 37 numeric and 14 refusal-token comparisons — have failed? On no row, and the cell says why not: the value is unchanged; the lie is confined to the *cert*ificate or registry that gold never reads; the erring side is the *verif*ier that gold never runs; or the gold *label* is itself what settled the reading. Both readings per row: [33].

Divergence	Cls	Right	Check fired	Value gold?
D_1 coverage domain gates before MC?	S	spec	V6b	✗label
D_2 β_v defined when both legs are NULL?	S	spec	V6b	✗verif.
D_3 prescribed anchor of an atomic metric	R	verif.	V0	✗cert
D_4 must a ratio carry a caliber route?	R	verif.	V4, V6c	✗cert
D_5 is `avg_handle_hours` a ratio?	R	spec	V3	✗label
D_6 must a rewrite emit w^*'s lower bound?	D_c	verif.	V6a	✗value
D_7 may an atom borrow another's binding row?	D_c	verif.	V3	✗cert
D_8 probe bound: arithmetic or literal window	D_c	verif.	V6b	✗cert
D_9 does a month marker denote the month?	S	comp.	V6c	✗verif.
D_{10} is a derived-table alias a read object?	D_v	comp.	V6a	✗verif.
D_{11} does a conformed-dim join leave closure?	R	verif.	V6a	✗cert
D_{12} replay of the interval-anchor predicate	D_v	comp.	V6a†	✗verif.
D_{13} where the re-anchoring $\bar{a}$ is carried	S	spec	V0	✗cert
D_{14} how a delta question presents $\vec{\omega}$	R	spec	V0	✗cert
D_{15} certificate spelling of an unreg. anchor	S	either	V0 (V2)	✗cert
5 S · 3+2 D · 5 R (25 rejections → 15 causes)				✗0/15

Independence, measured. Both sides read the same structured question, so a window read off it would restate what the certificate committed to; each role window is therefore re-derived from the registered as-of conventions over (G_v, D): the verifier derives 50 of 60 windows itself and, under a strict switch refusing every presented window, accepts exactly those 50/60 — the 10 rejections are the 8 questions whose window arrives as an explicit range request plus the 2 cross-window imperatives, where the presented coordinates *are* the declared input and the verifier fails closed; the mutation battery rejects a one-day shift of a certified window.

7.6 E6: What Independence Bought

The first cross-check of the two implementations — recorded during their integration on the production track that motivated this work (Table 5) — *rejected 25 of the 51 certificates then in force*, resolving into 15 distinct root causes: 5 implementation defects (3 compiler-side, 2 verifier-side), 5 registration gaps, 5 specification-reading divergences. Not one would have been caught by checking values against gold — the three compiler-side defects each returned the right answer under a non-re-derivable certificate claim. Two lessons shaped the public base. In every registration gap the load-bearing content lived in a free-text note or compiler source: *certificate checkability is a measurable proxy for governance registration completeness*, and §7.1's seed discipline is that lesson enforced. Three of the five specification divergences sit on when a guard's subject is undefined, each moving a refusal *class*; this suite's frozen guard-order adjudications descend from those rows.

7.7 Cost, and Threats to Validity

What certification costs. Counting work only (warm process, connections held; all timings on a 10-core arm64 macOS 15 machine; Python 3.9.6, DuckDB 1.4.4), verifying a decision costs a median 17.7× the answering query it certifies — 17.0 ms against 1.31 ms, per-question paired ratios over the 45 questions that emit SQL (the marginal medians' own quotient is 12.9×), spanning 2.5× to 44.4× (p90 36.1×); none is cheaper to verify than to answer, and the certificate is a paired median 10.2× the query's bytes (pretty-printed; 8.4× minified), the tail being coverage replay on the largest governed tables (0.18 s warm). The 60 per-certificate cold medians (median of 5 processes per certificate) sum to 7.60 s — median 116 ms, p90 144 ms, max 288 ms, 53 ms of it process floor; as whole cold processes the paired median falls to 1.8×, the fixed per-process cost dominating. §4's probe bound ([33]) covers the in-window probe term; the full-scan class it exempts has *no* instance here (AM(iv) audits are window-bounded), so we exhibit no scan-amplification regime and report no scalability sweep.

Threats. *Sixty questions is the unit count, not the evidence mass*: each unit is a governed instance carrying executed gold, a verifier-accepted certificate and an eight-arm response grid; leave-one-cluster-out moves no ordering: every fold keeps each plain baseline's error $\geq$0.40, governance below all four, the compiler at 0 [33]. *One sample per question is the protocol*: five reps of both headline arms (rep 1 the frozen run) hold the ordering in all five, flips $\leq$13.3% per arm [33]. *Nor does governance-blind temporal-SQL close it*: post-registered hand-built arms err 28/60 same-window — level with the governance-informed arm; 23/60 crediting five bare `NULL`s as refusals — 54/60 all-history, values 6/33 under both readings [33]. *Authored governance over real data*: the fact rows are shipped public data save `world_1`'s authored history, but the layer over them is ours; the mitigations are structural (non-degeneracy gates, witness pairs, distractor density, leak audit) and total transparency, not independence of authorship. *Scoring concessions run toward the baselines*: LLM refusals need no class, rewrites no declaration, a multiset amnesty admits shape-degenerate answers, and on the five hull-trim questions a window-blind aggregate equals gold by construction — the form split in Table 4 keeps the headline from hiding there (value errors 11/33 even with those freebies); and on one of the four version-flip pairs (`california_schools`) the two gold values differ by 0.48%, inside the 0.5% scoring tolerance, so that pair credits a version-blind answer. Running the other way, the compiler is handed the σ the LLM arms must recover from text (§7.4); a post-registered arm that extracts σ from the question text and hands it to the compiler errs 5/60 [33] — recovering σ is not where the difficulty lives. *One backbone* carries the variant and governance arms — conclusions about `claude-opus-4-6`; cross-family columns carry §7.1's gateway caveat. *The protocol is cross-lingual* (§7.1): the three Chinese-native families still err more than the backbone (71.7/73.3/76.7 against 60.0 — gateway-confounded columns, intervals too wide to rank families); a post-registered English run of both headline arms moves each arm one question (35/60, 29/60 against 36/60, 28/60), ordering and probe-side concentration intact [33]: language does not drive the gap. *The governance-informed comparator is bounded in our favour* (§7.3): full-layer packaging is an upper bound on what in-context governance buys, not a tuned opponent's floor; *prediction misses are published* (§7.3). *What no arm has* is *execution feedback*: every system answers from one statement — one pass cannot run the probes that decide the hard classes — and *prompt technique is one line deep* — no decomposition, self-consistency, in-domain few-shot or multi-turn arm.

8 RELATED WORK

The four nearest neighbours each hold one half of the problem (Table 6). *Schema-evolution robustness*: EvoSchema [36] perturbs schemas but asks a model to be *insensitive* to the version it faces;

Table 6: The four nearest neighbours and the closest temporal-database lineage, against the four properties as-of correctness needs: none binds a question to a point on both axes, none makes a refusal a checkable value.

	versioned sem. layer	valid-time as-of bind	typed refusal	checkable certificate
schema-evolution robustness [36]	∘	–	–	–
bitemporal agent memory [34]	–	✓	–	–
lakehouse time travel [2, 1]	–	∘	–	–
transaction time + schema evol. [24, 6]	∘	✓	–	–
static semantic layers [28]	–	–	–	–
this paper	✓	✓	✓	✓

∘ = partial: schema versions, not a governance layer; snapshot-level time travel without a semantic layer.

as-of correctness asks it to be *bound* to a declared one, saying so when no binding exists. *Bitemporal agent memory*: MemStrata [34] keeps `valid_from`/`valid_to`, supersession and as-of retrieval, but over retrieved facts, not compiled queries: no metric semantics, no compile-time decision. *Lakehouse time travel*: snapshot pins [2, 1] are a genuine as-of capability we consume, but answer "which bytes", not "which meaning", and return a number exactly where we refuse. *Static semantic layers*: a governed metric layer measurably improves LLM analytics accuracy [28] — our nearest neighbour and a natural baseline, whose in-context content §7.3 bounds — but a static document without versions, as-of binding or refusal. Substrates that own the version transition and disclaim metric correctness leave exactly the space this paper occupies; Theorem 2.9 makes the relationship precise.

Lineages we inherit. Temporal databases give us the data axis wholesale — valid and transaction time and the languages over them [29, 30, 4, 7, 16, 13], querying under schema evolution [24, 6], dataset and relational versioning [3, 11, 22] — but no second axis for a *governance* layer, no typed value for inability to answer — so a hand-built temporal-SQL implementation of the sixty queries is a point in the compiler's design space minus the refusal classes and the certificate. Post-registered, we built one [33]: it errs 28/60 — level with the governance-informed arm — all 15 refusals answered (23/60 crediting its five bare `NULL`s); an all-history twin errs 54/60; values 6/33 — the data axis is not the bottleneck. The NL2SQL lineage [38, 32, 35, 20, 25, 9, 12, 14, 8] treats the database as a fixed target — our evidence base takes nine of its databases [35, 20] and registers the governance layer binding needs; time-sensitive QA [15] scores answers, not queries.

Refusal, disclosure, certificates. Refusal is studied as completeness over incomplete data [26] and abstention on unanswerable questions [17]; ours are decidable predicates over the governance state (Proposition 3.2) carrying a class and a replayable witness. Fine-grained access control [27], purpose-limited disclosure [18] and anonymity thresholds [31, 21] are long-standing, our grain and mask lattices ordinary instances; §4 adds the joint statement *after* temporal binding. Certifying algorithms [23], provenance [10, 5] and authenticated query processing [19, 37] are our certificates' tradition; we add the coordinates they assume fixed and a certificate for a *refusal*, which none can express.

9 CONCLUSION AND AVAILABILITY

Conclusion. As-of correctness for governed analytical queries is a compile-time binding problem (compile-time relative to the model: the compiler decides by running probes against the store): we gave it a bi-temporal semantics whose refusals are values, a decidable four-way classification — under disclosure, a maximal legal rewrite or a complete structured refusal — and certificates an independent verifier re-derives from the governance tables and data. On a fully public base of 60 Chinese questions four frontier model families err on 60.0–76.7%, no prompt variant (53.3–66.7%) crosses the pre-registered elimination line, and the entire governance layer in context still leaves 46.7%, its refusal-side residue on decisions that need a data probe: under this protocol — one statement per question, no execution feedback — in-context governance yields no substantial elimination. Three findings carry forward: binding is an *execution-time* property no prompt content substituted for under this protocol — the informed model executed a disclosure policy on file zero times in three; disagreement between the two implementations was as often *registrational* as implementational — checkability measures how much of a layer is really governed; and the disagreements surviving the specification move refusal *classes* — the price of making refusal a first-class value.

What we did not prove. Answer soundness (Theorem 5.2(b)) rests on the still-open "syntax implies semantics" assumption (A-tmpl) over Definition 5.3's class, Theorem 4.1(a) on its *(A-emit)* bridge; the checking relation verifies a rewrite legal and its cut trace consistent, *not* maximal; continuous time, window bargaining, lag rules and the full maximal frontier are out of scope by (A6). The version axis is now exercised (two committed versions per database, four flip pairs, one off-diagonal pin), but three defence paths have no gold instance (refusal before the first governance commit, an uncommitted version pin, the certificate side of an alias-missing $\mathsf{MC}(i)$), and one adjudication is unresolved: whether a delta-period window needing a hull trim rewrites or refuses.

What the numbers license (§7.7). The 0% is a constructive upper bound over this suite — nothing follows about unseen questions or free-form language — and the 46.7% set against it bounds what in-context governance buys from a single statement; no execution-capable (agentic) arm was run — the probe-oracle ceiling (§7.3), execution-error residuals credited, bounds the recoverable share at 0.667.

Artifact availability. Everything needed to reproduce every number is at `https://github.com/anon-submit-0/asofgov-artifact`: compiler and independent verifier with the CI import-disjointness check; the nine warehouses with registries, per-row real/authored provenance and gold SQL; the 60 questions with gold values, refusal classes/subtypes, rewrite kinds and frozen scoring; the non-degeneracy gate reports; the arms pre-registration with predictions and both freeze manifests; the raw response cache for all 8×60 pairs *and* the voided `trivial_v2` first run; the 60 certificate envelopes, the forgery generator with its 11 bases and 34 forgeries, the cost measurements and the cluster-bootstrap script with its frozen seed; one driver re-runs the offline pipeline, the frozen cache reproducing every number without an API key. Not released: the enterprise development seeds — their only exhibits are §1's motivating rates and Table 5's ledger, provenance-marked in place; no public-suite number depends on them.

Disclosure of generative AI use. In accordance with the PVLDB and ACM publication policies on generative-AI tools, we disclose that large language models were used throughout the production of this work: in drafting and revising definitions, theorems and proof sketches; in implementing the compiler, verifier, forgery generators and figure pipeline; in building and running the evaluation harness; and in drafting and editing this manuscript. Large language models are also the object of study in §7, as evaluated systems. All definitions, theorems, proofs, experimental protocols, numbers and claims were reviewed and are vouched for by the authors, who take full responsibility for the entire content of this paper, including any errors.

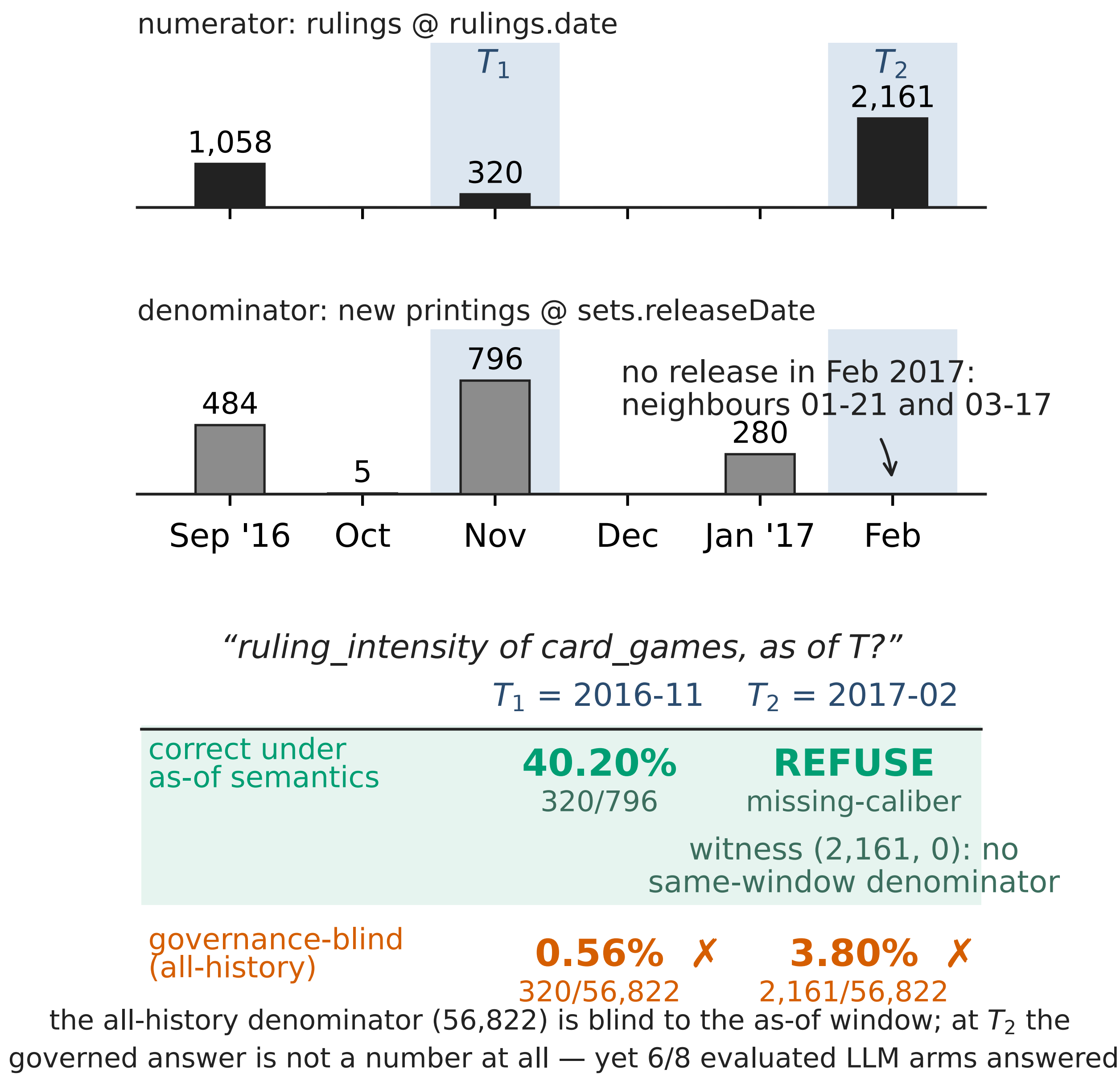

Figure 1: **The as-of gap.** Under the graph version in effect at the declared instant (v_1 of `card_games`; v_2 commits 2017-09-28), `ruling_intensity` anchors its numerator on `rulings.date` and its denominator on `sets.releaseDate`, bound by `same_valid_time_window`. At $T_2 = 2017\text{-}02$ the numerator leg holds 2,161 rulings, the same-window denominator is empty, and the denominator anchor's coverage hull $[1993\text{-}08\text{-}05,\ 2021\text{-}03\text{-}19]$ still contains February, so the request lies *inside* validity and $[\![q]\!]_{\mathcal{G},T} = \bot_{\mathsf{MC}}$, witnessed by a denominator probe re-evaluating to 0.

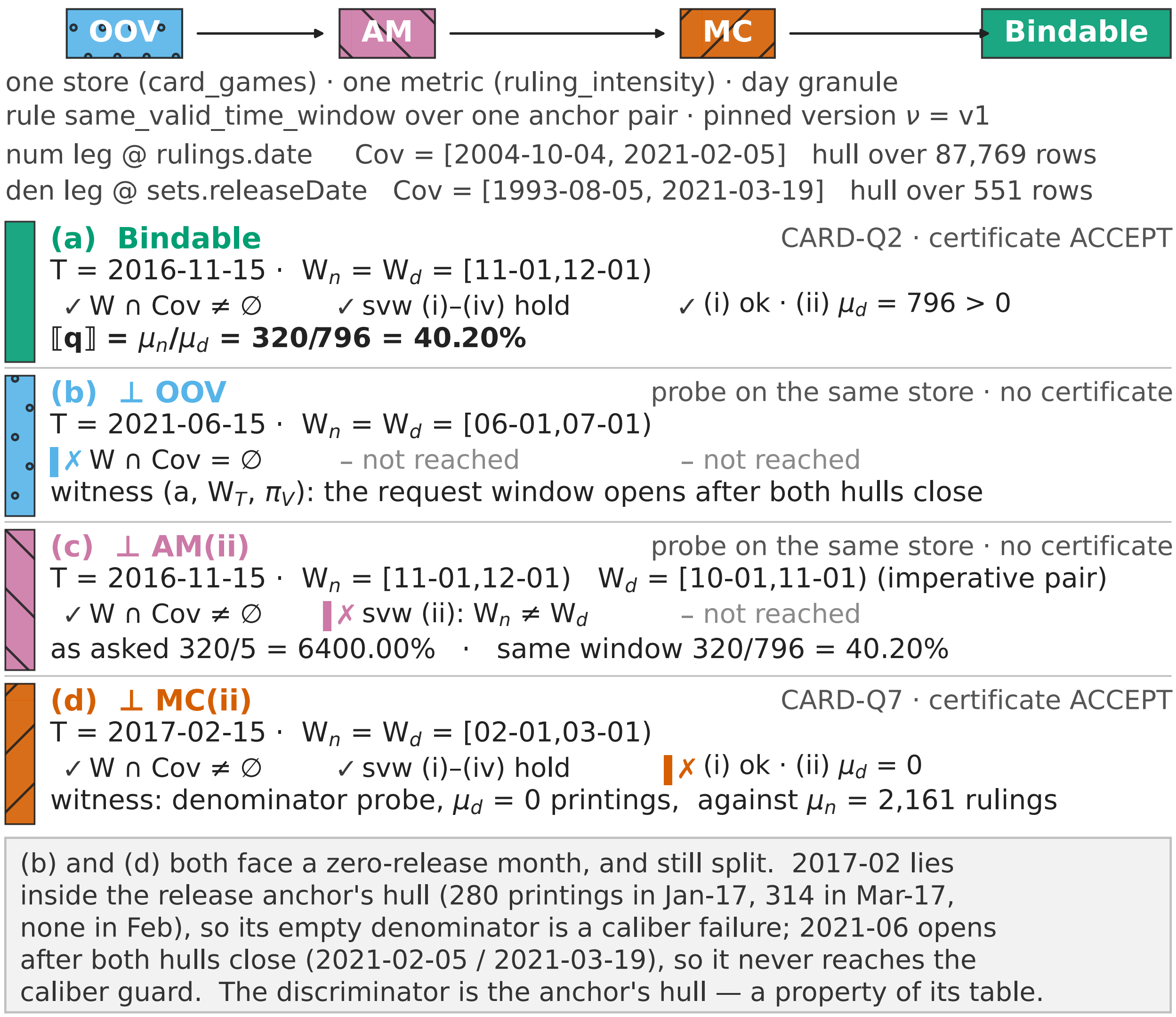

Figure 2: The partition on four sibling requests to the `card_games` store: only the coordinates move, so each row leaves the chain in a different column. (b) and (d) are the OOV/MC(ii) boundary — one metric, one empty same-month denominator, two classes — because Cov is the hull of the *anchor's* table: 2017-02 lies inside it, 2021-06 after it. (c) is what a refusal buys: the 6400% its cross-window binding would report against the 40.2% the same-window rule delivers in (a). (a) and (d) are suite questions with verifier-accepted certificates; (b) and (c) are probes on the same store, and say so.

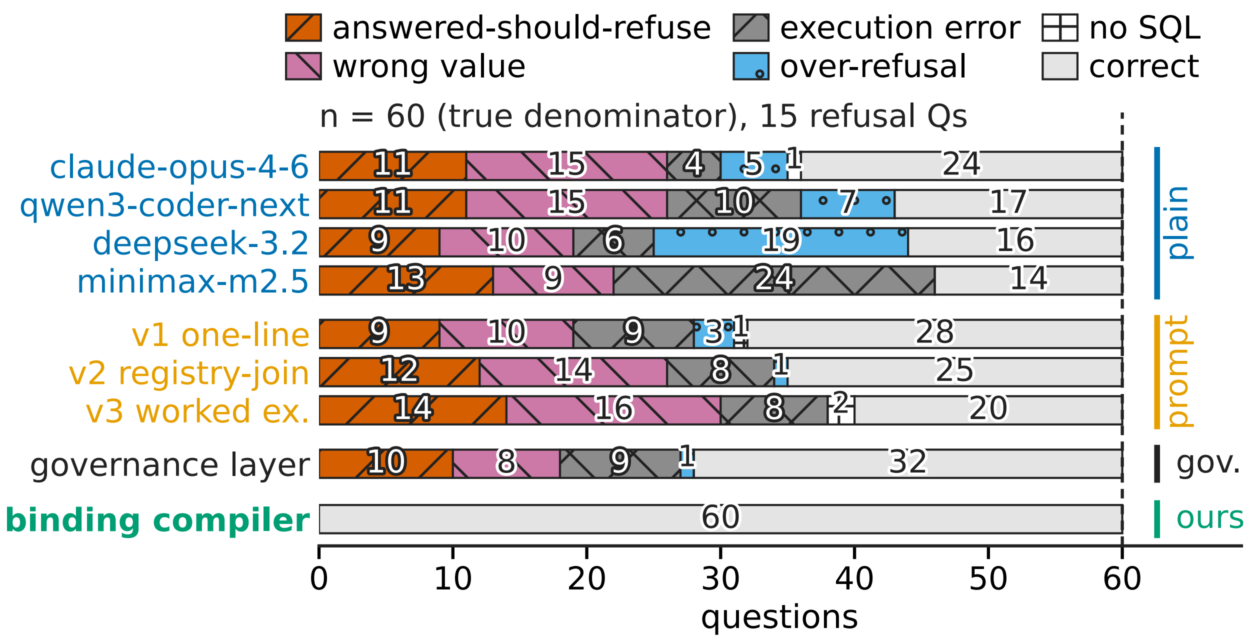

Figure 3: Failure taxonomy in the row order of Table 4: each bar stacks to the true denominator of 60 and every non-empty segment prints its count, so each arm's error segments sum to its E/60. Wrong value includes a wrong window or version; *no SQL* is neither an answer nor a refusal.

A Notation

Table 7: Notation used throughout the paper.

$\mathbb{T}, T$; $\mathbb{G}, g \sqcap g'$	time domain, as-of instant; granularities, commonly expressible
$\mathcal{W}, \mathcal{W}_g, \top$	window algebra; g-aligned sub-algebra; full window
$a, \mathrm{vt}_a, \mathrm{cm}_a$; $V_v(a), \mathrm{Cov}_v(a)$	anchor, valid-time map, coverage mode; marking set, coverage
G_v; M_v, R_v, A_v, β_v; ρ, K_ρ	graph version; metrics, routes, anchors, bindings; route, keys
$\mathcal{G}, t(\cdot), \mathrm{ver}(T)$; $q = (\sigma, T)$	version sequence, commit map, in-effect version; query
$\alpha_{q,v}$; Pop^r, μ^r	candidate anchor assignment; leg population, leg mass
$\mathsf{Bindable}$; $[\![q]\!]_{\mathcal{G},T}$	bindability; point-in-time semantics
$\bot_{\mathsf{OOV}}, \bot_{\mathsf{AM}}, \bot_{\mathsf{MC}}$; DB	refusal values; disclosure-blocked
$\Gamma_v, \mathbb{M}, \mathcal{D}_v, \pi$	grain lattice; mask lattice; policy set, policy
$b = \langle \alpha, w, v, g \rangle$, L; $\preceq_D, \prec_{\det}$	binding point, legal set; fidelity, selection order
$B = \langle \alpha, \nu, \rho, \delta \rangle$, C, Chk	binding quadruple; certificate; check relation

B The Running Example in Full

Example 2.1 of the body registers, over the public `card_games` database, `ruling_intensity` as a ratio of a rulings leg to a new-printings leg: the numerator routed on `rulings` (route `card.rt1`), the denominator attributed along the registered caliber route `cards`⋈`sets` (route `card.rt2`, join `setCode=code`, scoped caliber), both legs bound by `same_valid_time_window` (rows `B-CARD-RI-N`/`B-CARD-RI-D`) over the day-granular anchors `A-CARD-RUL` (`rulings.date`, hull coverage 2004-10-04..2021-02-05, 87,769 rows) and `A-CARD-SET` (`sets.releaseDate`, hull coverage 1993-08-05..2021-03-19, 551 rows). Four sibling requests against that one registration are worked out with every mass in the body's coverage-partition figure (Figure 2); the body's example points two further certified cases here — an OOV and a DISCLOSURE-BLOCKED listing, classes the `card_games` registration alone does not exercise on gold. All six are given below, each with the case of Theorem 3.1 (or the disclosure layer of §4) it falls in; every number is re-derivable from the released warehouses, and the four certified cases carry verifier-accepted certificates in `impl/certs2/`.

- **(a) — answer.** `CARD-Q2`: ruling intensity for the month of 2016-11 (as-of 2016-11-15, declared 2017-03-01). Both legs are populated on the same valid-time window under the prescribed anchors (320/796), hence case (a), Bindable, and $[\![q]\!]_{\mathcal{G},T} = 0.4020100502512563$ — the 40.2% of the figure. Suite question, certificate accepted.
- **(b) — OOV boundary probe.** The same metric probed at 2021-06: the month lies beyond both coverage hulls (numerator hull ends 2021-02-05, denominator hull 2021-03-19), the coverage guard fails on both legs, case (b), $\bot_{\mathsf{OOV}}$. A probe on the same store, printed in the figure and marked there as a probe, not a suite question.
- **(c) — AM cross-window probe.** Numerator window 2016-11 against denominator window 2016-10, both stated in the request: clause (ii) of svw fails, case (c), $\bot_{\mathsf{AM}}$ — the instance that also carries the *drop* α construction of Proposition 5.3. What the refusal buys is printed beside it in the figure: the cross-window binding would report $320/5 = 64.0$ — a 6400% "intensity" — against the 40.2% the same-window rule delivers in (a).
- **(d) — MC(ii).** `CARD-Q7`: the same metric at 2017-02 (as-of 2017-02-15, declared 2017-03-01). The numerator still carries mass (2,161 rulings, itself the gold value of the sibling `CARD-Q1`)

and 2017-02 lies inside both hulls, but no set release falls in the month, so the same-window denominator mass is 0: case (d), $\bot_{\mathsf{MC}}$ by mode (ii), witnessed by a denominator probe re-executing to 0 on the certified window. Suite question, certificate accepted. (b) and (d) are the caption's OOV/MC(ii) boundary pair: one metric, one empty same-month denominator, two classes, because Cov is the hull of the *anchor's* table — 2017-02 lies inside it, 2021-06 after it.

- **(certified OOV)** `F1-Q7` (`formula_1`): the month's count of grands prix as of 1949-05-15, declared 2011-03-01. The race anchor `A-F1-RACE` has hull coverage starting 1950-05-13 — the first world-championship race — so the requested month window [1949-05-01, 1949-06-01) misses the coverage domain entirely: case (b), $\bot_{\mathsf{OOV}}$, with the coverage probe (`MIN`/`MAX` over `races.date`, observed 1950-05-13..2017-11-26) recorded verbatim in the certificate and re-executed by the verifier. The seed's ND-1 witness pair flips the class by inserting a single 1949-05 race. Suite question, certificate accepted.
- **(certified disclosure-blocked listing)** `CODE-Q7` (`codebase_community`): list the `DisplayName` and raw `Location` of the ten users with the most accepted answers in 2013-06 — a per-user roster rigidly requesting raw values of masked columns. Binding succeeds, and the verifier replays $\neg$OOV $\wedge$ $\neg$AM first (V6b(0)); at the disclosure layer the mask policy `code.pi1` sends the rigid raw presentation to $\top_{\mathbb{M}}$ and the k-threshold policy `code.pi2` ($k = 5$, entity granule `user`) leaves no legal grain: the Suppmin transcript records the support probe whose re-execution observes a minimum cell of $1 < 5$, the blocking set $\{$`code.pi1`, `code.pi2`$\}$ is re-derived independently as Π, and $U_{\min} = \emptyset$ is asserted and replayed — the roster is refused as disclosure-blocked rather than delivered as an empty shell (§7.5). Suite question, certificate accepted.

The five compile-time stages, in words. A request ① — metric, scope, one window per role, optionally a *declared* anchor — pins the version in effect at T; binding compilation ② derives the anchor assignment α and the caliber routing ρ under the frozen order OOV $\to$ AM $\to$ MC, itself part of the semantics because two guards can hold at once (§3); the disclosure gate ③ climbs the grain lattice for the least legal grain instead of refusing whenever a legal query exists. Every outcome carries a certificate ④ that an independent verifier ⑤ re-decides from the governance tables and the raw data alone, sharing no project-internal module with the compiler. On the walk above, stage ② is where (b)–(d) and `F1-Q7` stop, each leaving its probe transcript; stage ③ is where `CODE-Q7` stops; and the binding rows exercised on the answering case are `B-CARD-RI-N`/`B-CARD-RI-D`.

C Disclosure-Constrained Rewriting: Full Development

§4 of the body states the policy model in one paragraph, the MLR problem, and the three guarantees (soundness, maximality with deterministic selection, refusal completeness). This section carries the development those statements summarize — the disclosure policy and its judgement relation, the binding-point coordinate semantics, Mlr-Compile itself, the structural lemmas the maximality argument consumes, and the complexity theorem. Proofs are in §D.

C.1 Grain lattice, mask lattice, disclosure policy

Two finite lattices are declared as part of G_v: the *grain lattice* $(\Gamma_v, \preceq_\Gamma)$ over a finite product of entity and time axes, with $g \preceq_\Gamma g'$ reading "g is finer than or equal to g'" and declared edges (e.g. `school` $\to$ `district` $\to$ `county`, `user` $\to$ `reputation_band`, `day` $\to$ `month` on the time axis) never inferred from data; and the *mask lattice* $(\mathbb{M}, \preceq_{\mathbb{M}})$ of presentation transforms, $\mu \preceq_{\mathbb{M}} \mu'$ reading "μ' discloses no more than μ", bottom id, top $\top_{\mathbb{M}}$.

Definition C.1 (disclosure policy and judgement)**.** A policy is $\pi = \langle P_\pi, \mathrm{sc}_\pi, \mu_\pi, \gamma_\pi, k_\pi \rangle$: a protected attribute set; a role branch $\mathrm{sc}_\pi = \langle \mathrm{sc}_\pi^{\mathrm{role}}, \mathrm{sc}_\pi^{\mathrm{row}} \rangle$ or $\perp$ pairing a context predicate with a row-ownership predicate; a minimum mask $\mu_\pi \in \mathbb{M} \cup \{\perp\}$ for raw presentation; a grain floor $\gamma_\pi \in \Gamma_v$; and a small-cell threshold k_π. Protected attributes propagate along declared carrying joins, $\bar{P}_\pi$ being the closure of P_π under that relation, computable by worklist in $O(|\rightsquigarrow| + |\mathrm{Att}|)$; $\mathcal{D}_v$ is finite and versioned with G_v. Let $R(q')$ be the read set of a candidate rewrite and $S_{\mathrm{raw}}(q')$ its *raw presentation set* (row-level original values; aggregates excluded), and call π *touched* when $\bar{P}_\pi \cap R(q') \neq \emptyset$. Then $q' \models_{\mathrm{ctx},I} \mathcal{D}_v$ holds when each touched π satisfies either **(D1)** $\mathrm{sc}_\pi \neq \perp$ and the context satisfies $\mathrm{sc}_\pi^{\mathrm{role}}$ — rows contributing $\bar{P}_\pi$ must then satisfy $\mathrm{sc}_\pi^{\mathrm{row}}$, grain and small-cell clauses are waived, the mask clause still applies — or **(D2)** the aggregate branch: (i) $\gamma_\pi \preceq_\Gamma g'$, (ii) every non-empty output cell in the window covers at least k_π distinct γ_π-entities under the instance I, and (iii) every $a \in \bar{P}_\pi \cap S_{\mathrm{raw}}(q')$ is presented at $\succeq_{\mathbb{M}} \mu_\pi$.

Legality is thus *instance-relative* (small-cell probe) and *context-relative* (role branch), which is why a certificate pins both the instance snapshot and a hash of the request context. Four of the nine public domains register disclosure policies (18 instance rows: 8 in `codebase_community`, 6 in `thrombosis_prediction`, 2 each in `california_schools` and `debit_card_specializing`) and five register none; a policy-free domain gets *absence means unconstrained* — the gate passes and the certificate carries an `ungoverned-disclosure` annotation, keeping "not governed" and "checked and passed" distinguishable — and every disclosure claim is scoped to the four governed domains. *Assumption (coarsening compatibility):* for candidate grains $g \preceq_\Gamma g'$ above the requested grain, $\mathrm{touched}(g') \subseteq \mathrm{touched}(g)$ and $S_{\mathrm{raw}}(\mathrm{cl}(g')) \cap \bigcup_\pi \bar{P}_\pi \subseteq S_{\mathrm{raw}}(\mathrm{cl}(g)) \cap \bigcup_\pi \bar{P}_\pi$ — coarsening touches no new policy and puts no protected attribute into raw presentation that was not already there, the second clause being needed because switching the group-by key on roll-up can bridge a new column into the output. The 18 policy-instance rows of the four governed domains are authored to satisfy both — each k-threshold row fixes one declared lattice on its own node, the mask and present-only rows are grain-independent, and the single carrying join (`code.pi1` onto `posts.OwnerDisplayName`) adds no attribute on roll-up — and V5's independent re-derivation of the legality and minimality clauses on every accepted certificate exercises the property on every certified path. Violating configurations lie outside the applicability domain of Lemma C.3, where the complexity bound survives and the minimal-frontier characterization does not.

C.2 The binding point space

Given q, T with version pin $p \in \mathcal{V} \cup \{*\}$ ($*$ = unpinned) and *commitment set* κ marking coordinates the question states as rigid, $X(q,T) = \{\langle \alpha, w, v, g \rangle : v \in V_{\mathrm{adm}},\ \alpha \in A_{\mathrm{adm}}(m, v),\ w \in W_{\mathrm{cand}}(\alpha, v),\ g \in \uparrow_{\Gamma_v} g_0\}$, the window coordinate being a leg-indexed vector with componentwise operations. The declared coordinate semantics delimit the model. **(S-A)** A_{adm} holds registered bindings compatible with κ: rewriting never moves a leg onto an unregistered anchor — when two systems' anchors do not line up the answer is a refusal, not a substitute. **(S-V)** V_{adm} holds versions compatible with p and κ; unpinned, it resolves *diagonally* to the version in effect at T, never the newest committed one, since recomputing history under the newest caliber is the documented mis-binding of §2; an explicit pin elsewhere is permitted and marks the certificate off-diagonal. **(S-W)** $W_{\mathrm{cand}}(\alpha, v) = \{w_0 \sqcap \mathrm{Env}(\alpha, v)\}$: the window is trimmed by *structural* endpoints alone — request endpoints, coverage endpoints, version boundaries — with Env per leg under the anchor's declared coverage mode. **(S-G)** candidate grains form the up-set $\uparrow g_0$. **(S-M)** the mask is not a search coordinate: each point takes the least legal closure $\mu^*(b, \mathrm{ctx})$, and a point whose closure sends a rigidly requested presentation attribute to $\top_{\mathbb{M}}$ is refused as *disclosure-blocked* (DB) rather than answered with an empty shell. A rewrite is the closure $q' = \mathrm{cl}(b, \mathrm{ctx}) = \langle m, \sigma, b, \mu^*(b, \mathrm{ctx})\rangle$, whose scope predicate absorbs the row-ownership predicate of each touched policy whose role branch the context satisfies. *Excluded by construction:* S-W does no data-driven window shrinking and no cross-leg bargaining, because small-cell legality is neither monotone nor convex along window containment — widening a window can *give birth* to new low-support cells — so joint window–grain bargaining would make the maximal elements depend on per-cell birth points. Structural trimming leaves one window candidate per slice, which is what our compilers do: insufficient population is a refusal, never a shrunk window.

For $q'_i = \mathrm{cl}(\langle \alpha_i, w_i, v_i, g_i\rangle) \in L$, $q'_1 \preceq_D q'_2 \iff \alpha_1 = \alpha_2,\ v_1 = v_2,\ w_1 \subseteq w_2,\ g_2 \preceq_\Gamma g_1$. $\preceq_D$ is a *fidelity* order, not a derivability order: for additive measures a coarse answer follows from a fine one by summation, but a coarse scoped ratio is in general not recoverable from fine rates; masks stay out of cross-point comparison for a related reason, "coarse but unmasked" and "fine but masked" being incomparable on information-theoretic grounds alone.

C.3 Structure, pruning, and the algorithm

Lemma C.2 (window structure and mass monotonicity)**.** *Fix a slice* (α, v) *and let* $w^* = w_0 \sqcap \mathrm{Env}(\alpha, v)$*. Then* (a) *every faithful, bindable rewrite in the slice has window* $w \subseteq w^*$*, so* w^* *is the greatest element of the slice's faithful bindable windows when that set is non-empty; and* (b) *if the denominator measure is non-negative,* $w \subseteq w' \Rightarrow \mu_{\mathrm{den}}(w) \le \mu_{\mathrm{den}}(w')$*, whence* $\mu_{\mathrm{den}}(w^*) = 0$ *propagates to every faithful window of the slice.*

Non-negativity carries (b) and is load-bearing: with refund-reversal columns the mass is not monotone along containment and zero-propagation pruning fails, and the algorithm's widened test $\mu_{\mathrm{den}} \le 0$ stays an honest guard there even though the pruning argument does not extend to it.

Lemma C.3 (grain-frontier structure)**.** *Fix* (α, v, w^*) *with* $\mu_{\mathrm{den}}(w^*) > 0$*, and let* $\mathrm{SUPPMIN}_\pi(g)$ *be the minimum, over non-empty output cells at grain* g*, of the number of distinct* γ_π*-entities covered. Then:* (a) $\{g \in {\uparrow}g_0 : \mathrm{SUPPMIN}_\pi(g) \ge k_\pi\}$ *is an up-set of* $({\uparrow}g_0, \preceq_\Gamma)$*;* (b) *for a fixed binding point and context the legal mask assignments form an up-set in the per-attribute order with least element* $\mu^*(b, \mathrm{ctx})(a) = \bigvee_{\mathbb{M}}\{\mu_\pi : \pi \text{ touched},\ a \in \bar{P}_\pi \cap S_{\mathrm{raw}},\ \mu_\pi \ne \bot,\ \text{clause applies}\}$ *(empty join* $= \mathrm{id}$*), so masks are derived, not searched; and* (c) *writing* $\mathrm{touched}_{\mathrm{D2}}(g, \mathrm{ctx})$ *for the touched policies whose D2 branch applies in the context and* $\mathrm{Legal}(g) \equiv \forall \pi \in \mathrm{touched}_{\mathrm{D2}}(g, \mathrm{ctx}) : \gamma_\pi \preceq_\Gamma g \wedge \mathrm{SUPPMIN}_\pi(g) \ge k_\pi \wedge \mu^*$ *keeps* S_{rig} *below* $\top_{\mathbb{M}}$*, under coarsening compatibility* $U = \{g \in {\uparrow}g_0 : \mathrm{Legal}(g)\}$ *is an up-set and* $\mathrm{cl}(\langle \alpha, w^*, v, g\rangle) \in L \iff g \in U$*.*

Restricting those clauses to $\mathrm{touched}_{\mathrm{D2}}$ is necessary, not tidy. The forcing instance is a development-track seed row carrying a role scope predicate *and* a grain floor with a threshold: when the context satisfies the role branch, a query over the requester's own rows is legal below that floor, and unrestricted quantification would reject it, roll the frontier up, and break Theorems 4.1 and 4.2 on that branch. The released public seeds register no role branch — the suite's request context is uniformly `external_analyst`, so D1 is vacuous there and every exercised disclosure decision runs the aggregate branch D2 (with its declared top-of-lattice exemption) — but the restriction stays in the statement because the judgement relation, not the suite, is what the theorems quantify over.

Corollary C.4 (pruning correctness)**.** *Traversing* ${\uparrow}g_0$ *bottom-up along covering edges, stopping at the first grain satisfying* Legal *and pruning its up-set, yields the set* $U_{\min}$ *of minimal elements of* U*;*

Algorithm 1 MLR-COMPILE$(q, T, \mathrm{ctx}; \mathcal{G}, I)$

out: ANSWER$\langle$plan, cert$\rangle$ or REFUSE$\langle$CT, cert$\rangle$; CT maps (v, α)-slices to $\{$MC, AM, OOV, DB$\}$

1: **if** unpinned **and** ver(T) undefined **then return** REFUSE$\langle\star\mapsto$OOV$\rangle$ ▷ version axis absent

2: $V_{\mathrm{adm}} \leftarrow (\text{pin } p\,?\,\{p\} \cap \mathcal{V} : \{v : \text{in-effect span} \cap T \neq \emptyset\}) \cap \kappa$

3: **if** $V_{\mathrm{adm}} = \emptyset$ **then return** REFUSE$\langle\star\mapsto$AM$\rangle$ ▷ commitment emptied

4: CT $\leftarrow \emptyset$; *Front* $\leftarrow \emptyset$

5: **for** $v \in V_{\mathrm{adm}}$ in $\prec_{\mathrm{det}}$ order **do**

6: $A_{\mathrm{adm}} \leftarrow$ ADMBINDINGS(m, v, κ); **if** $A_{\mathrm{adm}} = \emptyset$ **then** CT$[v, \star] \leftarrow (B_v[m]{=}\emptyset\,?\,MC:AM)$; **continue**

7: **for** $\alpha \in A_{\mathrm{adm}}$ in registration order **do**

8: $w^* \leftarrow w_0 \sqcap \mathrm{Env}(\alpha, v)$ ▷ S-W, per leg under declared coverage mode

9: **if** $w^* = \emptyset$ **or** (window rigid **and** a strict-member leg escapes its marking set **and** the pair-level clause-(iv) audit passes) **then** CT$[v, \alpha] \leftarrow$ OOV; **continue** ▷ on a distinct-anchor row the pair-level audit is adjudicated *ahead of* the per-leg coverage ruling (§3); both realisations empty $\Rightarrow$ the audit passes and OOV stands

10: **if** $\neg$RULESAT$(r_\alpha, \kappa, (w_r)_{r \in R(q)}; G_v, I)$ **then** CT$[v, \alpha] \leftarrow$ AM; **continue** ▷ clauses (i)–(iv) on the *per-role* windows α assigns — (ii) compares them and (iv) audits their realisations, so a single w_0 argument would not type; *not* the rule literal

11: **if** $\mu_{\mathrm{den}}(\alpha, v, w^*, \sigma) \leq 0$ **then** CT$[v, \alpha] \leftarrow$ MC; **continue** ▷ one probe; order OOV$\to$AM$\to$MC

12: $U_{\min} \leftarrow$ GRAINFRONTIER$(\alpha, v, w^*, \mathrm{ctx})$ ▷ bottom-up over $\uparrow g_0$: stop at Legal, prune its up-set (Cor. C.4)

13: **if** $U_{\min} = \emptyset$ **then** CT$[v, \alpha] \leftarrow$ DB **else** *Front* $\cup= \{\langle \alpha, w^*, v, g, \mu^*(\cdot, \mathrm{ctx})\rangle : g \in U_{\min}\}$

14: **end for**

15: **end for**

16: **if** *Front* $= \emptyset$ **then return** REFUSE$\langle$CT, Cert(CT, probes)$\rangle$

17: $\hat{b} \leftarrow$ SELECTMIN$_{\prec_{\mathrm{det}}}$(*Front*); **return** ANSWER$\langle$EMITPLAN$(\hat{b})$, Cert$(\hat{b}$, probes$)\rangle$

every pruned node is either illegal or sits above some member of $U_{\min}$ in $\preceq_D$. The traversal visits at most $|\uparrow g_0| \leq W_\Gamma(h_\Gamma + 1)$ nodes, for h_Γ, W_Γ the height and width of the grain lattice.

Lines 8–12 realize the report order OOV $\to$ AM $\to$ MC, DB following all three binding-layer reasons. Line 10 is load-bearing for soundness: it evaluates the *full* rule predicate, clauses (i)–(iv) of Definition 2.4, not the rule literal — the binding row of Example 3.3 registers `same_valid_time_window` with the same literal as its trivially compatible siblings, yet on the single-day window its two anchors realise 33 rows against none, so a literal-only check reaches the frontier and answers the request its clause-(iv) audit refuses, falsifying Theorems 4.1 and 4.2 directly. Every probe is an ordinary aggregate over the bound population. The five enterprise development-track compilers were degenerate instances of the algorithm — a single-point grain lattice, folded disclosure, one version; the public-track compiler runs it in full, with non-trivial grain lattices and versioned masks and thresholds in the four policy-registering domains and two committed versions in every domain.

C.4 Complexity

Theorem C.5 (complexity)**.** *The number of data probes is at most $|V_{\mathrm{adm}}||A_{\mathrm{adm}}|(2 + |\mathcal{D}_v| \min(|\uparrow g_0|, W_\Gamma(h_\Gamma + 1)))$, each probe one aggregate query over the bound population, and the symbolic work is $O(|V_{\mathrm{adm}}||A_{\mathrm{adm}}|[W_\Gamma(h_\Gamma+1)(|\mathcal{D}_v||\mathrm{Att}(q)| + c_\vee) + |E_{\mathrm{val}}|])$ with $c_\vee$ the mask-join cost and E_{val} the validity endpoint set. MLR is thus polynomial in the declared lattice and registry parameters; when $\uparrow g_0$ is a chain the per-slice grain probes drop to $O(|\mathcal{D}_v|\lceil \log_2(h_\Gamma + 2)\rceil)$ by threshold bisection,* Legal *being monotone along a chain.*

The bound rests on three assumptions we state rather than bury: extensions are finite, so probes evaluate; windows and anchor bands live in the calendar algebra, so trimming and emptiness testing are computable; and the denominator measure is non-negative, so Lemma C.2(b) holds. Drop any one and termination or the bound goes with it.

D Full Proofs

Standing assumptions (A1)–(A6) are those of §3: finite enumerable extensions with bounded per-fact predicate cost (A1); computable valid-time assignments and window operations over a finite meet-semilattice $\mathbb{G}$ (A2); a finite totally ordered, append-only version set with a computable strictly monotone commit map (A3); decidable binding-rule predicates (A4); per-role window derivations presented as total functions into $\mathcal{W}\uplus\{\uparrow\}$ (A5); and a discrete, effectively presented $\mathbb{T}$ with decidable order (A6). Throughout, compilation and verification read one (G_v, D) snapshot; cross-snapshot concurrency belongs to the versioned substrate.

D.1 Semantics (§2)

Lemma 2.2 (version-axis stability). *Let $\mathcal{G}'$ extend $\mathcal{G}$ by appending v_{n+1} with $t_{n+1} > t_n$. Then $\mathrm{ver}_{\mathcal{G}'}(T) = \mathrm{ver}_{\mathcal{G}}(T)$ and $[\![q]\!]_{\mathcal{G}',T} = [\![q]\!]_{\mathcal{G},T}$ for every $T < t_{n+1}$ and every query q declared at T.*

Proof. Write $\mathcal{G} = (\{G_{v_i}\}_{i\le n}, \preceq, t)$ with $t_i = t(v_i)$ and $t_1 < \cdots < t_n$. By definition $\mathrm{ver}_{\mathcal{G}}$ is the step function that sends $T \in [t_i, t_{i+1})$ to v_i, with $t_{n+1} = +\infty$, and is undefined below t_1. Appending v_{n+1} at $t_{n+1} > t_n$ replaces the last cell $[t_n, +\infty)$ of that partition by $[t_n, t_{n+1}) \uplus [t_{n+1}, +\infty)$ and leaves the cells $[t_i, t_{i+1})$, $i < n$, and the undefined region $(-\infty, t_1)$ untouched. Hence for every $T < t_{n+1}$ the cell containing T is the same in both partitions and carries the same version label, so $\mathrm{ver}_{\mathcal{G}'}(T) = \mathrm{ver}_{\mathcal{G}}(T) =: v$.

For the bracket, inspect its parameters. $[\![q]\!]_{\mathcal{G},T}$ is determined by $v = \mathrm{ver}(T)$, by the content of G_v, and by D: the candidate assignment $\alpha_{q,v}$ reads A_v, β_v and the override names of q; each leg population reads $[\![\cdot]\!]_v$, $R_v(m)$ and the anchor–window filter; each guard reads only these. Append-only (A3) states that the committed triple $(v, t(v), G_v)$ is neither deleted nor rewritten and that version numbers are not reused, so G_v is literally the same object in $\mathcal{G}$ and $\mathcal{G}'$. Every parameter of the bracket therefore agrees, and so do the values. Finitely many appends compose by induction on the number of appended versions. A rollback is not an exception: it is modeled as a fresh version number whose content may equal a historical G_{v_i}, which is an append. ☐

Proposition 2.7 (well-definedness). *When $\mathsf{Bindable}(q, \mathcal{G}, D)$ holds, $[\![q]\!]_{\mathcal{G},T}$ is determined by $(q, \mathcal{G}, D)$ and is independent of evaluation order.*

Proof. Four steps, each removing one source of choice. (i) $v = \mathrm{ver}(T)$ is a value of a function of T fixed by the commit map, and (b1) guarantees it is defined. (ii) $\alpha_{q,v}$ is a function of (q, v) by construction: each role takes its declared override if there is one and the prescribed anchor of $\beta_v(m)$ (or a_m, atomic case) otherwise, and the window is the value $\omega_r(T)$ of a total presentation (A5). No search occurs, so no tie-breaking is needed. (iii) For each role, $\mathrm{Pop}^r_{v,T}(q)$ is defined by a separating description — membership in $[\![o^r_m]\!]_v$, matching the scope after transfer along ρ's join keys, non-empty intersection of $\mathrm{vt}_{a_r}(x)$ with W_r — over a finite set (A1); a separating description determines a unique subset regardless of the order in which candidates are tested. (iv) $\mu^r_{v,T}(q)$ is a finite sum of rationals over that subset, and finite sums in $\mathbb{Q}$ are commutative and associative, hence order-independent. Finally (b4) makes $\mu^{\mathrm{den}} > 0$, so the quotient is a well-defined rational; in the atomic case no quotient is taken. The delta/multi-period expansion is by role $\times$ period into finitely many single-period queries in the declared order, and the lexicographic rule (declared period order, then guard order) selects a unique value, so single-valuedness is preserved. ☐

Theorem 2.9 (degeneration). *Let $\mathcal{Q}^{\downarrow}$ be the queries satisfying **(H1)** $T \geq t_n$, hence $\mathrm{ver}(T) = v_n$; **(H2)** $\omega_r(T) = \top$ for every role; **(H3)** $\bar{a} = \emptyset$. Then for every $q \in \mathcal{Q}^{\downarrow}$: (1) the evaluation satisfies SVRC, witnessed by v_n; (2) whenever $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q}$ it equals $[\![\sigma]\!]^{1}_{G_{v_n},D}$; (3) each failure class collapses to a non-temporal residue —* AM *to a failure of clause (iv) alone,* OOV *to $V_{v_n}(a) = \emptyset$ for some required anchor, and* MC *to a missing route or a non-positive domain-wide scoped denominator mass.*

Proof. (1) Read-consistency collapse. By (H1), $T \geq t_n$ lies in the last in-effect cell, so $\mathrm{ver}(T) = v_n$ and every governance object the evaluation consults — the metric record $m \in M_{v_n}$, the route $R_{v_n}(m)$, the anchors named by $\beta_{v_n}(m)$, the binding row itself — is read from G_{v_n}. The leg populations refer to $[\![\cdot]\!]_{v_n}$ and to D and to nothing else. Taking $v = v_n$ as the witness discharges the definition of single-version read consistency.

(2) Value collapse. Fix a role r. Under (H2) its window is $\top$. For a fact x with $\mathrm{vt}_{a_r}(x) \neq \emptyset$ we have $\mathrm{vt}_{a_r}(x) \cap \top = \mathrm{vt}_{a_r}(x) \neq \emptyset$, so the anchor–window filter of the bi-temporal semantics removes no such fact; and for a zero-length row, $\mathrm{vt}_{a_r}(x) = \emptyset$ meets every window emptily, so it is removed on the bi-temporal side and is likewise excluded on the single-version side, whose population retains the anchor-validity restriction $\mathrm{vt}_{a_r}(x) \neq \emptyset$. The two populations therefore contain exactly the same facts; since f^r_m and the scope predicate $\models_\rho s$ are the same in both readings, the masses agree and so does their quotient. On our prototype dimension the retained population is $281 = 392 - 111$, the 111 being the zero-length rows, which is why the restriction is stated rather than dropped. Whenever the bi-temporal value is a rational — i.e. Bindable holds — it therefore equals $[\![\sigma]\!]^{1}_{G_{v_n},D}$.

(3) Failure-class collapse. AM*:* (H3) means no override, so α_{q,v_n} takes the prescribed anchors and clause (i) of svw holds; (H2) gives $W_{\mathrm{n}} = W_{\mathrm{d}} = \top$, so clause (ii) holds; and $\top \in \mathcal{W}_g$ for every $g \in \mathbb{G}$, in particular for g^{cmp}, so clause (iii) holds. Hence $\neg P_{\mathrm{svw}}(\alpha_{q,v_n})$ reduces to the failure of clause (iv) alone. Clause (iv) is a call into the declared check library on $(a^*_{\mathrm{n}}, a^*_{\mathrm{d}})$ evaluated over (G_{v_n}, D); it mentions no T, so what remains of AM is a non-temporal anchor-pair audit. OOV*:* (H1) excludes the undefined-ver disjunct and (H2) the unresolvable-window disjunct; the coverage disjunct becomes $\top \cap \mathrm{Cov}_{v_n}(a) = \mathrm{Cov}_{v_n}(a) = \emptyset$. Under either coverage mode $\mathrm{Cov}_{v_n}(a) = \emptyset$ iff the marking set $V_{v_n}(a)$ is empty — $\mathrm{hull}(\emptyset) = \emptyset$ and $\mathrm{hull}(S) \neq \emptyset$ for $S \neq \emptyset$ — so the reduction is independent of cm_a. MC*:* mode (i) is already non-temporal (a lookup in M_{v_n}, R_{v_n}, β_{v_n}), and under (H2) mode (ii) probes μ^{den} over the whole scoped population, i.e. the domain-wide scoped denominator mass. ☐

Corollary (shadow of version-serializable reads; the body carries this as the prose following Theorem 2.9's pointer, not as a numbered display). *Every* Bindable *evaluation satisfies SVRC, witnessed by* $\mathrm{ver}(T)$*; and under the hypotheses of Theorem 2.9 the entire temporal content of the semantics collapses to SVRC.*

Proof. The first clause reuses the first half of part (1) above without (H1): Bindable includes (b1), so $v = \mathrm{ver}(T)$ is defined, and all governance reads and leg populations are taken at that v under the snapshot assumption. The second clause is parts (2) and (3): both the rational values and the refusal classes coincide with the non-temporal single-version reading, so the two degrees of freedom that vary with T — the anchor–window filter on the valid-time axis and the version selection on the version axis — are constant on $\mathcal{Q}^{\downarrow}$. We note again the declared strength: under the snapshot assumption the first clause is a construction-level invariant (the semantics does not manufacture cross-version reads), not a re-proof of a concurrency guarantee under version advance, whose statement needs a per-datum read-version function that a single-snapshot D cannot express. ☐

Remark (realizability on SQL:2011; sketch). A finite $\mathcal{G}$ satisfying (A1)–(A3) encodes into a SQL:2011 bi-temporal database so that the evaluation of Definition 2.6 is realized by generated `AS`

`OF` queries: governance tables are stored system-versioned with the commit map as system time, so that $\mathrm{ver}(T)$ becomes `FOR SYSTEM_TIME AS OF` T; interval anchors become application-time periods and snapshot anchors ordinary date columns; population filters become generated window predicates. Two gaps keep this a sketch: window-algebra operations and granularity alignment exceed the standard's temporal clauses and must be discharged in the compiler, and refusal values with their witnesses have no counterpart in the standard's value space. A faithfulness proof (the encoding preserves $[\![\cdot]\!]$) is an open obligation recorded in §6.

D.2 Binding Compilation (§3)

Theorem 3.1 (coverage partition, single period) — full proof. *For every well-formed single-period query q, under (A1)–(A6), the cases (a)* $\mathsf{Bindable}$*; (b)* OOV*; (c)* $\neg\mathsf{OOV} \wedge \mathsf{AM}$*; (d)* $\neg\mathsf{OOV} \wedge \neg\mathsf{AM} \wedge \mathsf{MC}$ *are pairwise disjoint and jointly exhaustive, and* $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q} \iff$ *(a),* $= \bot_{\mathsf{OOV}} \iff$ *(b),* $= \bot_{\mathsf{AM}} \iff$ *(c),* $= \bot_{\mathsf{MC}} \iff$ *(d).*

Proof. Disjointness. (b), (c), (d) are pairwise disjoint by the guard chain: (c) carries $\neg\mathsf{OOV}$ and (d) carries $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$. It remains to separate (a) from each. Obligation (b1) of $\mathsf{Bindable}$ *implies the negation of each disjunct* of OOV — $\mathrm{ver}(T)$ defined, every $\omega_r(T)$ defined, and $W_r \cap \mathrm{Cov}_v(a_r) \neq \emptyset$ for every role. The third conjunct is *per role* and therefore strictly stronger than the *pair-level* coverage disjunct of Definition 2.8(1), which on a paired ratio fires only when *both* leg realisations are empty; the implication runs in the direction disjointness needs, and only in that direction — so (a) implies $\neg$(b). Obligation (b3) asserts $P_{\mathrm{rule}}(\alpha_{q,v}; G_v, D)$ whenever $\beta_v(m)$ is defined, which is the negation of AM, so (a) implies $\neg$(c). Obligation (b2) negates mode (i) of MC ($m \in M_v$, and for a ratio metric $R_v(m)$ defined and recomputable and $\beta_v(m)$ defined) and (b4) negates mode (ii) ($\mu^{\mathrm{den}}_{v,T}(q) > 0$), so (a) implies $\neg$(d).

Exhaustiveness. Suppose (b), (c) and (d) all fail; we derive (a). Failure of (b) is $\neg\mathsf{OOV}$, which gives $\mathrm{ver}(T)$ defined and every $\omega_r(T)$ defined at once, and, on the coverage disjunct, exactly this: an unpaired (atom) role has $W_r \cap \mathrm{Cov}_v(a_r) \neq \emptyset$ — there the disjunct is already per role and the third conjunct of (b1) follows immediately — while a paired ratio has *not both* leg realisations empty.

The per-leg strengthening that (b1) needs on a ratio is *not* available from $\neg\mathsf{OOV}$ alone; it comes from $\neg\mathsf{AM}$ together with the well-formedness invariant on β_v (§2). Suppose, for contradiction, that exactly one leg had $W_\ell \cap \mathrm{Cov}_v(a_\ell) = \emptyset$. The pair's window realisation is then *mixed*: one leg realises rows in the requested window and the other realises none. By the invariant, a row pairing two *distinct* prescribed anchors declares a realisation-sensitive clause-(iv) member, and such a member is false on a mixed realisation; clause (iv) of Definition 2.4 therefore fails, so $P_{\mathrm{rule}}(\alpha_{q,v}; G_v, D)$ fails, so AM holds — which contradicts the failure of (c). (A row pairing one anchor with itself cannot produce a mixed realisation at all: the two legs share vt_a and the same window, so their realisation sets coincide.) Hence no leg's intersection is empty, which is the coverage half of (b1)'s per-role conjunct.

Realisation-sensitive clause-(iv) members. A member P of the clause-(iv) check library is *realisation-sensitive* when, on a binding row whose two prescribed anchors are distinct, (i) P implies that the two legs realise the *same* set over the compared window, and hence (ii) P is false whenever exactly one of the two realisation sets is empty. The marking-domain symmetric-difference audit is realisation-sensitive: its discriminant set is non-empty precisely when the realisations differ, and a mixed realisation differs on every date the non-empty leg realises. Interval containment alignment is realisation-sensitive for the same reason. The *trivially true* member is not, which is why the invariant forbids it on a distinct-anchor row. Two conventions make (i) checkable rather than rhetorical: the realisation set of a leg is taken at *day* granularity, and it is computed under the leg anchor's own declared coverage mode — the hull/strict_member distinction of §2 — which is exactly the accounting the compiler and the verifier both implement, so the certificate's clause-(iv) payload replays against the same sets the proof reasons about.

Because $\neg\mathsf{OOV}$ holds, we are inside the guard domain of AM, and Proposition 3.2 — whose proof nowhere uses the present theorem, so the appeal is not circular — makes AM decidable and therefore two-valued there. Failure of (c) then means $\neg\mathsf{AM}$, i.e. either $\beta_v(m)$ is undefined or $P_{\mathrm{rule}}(\alpha_{q,v})$ holds; both disjuncts imply (b3), which is a conditional with $\beta_v(m)$ defined as its antecedent. $\neg\mathsf{AM}$ also supplies the registration half of (b1)'s per-role conjunct: by Definition 2.3, an override resolving into RefA_v classifies the query $\perp_{\mathsf{AM}}$, so under $\neg\mathsf{AM}$ every a_r lies in A_v — which is why the two halves of that conjunct together are exactly what disjointness needs and no more. Likewise $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$ puts us inside the guard domain of MC, where Proposition 3.2 makes it two-valued, and failure of (d) gives $\neg\mathsf{MC}$, i.e. the negation of mode (i), which is (b2), together with the negation of mode (ii), which is (b4). The four obligations hold, so Bindable holds. The vacuity convention is what keeps this argument sound: a disjunct mentioning an object that is not nameable at v counts as false in OOV, so a missing object never enters through the temporal door — it is caught by AM(i) if it is an unregistered override name and by MC(i) otherwise.

Agreement with the semantics. Definition 2.6 is defined by cases: under Bindable it returns a rational (well defined by Proposition 2.7), and otherwise $\perp_c$ for the first guard c that holds, the enumeration order being OOV, AM, MC. Since the four cases are disjoint and exhaustive, each of the four biconditionals follows by reading the case definition in the direction of the case that holds. □

Proposition 3.2 (decidability and cost). *Under (A1)–(A6),* OOV*, and* AM *and* MC *on their respective guard domains, are decidable, with the constructive procedures and costs listed in §3.*

Proof. We show that each predicate is a finite Boolean combination of computable tests over finite structures, and read the cost off the procedure.

OOV*.* The first disjunct asks whether $\mathrm{ver}(T)$ is defined and, if so, which version it is. By (A3) the commit instants form a finite strictly increasing sequence $t_1 < \cdots < t_n$; by (A6) the comparisons $T < t_i$ are decidable; so binary search locates the unique i with $T \in [t_i, t_{i+1})$, or reports $T < t_1$, in $O(\log n)$ comparisons. The second disjunct is settled by evaluating $\omega_r(T)$, which by (A5) is presented as a total function into $\mathcal{W} \uplus \{\uparrow\}$; this is where (A5) is load-bearing, since from partial computability alone deciding $\omega_r(T) =\uparrow$ would be an instance of the halting problem and the disjunct would literally be undecidable. The third disjunct needs $\mathrm{Cov}_v(a_r)$: one scan of $[\![o_{a_r}]\!]_v$ taking min and max of the marking set for cm = hull, or a sort-and-merge of the marking intervals into a normal form union for cm = strict_member, with zero-length rows contributing $\emptyset$ and being skipped; both are computable by (A1), (A2). Intersecting with the normal form of W_r and testing emptiness is a merge over two finite interval unions, computable because request windows are capped to the calendar algebra. Summing: $O(\log n) + \sum_r O(|[\![o_{a_r}]\!]| \log |[\![o_{a_r}]\!]| + |W_r|)$.

AM*.* On its guard domain $\beta_v(m)$ is either defined or not, a lookup. If defined, P_{svw} is the conjunction of four clauses: (i) identifier comparison of the assigned and prescribed anchors; (ii) equality of window normal forms; (iii) for each maximal interval of W, a granule-boundary alignment test against g^{cmp}, computable because the calendar functions are; (iv) a call into the declared check library, which by (A4) denotes a decidable predicate — constant for the trivial variant, one scan of both anchor objects for the symmetric-difference audit, one merge for interval containment. Cost $O(|W| \log |W|) + O(|[\![o_{a^*_{\mathrm{n}}}]\!]| + |[\![o_{a^*_{\mathrm{d}}}]\!]|)$.

MC*.* Mode (i) is a key lookup in M_v and R_v plus a definedness test on β_v. Mode (ii) is one scan over $[\![o^{\mathrm{d}}_m]\!]_v$ accumulating f^{d}_m on facts passing the scope and anchor–window tests, each of bounded cost by (A1); the accumulated value is then compared with 0. Cost $O(|[\![o_m^{\mathrm{d}}]\!]|)$.

Decidability is preserved under finite Boolean combination, so each guard is decidable, hence two-valued on its guard domain — the fact consumed by Theorem 3.1. Finally, each procedure emits, as a by-product, the object the certificate needs: the located version, the failing clause label with its payload, or the probe query together with the assertion about its observed value. The decision procedure and the witness generator are the same object. □

Corollary 3.4 (per-period classification). *A delta or multi-period query expands by role × period into a finite family $\{q^{(j)}\}_j$ in the declared period order; Theorem 3.1 applies to each $q^{(j)}$, and the value of the whole query is fixed lexicographically.*

Proof. Each $q^{(j)}$ is a single-period query: its role set is $\{\mathrm{num}, \mathrm{den}\}$ or $\{\mathrm{atom}\}$, its windows are the declared windows of period j, and every other component is inherited from q. Theorem 3.1 applies verbatim. The value rule is Definition 2.6's lexicographic clause: if every period falls in case (a), the declared arithmetic combination of finitely many rationals is a rational; otherwise the first failing period j_0 in declaration order contributes its $\perp$. Single-valuedness holds because the period order is total and, within a period, the guard order is total.

Two side conditions are needed for this to be well posed, and both are forced by instances rather than chosen for tidiness. First, the guard predicates and P_{svw} must be evaluated on the single-period restriction $\alpha_{q,v}|_{R(q^{(j)})}$: the clauses of svw are defined on the two roles $\{\mathrm{num}, \mathrm{den}\}$, and the four-leg assignment of a two-period delta is not even well typed for them. Second, the existential quantifier over roles inside OOV must not range across periods. The instance is the one the body states after Corollary 3.4: `ruling_intensity` on the `card_games` store with the delta window pair [2017-02, 2021-06] — the two cells of Figure 2(b, d). Period 1 (2017-02) lies inside both anchors' coverage hulls and has zero same-window set releases against 2,161 rulings, so Theorem 3.1 classifies it $\perp_{\mathsf{MC}}$ by mode (ii); period 2 (2021-06) misses both coverage domains and satisfies the raw OOV condition. A cross-period quantifier would report $\perp_{\mathsf{OOV}}$ for the whole query, contradicting the lexicographic rule, which reports period 1's $\perp_{\mathsf{MC}}$ — and the latter is what our compilers and the certificate's first-failing-slice convention produce. □

The live guard-order collision (Example 3.3, in full). The three raw conditions of Definition 2.8 can hold at once — a window can miss a coverage domain *and* violate the binding rule — and the chain resolves the class to the first that holds, so where each guard's jurisdiction *begins* is itself part of the semantics. The suite's live collision is Example 3.3 (`EF2-Q6`): the numerator leg binds to the match-day anchor under hull coverage, the denominator leg to the weekly rating-snapshot anchor, a date-set anchor under strict_member whose marking dates surround the requested single day (2013-11-29, 2013-12-06) without containing it. On that one-day window the snapshot leg's strict_member coverage is empty, so a leg-at-a-time reading of OOV fires before the clause-(iv) audit ever runs and returns $\perp_{\mathsf{OOV}}$ — same witness replay, wrong reason class. The reading frozen before any model was called judges coverage vacuity for a pair-bound role at the *pair* level: both realisations empty is OOV, mixed realisation — here 33 match rows against no snapshot row, differing on exactly the discriminant date the certificate carries — is clause (iv)'s jurisdiction, and gold is $\perp_{\mathsf{AM}}$(iv). The development track recorded the same failure shape from the opposite direction: a ratio metric whose numerator anchor was overridden to another registered anchor, evaluated at an as-of instant outside every relevant coverage domain, where the raw OOV disjunct and the raw AM condition (clause (i), then discriminated by a marking-set symmetric difference of 57, an auxiliary audit the artifact still re-computes) held together; the frozen order $\mathsf{OOV} \to \mathsf{AM} \to \mathsf{MC}$ returns $\perp_{\mathsf{OOV}}$ there, while a rule-audit-first order returns $\perp_{\mathsf{AM}}$. In both instances the failure mode is a certificate whose *reason class* is wrong although its witness still replays against (G_v, D) — exactly the failure mode check V6b(0) of §5.1 was introduced to catch. The report order and the guard-domain adjudications are therefore part of the semantics, not an implementation convention, and a refusal certificate must replay the negation of every guard preceding its own class rather than merely its own definition body.

Remark (a double failure that does *not* witness the guard order). Not every simultaneously-true pair of raw conditions establishes the necessity of the report order. The instructive instance, from the development seeds: a metric registered as reference-only — its route declares no destination caliber, and the seed note records that without a denominator there is nothing on which to hang a valid-time anchor — carries no binding row at all. Under the vacuity convention its OOV coverage disjunct is false (the anchor it would mention is not nameable), its AM guard is false because $\beta_v(m)$ is undefined, and MC(i) holds — for *every* T, including instants far outside any coverage domain. Its value is $\perp_{\mathsf{MC}}$, so an older caliber-gate-first compiler that emits an MC verdict here is not wrong, and the instance cannot witness the necessity of the ordering. The public suite's MC(i) question (`CARD-Q6`, a reference-only route with `dst_caliber=none`) is the contrast case in the other direction: there both binding rows exist and both anchors are nameable, so $\neg\mathsf{OOV}$ and $\neg\mathsf{AM}$ hold genuinely rather than by vacuity, the failure is single, metadata-level, and decidable without a data probe — which is why the suite's live order-witness is Example 3.3 and not the MC family. A live counterexample must be one in which the OOV-side reading and the raw AM condition hold together, which is what the two instances above exhibit.

D.3 Maximal Legal Rewriting (§4)

Lemma C.2 (window structure and mass monotonicity). *Fix a slice (α, v) and let $w^* = w_0 \sqcap \mathrm{Env}(\alpha, v)$. Then (a) every faithful, bindable rewrite in the slice has window $w \subseteq w^*$, so w^* is the greatest element of the slice's faithful bindable windows when that set is non-empty; and (b) if the denominator measure is non-negative, $w \subseteq w' \Rightarrow \mu_{\mathrm{den}}(w) \leq \mu_{\mathrm{den}}(w')$, whence $\mu_{\mathrm{den}}(w^*) = 0$ propagates to every faithful window of the slice.*

Proof. (a) Faithfulness requires $w \subseteq w_0$ componentwise: a rewrite may trim the requested window but never widen it, since widening would answer a question about instants the requester did not ask about. The second inclusion is *structural* — a property of the candidate space, not a consequence of bindability. Errata (this edition). An earlier printing derived it from "bindability requires, per leg ℓ, that the leg window lie inside the anchor's coverage envelope". That reading is wrong: obligation (b1) of Definition 2.5 asks only for a *non-empty intersection* $w_\ell \cap \mathrm{Cov}_v(a_\ell) \neq \emptyset$, which does not entail containment — a window straddling the end of an anchor's coverage is bindable and not contained. The correct derivation is coordinate (S-W) of the structural candidate space $X(q, T)$: X offers exactly *one* window candidate per slice, namely $w^* = w_0 \sqcap \mathrm{Env}(\alpha, v)$, where $\mathrm{Env}(\alpha, v)$ is the componentwise coverage envelope — $V_v(a_\ell)$ for a $\mathsf{strict\_member}$ leg, $\mathrm{hull}(V_v(a_\ell))$ for a hull leg, empty days inside the span being lawful zeros — and no data-driven shrinking is performed. Every point the compiler considers therefore has $w \subseteq w^*$ by construction, and L is defined over the closures of points of X, so every member of L in this slice does too. When the set of faithful bindable windows is non-empty, w^* itself belongs to it (it satisfies both constraints by construction), so it is the greatest element. The lemma, and everything downstream of it, is accordingly a statement *relative to X*; widening the window space is the window-bargaining boundary §4 leaves open.

(b) $\mu_{\mathrm{den}}(w) = \sum\{f^{\mathrm{d}}_m(x) : x \in \mathrm{Pop}^{\mathrm{d}}(w)\}$ and $w \subseteq w'$ implies $\mathrm{Pop}^{\mathrm{d}}(w) \subseteq \mathrm{Pop}^{\mathrm{d}}(w')$, because the only clause of the population that mentions the window is the intersection test $\mathrm{vt}_a(x) \cap w \neq \emptyset$, which is monotone in w. A sum of non-negative terms is monotone under set containment, giving the inequality. Taking $w' = w^*$ and using (a): if $\mu_{\text{den}}(w^*) = 0$ then $\mu_{\text{den}}(w) = 0$ for every faithful window w of the slice.

Non-negativity is load-bearing and we flag its failure rather than hide it: with refund-reversal columns the denominator measure takes negative values, μ_{den} is not monotone along window containment, and the zero-propagation argument fails. The algorithm's widened test $\mu_{\text{den}} \leq 0$ remains an honest guard in that case — it still refuses a slice whose mass is non-positive — but the pruning justification does not extend to it. A second caveat of vocabulary: a row-count probe is equivalent to the mass probe only when $f_m^{\text{d}} \equiv 1$; a window containing exactly one row with `sales_qty`$= 0$ has row count 1 and mass 0. ☐

Lemma C.3 (grain-frontier structure), part (a) — support monotonicity along grain. *For fixed (α, v, w^*) and policy π, let $\mathrm{SUPPMIN}_\pi(g)$ be the minimum over non-empty output cells of the number of distinct γ_π-entities covered. Then $\{g \in \uparrow g_0 : \mathrm{SUPPMIN}_\pi(g) \geq k_\pi\}$ is an up-set of $(\uparrow g_0, \preceq_\Gamma)$.*

Proof. Let $g \preceq_\Gamma g'$ with $\mathrm{SUPPMIN}_\pi(g) \geq k_\pi$, and let c' be a non-empty output cell at g'. Since g' is coarser, c' is the union of the g-cells refining it, and because c' is non-empty at least one such g-cell c is non-empty. The set of distinct γ_π-entities covered by c' contains that covered by c, so $|\mathrm{ent}(c')| \geq |\mathrm{ent}(c)| \geq \mathrm{SUPPMIN}_\pi(g) \geq k_\pi$. Taking the minimum over all non-empty c' gives $\mathrm{SUPPMIN}_\pi(g') \geq k_\pi$. The argument uses set containment only — never additivity of support — so it is robust to entities repeated across cells. ☐

Lemma C.3, part (b) — least mask closure. *For a fixed binding point and context, the legal mask assignments form an up-set in the per-attribute order with least element $\mu^*(b, \mathrm{ctx})(a) = \bigvee_{\mathbb{M}}\{\mu_\pi : \pi \text{ touched},\ a \in \bar{P}_\pi \cap S_{\text{raw}},\ \mu_\pi \neq \bot,\ \text{clause applies}\}$ (empty join $= \mathrm{id}$).*

Proof. Every mask clause of Definition C.1 is a per-attribute lower-bound constraint of the form "a is presented at $\succeq_{\mathbb{M}} \mu_\pi$". The solution set of one such constraint is an up-set of the finite lattice $\mathbb{M}$; the solution set of the whole family is the intersection of up-sets, hence an up-set, and is non-empty because $\top_{\mathbb{M}}$ satisfies every lower bound. In a finite lattice the least common upper bound of a family of lower bounds is their join, and joins are computed componentwise across attributes because the constraints do not couple attributes. So μ^* as displayed is legal by construction (each clause is satisfied since $\mu^*(a) \succeq_{\mathbb{M}} \mu_\pi$) and least by the universal property of the join. Consequently the mask is a derived coordinate: it is a function of the binding point and the context, and adding it to the search space would only re-derive it. ☐

Lemma C.3, part (c) — up-set structure of per-grain legality. *Fix a slice with $\mu_{\text{den}}(w^*) > 0$. Under coarsening compatibility, $U = \{g \in \uparrow g_0 : \mathrm{Legal}(g)\}$ is an up-set, and $\mathrm{cl}(\langle \alpha, w^*, v, g \rangle) \in L \iff g \in U$.*

Proof. Let $g \preceq_\Gamma g'$ with $g \in U$. By coarsening compatibility (i), $\mathrm{touched}(g') \subseteq \mathrm{touched}(g)$; and whether a policy's D2 branch applies depends on $(\mathrm{sc}_\pi, \mathrm{ctx})$ only, not on the grain, so $\mathrm{touched}_{\mathrm{D2}}(g', \mathrm{ctx}) \subseteq \mathrm{touched}_{\mathrm{D2}}(g, \mathrm{ctx})$.

Grain clause. For $\pi \in \mathrm{touched}_{\mathrm{D2}}(g', \mathrm{ctx})$ we have $\pi \in \mathrm{touched}_{\mathrm{D2}}(g, \mathrm{ctx})$, hence $\gamma_\pi \preceq_\Gamma g \preceq_\Gamma g'$ by transitivity.

Small-cell clause. For the same π, part (a) gives $\mathrm{SUPPMIN}_\pi(g') \geq \mathrm{SUPPMIN}_\pi(g) \geq k_\pi$.

Non-empty disclosure. The set of mask obligations at g' is $\{(\pi, a) : \pi \in \mathrm{touched}(g'),\ a \in \bar{P}_\pi \cap S_{\text{raw}}(\mathrm{cl}(g'))\}$, and we claim it is a subset of the corresponding set at g. Clause (i) of coarsening compatibility handles the first component and clause (ii) the second; and clause (i) alone does *not* suffice, which is why (ii) is stated: rolling up can switch the group-by key and bridge a new column into raw presentation, so an already-touched policy can acquire at g', through its carrying closure, an obligation it did not have at g. With both clauses, the per-attribute join at g' is over a subset of the terms joined at g, hence $\mu^*_{g'}(a) \preceq_{\mathbb{M}} \mu^*_g(a)$ for every a. So if some rigidly requested presentation attribute were sent to $\top_{\mathbb{M}}$ at g', the same attribute would be at $\top_{\mathbb{M}}$ at g as well, contradicting $g \in U$.

All three clauses survive, so $g' \in U$ and U is an up-set.

Equivalence with membership in L. The other membership conditions of L have already been fixed by the slice's prefix checks: the scope, window, version, rule predicate and bindability do not depend on g. For a policy whose role branch the context satisfies (D1), the grain and small-cell clauses are waived by definition, its row-ownership predicate is folded into the scope predicate by the closure cl, and its mask obligation is carried by μ^*; what remains quantifies over $\text{touched}_{\text{D2}}$ only, which is exactly $\text{Legal}(g)$ together with the legality of the closure mask from part (b). Hence $\text{cl}(\langle \alpha, w^*, v, g\rangle) \in L \iff g \in U$.

The restriction to $\text{touched}_{\text{D2}}$ is necessary, not cosmetic: both governed domains contain policy rows carrying a role scope predicate *and* a grain floor with a threshold. When the context satisfies the role branch, a query over the requester's own rows is legal below that floor. An unrestricted quantification would declare it illegal, roll the frontier up, and falsify the maximality and refusal-completeness theorems on that branch. ☐

Corollary C.4 (pruning correctness). *Traversing $\uparrow g_0$ bottom-up along covering edges, stopping at the first grain satisfying* Legal *and pruning its up-set, yields the set $U_{\min}$ of minimal elements of U; every pruned node is either illegal or sits above some member of $U_{\min}$ in $\preceq_D$. The traversal visits at most $|\uparrow g_0| \le W_\Gamma(h_\Gamma + 1)$ nodes.*

Proof. An element of an up-set is minimal iff it belongs to U and none of its $\preceq_\Gamma$-predecessors does. A bottom-up traversal along covering edges visits every predecessor of a node before the node itself, so the first node at which Legal holds on any given chain has no finer member of U below it and is therefore minimal; conversely every minimal element is reached, because a node is expanded to its covers exactly when it fails Legal, so no minimal element can be skipped except by being pruned, and pruning only removes strictly coarser nodes. For a pruned $g'' \succeq_\Gamma g$ with $g \in U_{\min}$: if $g'' \notin U$ it is illegal and irrelevant; if $g'' \in U$ then $\text{cl}(g'') \preceq_D \text{cl}(g)$ — same slice, same window w^*, coarser grain — so it is dominated and cannot be maximal. Hence discarding it loses no maximal element. For the node bound, Mirsky's theorem partitions a finite poset of height h_Γ into at most $h_\Gamma + 1$ antichains, each of size at most the width W_Γ, so $|\uparrow g_0| \le W_\Gamma(h_\Gamma + 1)$. ☐

Theorem 4.1(a) (soundness). *If MLR-Compile returns* ANSWER*, then $\text{cl}(\hat{b}, \text{ctx}) \in L(q, T, \mathcal{G}, \text{ctx}, I)$ and the emitted plan is point-in-time correct in the sense of Definition 2.6.*

Proof. A binding point enters *Front* only after passing every step of the slice loop, and each step establishes one membership condition of L.

Version. The version step puts $v \in V_{\text{adm}}$, i.e. v is compatible with the pin p and with the rigid commitments κ; when unpinned, V_{adm} resolves diagonally to the version in effect at T.

Anchor. The binding step puts $\alpha \in A_{\text{adm}}(m, v)$: a registered binding compatible with κ. No unregistered anchor is ever introduced, so faithfulness on the anchor coordinate is immediate.

Window. The window step sets $w^* = w_0 \sqcap \text{Env}(\alpha, v)$ and rejects the slice when $w^* = \emptyset$ or a rigid strict_member leg escapes its marking set. Faithfulness ($w^* \subseteq w_0$) holds by construction. For a rigid strict_member leg the surviving window equals the committed one; for a hull leg the committed value is read with the hull-edge trimming exemption, so w^*_ℓ equals that committed value and the answer is delivered on the trimmed window, recorded in the certificate and disclosed as an edge trim.

Bindability. The window, rule and mass steps jointly discharge (b1)–(b4) of Definition 2.5: the window step gives the coverage clause under each anchor's declared coverage mode; the rule step evaluates the *full* predicate P_{rule}, clauses (i)–(iv) of Definition 2.4, including window–granularity expressibility and the instance-level anchor-pair audit; the mass step gives $\mu_{\text{den}}(w^*) > 0$; and membership of m, $R_v(m)$ and $\beta_v(m)$ in G_v is what admitted the slice at all. By Theorem 3.1 the point is $\mathsf{Bindable}$, so the plan emitted from it is point-in-time correct.

That the rule step evaluates the full predicate rather than the rule *literal* is load-bearing rather than defensive. The binding row of Example 3.3 registers the literal `same_valid_time_window` over an anchor pair whose realisations differ on the requested day (33 rows against none), and a development-track sibling registers the same literal over a marking-set pair with symmetric difference 57; under a literal-only check either row passes the rule step on a same-window request, reaches the frontier, and is answered — falsifying this theorem and Theorem 4.2 directly.

Disclosure. The frontier step returns $U_{\min} \subseteq U$ and the point is admitted only if $U_{\min} \neq \emptyset$; by Lemma C.3(c) each $g \in U$ makes the closure disclosure-admissible, and by Lemma C.3(b) the mask μ^* is legal. The row-ownership predicate of a role-matched policy is folded into the scope predicate by cl, so the emitted plan reads only rows the context is entitled to.

All membership conditions of L hold for $\text{cl}(\hat{b}, \text{ctx})$. $\square$

Theorem 4.1(b,c) (maximality and determinism). *Every element of Front is $\preceq_D$-maximal in $L/\approx_D$; the returned $\hat{b}$ is one of them; and the returned representative is a function of $\langle q, T, \mathcal{G}, \mathcal{D}, \text{ctx}, I\rangle$.*

Proof. Maximality. Let $\hat{q} = \text{cl}(\langle \alpha, w^*, v, g\rangle) \in$ *Front* and suppose $q'' = \text{cl}(\langle \alpha'', w'', v'', g''\rangle) \in L$ strictly dominates it. Since $\preceq_D$ relates only points sharing an anchor assignment and a version, $\alpha'' = \alpha$ and $v'' = v$. Domination requires $w'' \supseteq w^*$; every member of L in this slice is the closure of a point of $X(q, T)$, whose single structural window candidate for the slice is w^* (coordinate (S-W); Lemma C.2(a)), so $w'' \subseteq w^*$; hence $w'' = w^*$. Maximality is therefore maximality *within* X. Domination then requires $g'' \preceq_\Gamma g$ with $g'' \neq g$. By the equivalence of Lemma C.3(c), $q'' \in L$ gives $g'' \in U$; but g is a minimal element of U by Corollary C.4, so no strictly finer member of U exists — contradiction. The mask coordinate is determined by closure and so contributes no further comparison. Hence no strict dominator exists and $\hat{q}$ is maximal; $\hat{b}$ is selected from *Front*, so it is maximal too.

Determinism. Front is constructed without consulting $\prec_{\text{det}}$: the enumeration orders over versions and over registered bindings change the order of discovery, not the set of discovered points, and the frontier computation is a deterministic function of $(\alpha, v, w^*, \text{ctx}, I)$. All three keys of $\prec_{\text{det}}$ — version order, binding registration ordinal, and a declared linear extension of $\preceq_\Gamma$ — are versioned components of G_v, so $\textsc{SelectMin}_{\prec_{\text{det}}}$ is a function of the model inputs alone; determinism does not depend on file line order or database return order. Re-declaring $\prec_{\text{det}}$ changes which maximal element is returned, and neither the maximal set nor the $\mathsf{ANSWER}/\mathsf{REFUSE}$ verdict, since the emptiness test on *Front* is independent of it. $\square$

Theorem 4.2 (refusal completeness). *MLR-*$\textsc{Compile}$ *returns* REFUSE *iff* $L(q, T, \mathcal{G}, \text{ctx}, I) = \emptyset$*; on refusal* CT *carries a per-slice emptiness witness over* $\{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}, \mathsf{DB}\}$.

Proof. Necessity. If the algorithm returns ANSWER then by Theorem 4.1(a) the returned closure is a member of L, so $L \neq \emptyset$; contrapositively $L = \emptyset$ forces REFUSE. This direction is exactly where the full rule predicate is required: under a literal-only rule check our seeds contain an instance with $L = \emptyset$ on which the algorithm would return ANSWER.

Sufficiency. Suppose the algorithm returns REFUSE. Then every slice (v, α), including the two sentinel rows for an absent version axis and an emptied version commitment, carries a reason in CT, and we show slice by slice that it contributes no member of L.

OOV: either the version axis is absent at T under the diagonal resolution, in which case no version coordinate exists and the binding point space is empty; or the coverage check failed, i.e.

$w^* = \emptyset$ or a rigid strict_member leg escapes its marking set, in which case Lemma C.2(a) says the slice has no faithful bindable window at all.

AM: either the version commitment is emptied (a pin to a non-existent version, or κ incompatible with every resolvable version) — note this shape is *not* OOV, since resolvable versions may exist and it is the commitment that removes them; or κ empties A_{adm}, so no registered binding is compatible with the commitments; or the rigid commitments and the full rule predicate cannot hold together *on the slice's one structural window candidate* w^*. In each case a member of L would have to satisfy faithfulness and the full rule predicate simultaneously, and since L ranges over the closures of points of $X(q,T)$, which offers no window in the slice other than w^*, the slice makes that impossible. The sufficiency direction is thus, like Lemma C.2(a) it rests on, relative to X: it says the slice is unsatisfiable *in the candidate space the compiler searches*, not that no window whatsoever could have been bound.

MC: either m has no registered binding at v, so the slice offers no binding point; or $\mu_{\mathrm{den}}(w^*) \leq 0$, in which case Lemma C.2(b) propagates zero mass to every faithful window of the slice under the non-negativity assumption, so (b4) fails everywhere in it.

DB: GrainFrontier terminated with $U_{\min} = \emptyset$. By the traversal invariant of Corollary C.4, a node is expanded to all its covers whenever it fails Legal, so every member of U is discovered through one of its minimal elements; $U_{\min} = \emptyset$ therefore implies $U = \emptyset$, and Lemma C.3(c) turns that into: no grain in $\uparrow g_0$ yields a member of L in this slice — including the finer, role-branch-legal grains, which $\mathrm{touched}_{\mathrm{D2}}$ keeps inside U rather than discarding.

Taking the union over the finitely many slices gives $L = \emptyset$. The decomposition statement reads off the trigger of each step. Finally, when $\mathcal{D}_v = \emptyset$ nothing is touched, $\mathrm{Legal}(g_0)$ holds vacuously, the frontier returns $\{g_0\}$ in one step, and DB is unreachable, so the theorem reduces to binding-layer completeness. □

Theorem C.5 (complexity). *The number of data probes is at most* $|V_{\mathrm{adm}}| \cdot |A_{\mathrm{adm}}| \cdot (2 + |\mathcal{D}_v| \cdot \min(|\uparrow g_0|, W_\Gamma(h_\Gamma + 1)))$*, each probe one aggregate query over the bound population; the symbolic work is* $O(|V_{\mathrm{adm}}||A_{\mathrm{adm}}|[W_\Gamma(h_\Gamma + 1)(|\mathcal{D}_v||\mathrm{Att}(q)| + c_\vee) + |E_{\mathrm{val}}|])$.

Proof. Count probes per slice. The rule step issues at most one data probe: clauses (i)–(iii) are symbolic, and only clause (iv) may query the instance (the marking-set symmetric difference). The mass step issues exactly one μ_{den} aggregate. The frontier visits at most $\min(|\uparrow g_0|, W_\Gamma(h_\Gamma+1))$ nodes by Corollary C.4 and Mirsky's bound, and at each visited node issues at most one $\mathrm{SUPPMIN}_\pi$ query per touched policy, hence at most $|\mathcal{D}_v|$ of them. Multiplying by the number of slices, at most $|V_{\mathrm{adm}}||A_{\mathrm{adm}}|$, gives the stated product. Symbolically, each visited node performs the grain-clause comparisons and the per-attribute mask join, costing $|\mathcal{D}_v||\mathrm{Att}(q)|$ comparisons and $c_\vee$ join operations, and the envelope trimming scans the validity endpoint set E_{val} once per slice. Summing gives the displayed bound, which is polynomial in the declared lattice and registry parameters.

The bound rests on three assumptions we state rather than bury: the relevant extensions of the instance snapshot are finite, so the probes evaluate; request windows and anchor bands live in the calendar window algebra, so trimming and emptiness testing are computable; and the denominator measure is non-negative, so Lemma C.2(b) holds. Drop any one and termination or the bound goes with it.

Chain refinement. If $\uparrow g_0$ is a chain, Legal is monotone along it by Lemma C.3(c), so the up-set is a threshold set $\{g : g \succeq g^\dagger\}$ or empty. Binary search maintaining the invariant "the lower endpoint is illegal, the upper endpoint is legal or not yet refuted" converges in $\lceil \log_2(h_\Gamma + 2) \rceil$ evaluations of Legal, each costing at most $|\mathcal{D}_v|$ probes, plus one evaluation at the top of the chain to decide emptiness. □

Proposition (one-dimensional degeneration; the substrate boundary). *Collapsing the* *valid-time axis to a single instant, the grain lattice to one point and the policy set to empty turns MLR-Compile into a version selector whose emitted plan reads every leg at one version. The correspondence runs in both directions inside the registered closure only.*

Proof. Under the collapse, W_{cand} is the single rigid window $\{\text{now}\}$ per leg, $\uparrow g_0 = \{g_0\}$, and touched $\equiv \emptyset$ so $\text{Legal}(g_0)$ holds vacuously and DB is unreachable. What remains of the loop is the choice of $v \in V_{\text{adm}}$ and of a registered α, followed by the rule and mass steps; the emitted plan reads every leg at that single v, which is the algorithmic counterpart of Theorem 2.9. Conversely, a single-version read of two tables satisfies the version-equality condition but has a preimage under the collapse only if those tables are *registered* with an anchor and a binding: the anchor coordinate admits registered bindings only (no anchor repair), so an unregistered pair has no preimage and the algorithm refuses. Identifying the governance graph version with a data or materialization version is an additional postulate needed to align the two vocabularies, not a consequence; outside the degenerate setting the two axes are independent, since a graph version pin does not filter the in-effect rows of a type-2 dimension. □

D.4 Certificates (§5)

Witness content per refusal class (table moved from §5.1). Every row of Table 8 is a finite decision replayable on (G_v, D, ctx); the replay *mode* is fixed by a machine-readable declaration, never by free text.

Table 8: Witness content per refusal class.

r	witness w
MC	*(i)* routing lookup (m, κ) showing no recomputable route under the caliber key, reference-only routes included; or *(ii)* denominator probe (m, p, W) with $\text{eval}_D(p)$ in the widened empty set — empty, null, or $\mu_{\text{den}} \leq 0$ — the observed value recorded verbatim.
AM	$(c, \text{payload})$ with the failing clause $c \in \{\text{i–iv}\}$ of Definition 2.4: (i) declared and prescribed anchors with the lookup assertion; (ii) the offending window pair; (iii) the declared comparison granularity, the window and a misaligned boundary point — replayed against $\mathcal{W}_{g^{\text{cmp}}}$, not a single anchor's granularity, which would be a stronger test able to issue a spurious clause-(iii) witness; (iv) a discriminant date in the symmetric difference of the marking sets, under the declared admissibility variant. For the *emptied version commitment* shape — a pin to a non-existent version, or a caliber key compatible with no resolvable version — the payload is $(p, \kappa, V_{\text{res}})$: the pin, the key, and the enumerated resolvable versions the commitment excludes. This is the home of that shape's CT sentinel row (§D.3, sufficiency step): it is *not* OOV, since resolvable versions exist and it is the commitment that removes them, and it replays as one registry lookup. It parallels the three widened well-formedness variants of §5.1 without adding to them — those widen *well-formedness*, this one names a witness payload.
OOV	(a, W_T, π_V): anchor, request window and a coverage profile asserting $W_T \cap \text{Cov}_v(a) = \emptyset$ under the declared coverage mode; degenerate variants cover an unresolvable as-of and an undefined $\text{ver}(T)$.
DB	$(\text{CT}_{\mathsf{DB}}, \Pi_{\text{blk}}, \text{probes})$: blocked slices, triggering policies and the support-probe transcript with its snapshot version, jointly asserting $U_{\min} = \emptyset$.

The checking relation Chk (table moved from §5.1). Table 9 tabulates the ten obligations the body states as running text. V6b(0) and V6c are the two checks a purely definition-body verifier lacks; both were added because forgeries pass without them.

Table 9: The checking relation Chk.

check	obligation
V0	re-derive $\alpha_{q,v}$ from (q, T, G_v) through an independently written window arithmetic and governance-table lookup, and compare role by role; exemptions only by certificate record (machine-readable override, cut-trace-backed $\mathsf{REWRITE}$).
V1	resolve ν to a unique typed commit, check the tables are readable at that version, re-resolve $\mathrm{ver}(T)$, check diagonality; an off-diagonal pin without the marker is rejected.
V2	each anchor exists at v; each window is well typed and compatible with the anchor type.
V3	look the binding up *by identifier* — $(\mathrm{domain}, \mathrm{metric})$ is not a functional key, one seed registering a positive and a negative row for the same metric — read the declared admissibility variant, re-execute the rule on α's windows, and match a three-valued table: answers demand a passing replay, an AM refusal a failing replay whose clause matches the witness, other classes exempt.
V4	every routing edge exists at v with matching join keys and attribution alignment, and the path is continuous.
V5	recompute which policies should be in force from the touched tables and columns and compare with Π; replay grain and mask clauses against $\delta.g, \delta.\mu^*$; evaluate role branches against ctx; re-run small-cell probes on D; on a policy-free domain check the annotation is present, since absence is not passing.
V6a	the emitted text touches only objects certified by α, ρ or reachable along declared derivation edges; every time predicate *denotes* a subset of the certified window (a half-open upper bound may itself lie outside it); the ratio shape aggregates each leg before dividing.
V6b(0)	replay the negation of every guard preceding r in the frozen order: an MC refusal must replay $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$, an AM refusal $\neg\mathsf{OOV}$.
V6b(1)	replay the witness of Table 8 under the declared modes, scoped to the selected first failing slice; other CT rows travel as annotations and their row-by-row replay is a declared non-goal.
V6c	for every answer, and each retained slice of a rewrite, replay the negation of *all* refusal guards: coverage intersection non-empty per role, rule replay passing, route recomputable, and, for ratios, the denominator probe evaluating outside the empty set.

Theorem 5.1 (verification completeness). *Under the snapshot and commit assumptions, (A1)–(A6), and a compiler each of whose certificate fields is derived from facts or probe results over* (G_v, D, ctx) *using the guard order and replay modes declared in* G_v*: every well-formed certificate the compiler emits, degenerate variants included, satisfies* $\mathsf{Chk}(C, G_v, D, \mathrm{ctx}) = \mathsf{ACCEPT}$*.*

Proof. Check by check. V0 re-derives $\alpha_{q,v}$ from (q, T, G_v); the compiler derived the same assignment from the same inputs by Definition 2.3, and the only permitted differences — a declared anchor override, a $\mathsf{REWRITE}$ narrowing — are recorded machine-readably in the certificate and are therefore granted as exemptions. V1 resolves ν: the commit axioms make the typed commit identifier unique and fresh, the governance tables are readable at v by construction, and $\mathrm{ver}(T)$ is re-resolved by the same computable commit map, so diagonality is confirmed or the off-diagonal marker is present by the premise. V2 holds because the compiler only ever assigns anchors from A_v with windows in the calendar algebra typed to the anchor. V3 looks the binding up by identifier, which the certificate carries; the replay re-executes the same decidable predicate on the same windows over the same snapshot, so it reproduces the compiler's outcome, and the three-valued table matches by the premise that the compiler used the declared replay mode. V4 holds since the routing path was read from E_v and its continuity is a syntactic property of the recorded sequence. V5 recomputes the touched set from the same tables and columns, and the premise that Π, g and μ^* were derived by the declared clauses makes the comparison succeed; on a domain with no policy row the required annotation is present by the same premise. V6a holds because the emitted text is instantiated from the certified assignment by a template whose time predicates denote subsets of the certified window and whose ratio shape aggregates before dividing. V6b(0) holds because the compiler reports the first guard that holds in the frozen order, so all preceding guards fail and their negations replay; V6b(1) holds because the witness was produced by the decision procedure of Proposition 3.2 as a by-product of the very test that failed. V6c holds on answers and retained slices because the compiler emitted them only after all four guards were negated. The widened well-formedness variants — a metric with no assignable anchor, an unresolvable as-of, an undefined $\mathrm{ver}(T)$ — are covered because each is a declared degenerate shape of the same fields rather than a missing field, so the corresponding checks read the declared variant instead of failing on absence. □

Theorem 5.2 (verification soundness). *Suppose* $\mathsf{Chk}(C, G_v, D, \mathrm{ctx}) = \mathsf{ACCEPT}$. *(a) If C is a refusal certificate, its reason holds at the coordinate (T, v, D) under the guard semantics of §2 for $r \in \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}\}$ and under the disclosure semantics of §4 for $r = \mathsf{DB}$. (b) If C is an answer certificate whose emitted text belongs to $\mathcal{Q}_{\mathrm{tmpl}}$, that text is point-in-time correct at (T, v); for a rewrite, correctness is relative to the trimmed request.*

Proof. (a) By Theorem 3.1, $[\![q]\!]_{\mathcal{G},T} = \bot_r$ holds exactly when every guard preceding r in the frozen order fails and r's own body holds. V6b(1) establishes the second conjunct by replaying the witness under the declared mode: an MC witness is a routing lookup returning no recomputable route, or a denominator probe whose observed value lies in the widened empty set; an AM witness is a failing clause with its payload, re-executed against the binding looked up by identifier; an OOV witness is a coverage profile asserting empty intersection under the anchor's declared coverage mode, or a degenerate variant for an unresolvable as-of or an undefined $\mathrm{ver}(T)$; a DB witness is the blocked-slice table with its support-probe transcript asserting $U_{\min} = \emptyset$, whose sufficiency for $L = \emptyset$ on that slice is Theorem 4.2. V6b(0) establishes the first conjunct by replaying the negation of each preceding guard. Both are needed: replaying the body alone would give only the weaker second conjunct, and the two come apart on the instance of §3, where a request window outside every coverage domain and a mismatched declared anchor make the raw OOV and AM conditions true together, so a certificate claiming AM carries a witness that replays while its class is wrong. V0 additionally pins the replayed windows to those of q, so a witness computed on some other window cannot stand in for the one the question asked about.

V3/V6c and the well-formedness invariant. The verifier neither *assumes* the invariant nor inherits the declaration that would discharge it. The nine-domain seeds carry no `adm_check_mode` column at all, so a verifier that read the admissibility variant off the row would in fact be reading it off the certificate; V3 therefore re-derives the variant from β_v itself, refusing to replay clause (iv) by its trivial member on any binding row whose two prescribed anchors differ, and V6c replays the guards under that same re-derivation. Clause (iii)'s comparison granularity is re-derived the same way, from the row's registered window granularity rather than from the certificate's `g_cmp`. The invariant is thus a verifier obligation and not only a one-line registration audit over β_v: it says a distinct-anchor row must not discharge clause (iv) trivially, and Chk now enforces it rather than replaying the declaration as given. The nine-domain seeds satisfy it — of the 24 two-leg binding groups (12 metrics × 2 committed versions) exactly four are distinct-anchor, namely `card.ruling_intensity` and `ef2.rating_weighted_win_rate` in both versions, and for those our compilers derive the `window_realization_symdiff` member from the anchor pair rather than reading it off the row — which is exactly what the verifier now re-derives for itself — with the pair-level audit ordered ahead of the per-leg coverage adjudication.

(b) In two layers. The *semantic-coordinate* layer is V0 and V6c: V0 fixes that the emitted text was instantiated with q's own anchor assignment at the certified version, and V6c replays the negation of all four refusal guards at that coordinate — coverage non-vacuous, rule replay passing, route recomputable, denominator probe outside the widened empty set. On a paired ratio the first of those replays is pair-level, exactly as Definition 2.8(1) states it, so it does not by itself deliver

the per-role conjunct of (b1); the same two-step of the exhaustiveness argument above supplies it. $\neg\mathsf{AM}$ replayed by V6c means clause (iv) holds, and on a distinct-anchor row the invariant makes that member realisation-sensitive, so the realisation cannot be mixed; $\neg\mathsf{OOV}$ then rules out both realisations being empty; the two together give a non-empty intersection on each leg. By Theorem 3.1 these four negations are exactly $\mathsf{Bindable}$, so $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q}$ and the governed value at the coordinate is the ratio (or the atomic mass) of Definition 2.6. The *syntactic* layer is V2–V4 and V6a: the anchors exist at v with well typed windows, the binding rule is satisfied, the denominator is attributed along the certified route with matching join keys and attribution alignment, every time predicate of the text denotes a subset of the certified window, and the aggregation shape aggregates each leg before dividing. On the template class $\mathcal{Q}_{\text{tmpl}}$ — each leg a single-block aggregate over a certified object with a window predicate, joined along the certified route, with the ratio recomputed at the end — these syntactic conditions force the evaluated population of each leg to coincide with $\mathrm{Pop}^r_{v,T}(q)$ and the evaluated measure with $\mu^r_{v,T}(q)$, so evaluation agrees with Definition 2.6. For a rewrite, V5 and the cut trace replace the requested window and grain by the delivered ones, and the same argument runs against the trimmed request.

Declared scope. V6a is syntactic containment, not semantic equivalence; equivalence for arbitrary SQL with arithmetic and aggregation is undecidable, so (b) is asserted on $\mathcal{Q}_{\text{tmpl}}$ and not beyond. The template class needs its own formal definition and a "syntax implies semantics on the class" lemma; we record this as an open obligation, and note that a delta template subtracting two rates already sits outside the single-window shape covered here.

Off-diagonal scope. Theorem 5.2 is stated on the *diagonal* $T_{\text{vt}} = T_{\text{ver}} = T$ that Definition 2.3 defaults to. The two-parameter bracket is

$$[\![q]\!]^2_{\mathcal{G},(T_{\text{vt}},\,T_{\text{ver}})} \;=\; [\![(\sigma, T_{\text{vt}})]\!]_{\mathcal{G},\,\mathrm{ver}(T_{\text{ver}})},$$

the valid-time parameter selecting the role windows $\omega_r(T_{\text{vt}})$ and the version parameter selecting the graph $G_{\mathrm{ver}(T_{\text{ver}})}$. *Two independent relations are at stake and this report has used one word for both; we separate them here.* The *parameter* diagonal is $T_{\text{vt}} = T_{\text{ver}}$, which Definition 2.3 defaults to; our evaluated suite is *parameter-off-diagonal* throughout, since each question carries an as-of instant and a distinct declaration instant. The *pin* relation is separate: a *pin-off-diagonal* read is a $p \neq \mathrm{ver}(T_{\text{ver}})$, which a certificate must declare as $\mathrm{offdiag} = (p, \mathrm{ver}(T))$. Below, "off-diagonal" unqualified means the pin relation. What carries over, and what does not:

- Definition 2.5, Theorem 3.1 (coverage partition) and Proposition 3.2 (decidability and cost) generalise verbatim, reading v as p and W_r as $\omega_r(T_{\text{vt}})$: every guard is evaluated at a *fixed* (v, W) pair, and the two parameters only change how that pair is selected.
- Theorem 5.1 (verification completeness) generalises verbatim once ν carries offdiag, since V1 tests exactly the pin-versus-ver(T) relation the marker declares.
- Theorem 5.2(a) and (b) generalise for a *certified* off-diagonal read — one whose marker is present and verified — with (T, v) read as (T_{vt}, p); the proofs above never use $p = \mathrm{ver}(T)$, only that both sides resolve the *same* G_p through $C.\nu$.
- What does *not* follow from the bracket is *admissibility*: whether a given off-diagonal pair may be answered at all is a disclosure decision on the governance side, not a property of $[\![\cdot]\!]^2$, and we take no position on it here.
- Lemma 2.2 (version-axis stability) concerns the version parameter alone and is unaffected.

The suite's single *pin*-off-diagonal certificate is of the certified kind, and the forgery battery strips its marker (§7.5); every one of the sixty is parameter-off-diagonal, which is why the bracket above is the object the evaluation instantiates rather than a generalisation left for later. $\square$

Proposition 5.3 (field non-redundancy). *For each $x \in \{\nu, \alpha, \rho, \delta, w\}$, let* Chk_{-x} *weaken* Chk *by dropping the checks that consume x. Then there is an instance that* Chk_{-x} *accepts although point-in-time correctness fails.*

Proof. One construction per field; each exhibits a certificate accepted by the weakened relation together with a point-in-time correctness violation at the same coordinate.

Drop ν (version swap). Without the version pin, V1 disappears and the verifier replays against whatever graph it currently reads. Let a caliber be redefined between v_1 and v_2 — say the denominator route of a ratio metric is re-scoped — and take a certificate correct at v_1. Its metric name, anchor names, binding row name and routing edge names all persist at v_2, so V0, V2, V3, V4 and V6c replay successfully at v_2 and the certificate is accepted; yet its numeric answer is the v_1 value, which is the value of no query under the v_2 semantics. Point-in-time correctness at the coordinate fails. On the public base this construction is no longer only constructive: every domain registers two committed versions, four question pairs flip their value under the version change alone, and the mutation battery of §7.5 builds exactly this forgery — a certificate re-pinned to the other committed version, and separately an off-diagonal commitment marker stripped — with the version-pin check rejecting both.

Drop α (window swap). Without the anchor assignment, V0 and V2 disappear and the verifier sees only the emitted text. Take probe (c) of the running example (§B): a numerator window of 2016-11 against a denominator window of 2016-10 on `ruling_intensity`. Each side is syntactically impeccable — a well formed aggregate with a well formed window predicate — and no remaining check compares the two, so the certificate is accepted; but clause (ii) of svw fails, the governed value is $\perp_{\mathsf{AM}}$, and the emitted number is a ratio of two different months — 64.0 where the same-window rule delivers 0.402. Restoring α makes the violation replayable by a third party, and it is the recording of *both* role assignments that does it.

Drop ρ (caliber-blind substitution). Without the routing path, V4 disappears. Anchors and windows can be correct while the denominator is attributed along a different route: on a production governance line the same (m, s, T) yields 16.2% under `rma_problem_to_sales` (denominator: scoped total sales) and 3.8% under a control route whose denominator is the complaint base itself — same anchors, same window, a 4.2× difference. The public base carries the same phenomenon as frozen build evidence: the `financial` penalty-transaction rate reads 0.14310% under the registered caliber-aware route against 2.17426% under the caliber-blind reading — 15.19× apart on public data. Both certificates pass $\mathsf{Chk}_{-\rho}$ and at most one of them can be point-in-time correct.

Drop δ (disclosure laundering). Without the disclosure decision, V5 disappears, in two shapes. First, a query touching policy-restricted columns is presented in full because the verifier cannot recompute which policies should have applied, so an unlawful presentation is certified. Second, a legally trimmed answer is presented as an answer to the untrimmed question: the cut trace is gone, the delivered grain is not compared with the requested one, and a lawful rewrite becomes a caliber error wearing a certificate.

Drop w (unfalsifiable refusal). Without the witness, V6b disappears and a refusal reduces to the compiler's say-so: over-refusal and lawful refusal become indistinguishable, and the four reason classes can be interchanged at will, since nothing consumes them. With witnesses, a misreported reason fails a governance-table lookup, or lands outside the asserted probe result, or offers a discriminant date absent from the symmetric difference computed under the declared replay mode. □

Boundary of the minimality claim. Proposition 5.3 establishes per-field non-redundancy under Chk. It does not assert that this field set is the smallest conceivable encoding: other encodings of equal discriminating power may exist — folding the route into an extended anchor, say — but any such encoding has to discharge the five discriminating duties above. Symmetrically, the maximality of a rewrite is a declared non-goal of Chk: verifying it would mean re-implementing the search of §4 inside a verifier forbidden to share code with the compiler, so we name the gap instead of relaxing the non-circularity rule.

E System Implementation Detail

E.1 Compiler module layout

The binding compiler is about 3.8k lines of Python: a shared core plus one adapter layer per evidence base, as the body states. `core` (711 lines) holds version resolution, the candidate anchor assignment, the guard evaluator in the frozen order $\mathsf{OOV} \to \mathsf{AM} \to \mathsf{MC}$, and the rewriting search of §4. `certificate` (180 lines) builds the envelope field by field from the decision record produced by `core`, never by re-inspecting the emitted SQL — the *no post-hoc narration* rule of §6. The adapter layers are thin per-domain maps whose only job is to name the physical tables behind the governance objects of G_v (`adapters`, 1,392 lines, for the five enterprise development domains and the legacy public track; `adapters_pilot2`, 1,178 lines, for the nine public databases); an adapter is deliberately incapable of expressing metric semantics, so that a domain cannot smuggle a caliber decision into code the verifier is forbidden to read. Two acceptance harnesses close the loop in one pass each: the public harness re-runs the 60 suite questions against the frozen gold with certificate emission and the well-formedness self-check, and the development harness keeps the legacy 51-question suite as a regression.

E.2 Steps elided in the body

Two steps of the compiler walk-through are summarised in one clause in §6 and are given here in full.

Delta expansion. A question whose specification declares a difference of two periods ("how did the rate move from T_1 to T_2") is expanded into the declared period sequence *before* guard evaluation, and each period is bound independently. A delta is therefore answerable only if every constituent period binds; if any one of them yields $\perp_c$, the delta inherits that class rather than reporting a difference against a missing leg. This is also why the answer-soundness statement (Theorem 5.2(b)) does not cover delta templates: subtraction of two certified rates is outside the template class for which the syntax-implies-semantics argument is discharged.

Disclosure gate. After binding succeeds, the gate climbs the grain lattice from the requested presentation grain until it reaches a minimal legal one, then closes the mask set under the policy's implications, and records both the climb and the closure as the per-slice cut trace. The climb is bounded by the lattice height and the closure by the number of binding rules, which is the probe bound proved in §4; the trace is what lets a verifier decide legality and cut-trace consistency without re-running the search.

E.3 Envelope and deviation ledger

A certificate travels as a JSON envelope: the produced artifact (SQL text, or refusal reason plus witness), the certificate body of §5.1 with its probe transcripts, and a machine-readable deviation ledger attached entry by entry where a departure exists. Each ledger entry names the clause of the frozen specification it departs from, the scope over which the departure applies (a domain, an anchor, or a whole track), and whether it weakens or merely annotates a check. The legacy public track of the development suite contributes the canonical example, its `version_table_absent` marker: that substrate has no version registry, so the version pin cannot be resolved back to a committed graph, and every certificate from that track carries the shortfall rather than presenting an unresolvable pin as a resolved one — 33 of the 51 development-suite certificates carry at least one such entry. On the nine public databases the registries the frozen specification expects are all present and committed, and no certificate of the released 60 carries a deviation entry; the guard-domain adjudications frozen for that suite (the pair-level coverage-vacuity reading of §3 among them) are registered in the artifact's port and adjudication records rather than per certificate, and §7.6 is the measurement of what unregistered adjudications of exactly that kind cost.

F Evaluation Detail

F.1 The cross-implementation divergence ledger in full

§7.6 of the body reports that the first cross-check of the compiler and the independent verifier — recorded during their integration on the production track that motivated this work, before the public suite existed — rejected 25 of the 51 certificates then in force, resolving into 15 distinct root causes in three classes of five: five implementation defects (three compiler-side, two verifier-side), five governance-registration gaps, and five divergent readings of the frozen specification (Table 5). The table's caption promises both readings per row; this subsection carries them — the complete ledger, each entry naming the disagreement, which side was right, and where it landed. Its provenance is the one the body's table declares: specification-level content — readings, checks, guard order — carrying no production data value; the machine-readable form, with the adjudication that closed each row, ships with the artifact as `impl/INTEGRATION_REPORT.md`.

Five implementation defects (D6–D8 compiler-side, D10 and D12 verifier-side). *(D6)* A REWRITE's emitted SQL kept the *requested* window's upper bound while the certificate asserted a trimmed one; the values agreed only because no earlier rows existed, so the certificate's lower bound was syntactically uncheckable — the window-shift attack of §7.5. *(D7)* An atomic metric took its anchor by following another metric's binding row, which governs a different metric, so V3's lookup-by-identifier could not confirm it. *(D8)* A probe window was emitted as a date-arithmetic expression (`recorded_at < DATE 'd' + INTERVAL 1 DAY`); no syntactic reading pins its denotation down, so the verifier read the denoted set as $\top$ and the probe window as blank. *(D10)* The verifier's SQL scan counted a derived-table correlation name (`FROM (SELECT …) n`) as a touched object, reporting a spurious closure escape. *(D12)* The verifier's replay of an interval-anchor point predicate was never implemented and raised at replay time. All three compiler-side defects share one shape: the value was right and the certificate's claim was not re-derivable. Value-level gold testing passed them 51/51; only independent re-derivation failed them.

Five registration gaps (D3, D4, D5, D11, D14). In each, both sides read the specification the same way and the governance content simply was not registered where the verifier, restricted to (G_v, D), could read it. *(D3)* Atomic metrics took their prescribed anchor from a dictionary inside the compiler rather than from an anchor assignment carried by A_v; closed by a `metrics` anchor column plus a registered-assignment branch in the checker. *(D4)* A same-anchor, same-window count ratio was allowed to travel without a routing path, where Definition 2.5(b2) requires $R_v(m)$ defined and recomputable for every ratio; closed by registering a routing-ownership column and an explicit identity route. *(D5)* A ratio metric whose leg alignment lived only in a prose routing note carried no binding row at all, so the verifier's reading (a ratio with $\beta_v(m)$ undefined is $\perp_{\mathsf{MC}}$(i)) collided with a gold value question; closed by registering the binding row that prose already described. *(D11)* A conformed dimension was joined to resolve a scope predicate but had no `dimension_of` edge, so the join left the certified closure; closed by registering the edge. *(D14)* A delta question presented one window where the semantics expands role × period, so the verifier saw a window mismatch and a surplus role; closed by presenting the period sequence in the question specification.

Five specification-reading divergences (D1, D2, D9, D13, D15). All five sit on when a guard's subject is undefined, and each moves a refusal *class* rather than a wording. *(D1)* How a coverage domain is computed when no mode is declared: reading Cov under the default hull makes the requested window miss coverage and OOV fire first, while treating the leg's envelope as $\top$ leaves MC(ii) as the first guard — the two readings disagree on the class of the same question, and the frozen gold pins MC(ii). *(D2)* When a binding prescribes anchors: "a binding row exists" and "the rule prescribes both legs" are both defensible, and a row registered with two null anchor legs is $\beta_v(m)$ defined under the first and undefined under the second, moving the question between AM and MC(i). *(D9)* Whether a coarse-granule marker denotes its granule: a month-level row keyed by its first day does not textually equal the month it denotes, so literal denotation equality on probe windows rejects honest certificates while the granule reading accepts them. *(D13)* Where an anchor override $\bar{a}$ is carried: the frozen text requires it machine-readable in the *certificate* and is silent on the question side, and a certificate-only override is a self-signed exemption (forgery F3, §G.1). *(D15)* The certificate spelling for an unregistered anchor reference, which the frozen schema leaves underdetermined; both spellings are now accepted and the verifier re-checks the declaration rather than the spelling.

On the E2 prompt-variant quantities. The released suite's E2 result (§7.2) carries every prompt-variant quantity — per-arm error rates, cluster bootstrap intervals, answered shares and correct-refusal counts — in body Table 4, with the frozen figure input `figures/fig_data_pilot2.json` and the bootstrap seed shipping in the artifact; this report adds the per-question answer/refusal table as its own appendix rather than a redundant panel. An earlier development-track ablation panel over the 51-question enterprise suite exists in the artifact's development record (`figures/fig4_ablation.*`); its numbers are development-track history, not the released benchmark's, and are therefore not reproduced here.

G Evaluation Detail

G.1 Forgeries Built Against Our Own Relaxations

Section 7.5 of the body reports the released battery: 30 forged certificates over 10 unmodified compiler bases, all 30 rejected and all 10 bases accepted, each rejection pinned on its mutation. The artifact retains, as a permanent regression alongside it, the *development-track* battery that preceded it — 16 forgeries over 3 unmodified controls, still rejected 16/16 with the controls accepted 3/3 on every re-run, each rejection firing on the check its design predicts. Two of the sixteen deserve the detail this appendix carries, because they were not built against the design as published — they were built against the two places where integration *weakened* the checking relation. Each relaxation was paired with an attack before it was accepted; both attacks are in the released forgery generator, and the second discipline is re-attacked directly on the public track — the self-granted anchor override of the released battery's semantic-lie family (§7.5).

Narrowed probe (F3). The refusal witness for a missing-caliber refusal is a denominator probe that must re-evaluate to 0 on the certified window. During integration the verifier stopped demanding literal denotation equality on coarse-granule objects, because a marker value standing for a granule (a month-level row keyed by its first day, say) does not textually equal the window it denotes, and insisting on textual equality rejected honest certificates. The attack exploits exactly that slack: it shrinks the probe window recorded in the witness until the denominator looks empty, leaving a certificate whose refusal is "justified" by a probe narrower than the question asked. The syntactic check passes. The semantic replay does not: re-querying the marked rows under the certified window and under the witness's window returns different counts, and the verifier rejects on that mismatch rather than on the text of the window.

Self-signed override (F3). A question may declare an anchor override, and a certificate that carries one is exempt from the ordinary anchor-assignment derivation. The attack writes the override into the certificate itself, so that the certificate authorises its own exemption. The verifier rejects because the override credential is cross-checked against the question specification supplied by the auditor, never read from the certificate that would benefit from it — the same discipline, at the field level, that the anti-certification-loop rule imposes at the module level (§6).

Why these two are reported. Both relaxations make the checking relation strictly weaker than the one stated in §5, and weakenings introduced under integration pressure are precisely the ones a reader has no way to audit from the theory. We therefore state them, attack them, and report that the attacks are caught by a check that survives the relaxation — semantic replay in the first case, cross-checked provenance of the credential in the second.

H The 60-Question Suite, Question by Question

The answer/refusal table the body's certificate section points to: all 60 questions of the public suite with their bi-temporal coordinates, governed metric, gold outcome class and gold value or typed-refusal class. The deterministic compiler matches all 60 gold outcomes (§7.4) and the independent verifier accepts all 60 certificates (§7.5); the questions marked *window-request* or *cross-window* in their coordinate notes are the 10 the strict verifier track refuses by design (fail-closed, §7.5). Question text is reproduced verbatim from the frozen gold files (questions are authored in Chinese; the metric and coordinate fields are the language-neutral specification the compiler consumes).

Table 10: The 60-question public suite, question by question. For each question: declared time point, as-of instant, governed metric, gold outcome class, and gold value or typed-refusal class (with clause subtype where registered). The second line of each entry is the question text as frozen in `questions.json`, followed by any non-default request coordinates. The deterministic compiler matches all 60 gold outcomes and the independent verifier accepts all 60 certificates (§7.4, §7.5).

QID	declared	as-of	metric	expected	gold
financial (cluster `fin`, 8 questions)					
`FIN-Q1`	1997-06-15	1996-06-30	`fin.problem_loan_rate`	value	`0.13675213675213677`
以 1997-06-15 为声明时点，按当时生效口径计算 1996 自然年（as-of 1996-06-30 所在年窗）的问题贷款率。					
`FIN-Q2`	1998-09-15	1996-06-30	`fin.problem_loan_rate`	value	`0.08547008547008547`
以 1998-09-15 为声明时点，按当时生效口径计算 1996 自然年（as-of 1996-06-30 所在年窗）的问题贷款率。					
`FIN-Q3`	1998-09-15	1997-06-30	`fin.penalty_trans_rate`	value	`0.0014310376957128643`
以 1998-09-15 为声明时点，1997 自然年（as-of 1997-06-30 所在年窗）的罚息交易占比是多少（治理路由口径）?					
`FIN-Q4`	1998-09-15	1997-05-15	`fin.loan_count_month`	value	`16`
以 1998-09-15 为声明时点，1997 年 5 月（as-of 1997-05-15 所在月窗）当月新增放贷笔数是多少?					
`FIN-Q5`	1997-06-15	1996-06-15	`fin.trans_count_month`	value	`15885`
以 1997-06-15 为声明时点，1996 年 6 月（as-of 1996-06-15 所在月窗）当月交易笔数是多少?					
`FIN-Q6`	1998-09-15	—	`fin.trans_count_range`	rewrite (hull_trim)	`151237`
以 1998-09-15 为声明时点，统计 1998-07-01 至 1999-06-30 的区间交易笔数。 [window-request]					
`FIN-Q7`	1998-12-15	1999-06-15	`fin.trans_count_month`	refusal	$\perp_{\mathsf{OOV}}$
以 1998-12-15 为声明时点，查询 as-of 1999-06-15 所在月窗的当月交易笔数。					
`FIN-Q8`	1998-09-15	1997-05-15	`fin.penalty_trans_rate`	refusal	$\perp_{\mathsf{AM}}$(ii)
计算罚息交易占比，但分子取 1997 年 5 月、分母取 1997 年 4 月（声明时点 1998-09-15）。 [cross-window]					
formula_1 (cluster `f1`, 8 questions)					
`F1-Q1`	2009-12-01	2009-07-01	`f1.driver_season_points`	value	`100.0`
以 2009-12-01 为声明时点，按当时生效积分口径，车手 button 2009 赛季（as-of 2009-07-01 所在年窗）的赛季车手总积分是多少?					
`F1-Q2`	2011-03-01	2009-07-01	`f1.driver_season_points`	value	`252.0`
以 2011-03-01 为声明时点，按当时生效积分口径，车手 button 2009 赛季（as-of 2009-07-01 所在年窗）的赛季车手总积分是多少?					
`F1-Q3`	2017-06-01	2009-07-01	`f1.driver_season_points`	value	`100.0`
以 2017-06-01 为声明时点，显式钉定治理版本 v1（2009 当时口径），回算车手 button 2009 赛季（as-of 2009-07-01 所在年窗）的赛季车手总积分。 [pinned ν=`v1`]					
`F1-Q4`	2011-03-01	—	`f1.season_points_delta`	value	`-38.0`
以 2011-03-01 为声明时点，按当时生效积分口径，车手 button 2010 赛季与 2009 赛季的两季车手积分差（2010 − 2009）是多少? [multi-period]					
`F1-Q5`	2011-03-01	2009-07-01	`f1.entry_count`	value	`34`
以 2011-03-01 为声明时点，车队 brawn 2009 赛季（as-of 2009-07-01 所在年窗）的赛季参赛记录条数是多少?					
`F1-Q6`	2017-12-15	—	`f1.races_count_range`	rewrite (hull_trim)	`6`
以 2017-12-15 为声明时点，统计 2017-10-01 至 2018-03-31 的区间大奖赛场数。 [window-request]					
`F1-Q7`	2011-03-01	1949-05-15	`f1.races_count_month`	refusal	$\perp_{\mathsf{OOV}}$
以 2011-03-01 为声明时点，查询 as-of 1949-05-15 所在月窗的当月大奖赛场数。					
`F1-Q8`	2011-03-01	2010-07-01	`f1.points_per_entry`	refusal	$\perp_{\mathsf{MC}}$(ii)
以 2011-03-01 为声明时点，车队 brawn 2010 赛季（as-of 2010-07-01 所在年窗）的车队场均积分是多少?					
card_games (cluster `card`, 7 questions)					
`CARD-Q1`	2017-03-01	2017-02-15	`card.monthly_rulings`	value	`2161`
以 2017-03-01 为声明时点，2017 年 2 月（as-of 2017-02-15 所在月窗）的当月裁定条数是多少?					

(continued)

QID	declared	as-of	metric	expected	gold
CARD-Q2	2017-03-01	2016-11-15	`card.ruling_intensity`	value	`0.4020100502512563`
以 2017-03-01 为声明时点，2016 年 11 月（as-of 2016-11-15 所在月窗）的裁定强度是多少?					
CARD-Q3	2021-03-15	2021-03-10	`card.standard_card_count`	value	`2339`
以 2021-03-15 为声明时点，按当前轮替口径，截至 as-of 2021-03-10 的轮替合法卡牌总数是多少?					
CARD-Q4	2015-09-01	2015-06-22	`card.card_rulings_count`	value	`32`
以 2015-09-01 为声明时点，Kytheon, Hero of Akros // Gideon, Battle-Forged（ORI 印次）在 2015 年 6 月（as-of 2015-06-22 所在月窗）的单卡当月裁定条数是多少?					
CARD-Q5	2021-03-15	—	`card.rulings_range`	rewrite (hull_trim)	`7542`
以 2021-03-15 为声明时点，统计 2020-10-01 至 2021-06-30 的区间裁定条数。 [window-request]					
CARD-Q6	2017-03-01	2017-02-15	`card.collector_premium`	refusal	$\perp_{\mathsf{MC}}$(i)
以 2017-03-01 为声明时点，2017 年 2 月的保留卡收藏溢价率是多少?					
CARD-Q7	2017-03-01	2017-02-15	`card.ruling_intensity`	refusal	$\perp_{\mathsf{MC}}$(ii)
以 2017-03-01 为声明时点，2017 年 2 月（as-of 2017-02-15 所在月窗）的裁定强度是多少?					
codebase_community (cluster `code`, 7 questions)					
CODE-Q1	2011-02-01	2010-12-15	`code.accepted_rate`	value	`0.6808510638297872`
以 2011-02-01 为声明时点，2010 年 12 月（as-of 2010-12-15 所在月窗）的提问采纳率是多少?					
CODE-Q2	2014-02-01	2013-12-15	`code.accepted_rate`	value	`0.30677966101694915`
以 2014-02-01 为声明时点，2013 年 12 月（as-of 2013-12-15 所在月窗）的提问采纳率是多少?					
CODE-Q3	2011-02-01	2010-12-15	`code.monthly_votes`	value	`3208`
以 2011-02-01 为声明时点，2010 年 12 月（as-of 2010-12-15 所在月窗）的当月投票总数是多少?					
CODE-Q4	2013-08-01	2013-06-15	`code.user_accepted_report`	rewrite (granularity_rollup)	`[["[0,100)",13.0],["[100,1000)",56.0],["[1000,10`
以 2013-08-01 为声明时点，给出 2013 年 6 月各用户（用户级）的采纳答案数报表。 [grain=`user`]					
CODE-Q5	2013-08-01	2013-06-15	`code.daily_questions`	rewrite (granularity_rollup)	`942`
以 2013-08-01 为声明时点，给出 2013 年 6 月的逐日新提问数报表（日粒）。 [time-grain=`day`]					
CODE-Q6	2013-08-01	—	`code.user_location`	rewrite (mask)	`"UK"`
以 2013-08-01 为声明时点，用户 Id=22047 的用户所在地是什么（按现行披露策略呈现)?					
CODE-Q7	2013-08-01	2013-06-15	`code.top_answerer_roster`	refusal	$\perp_{\mathsf{DB}}$
以 2013-08-01 为声明时点，列出 2013 年 6 月采纳答案数前十用户的 DisplayName 与 Location 原值。 [grain=`user`]					
debit_card_specializing (cluster `deb`, 7 questions)					
DEB-Q1	2013-09-15	2013-08-15	`deb.avg_consumption`	value	`15626.138213557568`
以 2013-09-15 为声明时点，2013 年 8 月（as-of 2013-08 月粒）CZK 结算客户的月均客户消费额是多少?					
DEB-Q2	2013-09-15	2013-08-15	`deb.avg_consumption`	value	`1166.1530437665776`
以 2013-09-15 为声明时点，2013 年 8 月（as-of 2013-08 月粒）EUR 结算客户的月均客户消费额是多少?					
DEB-Q3	2013-09-15	2013-06-15	`deb.consumption_sum`	value	`121345577.66`
以 2013-09-15 为声明时点，2013 年 6 月（as-of 2013-06 月粒）SME 细分客户的月消费总额是多少?					
DEB-Q4	2012-09-15	2012-01-15	`deb.consumption_sum`	value	`44080288.83999992`
以 2012-09-15 为声明时点，2012 年 1 月（as-of 2012-01 月粒）全体客户的月消费总额是多少?					
DEB-Q5	2013-09-15	—	`deb.segment_avg_trans`	rewrite (granularity_rollup)	`[["all",23.87878787878788]]`
以 2013-09-15 为声明时点，统计 2012-08-23 至 2012-08-26 间 EUR 结算 LAM 细分（细分级）的细分平均单笔交易金额。 [window-request, grain=`Segment`]					
DEB-Q6	2013-11-15	2013-12-15	`deb.avg_consumption`	refusal	$\perp_{\mathsf{OOV}}$
以 2013-11-15 为声明时点，查询 as-of 2013-12（月粒）的全体客户月均客户消费额。					
DEB-Q7	2013-09-15	—	`deb.avg_consumption`	refusal	$\perp_{\mathsf{AM}}$(iii)
以 2013-09-15 为声明时点，统计自 2013-03-04 起一周的月均客户消费额。 [window-request]					
california_schools (cluster `ca`, 6 questions)					
CA-Q1	2015-05-01	2015-04-15	`ca.free_meal_rate`	value	`0.37985585826806795`
以 2015-05-01 为声明时点，按当时默认口径，Alameda 县 2014-2015 学年（as-of 2015-04-15）的免费餐率是多少?					
CA-Q2	2015-09-01	2015-04-15	`ca.free_meal_rate`	value	`0.37804549460887515`
以 2015-09-01 为声明时点，按当时默认口径，Alameda 县 2014-2015 学年（as-of 2015-04-15）的免费餐率是多少?					
CA-Q3	2015-05-01	2015-01-01	`ca.schools_in_effect`	value	`10629`
以 2015-05-01 为声明时点，as-of 2015-01-01 时点在效学校数是多少?					
CA-Q4	2015-05-01	2000-09-01	`ca.schools_in_effect`	value	`9270`
以 2015-05-01 为声明时点，as-of 2000-09-01 时点在效学校数是多少?					
CA-Q5	2015-09-01	2015-04-15	`ca.school_frpm_report`	rewrite (granularity_rollup)	`[["Alameda",0.37804549460887515]]`
以 2015-09-01 为声明时点，给出 Alameda 县 2014-2015 学年各学校（校级）的校级免费餐率报表。 [grain=`school`]					
CA-Q6	2015-09-01	2015-04-15	`ca.free_meal_rate`	refusal	$\perp_{\mathsf{AM}}$(i)
以 2015-09-01 为声明时点，按特许授权日期开窗统计 2014-2015 学年的免费餐率。[$\bar{a}$=`charter_authorization_date`]					
european_football_2 (cluster `ef2`, 6 questions)					
EF2-Q1	2016-07-01	2015-09-21	`ef2.avg_overall_rating`	value	`69.97529917152501`
以 2016-07-01 为声明时点，as-of 2015-09-21 快照日的快照日球员平均综合评分是多少?					

(continued)

QID	declared	as-of	metric	expected	gold
`EF2-Q2`	2016-07-01	—	`ef2.win_rate`	value	`0.4131578947368421`
以 2016-07-01 为声明时点，England Premier League 2015/2016 赛季（窗 2015-08-01 至 2016-05-25）的主场胜率是多少？ [window-request]					
`EF2-Q3`	2016-07-01	2015-06-15	`ef2.pa_month_count`	value	`681`
以 2016-07-01 为声明时点，2015 年 6 月（as-of 2015-06-15 所在月窗）的当月球员属性快照条数是多少？					
`EF2-Q4`	2016-07-01	—	`ef2.matches_range`	rewrite (hull_trim)	`915`
以 2016-07-01 为声明时点，统计 2016-03-01 至 2016-08-31 的区间比赛场数。 [window-request]					
`EF2-Q5`	2014-02-01	2013-12-25	`ef2.avg_overall_rating`	refusal	$\perp_{\mathsf{OOV}}$
以 2014-02-01 为声明时点，as-of 2013-12-25 快照日的快照日球员平均综合评分是多少？					
`EF2-Q6`	2014-02-01	2013-12-01	`ef2.rating_weighted_win_rate`	refusal	$\perp_{\mathsf{AM}}$(iv)
以 2014-02-01 为声明时点，as-of 2013-12-01 单日窗的评分归一当日胜率是多少？					
thrombosis_prediction (cluster `th`, 6 questions)					
`TH-Q1`	1998-06-01	1997-06-30	`th.abnormal_lab_rate`	value	`0.01901743264659271`
以 1998-06-01 为声明时点，1997 自然年（as-of 1997-06-30 所在年窗）的异常化验率（GOT≥60 口径）是多少？					
`TH-Q2`	1996-06-01	1995-06-30	`th.exam_count`	value	`207`
以 1996-06-01 为声明时点，1995 自然年（as-of 1995-06-30 所在年窗）的年度检查记录数是多少？					
`TH-Q3`	1998-06-01	1997-06-30	`th.patient_abnormal_report`	rewrite (granularity_rollup)	`[["all",12.0]]`
以 1998-06-01 为声明时点，给出 1997 自然年各患者（患者级）的患者异常化验报表（GOT≥60）。 [grain=`patient`]					
`TH-Q4`	1998-06-01	—	`th.patient_birthday`	rewrite (mask)	`"1934"`
以 1998-06-01 为声明时点，患者 ID=2110 的患者出生日期是什么（按现行披露策略呈现）?					
`TH-Q5`	1998-06-01	1997-06-30	`th.patient_roster`	refusal	$\perp_{\mathsf{DB}}$
以 1998-06-01 为声明时点，列出 1997 年有异常化验（GOT≥60）患者的 ID、出生日期与诊断原值。[grain=`patient`]					
`TH-Q6`	1998-06-01	1997-06-30	`th.patient_roster`	refusal	$\perp_{\mathsf{DB}}$
以 1998-06-01 为声明时点，给出 1997 年异常化验患者名单（患者级逐行呈现）。 [grain=`patient`]					
world_1 (cluster `w1`, 5 questions)					
`W1-Q1`	2026-06-15	2026-05-10	`w1.asia_population_sum`	value	`4068933086`
以 2026-06-15 为声明时点，按当时生效人口口径，as-of 2026-05 月粒的亚洲人口总和是多少？					
`W1-Q2`	2026-08-15	2026-05-10	`w1.asia_population_sum`	value	`4313069067`
以 2026-08-15 为声明时点，按当时生效人口口径，as-of 2026-05 月粒的亚洲人口总和是多少？					
`W1-Q3`	2026-06-15	2026-03-10	`w1.population_share`	value	`0.6075330810301713`
以 2026-06-15 为声明时点，as-of 2026-03 月粒的亚洲人口占比（亚洲/全球）是多少？					
`W1-Q4`	2026-08-15	—	`w1.history_rows_range`	rewrite (hull_trim)	`717`
以 2026-08-15 为声明时点，统计 2026-10 至 2027-03 各月的区间历史记录条数。 [window-request]					
`W1-Q5`	2026-08-15	2026-05-10	`w1.population_share`	refusal	$\perp_{\mathsf{AM}}$(ii)
计算亚洲人口占比，但分子取 2026-05、分母取 2026-04（声明时点 2026-08-15）。 [cross-window]					

I Complete Forgery Enumeration and Acceptance Record

The complete forgery enumeration the body's E5 section summarises — 34 forgeries over 11 unmodified bases (families F1–F5), each row naming the mutation and the check that rejects it — together with the full acceptance record it belongs to: gold acceptance, both verifier tracks with the strict-track per-question list, the disclosure-class replay audit, cost re-measurement, scale numbers and the CI gates. What follows is a mechanical typesetting of the frozen `pilot2/ACCEPTANCE_REPORT.md` (2026-08-04, in Chinese); section numbering below is the report's own. The forgery table is §3; the honest-boundary ledger is §7.

Reading the two dates. The frozen report is reproduced *unamended*, so it states the battery as it stood on 2026-08-04: 30/30 over 10 bases, families F1–F4. A fifth family was added on 2026-08-10, when a review pass found two faces of the guard chain that no forgery landed on and the verifier was repaired until the new rows rejected; the body's 34/34 over 11 bases is that current battery, and §3 below is its 2026-08-04 subset, unchanged. Two further rows of the frozen report are likewise dated rather than current, and we flag them here rather than silently re-typesetting a record: its *development-track* regression line (`old51`, the enterprise pilot that contributes motivation only and no result in the body) read 51/51 and 49/51 on 2026-08-04 and reads 45/51 on both tracks today — the verifier has since been tightened three times, each time in the rejecting direction, and the compiler-side defect those rejections expose (a clause-(iv) variant hard-coded to `trivial_true` on distinct-anchor pairs) was left unfixed because it is outside the public corpus the paper measures. Every pilot2 number in the frozen report — gold 60/60, verifier 60/60 and the strict 50/60 — is unchanged and was re-run on 2026-08-10.

The four added rows, and the two verifier repairs they forced, in this report's own voice, so that the enumeration above is complete:

F5 row	base	rejected by	the mutation, and why it must fail
`F5a` oov class substitution, honest leg	EF2-Q6	V6b	An AM(iv) refusal re-issued as OOV, witnessed on the leg where the vacuum claim is *true*: 2013-12-01 is no `Player_Attributes` marker day. The pair is mixed — the match leg meets its hull — and the coverage disjunct of Definition 2.8(1) is adjudicated pair-wise, so the denotation is $\perp_{\mathsf{AM}(iv)}$. A single-anchor witness must not carry a class the semantics decides on both legs.
`F5b` oov class substitution, covered leg	EF2-Q6	V6b	The same substitution witnessed on the covered leg, false at leg level too. This is the half the pre-repair verifier already caught; it is kept as the control that isolates what F5a adds.
`F5c` clause-(iv) variant swap, fabricated refusal	CARD-Q2	V3	A question the compiler *answers*, turned into a refusal by self-declaring the unwindowed `symdiff_audit` variant, under which the two marker sets do differ. β_v registers a coverage mode on both anchors, which fixes the window-realisation audit — and that audit passes. The payload is arithmetically honest; what is forged is the choice of replay mode.
`F5d` clause-(iv) variant swap, genuine refusal	EF2-Q6	V3	Right verdict, self-chosen reason: a genuine AM(iv) refusal re-declared under a variant β_v does not register for this pair, with the payload computed honestly *under that variant*. Catching this one requires re-deriving the variant, not re-checking the numbers.

Neither row rejected before the repairs: F5a and F5c were **accepted** by the 2026-08-07 verifier, and F5d rejected on the wrong check. The two repairs are verifier-side only — no certificate byte, no seed and no compiler file moved, and the 60 frozen certificates re-emit identically. (i) **V6b(0)** now replays the coverage disjunct for OOV as well as $\mathsf{AM}/\mathsf{MC}/\mathsf{DB}$, pair-wise, whenever the witness is the coverage-vacuity triple, so a refusal cannot be relabelled into a class that does not hold at the level its disjunct is adjudicated. (ii) **V3** now re-derives the clause-(iv) variant from β_v alone and treats the certificate's `adm_check_mode` as a *claim to audit* rather than an instruction to obey; a contradiction is a rejection. The honest cost of (ii) is recorded above: it is what moved the development track from 46/51 to 45/51.

I.1 ACCEPTANCE_REPORT — pilot2 全量验收（九公开库 · 60 题 · 双侧闭环）

日期：2026-08-04。范围：任务书六项验收（金标 / 证书两轨 + 披露类首验 / 伪造族 F1–F4 / 代价重测 / 规模数字 / ci 零交集）+ 泄漏闸复跑。一切数字由本次验收现场复算；命令均可复跑（§8）。前置基线：`pilot2/BUILD_REPORT.md`（建仓）、`impl/PORT_REPORT.md`（移植）。

I.1.1 0. 终局判定矩阵

#	验收项	结果	证据
1	金标验收（编译器 vs 冻结金标）	**60/60 全过**，良构自检 0 错	`acceptance_pilot2.py` 实跑；证书 60 份重发射逐字节稳定（聚合 sha1 `60dbd2560450f157cb57b2e7b8bb9450180` = PORT 基线）
2	证书验收（独立校验器常规轨）	**ACCEPT 60/60**（window_source: derived 50 / declared 8 / declared-override 2）	`runall.py p2` 实跑
2′	严格轨 `--no-declared-windows`	**ACCEPT 50/60**；10 个 REJECT 全部 V0、恰为呈现窗坐标题（8×window_request + 2×cross_window），fail-closed	逐题清单 §2.2
2″	披露类证书首次验收	DB 拒答 ×3 / 粒度上卷 ×5 / 掩码 ×2 / 离对角 ×1 全部重演闭环（V5 Π+ 条款独立重算、V6b 笔录重执行）	§2.3
3	伪造族（新基座全套）	**30/30 REJECT 且拒因命中**（F1×6 F2×5 F3×9 F4×10）；10 个真证书基座全 ACCEPT；老族 16/16 + aux symdiff-57 复算 PASS	`forge_p2.py` / `forge.py` 实跑
4	代价重测（认输轴更新）	60 证书冷校验总 **7.60 s**；warm 校验/回答比中位 **17.7×**（p90 36.1×）；证书中位 **2,600 B**	`impl/cost_p2.json`
5	规模数字	Σreal **3,830,036** 行（trans **1,056,320** ≥ **10**^{5} 零合成 → P6）；gov 种子 **847** 行；版本 **2/库 ×9**；策略实例 **18** 行	§5，全部由仓库现查，provenance 零失配
6	ci_check 零交集	**PASS**：failures=[]、shared_internal_roots=[]（含新增 forge_p2.py 后复跑）	§6
6′	pilot2 泄漏闸	leak_check **PASS**（n=60, problems=0, inter=[]）	§6

I.1.2 1. 金标验收（60/60）

`python3 impl/asof_compiler/acceptance_pilot2.py` → `questions=60, gold_match=60/60, certificates_written=60` → `impl/certs2, wellformed_selfcheck_errors=0, ACCEPTANCE-P2: ALL OK`。

构成与规格逐格全同（由 questions.json 金标侧现场重 tally）：

轴	分布
expected_kind	value 33 / rewrite 12 / refusal 15
决策	ANSWER 33 / REWRITE 12 / REFUSE 15（SQL 出码 45 = 33+12）
拒因	out-of-validity 4 / anchor-mismatch 5 / missing-caliber 3 / disclosure-blocked 3
拒因子型	am_i 1, am_ii 2, am_iii 1, am_iv 1（四子型全）; mc_i 1, mc_ii 2（两型全）
rewrite kind	granularity_rollup 5 / hull_trim 5 / mask 2
簇预算	fin 8 / f1 8 / card 7 / code 7 / deb 7 / ca 6 / ef2 6 / th 6 / w1 5

证书重发射幂等：本次两轮 `acceptance_pilot2.py` 重写 60 份后 `cat [A-Za-z]*.json | shasum -a 1` = `60dbd2560450f157cb57b2e7b8bb94501801edcd`，与 PORT_REPORT 冻结值逐字节一致（AppleDouble `._*` 过滤口径同 PORT §0）。

I.1.3 2. 证书验收（独立校验器）

2.1 四轨总表（`python3 impl/asof_verifier/runall.py`）

轨	ACCEPT	window_source
p2 / declared-ok	**60/60**	derived 50, declared 8, declared-override 2
p2 / no-declared-windows	**50/60**	derived 52,（拒于窗推导前）8
old51 / declared-ok（回归）	51/51	derived 49, declared-override 2
old51 / no-declared-windows（回归）	49/51	derived 51

老轨两数与移植前 INDEPENDENCE/PORT 基线一致，回归不破。

2.2 严格轨 10 题逐题（全部 V0 拒，fail-closed 属设计后果） 8×window_request：CARD-Q5、DEB-Q5、DEB-Q7、EF2-Q2、EF2-Q4、FIN-Q6、F1-Q6、W1-Q4（拒详情均为 I2′(b) 非退化变体缺失——`range_request` 登记语义把呈现窗当声明输入，独立推导如实弃权）；2×cross_window：FIN-Q8（num 窗 [1997-05,1997-06) ≠ 推导 [1997-01,1998-01)）、W1-Q5（den 窗 [2026-04,2026-05) ≠ 推导 [2026-05,2026-06)）。与 PORT §4 裁定 4 预言集完全一致。

2.3 披露类证书首次验收（逐张核对校验器 check 明细）

路径	证书	校验器重演闭环（实测 check 详情）
披露门 DISCLOSURE-BLOCKED	CODE-Q7 / TH-Q5 / TH-Q6	V5 PASS：II 独立重算（code=[pi1,pi2]; th=[pi1,pi2,pi3]），REFUSE 条款重演委派 V6b；V6b PASS：¬OOV/¬AM 先行重演 + 阻断策略在册核验 + SUPPMIN 笔录 SQL **重执行**低于阈值（实体粒 observed=1 < k：code 5 / th 10）+ U_min=∅ 断言

路径	证书	校验器重演闭环（实测 check 详情）
粒度爬升 REWRITE	CODE-Q4 (user→reputation_band, 胞 13≥5)、CA-Q5（school→county）、TH-Q3（→all 顶级豁免）、DEB-Q5（Segment→all，v2 治下 k=20>16）、CODE-Q5（time_floor day→month）	V5 PASS："k/lattice, mask and time-floor clauses re-established from (G_v, D)"——`_p2_min_cell` 不读证书笔录、自装胞普查 SQL 逐级重算，同时压最小性与合法性两向
掩码闭包 REWRITE	CODE-Q6（Location generalize_last_component）、TH-Q4（Birthday year_only@v2 掩码升级）	V5 PASS：μ^* 覆盖 mask_class + 产出 SQL 掩码变换签名双查；V0 把 $\mu^*\neq$id 计入 REWRITE 合法收窄轴
离对角版本钉	F1-Q3（pinned v1, ver(declared_at)=v2）	V1 PASS："off-diagonal read explicitly committed: pinned 'v1', ver(T)='v2' "

I.1.4 3. 伪造族（新基座全套 + 老族回归）

新增校验器侧 `impl/asof_verifier/forge_p2.py`（仅 import chk；输出 `forge_p2_out/`）。与老 forge.py 的手搭基座不同，p2 基座直接取 **10 张真编译器证书**（FIN-Q1、F1-Q3、CARD-Q7、EF2-Q6、CA-Q5、CODE-Q4、CODE-Q6、CODE-Q7、TH-Q4、TH-Q5），未变异全 ACCEPT——每个 REJECT 钉在变异本身。**FORGE-P2: PASS (10 bases, 30 forgeries)**：

族	伪造（基座）	期望 → 实测
F1a	整年错窗 ANSWER (FIN-Q1→1995，SQL+α+ 探针连贯移)	V0 ✓
F1b	空分母拒答翻转成 ANSWER（CARD-Q7，DEN_POP 记 490 谎值）	V6c ✓（重执行 μ_den=0∈$\mathcal{Z}$）
F1c	MC 拒答窗移位（CARD-Q7→2016-12 同为空月，重演自洽）	V0 ✓（α 非 α_{q,v}）
F1d	同窗比率捏造 AM(ii) 窗对拒答 (FIN-Q1)	V0 ✓
F1e	离对角承诺标记删除（F1-Q3）	V1 ✓（GOLD-1 零痕迹镜像）
F1f	版本换轨（FIN-Q1 ν→v2, ver(declared_at)=v1）	V1 ✓
F2a	删 ν（FIN-Q1）	V0 ✓ **且断言 V1 同失**（p2 别名层随版本作用域，V0 别名重解析先失；V1 以经典 "graph_pin missing" 同报）
F2b	删 α	V0 ✓
F2c	删 ρ	V4 ✓
F2d	删 δ	V5 ✓
F2e	删拒答见证（CARD-Q7）	V6b ✓
F3a	悬空版本钉 v9	V0 ✓ **且断言 V1 同失**（"0 row(s)"）
F3b	无策略域漏 D5 标注	V5 ✓
F3c	sum(rate) 形（AVG 日率）	V6a ✓

族	伪造（基座）	期望 → 实测
F3d	AM(iv) 载荷窗造假（EF2-Q6 num→2013-11-30）	V6b ✓
F3d2	AM(iv) symdiff_count 篡改（1→3）	V6b ✓
F3e	域外表（FIN-Q1 SQL 读 “card”）	V6a ✓
F3f	证书自颁改锚豁免（→A-FIN-TRANS + declared_override）	V0 ✓
F3g	路由 hop via_table 篡改（→trans）	V4 ✓
F3h	别名层度量替换（→fin.penalty_trans_rate）	V0 ✓
F4a	**细粒度伪 ANSWER**（CA-Q5 school 粒直答，剥上卷痕迹）	**V5 ✓**（k 条款独立重算：认证粒 min cell 1<10）
F4b	**伪造上卷痕迹 · 过卷**（CODE-Q4 band→all，band 本已合法 13≥5）	**V5 ✓**（SelectMin 最小性两向压制）
F4c	**伪造 SUPPMIN 笔录**（CODE-Q7 笔录 SQL→SELECT 999）	**V6b ✓**（重执行 999≥ 阈值，拒答不重演）
F4d	**策略集 Π 清空**（CA-Q5 policy_ids=[]）	**V5 ✓**（Π 独立重导出 ≠[]）
F4e	**受治域谎报无治理**（CODE-Q6 全洗白 + 原值列直出）	**V5 ✓**
F4f	**DB 见证阻断集清空**（CODE-Q7 blocking=[]）	**V6b ✓**
F4g	**DB 见证引用未登记策略**（TH-Q5 +th.pi99）	**V6b ✓**
F4h	**掩码闭包缺项**（CODE-Q6 μ^*=[] 而 dec=REWRITE）	**V0 ✓**（C4 Def 4.10 映射：无收窄轴则须 ANSWER，静默收窄面封死）
F4i	**掩码变换剥除**（μ^* 仍宣称，SQL 原值直出 Location）	**V5 ✓**（掩码签名双查）
F4j	**掩码强度降级**（TH-Q4@v2 year_only→year_month）	**V5 ✓**（版本化掩码语义重演）

老族回归：`forge.py` → **PASS（3 基座 + 16 伪造全拒）**，aux symdiff-57 独立复算 PASS。

实测边界（诚实披露）：DB 见证 `blocking_policy_ids` **部分缺项**（删 code.pi2 仅留 code.pi1，非空）当前 ACCEPT——该字段在非空 + 全在册之外的完备性不承载判定（拒答本身及拒因均仍正确，$\delta.\Pi$ 另由 V5 独立重算钉住）；候选加固：V6b 将引用集与独立重算的阻断集对账。列入 §7 待办。

I.1.5 4. 代价重测（`impl/cost_p2.json`，认输轴数字更新）

`python3 impl/measure_cost.py --pilot pilot2 --certs impl/certs2 --out impl/cost_p2.json`（measure_cost.py 增 `--pilot`/`--certs` 形参，缺省行为与老语料逐字节同路径；老 `impl/cost.json` 未触碰）。

量	pilot2（60 证书 =45 SQL+15 拒答）	pilot-51 基线（cost.json, 2026-08-03）
校验冷总时	**7.60 s**（60 进程）	5.04 s（51 进程）
校验冷 med/p90/max	0.1158 / 0.1438 / 0.2877 s	0.0965 / 0.1067 / 0.1156 s
冷进程地板（import duckdb）	0.0526 s	0.0521 s

量	pilot2（60 证书 =45 SQL+15 拒答）	pilot-51 基线（cost.json, 2026-08-03）
校验热 med/p90/max	**0.01696 / 0.04594 / 0.17838 s**	0.00615 / 0.01199 / 0.02132 s
回答热 med	0.00131 s	0.00024 s
比值 (热) med/p90/max	**17.7**× / **36.1**× / **44.4**×（n=45）	22.0× / 43.4× / 77.9×（n=37）
比值 (冷) med/max	1.79× / 4.40×	1.47× / 1.72×
证书字节 med/min/max	**2,600 / 1,760 / 4,725 B**	3,106 / 1,587 / 5,214 B
SQL 字节 med/min/max	204 / 88 / 810 B	—

注：① 60 张测量期全 ACCEPT；② p2 无 `symdiff_audit` 全集扫描证书（AM(iv) 均为窗限 `window_realization_symdiff`，受请求窗界定——Prop 5.11 声明的全扫逃逸类在 p2 未被实例化，scan-amplification 表为空；老语料 cost.json 仍是该类的参考件）；③ 热校验尾部（CODE-Q1/Q2 ≈0.18 s、CODE-Q4 0.107 s）来自大表（posts 92,987 行自连接）上的覆盖/胞普查重演，与 Prop 5.11 的治理表项界一致。

I.1.6 5. 规模数字（RC-7 / RC-8 / P6 证据；全部由仓库现查，provenance 零失配）

5.1 逐库行数（real=BIRD/Spider 原值拷贝；authored= 治理所需最小作者件）

库	real 行	authored 行	最大真实表	版本行 (distinct)	策略实例行 (ids)
financial	**1,079,680**	0	**trans 1,056,320**	2 (2)	0
card_games	803,445	0	legalities 427,907	2 (2)	0
codebase_community	740,646	0	postHistory 303,155	2 (2)	8 (4)
formula_1	514,287	18	lapTimes 420,369	2 (2)	0
debit_card_specializing	423,050	0	yearmonth 383,282	2 (2)	2 (1)
european_football_2	222,796	0	Player_Attributes 183,978	2 (2)	0
california_schools	29,941	0	schools 17,686	2 (2)	2 (1)
thrombosis_prediction	15,952	0	Laboratory 13,908	2 (2)	6 (3)
world_1	239	20,076	country 239	2 (2)	0
Σ	**3,830,036**	**20,094**		**18 行 / 9×2**	**18 行 / 9 ids**

- **P6**：financial.trans 1,056,320 行 $\geq 10^5$ 且 authored=0（零合成承接 BIRD 原值；provenance 源 sha256 在册）。authored 全账：w1 country_history 20,076 = 239×84 月 + f1 points_scheme 18（真实历史积分口径的关系原子化），逐表与 provenance.json 声明比对**零失配**。
- **gov_* 种子 847 行**（DB 现查 = 种子文件行数 =847，90 文件）：version 18 / node 112 / metric 122 / measure_def 151 / alias 122 / routing 96 / anchor 38 / binding 130 / gran_edge 40 / disclosure 18。
- **RC-7（版本轴非平凡）**：9 库各 2 提交版本（timestamps 提交映射，ver(declared_at) 首次非平凡）；离对角承诺 F1-Q3（pinned v1 @ ver(T)=v2）由 V1 重演、其删标即拒由 F1e 伪造实证。
- **RC-8（治理变更翻转值）**：4 组同题双 T flip 全部落在本次 60/60 金标实跑内—— FIN-Q1/FIN-Q2（同 as_of=1996-06-30，declared_at 1997-06-15 vs 1998-09-15）= 0.13675213675213677 vs 0.08547008547008547；F1-Q1/F1-Q2 = 100.0 vs 252.0（2009 积分 10-8-6-5-4-3-2-1 vs 2010 25-18-…）；CA-Q1/CA-Q2 = 0.37985585826806795 vs 0.37804549460887515；W1-Q1/W1-Q2 = 4,068,933,086 vs 4,313,069,067（×1.06 常住口径）。

- **RC-7×RC-8 交互**：DEB-Q5 v2 治下 k 10→20 → Segment 胞 16<20 上卷 REWRITE（本次金标 +V5 重算双验）；建仓阴性对照 v1 治下同题 ANSWER [['LAM', 26.069]]（BUILD_REPORT）。
- **双路径口径对照**（建仓冻结证据，aware 进金标/blind 只入 notes）：fin 罚息率 0.14310% vs 2.17426%（15.19×）；code 采纳率 0.68085 vs 0.29038（2.34×）；debit 双币 CZK 15,626.14 vs EUR 1,166.15（13.4×）。dualpath_report.json：n=60, agree=60, mismatch=0。

I.1.7 6. CI 闸复跑

- `python3 impl/asof_verifier/ci_check.py` → **CI-CHECK: PASS**，`failures=[]`、`shared_internal_roots=[]`；校验器文件集 now={chk, forge, ci_check, runall, **forge_p2**}，import 根 ⊆ stdlib ∪ {duckdb} ∪ 校验器自身模块；A2/A4 无 compiler/pilot 向 sys.path/动态 import 逃逸。零共享红线在新增伪造件后继续成立。
- `python3 pilot2/ci/leak_check.py` → **PASS**(n=60, problems=0, importsA=[], importsB=[datetime,json], inter=[])。
- 建仓期 ND 闸证据未动（ci/nondegeneracy_report.json：ND-2 90 文件 847 行 0 命中、ND-3 干扰行 57.1%；ci/witness_report.json：ND-1 见证对 8/8）。
- 冻结面纪律：本任务对 `theory/`、`pilot/`、`pilot2/domains/`、`pilot2/build/`、`pilot2/ci/`（除 leak_report.json 为 leak_check 自身重写）零写入；pilot2/ 新增仅本报告。内容级幂等交叉核：certs2 重发射聚合 sha1 与 PORT 冻结值同；gov 种子文件 ↔ 库行数同 847。（建仓自报的整链 sha256 `d3258d3e`⋯属建仓脚本内部配方；本验收另立可复算配方：9×questions.json + 90×gov_seed.jsonl 排序串接 sha256 = `ae097d058e6200d98a16fe971aa19e8bc8b5e9d1c123b6a1835da44bd3aba3ac`，供后续对账。）

I.1.8 7. 诚实边界与待办

1. **DB 见证阻断集部分缺项 ACCEPT**（§3 边界）：非空 + 全在册即过；候选加固为 V6b 与独立重算阻断集对账。不影响判定正确性（决策与拒因均由其他检查独立钉住）。
2. **严格轨 50/60 属登记语义设计后果**：`range_request`/跨窗祈使把呈现窗当声明输入；若补机器可读窗形登记列可再收窄（承 PORT §4 裁定 4）。
3. **ND-4（治理知情臂原协议重挂 pilot2）预注册仍待执行**——本报告是证书学 V0–V6c 验收，不是评测臂结论。
4. **未激发的防御路径**（60 题无位点，代码在）：ver(T)↑ 前治理期 OOV、pinned_version_uncommitted、p2 别名缺失 MC(i) 证书侧；F1-Q4 型 delta 期窗 hull 裁剪的 REWRITE-vs-OOV 裁定分歧继续留档。
5. **Prop 5.11 全扫逃逸类在 p2 未实例化**（AM(iv) 均窗限模式）；该类代价证据以老语料 cost.json 为准。
6. 伪造电池为族级覆盖（每类检查面 ≥1 落点、30 例），非逐题 × 逐字段全笛卡尔。

I.1.9 8. 复跑清单

```
cd "/Volumes/SSD 1/explore_opportunity_cc/impl"
python3 asof_compiler/acceptance_pilot2.py      # §1 金标 60/60 + 证书重发射
python3 asof_verifier/runall.py                 # §2 四轨 (old51+p2 × 默认/严格)
python3 asof_verifier/forge_p2.py               # §3 新基座 10 基 + 30 伪造
python3 asof_verifier/forge.py                  # §3 老族 3 基 + 16 伪造 + symdiff-57
python3 measure_cost.py --pilot "../pilot2" --certs certs2 --out cost_p2.json   # §4
python3 asof_verifier/ci_check.py               # §6 零共享红线
python3 ../pilot2/ci/leak_check.py              # §6 泄漏闸
```

J Pre-Registration Protocol: the Eight pilot2 LLM Arms

The operative pre-registration for every model-arm number in the body (Table 4): frozen 2026-08-04, before any pilot2 LLM call, with prompt-pack hashes, scoring rules, the 40% elimination line, and the paper-rewrite rule for each possible outcome. Mechanical typesetting of the frozen `pilot2/PREREG_pilot2_arms.md` (Chinese); its §3 re-hangs the governance-informed arm protocol of Appendix K.

J.1 PREREG · pilot2 LLM 臂全集（8 臂 × 60 题，公开基座终局实证）

> **状态：FROZEN 2026-08-04，写于任何针对 pilot2 的 LLM 调用之前**。本文件写就时 `pilot2/runs/` 目录**不存在**，8 个 LLM 臂一次调用也未发生；本预注册阶段（含提示物料打包、泄漏审计、哈希冻结）**LLM 调用次数 = 0**。任务来源：作者裁定”论文只用公开数据证据”后，重跑 LLM 各臂——论文实证节最后一块。冻结的目的：8 臂结果直接决定论文 Table 2 与头条对照；若不先写死”每臂喂什么 / 怎么评分 / 三种结果各自怎么改写论文”，事后任何调整都等于调参到想要的结论。
>
> **上游冻结面（本协议只读引用，不改一字节）**：`pilot2/DESIGN_SPEC.md`（基座规范）、`pilot2/BUILD_REPORT.md`（建仓）、`pilot2/ACCEPTANCE_REPORT.md`（验收：编译器金标 60/60、校验器 ACCEPT 60/60、严格轨 50/60、30 伪造全拒、certs2 已在册）、`pilot/PREREG_governance_arm.md`（B6 原协议——本文件 §3 是其 ND-4 重挂）。**本文件与工件冲突时以工件为准**：§2/§3 的装配行为由 `prompt_pack/build_prompt_pack.py` 定义并已产出冻结包，本文是其可读镜像。

J.1.1 0. 冻结基线与哈希（全部由本机 sha256 复算，非手抄）

<table>
<tr><th>工件</th><th>sha256</th></tr>
<tr><td><code>pilot/run_pilot.py</code>（冻结 runner，本轮不修改一个字节；与 B6 §0 冻结值逐字节一致）</td><td><code>fecda681ccce203fa08e1a8b28a8ff722093a50ce656c62e86d921b594</code></td></tr>
<tr><td><code>~/.claude/skills/llmhub/bin/llmhub.py</code>（与 B6 冻结值一致）</td><td><code>f3dba7e05b2c089ebd1675b1e7b075c3813136c974192b791db27c6850</code></td></tr>
<tr><td><code>~/.claude/skills/llmhub/channels.json</code>（与 B6 冻结值一致）</td><td><code>d8d49e05decee5f517f41ee92f123a5a8560637bb49fe4b48e80fab8c4</code></td></tr>
<tr><td><code>pilot2/prompt_pack/build_prompt_pack.py</code>（本次新增并冻结）</td><td>见 <code>FREEZE_pilot2_arms.json</code></td></tr>
<tr><td><code>pilot2/prompt_pack/MANIFEST.json</code></td><td><code>c3618be218d922a62fa05d439c37c46ff48bd40a0c6373ae104e36ad48</code></td></tr>
<tr><td>9 库 <code>questions.json</code> + 90 份 <code>gov_seed/*.jsonl</code> 排序串接聚合</td><td><code>d3258d3e83d0d2d01555382283be754a7353087b1d76d0e39a046bc67b</code>
(= 建仓自报整链值，逐字节复现；配方：<code>cat $(ls domains/*/questions.json domains/*/gov_seed/*.jsonl \| sort) \| shasum -a 256</code>)</td></tr>
<tr><td>9 库 <code>warehouse.duckdb</code>、9+90 份逐文件哈希、18 份 prompt_pack 包</td><td>逐项见 <code>pilot2/FREEZE_pilot2_arms.json</code></td></tr>
<tr><td>本协议自身</td><td>写入 <code>FREEZE_pilot2_arms.json</code>（该文件在本文最后一次编辑之后生成，用作篡改留痕：本文此后若被修改，该项即不再匹配）</td></tr>
</table>

> **完整哈希清单一律以 `pilot2/FREEZE_pilot2_arms.json` 为准**——脚本生成，不经人手转抄；该 JSON 亦记录 `llm_calls_at_generation_time = 0` 与 `runs_dir_exists_at_freeze = false`。

对照锚点（全部取自已冻结的 `ACCEPTANCE_REPORT.md`，本轮**不重跑不重评**）：mechanism（确定性编译器，certs2）金标 60/60、校验器 ACCEPT 60/60 → **error_rate = 0/60**；题集构成 value 33 / rewrite 12 / refusal 15；拒因 OOV 4 / AM 5 / MC 3 / DB 3；簇预算 fin 8 / f1 8 / card 7 / code 7 / deb 7 / ca 6 / ef2 6 / th 6 / w1 5（9 簇 = 60 题）。旧 pilot（51 题企业 + 旧公共轨）的全部数字自本轮起**只作动机与附录**，不进结果表。

J.1.2 1. 臂表与逐臂参数 —— 一切与 B6/旧协议一致处直说继承

1.1 臂表（8 个 LLM 臂 × 60 题 = 480 次入账调用；mechanism 零调用）

臂	模型	渠道（llmhub 注册表首命中；跑前 A1 断言）	`{extra}` 槽位
`baseline_claude`	`claude-opus-4-6`	`tuzi` (`['tuzi','tokens1688']` 首位）	`""`
`baseline_qwen`	`qwen3-coder-next`	`gateway`	`""`
`baseline_deepseek`	`deepseek-3.2`	`gateway`	`""`
`baseline_minimax`	`minimax-m2.5`	`gateway`	`""`
`trivial_claude`	`claude-opus-4-6`	`tuzi`	`TRIVIAL_NOTE`（115 字符，sha16 `c3dee63a1f45c5a9`）
`trivial_v2`	`claude-opus-4-6`	`tuzi`	`TRIVIAL_V2_NOTE`（226 字符，sha16 `6de5f288aa4448fb`）
`trivial_v3`	`claude-opus-4-6`	`tuzi`	`TRIVIAL_V3_NOTE`（187 字符，sha16 `775f3f5dda2611eb`）
`governance_informed`	`claude-opus-4-6`	`tuzi`	`TRIVIAL_V2_NOTE + <db>.gov.txt`（§3；ND-4 重挂）
`mechanism`	确定性编译器	—（零 LLM 调用）	—（结果 = certs2 验收，§0 锚点）

三个提示变体的 note 文本**逐字节继承** `run_pilot.py` 顶部常量（不为 pilot2 改写一字—— 它们提到的 `gov_valid_time_anchor` / `gov_temporal_binding` 表在 pilot2 十表中同名存在，语义兼容）。臂名与旧 pilot 完全一致，论文表格逐列可对映。

1.2 公共参数（8 个 LLM 臂逐项相同；除标注外全部继承自 `run_pilot.llm()` 与 llmhub 缺省）

项	取值	继承性
调用方式	`source .work/env.sh` 后 `python3 ~/.claude/skills/llmhub/bin/llmhub.py chat --model <M> --prompt <P>`，经 `run_pilot.llm()` **import 原样调用**	与 B6 逐字节相同

项	取值	继承性
temperature	**不发送**（请求体只含 `model` / `max_tokens` / `messages`，llmhub.py `call()` 实测如此），由渠道默认值决定	继承
max_tokens	**512**（llmhub `chat` 缺省；不传 `--max-tokens`）	继承
HTTP 超时	**60 s**（llmhub 缺省 `--timeout`）	继承
子进程超时	**180 s**（`run_pilot.llm()` 的 `subprocess.run(timeout=180)`）	继承
重试	`retries=2` → 最多 **3 次尝试**；仅在输出为空（超时/空文本）时重试，**不因内容不满意重试**	继承
空响应	三次仍空 → `raw=""` → `kind="no_sql"` → 计错	继承
并发	臂内 `ThreadPoolExecutor(max_workers=4)`；臂间**串行**，顺序 = §1.1 表序（素基线 4 → 变体 3 → 治理臂）	继承（臂序为本轮新冻结项）
回合数	**单回合**：首个响应即终局，无执行反馈、无修复轮	继承
提示模板	`run_pilot.PROMPT` 原样（217 字符，sha16 `4a2aaf1d0b010c56`，含”可拒答”条款；所有臂共享同一模板与同一拒答许可——公平性条款）	继承
题目集与枚举序	`pilot2/domains/` 下 `sorted()` 9 目录 × 各自 `questions.json` 数组序（60 题；无 `public/` 目录）	结构继承，题集为 pilot2
数据库	各域 `warehouse.duckdb`，`duckdb.connect(read_only=True)`，经 `run_pilot.run_sql()`（数值金标）或 §4.2 行集取数（6 道非数值金标）	继承 + §4.2 扩展
响应缓存	`pilot2/runs/<arm>/<qid>.json`：`{qid, system, kind, value, verdict, raw[:2000], sql, prompt_sha256, prompt_chars, empty_response}`——**prompt_sha256/prompt_chars 对全部 8 臂记录**（B6 只对治理臂记，本轮推广到所有臂；不改变任何评分）	继承 + 留痕推广
RAW_CAP / REL_TOL	2000 / 0.005	继承
运行器	`pilot2/run_pilot2_arms.py`（运行阶段新写）：**import `run_pilot`，不复制其任何函数**；只提供 pilot2 路径装配、§4.2 扩展评分、`runs/` 写入	结构同 `run_gov_arm.py`
调用预算	**480 次入账**（8×60）；另许每个不同模型 ≤1 次两字连通性探针（≤4 次，不写任何缓存）；重试不增加入账题数	B6 同款探针条款

不得触碰：旧 pilot/runs/、pilot/pilot_summary.json、pilot/runs_gov/、impl/certs2/、pilot2/domains|build|ci|prompt_pack/ ——全部只读（§7 A6 跑后哈希断言）。磁盘纪律：一切写盘在 /Volumes/SSD 1/explore_opportunity_cc 下，每个 shell 先 source .work/env.sh。

J.1.3 2. 素基线与变体的 prompt 装配（含禁投字段清单）

2.1 装配式（对 8 个臂统一，唯一变量是 {extra}）

```
prompt = run_pilot.PROMPT.format(
    schema  = prompt_pack/<db>.schema.txt,     # 本域全部题共享，逐字节相同
    extra   = §1.1 表列值,                      # ""|NOTE|V2|V3|V2+gov
    question= q["question_zh"])                 # 题面唯一入口，逐字原样
```

2.2 schema 包（<db>.schema.txt，已产出冻结） 从该域 warehouse.duckdb 确定性导出：main schema **全部表**（业务表 + gov_* 十张登记表）按表名字典序，每表一行 CREATE TABLE main."<t>"("<c>" <TYPE>, …);（标识符一律双引号，列按物理序）+ 一行行数注记 -- rowcount main."<t>" = N，文件头两行来源注释。**gov_* 表入 schema 是继承而非新让利**：旧 pilot 的 schema.txt 同样列出治理表，素基线在两代基座下拥有同等的" 自行 SQL 查询治理表" 权利（单回合内）。9 份尺寸 2,833–7,371 字符，哈希见 MANIFEST.json。

2.3 禁投字段清单（A2 泄漏闸，适配 pilot2 字段分区） pilot2 questions.json 的 26 个字段按 ci/leak_check.py 冻结分区为评测可见集（16）与金标侧集（10），两集不交。**进入提示的只有 question_zh 一个字段**；其余逐条：

(a) 金标侧 10 字段 —— 一律不得进入提示、不得参与选行、不得影响任何分支：

1. gold_value —— 金标值
2. gold_sql —— 金标 SQL
3. expected_kind —— 应答/应改写/应拒
4. refusal_reason —— 拒因类（out-of-validity / anchor-mismatch / missing-caliber / disclosure-blocked）
5. refusal_subtype —— 拒因子型（am_i–iv / mc_i–ii）
6. rewrite —— 改写金标（kind / requested / effective）
7. windows —— 显式窗口坐标 ω_r（C3 A5 一等字段，只供校验器比对）
8. windows_note
9. notes —— 建仓注记（含双路径 blind 值等）
10. metric —— σ 的指标键（pilot2 把它划入金标侧；**也不得用作治理包选行谓词**，§3.2）

(b) 评测可见但非题面的 15 字段 —— 不得以独立字段形式注入提示：qid，domain，as_of，declared_at，metric_alias，scope，pinned_version，cross_window，anchor_override，window_request，requested_granularity，requested_time_gran，presentation，ctx_role，periods。这些是编译器/校验器消费的结构化意图 σ 与机器侧登记；声明时点/as-of/跨窗祈使/钉版本等信息**本来就写在 question_zh 的自然语言里**，那是 8 臂共见的同一段文本——LLM 臂只许从题面自然语言里读它们，与 mechanism 读结构化 σ 形成" 同信息、异形态" 的公平对照（继承 B6 §2.5 第 10 条的裁定）。

(c) 环境侧禁投：impl/certs2/*（证书）、impl/asof_compiler/*（含 QSPEC/适配器）、旧 pilot 任何 runs*/、任何既有臂表现摘要、pilot2/build/questions_def.py 等出题定义件。

2.4 A2 字符串级泄漏断言（已于冻结前实跑，跑时复跑） 对每题以**最大超集提示**（governance_informed 的完整装配）做逐字符串比对：gold_sql、json.dumps(windows)、windows_note、notes、json.dumps(rewrite)、refusal_reason、str(gold_value) 不得出现；长度 <8 的串跳过并记录（单个数字如 16 当然会出现在行数注记里，非泄漏——B6 同款护栏）。**2026-08-04 冻结前实测：60 题 × 7 字段命中 0；数值金标字面（≥8 位）弱检查命中 0**；短串跳过 22 项（全部为数值金标短字面），如实记录于本条。**较 B6 的强化**：refusal_reason 本轮**列入**断言——pilot2 种子经 ND-2 闸保证不含裁定类别 token（grep 四类拒因串于 90 份种子 = 0 命中，冻结前实测），故不存在 B6 时" 治理 note 原文即含 REFUSE 判据" 的豁免理由。

J.1.4 3. 治理知情臂：治理包构建规则（ND-4 —— B6 原协议重挂 pilot2）

重挂声明：臂定义、唯一变量（extra = TRIVIAL_V2_NOTE + 治理块）、二因子对照读法（trivial_v2 − baseline_claude = 指令；governance_informed − trivial_v2 = 治理内容）、评分与”只跑一次”纪律全部继承 pilot/PREREG_governance_arm.md §1/§4/§6；本节只重定义**治理块的来源与序列化**（旧包读企业 warehouse 的 gov 表，新包读 pilot2 种子）。

3.1 来源与打包单位

- 来源 = pilot2/domains/<db>/gov_seed/*.jsonl **十表全量**（DESIGN_SPEC §3.1 schema 法登记表；jsonl 即种子真源，与 warehouse 内 gov_* 表已由验收对账 847 = 847 行）。
- 打包单位是**库（目录）**，不是题目：同库所有题拿到逐字节相同的治理块，与 schema 包共享方式一致。域越界守卫：任何带 domain 字段的行其值必须等于目录名，否则打包器 SystemExit（实测：零越界）。
- **不按 metric/metric_alias 选行**（继承 B6 §2.1 裁定）：按题选行等于把 σ 的路由能力搬进本臂；路由留给模型在 122 条 metric、130 条绑定行里自己做——比按题喂更难，偏差方向对我方不利。

3.2 块清单与固定序（P0→P9 = 解析链序；截断丢块从 P9 反向开始）

序	登记表	内容（全部为 DESIGN_SPEC §3.1 白名单语汇：标识符/列名指针/闭枚举/谓词原子/披露参数/commit 元数据）
P0	gov_semantic_graph_version	版本轴：(committed_at, commit_seq) 全序，T→ver(T) 的唯一依据
P1	gov_metric_alias	表面词 →metric（**随版本变**，取错版本即撞干扰行）
P2	gov_metric	指标登记
P3	gov_measure_def	度量/谓词关系原子（口径本体；无可粘贴 SQL）
P4	gov_caliber_routing	口径路由（via 对象 + join 键列名 + 归因对齐）
P5	gov_valid_time_anchor	有效时间锚：effective_col/vf_col/vtc_col **列名指针** + coverage_mode（**无覆盖区间字面量**——覆盖域不物化公理）
P6	gov_temporal_binding	绑定规则（腿 → 锚 + rule_id 如 same_valid_time_window）
P7	gov_semantic_node	节点 → 物理表
P8	gov_granularity_edge	粒度格（实体/时间轴上卷边）
P9	gov_disclosure_policy	披露策略（π/k/mask/γ；非治域文件为空 → 块如实呈现 rows=0）

3.3 序列化（逐字节确定；build_prompt_pack.py 为准）

```
\n## 治理元数据（本库 gov_* 登记表全量导出；与具体问题无关，同库所有问题看到的内容完全相同）\n
-- 来源：<db>/gov_seed/*.jsonl（DESIGN_SPEC §3.1 十表 schema 法）；版本轴见 gov_semantic_graph_version\n
-- 每个块 = 一张登记表，一行一个 JSON 对象，行按文本字典序排列\n
\n-- TABLE <t>  (rows=<n>)\n<行…>\n … （块按 P0→P9）
```

行渲染 json.dumps(obj, ensure_ascii=False, sort_keys=True, default=str)，行按文本字典序 sorted()；重跑打包器已验证**逐字节一致**。

3.4 截断规则 GOV_CAP（确定性，非临场判断；继承 B6 §3 全文） `GOV_CAP = 32000`（只约束治理块；schema 与 TRIVIAL_V2_NOTE 不计入）。超限时先整块丢弃最低优先级（P9→P1，记 `dropped_blocks`），只剩一块仍超限则按既定字典序保留整行至 `GOV_CAP − 表头 − 300`，末尾追加固定标记 `-- [TRUNCATED@GOV_CAP=32000] dropped_blocks=[⋯]; kept_lines_in_last_block=<k>/<N>`。**实测：9 库全部未触发**（最大 `codebase_community` 23,132 字符 = 上限的 72.3%；`MANIFEST.json` 9 项 `truncated` 全 `false`）。上限继承 B6 数值且仍留 ≥27% 余量，选定依据是本地输入尺寸，与任何模型输出无关。

3.5 实测包尺寸与提示长度（`MANIFEST.json`；§7 A4 断言基准）

库	治理块字符	块数	schema 字符	治理臂提示 min–max
california_schools	12,530	10	4,946	17,941–17,970
card_games	18,295	10	5,194	23,949–24,029
codebase_community	23,132	10	4,660	28,256–28,279
debit_card_specializing	15,932	10	3,458	19,854–19,886
european_football_2	17,438	10	7,371	25,279–25,317
financial	19,211	10	3,825	23,505–23,523
formula_1	21,416	10	5,236	27,122–27,163
thrombosis_prediction	16,034	10	3,892	20,390–20,415
world_1	12,874	10	2,833	16,176–16,181

60 题提示总长（字符）：素基线四臂各 **292,332**；trivial_claude **299,232**；trivial_v2 **305,892**；trivial_v3 **303,552**；governance_informed **1,372,803**。逐包 sha256 见 `MANIFEST.json` / `FREEZE_pilot2_arms.json`。

3.6 已知让利与机理声明（跑前写死，论文必须披露）

- **结构上已消除的旧让利**：pilot2 种子经 ND-2 不含 `select_expr`/`where_expr`/ `snapshot_table`/覆盖区间字面量/裁定类别 note——B6 时代 public 簇"绑定给表达式、锚给快照表"的退化查表路径**不可表达**；这正是 ND-4 要检验的设计主张。
- **仍然让利的部分（刻意，如实自曝）**：绑定规则、锚列名指针、版本轴、别名映射、谓词原子、披露参数（k 值/掩码类/粒度格）全部原样给出，并配 `TRIVIAL_V2_NOTE` 显式使用指令。治理臂成绩应读作"上下文内治理 + 指令"的**上界**，不是未调优对手。
- **机理预测的依据**（DESIGN_SPEC §3.3）：正确作答须走通版本解析 → 别名 → 度量 → 路由 → 锚/窗推导的 4–6 步符号间接解析（每步有版本过滤与 ND-3 干扰行，全库占比 57.1%），以及对 OOV 判空/MC(ii) 分母质量/披露小胞 k 计数/AM(iv) 对称差的**数据探针**—— 后者在单回合无执行下原理上不可判（ND-1 见证对 8/8 已物化：同种子异数据翻转标签）。

J.1.5 4. 评分 —— 对全部 9 臂（8 LLM + mechanism）统一，不为任何臂另立标准

4.1 判定链（继承，import 不复制） `is_refusal()`（2026-08-03 修复版：剥围栏、容前置散文 ≤6 行、装饰符/全半角冒号/任意大小写）→ `extract_sql()` → REFUSE / 第一条语句执行（read_only）/ 无 SQL；六分类 verdict：`correct | wrong_value | execution_error | answered_should_refuse | refused_should_answer | no_sql`。LLM 臂拒答**不要求理由匹配**（继承；mechanism 的理由匹配已在 certs2 验收里完成）。

4.2 金标形态分派（本轮唯一评分扩展；60 题逐题冻结，运行期不得改派） pilot2 金标三形态。数值金标沿用 `run_pilot.score()` 原文；新增两形态只覆盖 **6 道题**：

形态	题数	题号	取数	判对规则
数值（value 33 + rewrite 6）	39	其余全部	`run_pilot.run_sql()` 首行首列	`run_pilot.score()` 原文：gold==0 时 `abs`≤`1e-9`，否则相对误差 ≤ REL_TOL=0.005
行集（rewrite/rollup）	4	CA-Q5, CODE-Q4, DEB-Q5, TH-Q3	`fetchall()` 全行	**多重集合相等，行序无关**；逐格：可数值化格按 REL_TOL 比，其余格 `str.strip()` 精确等；行元数须一致。**宽赦条款**：金标恰 1 行且恰 1 个数值格时，返回 1×1 标量命中该数值亦判对
字符串（rewrite/mask）	2	CODE-Q6, TH-Q4	首行首列	双方可数值化则按 REL_TOL 比（TH-Q4 “1934”），否则 `str.strip()` 精确等（CODE-Q6 “UK”）

数值化谓词（冻结）：`float(str(x))` 成功即数值。取数异常 → `execution_error`；空行集 → `execution_error`；`REFUSE` 于非 refusal 题 → `refused_should_answer`。缓存重评分（`rescore_cached` 语义）对 6 道扩展题由 `run_pilot2_arms.py` 以同一分派实现——评分是缓存 raw 的纯函数，修分永不重询模型。

让利声明（跑前写死，论文披露）：(a) rewrite 12 题**不要求 LLM 宣告改写**——只按改写后金标值判对；mechanism 则须出具含改写披露的证书才算数（certs2 已验）。方向 = 加强对手。(b) 行集宽赦条款使朴素全局聚合在 DEB-Q5/TH-Q3/CA-Q5 上无标签也可判对。方向同上。(c) **hull_trim 5 题（CARD-Q5, EF2-Q4, FIN-Q6, F1-Q6, W1-Q4）对朴素作答者天然易**：hull 外无数据行，朴素全窗聚合与裁剪窗聚合数值相等。此为如实预告，不改题不改分—— 但 §4.4 的按形态分片报告为**强制**，头条数字不得被这 5 题的白送分粉饰。

4.3 拒答统计（pilot2 分母）

- `correct_refusals` = #{expected_kind=refusal 且 verdict=correct}，**分母 15**；按拒因分报 OOV 4 / AM 5 / MC 3 / DB 3。
- `over_refusals_on_answer_questions` = #{expected_kind≠refusal 且 verdict=refused_should_answer}，**分母 45**（= value 33 + rewrite 12）。
- `coverage.answered` = #{kind∈{value,error}}/60；`coverage.refused` = #{kind=refuse}/60。

4.4 统计与对照（跑前写死）

- **簇自助**：**9 簇**（= 9 库）整簇有放回重采样，`B=2000`，每臂各自新建 `random.Random(20260731)` → 各臂抽样序列**逐次相同**（公共随机数，配对比较友好；与 B6 单臂做法一致的种子纪律）。CI95 = 排序后 `[rates[49], rates[1949]]`。
- **强制分片报告**：逐簇（9）× 逐金标形态（value/rewrite/refusal）× 逐拒因（4）—— 三张分片表全部落 `pilot2_arms_summary.json`，不许只报总数（§4.2(c) 与 §3.6 的原因）。

- **对照参考** = `baseline_claude`；对每臂报 `elim_<arm>` = 该臂答对的 reference 错题比例；对 `governance_informed` 另报与 `baseline_claude`、`trivial_v2` 的配对不一致计数 b/c（**只报计数，不报 p 值**；n=60、9 簇，不主张显著性）。
- **mechanism 0/60 直接取自 certs2 验收**，本轮零调用零重评；`elim_mechanism = 1.0` 恒等。
- **公开基座三判据（新预注册的描述性判据，不是旧 A/B/C 闸的重跑**——旧闸裁定已冻结于旧 pilot，不因本轮任何结果变更）：
  - **A′**：min(4 个素基线 error_rate) ≥ 0.30；
  - **B′**：mechanism error_rate = 0 且 elim = 1.0（已由 certs2 保证，只作陈列）；
  - **C′**：max(3 个 trivial 变体 elim) < 0.40（0.40 = 冻结 C 判据同一条线）。
- 汇总产物：`pilot2/pilot2_arms_summary.json`（唯一新增汇总文件；含逐题表、空响应计数、prompt 字符账、scorer 溯源三元组 run_pilot.py sha / REL_TOL / RAW_CAP）。

J.1.6 5. 预测（跑前写下：逐臂点预测 + 三情形机械判定 + 论文改写）

5.1 逐臂点预测与 80% 区间（错误数 /60）

臂	点预测	80% 区间	依据摘要
`baseline_claude`	**25（41.7%）**	18–32	15 拒答题多数漏拒（预测 correct_refusals 3/15，区间 1–6）+ 版本 flip 8 题错约半 + 口径路由/披露题散错
`baseline_qwen`	30	22–38	旧基座上与 claude 相近偏弱
`baseline_deepseek`	31	23–39	旧基座 0.529 的相对位次外推
`baseline_minimax`	36	27–45	旧基座 0.686 最弱位次外推
`trivial_claude`	25	18–33	一行提示在新基座边际益小
`trivial_v2`	24	17–32	指令臂略优于素基线
`trivial_v3`	25	18–33	通用示例迁移弱
`governance_informed`	**14（23.3%）**	8–22	见 5.2 分解
`mechanism`	**0（已定）**	—	certs2 60/60（非预测，锚点）

判据预测：**A′ PASS**（min 素基线 ≥ 0.30；点预测 25/60=41.7%）；**C′ PASS**（trivial elim 点预测 ≈0.15，区间 0.05–0.35）。若 A′ 实测 FAIL，处置见 §5.4(iv)。

5.2 治理臂分解预测（三数自洽：任一被推翻，另两个须同时被推翻）

- `elim_gov` 点 **0.55**（区间 0.35–0.75）⇒ 修掉 baseline_claude 约 25 错中的 ~14 道 ⇒ 残留 11；新引入错（“爱拒答”）3（`over_refusals` 由素基线的 ~1/45 升至 ~4/45，区间 1–9）⇒ **E_gov = 14**。
- `correct_refusals` 点 **9/15**（区间 6–12）：元数据可判 5 题（CA-Q6 AM(i)、FIN-Q8/W1-Q5 AM(ii)、DEB-Q7 AM(iii)、CARD-Q6 MC(i)）预期大多修复；探针专属 7 题见下。
- **决定性探针集（元数据不可判、ND-1 见证对已物化）= 7 题**：OOV×4 {FIN-Q7, F1-Q7, DEB-Q6, EF2-Q5}（覆盖域不物化，判空须对 D 现算）、MC(ii)×2 {CARD-Q7, F1-Q8}（同窗分母质量，`_den_present()` 型探针）、AM(iv)×1 {EF2-Q6}（锚对对称差审计）。**预测治理臂在其中答错 ≥3/7**（区间 3–6）；DISCLOSURE-BLOCKED 3 题不入决定集（策略在册即强提示 REFUSE，属半让利位点）。
- 按簇错误数点预测（合计 =14 自洽）：fin 2/8、f1 2/8、card 2/7、code 2/7、deb 2/7、ca 1/6、ef2 2/6、th 1/6、w1 0/5。

5.3 情形机械判定表（E = governance_informed 错误数/60；不许事后解释）

判定	条件	情形
ND-4 FAIL	$E = 0$	基座缺陷：新种子仍退化
(ii) 近退化	$1 \leq E \leq 2$	ND-4 名义通过但设计意图未达，按缺陷处理
(i) 显著改善但远大于 0	$E \geq 3$ 且 elim_gov ≥ 0.40	预期最可能
(iii) 未见实质改善	$E \geq 3$ 且 elim_gov < 0.40	—

0.40 即冻结 C 判据同一条线（`run_pilot.py` 第 334 行），与 B6 §5.3 同尺。

5.4 三情形（+ 一应急）各自怎么改写论文（防叙事漂移；先写死后看数） **(i) 显著改善但仍远大于 0 —— 预期最可能** 1. Table 2 以 pilot2 四组呈现（plain ×4 / prompt-variant ×3 / governance-informed / compiler），企业 51 题全部数字退出结果节，只留动机段（作者裁定的执行落点）。2. 头条重述："0% vs X%（治理知情后）于 60 题公开基座"；A′ 成立则保留" 前沿模型在治理绑定上系统性失败" 的限定表述（限定语 = 未获治理内容的臂）。3. **新增按族分解图/表**（本轮科学承重）：残余错误按 OOV/AM/MC(i)/MC(ii)/DB/值题分族，决定性探针集 7 题逐题列裁定——把" 探针类判断提示层原理上够不着（ND-1）"摆成可核结论。4. ND-4 作为**基座有效性证据**入 § 评测方法：治理知情臂 >0% 错误 ⇒ pilot2 修复了旧公共轨 0/20 退化（引 B6 数字作对照），" 非退化是设计出来的且被对抗性检验过"。5. 数字门与红线脚本同步换 pilot2 数字；旧" 无对照臂见过治理内容" 表述已在 B6 后删改，本轮确认不回潮。

(ii) 近退化（$E \leq 2$，含 $E=0$ 的 ND-4 FAIL）—— 必须如实报告的基座缺陷 1. **禁止庆祝**：这不是"LLM 很强" 的证据，而是 pilot2 种子仍在泄漏标签可判性—— 与 B6 公共轨 0/20 同病。论文**不得**以 pilot2 治理臂数字支撑" 提示层不够" 的主张。2. 立即做逐题根因剖检（哪些题、哪张登记表、哪行种子代答了标签）并写入 `pilot2_arms_summary.json.nd4_postmortem`；结果节改以 mechanism 证书支柱 + 素基线/变体对照为主，治理臂数字降级为" 基座局限" 段如实披露。3. 唯一允许的后续：**另行预注册**修订种子后的重建重跑（新协议新冻结），本轮数字保留在案。4. 摘要与贡献重排：证书可核验性提为第一贡献；"绑定正确性 LLM 不可达" 降格为 "在非退化性可验证的前提下开放"。

(iii) 未见实质改善（$E \geq 3$ 且 elim < 0.40） 1. 按原样报告，头条不变；**必须自查并在文中列出三个替代解释**：(a) 长上下文稀释（治理臂提示 16k–28k 字符，报逐题长度与错误的相关）；(b) 传输故障（`empty_responses` 计数与 §6 作废规则）；(c) 顺从性（AM(ii) 祈使句两题的逐题裁定）。2. 只可写" 在本协议（单回合、512 输出上限、库级全量治理内容 + 指令）下无效"，不得外推为" 提示工程永远无效"。3. ND-4 仍按 $E>0$ 判为基座非退化成立，(iii) 与 (i) 的差异只改变" 治理内容在上下文里有多大用" 的措辞档位，不改变机制主张。

(iv) 应急：A′ FAIL（min 素基线 < 0.30） 不重跑、不换 prompt、不换模型。论文把" 前沿系统性失败" 降档为实测错误率区间陈述，并把重心移到 refusal/rewrite 分片（预测素基线 correct_refusals ≤ 6/15——该分片几乎不可能与 A′ 同时翻车；若连它也翻车，如实报告并承认公开基座上素基线已够用，主张退守证书可核验性）。

5.5 页面预算 pilot2 数字**替换**同位置旧数字（Table 2 / 摘要 / §8 文本），零新增页面；按族分解表替换旧企业逐题附录的等量篇幅。不得压缩认输面（F 族/让利表）腾地方。

J.1.7 6. 只跑一次（继承 B6 §6，臂级适配）

1. **每臂只跑一次**。不得因结果不合意重跑、换提示、换模型、换渠道、调温度、加轮次、改截断、改选行、改评分分派。
2. **缓存只写不删**：`pilot2/runs/<arm>/<qid>.json` 一旦写入即终局；删除后重跑 = 换汤重试，明令禁止。断点续跑只补**从未写入**的 qid。
3. **空响应即错误**（3 次尝试仍空 → no_sql 计错）。
4. **臂级作废条款**：某臂空响应 ≥ 3 题（≥5%）判传输故障，**该臂整臂作废重跑**并在论文与本文件追记披露两次运行；不许只重跑失败题；其余臂不受牵连。

5. 本文件冻结后若需修改 §1–§4 任一条，必须在 §8 追记标注时间、内容、当时是否已见任何模型输出；见输出后的修改一律视为调参并须在论文披露。

J.1.8 7. 运行阶段验收断言（跑前 A0–A4/A7 全过才发第一次调用；跑后 A5/A6/A7 复验）

#	断言	失败处置
A0	`FREEZE_pilot2_arms.json` 全部 sha256 复算一致（runner/llmhub/channels/9 questions/90 seeds/9 warehouses/18 包/MANIFEST/本协议）	中止；如实报告漂移
A1	渠道解析：`claude-opus-4-6→tuzi`（首命中）、`qwen3-coder-next/deepseek-3.2/minimax-m2.5→gateway`	中止（渠道变 = 对手变）
A2	§2.4 泄漏断言复跑：60 题 × 7 字段 × 最大超集提示，0 命中	中止
A3	`build_prompt_pack.py` 重建与冻结包逐字节一致（18 份 sha256）	中止
A4	逐臂提示总长与逐库 min–max 与 §3.5/MANIFEST 精确相等（8 臂：292,332×4 / 299,232 / 305,892 / 303,552 / 1,372,803）	中止
A5	运行器 import `run_pilot`（不复制函数）；`run_pilot.py` sha256 跑后不变	中止
A6	写入路径只有 `pilot2/runs/**` 与 `pilot2/pilot2_arms_summary.json`；`pilot/`、`impl/`、`pilot2/{domains,build,ci,prompt_pack}/` 跑后 sha256/mtime 零变动	中止
A7	60 题、qid 唯一；expected_kind 计数 = value 33 / rewrite 12 / refusal 15；拒因计数 4/5/3/3；9 目录	中止

产出物（新增，不覆盖任何既有文件）：`pilot2/run_pilot2_arms.py`、`pilot2/runs/<8 臂 >/<qid>.json`（480 份）、`pilot2/pilot2_arms_summary.json`。

J.1.9 8. 追记

（留白。冻结后的任何偏离写在这里，注明时间、内容、当时是否已见模型输出。）

- 2026-08-04 预注册写就；`prompt_pack/` 18 包 + MANIFEST 生成并二次重建验证逐字节一致；A2 泄漏审计 60×7 零命中（含数值弱检查零命中）；ND-2 拒因 token 于种子 grep 零命中；`runs/` 不存在；**LLM 调用次数：0**。

- 2026-08-04 14:1x 追记（**写于全部 480 份缓存终局、结果已见之后**；由汇总代理执行；本条不改 §1–§7 任何一字，仅披露。本追记使本文件现哈希偏离 FREEZE 内自哈希—— 冻结字节哈希 `86e6a42e`⋯仍在 FREEZE_pilot2_arms.json，diff 仅本节，篡改留痕机制照常工作）：

  1. **trivial_v2 整臂作废重跑**（§6.4 执行，见输出后依条款行事）：run1 空响应 4 题（F1-Q1/FIN-Q2/W1-Q2/W1-Q3）≥3 阈 → 作废，原样保留于 `runs/trivial_v2__VOID_RUN1_2026-08-04/`（61 份，含 VOID_DISCLOSURE.json）；run2 空响应 0 为有效运行。两次运行（27/33 与 25/35）论文双披露。
  2. **A 组中断与续跑**（§6.2 条款内）：A 组运行器 13:12 后失联（baseline_claude 21/60 已写盘，成因不可远程判定），13:46 由汇总代理以冻结运行器断点续跑（跑前 A0–A4/A7 全 PASS 后发首调；只补从未写入 qid，缓存零删除零重询），14:08 四臂完成 exit 0；全程输出存 `runs/_groupA_resume_20260804_1346.log`。
  3. **连通性探针**：汇总阶段对 qwen3-coder-next/deepseek-3.2/minimax-m2.5 各做 1 次两字探针（不写缓存不入账）；无法查证 A 组中断前是否已各做 1 次，单模型累计可能为 2（条款字面” 每模型 ≤1”），如实存疑。
  4. **汇总阶段新增写盘（A6 清单之外，冻结面零触碰，跑后 –dry A0/A3 复验 PASS）**：`make_pilot2_summary.py`、`pilot2_arms_summary.json`（§4.4 指名件）、`pilot2_summary.json`（旧结构对齐版）、`ARMS_REPORT.md`、上述续跑日志。结果速记（详见 ARMS_REPORT.md）：E_gov=28>0 → **ND-4 PASS**；elim_gov=0.361<0.40 → 情形 **(iii)**；A′ PASS（min 素基线 0.600）；C′ PASS（max trivial elim 0.222）。

K Pre-Registration Protocol: the Governance-Informed Arm (B6)

The original pre-registration of the governance-informed arm (the "handed the governance layer verbatim" arm), written against the development suite and re-hung onto the public suite by Appendix J §3. Mechanical typesetting of the frozen `pilot/PREREG_governance_arm.md` (Chinese).

K.1 PREREG · B6 治理知情臂 `governance_informed`

> **状态：FROZEN 2026-08-03，写于任何 LLM 调用之前**。本文件在 `runs_gov/` 目录尚不存在、`governance_informed` 臂一次调用也未发生时写就。冻结的目的只有一个：治理知情臂**很可能显著好于素基线 39.2%**，若不先写死" 喂什么 / 怎么截断 / 怎么评分 / 三种结果各自怎么改写论文"，事后调整任何一项都等于调参到想要的结论。
>
> 任务来源：`paper/EXEMPLAR_GAP_PLAN.md` §B6（作者已批准，含明示批准消耗 API 调用）。标尺：`~/.claude/skills/vldb-exemplar-sop/`。判据：审稿人能从展品恢复出多少科学内容。
>
> **本文件与工件冲突时以工件为准**：§2/§3 的行为由 `gov_pack/build_gov_pack.py` 定义，本文只是它的可读镜像；若二者不符，以脚本实际行为为准并在此如实更正。

K.1.1 0. 冻结基线与哈希（全部由本机 sha256 复算，非手抄）

工件	sha256
`pilot/run_pilot.py`（冻结 runner，**本臂不修改一个字节**）	`fecda681ccce203fa08e1a8b28a8ff722093a50ce656c62e86d921b594`
`pilot/gov_pack/build_gov_pack.py`（治理打包器，本次新增并冻结）	`fdd680cc983facc27d285fb7c96ac45c5f3fa13ce7a2562ad5e2a0a5f5`
`pilot/gov_pack/MANIFEST.json`	`b51a3c6501bc2d8ffec7fe3b787d7bac74c26b75ae78a21a75afcf05ab`
`~/.claude/skills/llmhub/bin/llmhub.py`	`f3dba7e05b2c089ebd1675b1e7b075c3813136c974192b791db27c6850`
`~/.claude/skills/llmhub/channels.json`	`d8d49e05decee5f517f41ee92f123a5a8560637bb49fe4b48e80fab8c4`
6 个目录的 `questions.json` / `schema.txt` / `warehouse.duckdb`（18 份）	见 `gov_pack/FREEZE_HASHES.json`
6 个治理包 `gov_pack/*.gov.txt`	见 `gov_pack/FREEZE_HASHES.json`（与 §2.8 表一致）

> **完整 64 位哈希一律以 `pilot/gov_pack/FREEZE_HASHES.json` 为准**——该文件由脚本生成，不经人手转抄；本文表格只重复 4 个关键脚本的全值以便肉眼比对。该 JSON 亦记录了它自身生成时 `llm_calls_at_generation_time = 0`，并含 `PREREG_governance_arm.md` 自身的哈希（在本文最后一次编辑之后生成，用作篡改留痕：本文此后若被修改，该项即不再匹配）。
>
> 对照锚点（取自 `pilot/pilot_summary.json`，B3/B5 修分后的当前值，未改动）：`baseline_claude` 错误率 **0.39215686…（20/51）**、`baseline_qwen` 0.37254902（19/51）、`trivial_claude` 0.49019608（25/51）、`trivial_v2` = `trivial_v3` 0.45098039（23/51）、`baseline_deepseek` 0.52941176、`baseline_minimax` 0.68627451、`mechanism` **0.0（51/51 correct）**。`refusal_stats.baseline_claude.correct_refusals = 3/14`；`mechanism = 14/14`。

K.1.2 1. 臂的定义 —— 唯一变量是”是否看到治理内容”

名称：governance_informed（论文 Table 2 中列为第三组”governance-informed”，与 plain / prompt-variant 两组并列）。

逐项取值（左列若与既有臂不同即为违约）：

项	取值	与既有臂的关系
backbone	claude-opus-4-6	同 baseline_claude / trivial_claude / trivial_v2 / trivial_v3
渠道	tuzi (https://api.tu-zi.com/v1)	同上。**注册表解析**：claude-opus-4-6 同时存在于 ['tuzi','tokens1688']，llmhub 在未给 --channel 时取注册表中**第一个**命中渠道 = tuzi；本臂与既有臂一样**不传 --channel**，跑前必须断言解析结果仍为 tuzi（§7 A1）
调用方式	python3 ~/.claude/skills/llmhub/bin/llmhub.py chat --model claude-opus-4-6 --prompt <P>，经 run_pilot.llm() **原样调用**（import，不复制不改写）	逐字节相同
temperature	**不发送**（请求体只含 model / max_tokens / messages），由渠道默认值决定	与全部 8 臂完全一致；本臂不引入任何采样参数
max_tokens	**512**（llmhub 默认；run_pilot 不传 --max-tokens）	同上
HTTP 超时	**60** s（llmhub 默认 --timeout）	同上
子进程超时	**180** s（run_pilot.llm() 的 subprocess.run(timeout=180)）	同上
重试	retries=2 → **最多 3 次尝试**；仅在”输出为空”时重试（超时或空文本），**不因内容不满意重试**	同上
空响应	三次仍空 → raw="" → extract_sql 返回 None → kind="no_sql" → 判 no_sql（计为错误）	同上
并发	ThreadPoolExecutor(max_workers=4)	同 LLM 臂（mechanism 才是 1）
题目集	同一 51 题，同一枚举顺序 (sorted(domains/*) + public)	同上
数据库	同一 warehouse.duckdb，read_only=True，同一 run_sql()	同上
回合数	**单回合**：首个响应即终局，无执行反馈、无修复轮	同上（08-eval.tex:50-51）
提示模板	run_pilot.PROMPT **原样**（217 字符，含”可拒答”条款）	同上；**只有 {extra} 的内容不同**
响应缓存	runs_gov/governance_informed/<qid>.json 记录 {qid, system, kind, value, verdict, raw[:2000], sql} + 附加 prompt_sha256 / prompt_chars	记录形状兼容 rescore_cached()；**写入新目录，不触碰任何既有 runs/**

项	取值	与既有臂的关系
RAW_CAP	2000	同上

唯一变量：`{extra}` 槽位的内容。- `baseline_claude`：`extra = ""` - `trivial_v2`：`extra = TRIVIAL_V2_NOTE`（226 字符，2026-07-31 冻结）- `governance_informed`：`extra = TRIVIAL_V2_NOTE + < 本目录治理块 >`

这给出一个干净的二因子对照，**两条差分各只动一个变量**：

```
trivial_v2          − baseline_claude = 指令（已测：45.1% vs 39.2%）
governance_informed − trivial_v2      = 治理内容（本次要测的量）
```

选择”指令 + 内容”而非”裸内容”的理由（**跑前写死**）：`trivial_v2` 已经单独测过”只给指令不给内容”；若本臂只给内容不给指令，审稿人会说”你没告诉它怎么用这些表”。本臂设计为**同时支配两条平凡臂**（内容 ∪ 指令），使”仍然失败”这一结论无法用”提示写得不好”解释。

K.1.3 2. 喂什么 —— 打包单位是目录，不是题目

2.1 打包单位与筛选谓词 治理块 = 该题所属**目录**（5 个企业域 + `public`）的治理表导出。**同一目录下的所有问题拿到逐字节相同的治理块**——与 `schema.txt` 的共享方式完全一样。

- 企业域谓词：`WHERE domain = '< 目录名 >'`（`gov_business_domain` 用 `domain_key`）。
- `public` 谓词：`WHERE domain LIKE 'public:%'`（该目录含 4 个子域，全部纳入）。
- **该谓词是空操作**：仓库本身按域切分，实测每张带 `domain` 列的表其取值恒等于目录名（`public` 恒为 `public:*`）。打包器**断言**而非静默过滤：一旦出现越界取值即 `SystemExit` 中止（`build_gov_pack.py::_guard_domain`）。
- **不按 `metric` 选行 —— 这是刻意的**。`q['metric']` / `q['binding_id']` 属于结构化意图 σ，是编译器消费而基线不消费的输入；若按 metric 过滤，等于把 σ 的路由能力搬进本臂，“唯一变量是治理内容”就不再成立。**路由留给模型自己做**（指标名多数已出现在题干里）。代价是本臂要在 12 条 `public` 绑定里自己找对那一条——这比按 metric 喂**更难**，即偏差方向对我方不利，符合”宁可强化对手”的原则。

2.2 块清单与优先级（`build_gov_pack.py::PRIORITY`）

序	表	内容	计划 §B6 点名
P0	`gov_semantic_graph_version`	版本轴（graph version 钉）	是（“版本轴”）
P0	`gov_business_domain`	域定义/主题域（`public` 无此表）	是（“gov_* 表”）
P1	`gov_temporal_binding`	**绑定规则**（企业域：分子/分母锚 + `rule` + note；`public`：`semantic_object/select_expr/where_expr/binding_rule`）	是
P2	`gov_valid_time_anchor`	有效时间锚（企业域：`anchor_type/effective_date/valid_from_col/valid_to_col`；`public`：`snapshot_table` + `valid_from/valid_to`）	是
P3	`gov_caliber_routing`	**口径路由**（via_table / join_keys / attribution_alignment）	是（“口径文档”）

序	表	内容	计划 §B6 点名
P4	`gov_semantic_edge`	`numerator_of` / `denominator_of` / `aggregate_of` 语义边	是
P5	`gov_semantic_node`	节点 → 物理表 FQN、层、ai_usable	是
P6	`gov_disclosure_policy`	结构化拒答披露策略（**仅 aibuy/email 存在**）	超出点名，见 2.3
P7	`gov_semantic_test_case`	治理断言（含 anchor_alignment 断言）	超出点名，见 2.3
P8	`gov_physical_coverage`	表 → 层覆盖	超出点名，见 2.3
P9	`gov_physical_column`（**过滤**）	列注释中与时态/口径有关的行	超出点名，见 2.3

P9 的谓词冻结如下（不得按结果调整）：

```
SELECT * FROM main.gov_physical_column
WHERE layer = 'ADS'
   OR regexp_matches(coalesce("comment", ''),
        '分子|分母|口径|caliber|锚|有效时间|快照|snapshot|版本|version')
```

理由与实测：全量 `gov_physical_column` 为 123–341 行 / 32,821–77,393 字符，是**其余全部治理内容之和**的 **2.33**×–**6.45**×（aibuy 2.38 / domestic_newprod 2.92 / email 3.94 / quality_voc 2.33 / rma 6.45），全量纳入会 (a) 把绑定规则挤出预算或 (b) 把提示灌到 ~110k 字符、显著改变延迟与”大海捞针”特性——两者都会让本臂**变弱**，即偏差指向我方结论，不可接受。同时实测 **只有 8.5%–13.1% 的注释是列名回声**，其余含真实业务语义，其中 6–19 行/域直接含“分子/分母/口径/caliber/锚”字样（如 `problem_rate` 分子、`problem_rate` 重算（`caliber=rma_problem_to_sales`））。故取”结构位（ADS 层）∪ 词法命中”这一确定性谓词，把有绑定价值的行留下、把 300 行样板挡在外面。

2.3 为什么纳入 P6–P9（超出计划点名的 4 张表） 计划 §B6 点名 5 张表 + schema。本预注册**多喂**了 4 张。方向是**加强对手**，理由：审稿人的第一问是”要是把治理元数据喂给它呢”，最强的回答是”我们把该域**全部**治理内容都给了它”。`gov_physical_column` 是唯一被过滤的表，理由见 2.2；被完全排除的表：**无**。

2.4 序列化格式（逐字节确定）

```
\n## 治理元数据（本库 gov_* 导出；与具体问题无关，同库所有问题看到的内容完全相同）\n
-- 来源：<dir>/warehouse.duckdb 的 main schema；版本轴见 gov_semantic_graph_version\n
-- 每个块 = 一张治理表，一行一个 JSON 对象，行按文本字典序排列\n
\n
-- TABLE main.<t>  (rows=<n>)\n<行1>\n<行2>…
\n\n
-- TABLE main.<t2> …
```

- 行渲染：`json.dumps(dict(zip(cols,row)), ensure_ascii=False, default=str)`（保留 NULL 为 `null`，保留全部列，不裁字段）。
- **行序**：渲染后按文本字典序 `sorted()`——与 DuckDB 扫描顺序无关，重建逐字节可复现（已验证）。
- **块序**：P0→P9 固定。

2.5 禁投清单（逐条；违反任一条即本臂作废） **只有 `q['question_zh']` 一个字段进入提示，逐字原样**。以下一律不得进入提示、不得参与选行、不得影响任何分支：

1. `q['gold_value']` —— 金标值
2. `q['gold_sql']` —— 金标 SQL

3. `q['naive_value']` / `q['naive_sql']` —— 朴素对照值与 SQL
4. `q['expected_kind']` —— 该题应答还是应拒
5. `q['refusal_reason']` —— 拒答类别（missing-caliber / anchor-mismatch / out-of-validity）
6. **`q['windows']`** —— 显式窗口坐标 ω_r（C3 A5 的一等字段）
7. `q['windows_note']`
8. `q['binding_id']` —— 结构化意图 σ 的绑定键
9. `q['metric']` —— σ 的指标键（**也不得用作选行谓词**，见 2.1）
10. `q['as_of']` / `q['domain']` / `q['params']` / `q['notes']` / `q['qid']` 作为**独立字段**（as-of 时点本来就出现在 `question_zh` 的自然语言里，那是所有 8 臂共见的同一段文本）
11. `impl/certs/*.json` —— 证书（α 物化窗、探针、见证）
12. `<dir>/compiler.py` 及其 `QSPEC` —— 编译器的每题结构化参数
13. `pilot_summary.json` / `runs/*` / `RESCORE_DIFF.md` —— 任何既有臂的表现
14. 任何**按题**过滤治理行的做法（打包单位恒为目录）

结构性保证优于事后检查：`build_gov_pack.py` 只打开 `warehouse.duckdb`，源码中**不出现** `questions.json`、`certs`、`compiler`、`runs` 任何路径；提示装配函数的入参只有 `(dir, question_zh)`。§7 A2 另加字符串级泄漏断言作为第二道闸。

2.6 泄漏审计（已跑，2026-08-03，跑前） 对 6 个包 × 51 题做逐字符串比对：`gold_sql` / `naive_sql` / `windows` 的 JSON / `windows_note` / `notes` / `question_zh` **命中 0 次**。朴素子串测试另报 3 例假阳性（`PUB-C-03` 金标值 3、`PUB-E-03`/`PUB-O-03` 金标值 4—— 单个数字字符当然会出现在行数注记里），非泄漏，如实记录。

2.7 已知让利（必须在论文里披露，不得隐藏） 治理内容本身就写着拒答规则，喂它等于把大量拒答判据直接交给对手。**这是刻意的**，但必须逐条自曝，否则读者会以为本臂是靠自己推出来的：

让利	出处	影响
`gov_caliber_routing['reco_item` **逐字写着”（AIBUY-Q5=MC(i) 不变）“**	`gov_packs/aibuy.gov.txt:26`	**1/51 题的裁定类别被点名**。仍照喂（删改治理内容等于对我方有利的裁剪）。该题在逐题附录里打 † 标；注：`baseline_claude` 本来就答对该题，故不改变对照结论
`rma` 绑定 note 写着”536-SKU 无同月销量分母 → missing-caliber REFUSE”	`rma.gov.txt:13`	`rma_q5` 可被 note 直接模式匹配。故 `rma_q5` **不计入** §5 的” 决定性 MC(ii) 预测集”
`email` 绑定 note 写着” 以 dwd__email__message__di(MailDate 锚) 的邮件量充当分母 = 跨锚跨口径 → anchor-mismatch REFUSE”	`email.gov.txt:15`	与 `EMAIL-ASOF-05` 的构造逐字对应
`gov_semantic_test_case` 的 `*_A1/A2/S7/S8/S9/S10` 断言逐条写明” 否则 REFUSE (anchor-mismatch)”	各域 P7 块	四道 anchor-mismatch 题的判据被直接给出
`public.gov_temporal_binding` 含 `select_expr` / `where_expr`（如 `SUM("Population") + "Continent" = 'Asia'`），`gov_valid_time_anchor` 含 `snapshot_table` 与 `valid_from`/`valid_to`	`public.gov.txt`	**`public` 20 题近乎退化为查表**：绑定给表达式、锚给快照表与有效区间。这是治理内容自带的形态（编译器读的正是这两张表），不做裁剪；后果是本臂在 public 簇上**预期大幅领先**，必须按簇分报（§4.4），不许只报总数

读法（跑前写死）：本臂的成绩应当被读作” 上下文内治理 + 指令” 的**上界**，而不是” 一个未调优的对手”。若它在这种让利下**仍然**留下错误，该结论比不让利时更强。

2.8 实测包尺寸与哈希（gov_pack/MANIFEST.json）

目录	治理块字符	sha256	块数	schema.txt 字符	提示总长 min–max
aibuy	21,660	aed6ee958c68925f164fe32fe448af5a8b965ab71[illegible]ab3edd[illegible]83b7804540e	1	6,131	25,396–28,390
domestic_newprod	30,362	787a69fbb976944f40403e60f15[illegible]bad7e9e7fa75[illegible]a14f7[illegible]e0af59099e8	10	10,251	41,080–41,153
email	25,039	a22f71a419aaae1197821d115e1[illegible]8dfc07d3ce36a[illegible]b5b1[illegible]def7cd124dc	1	9,439	34,376–34,745
quality_voc	20,398	107df7d54affd6ee85e13364f82[illegible]e6d8dbc8d2bed[illegible]1d[illegible]4899fe5a8af	18	7,942	27,884–27,904
rma	15,495	071b6c08c2cd7a5b0747ec6adc1[illegible]1d2e1b3d93a2[illegible]2de[illegible]d61ec5adcb	15	13,803	34,700–34,845
public	6,685	dc955a0c839d07991a6c585cddb[illegible]5171b903a2ef0[illegible]ba[illegible]529eaae	3	2,019	9,147–9,189

51 题提示合计 **1,225,915 字符**。`TRIVIAL_V2_NOTE` 226 字符（sha256 前 16 位 `6de5f288aa4448fb`）。`PROMPT` 模板 217 字符。**跑前必须校验这 6 个 sha256**（§7 A3）。

K.1.4 3. 截断规则（确定性，非临场判断）

```
GOV_CAP = 32000            # 只约束治理块；schema.txt 与 TRIVIAL_V2_NOTE 不计入
while len(pack) > GOV_CAP and 块数 > 1:
    丢弃**最低优先级的整块**，记入 dropped_blocks
if len(pack) > GOV_CAP:    # 只剩一块仍超限
    保留该块的表头行 + 按既定字典序尽量多的整行，直到 (GOV_CAP − 表头 − 300) 用尽
末尾追加固定标记:
-- [TRUNCATED@GOV_CAP=32000] dropped_blocks=[…]; kept_lines_in_last_block=<k>/<N>
```

- 丢弃顺序 = P9 → P8 → … → P1，**永远先丢最不绑定相关的**。
- 截断只发生在**整行**边界，绝不切半个 JSON。
- **实测：6 个目录全部未触发**（最大 30,362 = 上限的 94.9%，`domestic_newprod`）。`MANIFEST.json` 中 6 项 `truncated` 全为 `false`。
- 上限如何选定的**如实交代**：先本地实测 6 个包的尺寸（30,362 为最大），再取 32,000（千位上取整 + ~5% 余量），**目的就是不触发截断**——因为截断只会削弱本臂。该选择基于**本地输入尺寸**，在看到任何模型输出之前完成；预注册要防的是按**结果**调参，不是禁止按输入体量设预算。此段落即为该选择的公开留痕。
- 响应侧截断沿用既有规则：缓存 `raw` 截至 **2000 字符**（`RAW_CAP`），`rescore_cached()` 对达到上限的记录改用写入时提取的 `sql` 字段——与 8 臂同一段代码。

K.1.5 4. 评分规则 —— 沿用，不为新臂另立标准

4.1 一律复用 `run_pilot.py` 的函数（import，不复制） `is_refusal()` / `extract_sql()` / `run_sql()` / `score()` / `rescore_cached()` / `llm()` / `PROMPT` / `REL_TOL=0.005` / `RAW_CAP=2000`。**不得复制粘贴其中任何一行**，避免出现”新臂专用副本”。

4.2 判定链（与 LLM 臂逐字相同）

1. raw → extract_sql()：先剥 ``` 围栏，再判 REFUSE（2026-08-03 修复版 is_refusal()：容忍前置散文 ≤6 行（_PROSE_LOOKAHEAD=6）、行首装饰符 > * _ # - • · 与反引号、半角/全角冒号、任意大小写、未闭合围栏；一旦遇到 WITH / SELECT / (开头即判定该块是作答）。
2. REFUSE → kind="refuse"；否则取第一条语句执行：有值 → kind="value"，None/异常 → kind="error"；无 SQL → kind="no_sql"。
3. score(q, kind, value)：
   - expected_kind == "refusal"：refuse → **correct**，否则 **answered_should_refuse**。
   - 值题：refuse → **refused_should_answer**；error → **execution_error**；no_sql → **no_sql**；否则 gold==0 时 |v|≤1e-9、否则相对误差 ≤0.5% → **correct**，不然 **wrong_value**。
4. **拒答理由不作要求**：run_pilot 只对 mechanism 分支要求 refusal_reason 子串匹配；本臂是 LLM 分支，**不施加该要求**——给对手同样的宽松，也避免新标准。

4.3 五类分型 + correct 的计法

taxonomy = {correct, wrong_value, execution_error, answered_should_refuse, refused_should_answer, no_sql}（六键，取值为 0 的键在 summary 中省略，与既有 summary 写法一致）。

- correct_refusals = #{qid : expected_kind=="refusal" 且 verdict=="correct"}，分母 **14**。
- over_refusals_on_value_questions = #{qid : expected_kind!="refusal" 且 verdict=="refused_should_ans
  分母 **37**。
- coverage.answered = #{kind ∈ {value,error}} / 51；coverage.refused = #{kind=="refuse"} / 51。
- error_rate = #{verdict != "correct"} / 51。

4.4 统计与对照（跑前写死）

- **簇自助**：与既有臂同一程序——6 个来源簇（5 域 + public）整簇重采样，B=2000，random.Random(20260731)。因本臂单独跑，抽样序列是**独立的一次 2000 次抽取**（不与 8 臂那轮共享 rng 状态）；这不改变任何既有数字，**pilot_summary.json 一个字节都不动**。
- **必须按簇分报**：6 个簇各自的 error/correct_refusals 必须落在产出里，理由见 §2.7 最后一行（public 簇的绑定表自带表达式，只报总数会误导）。
- **配对对照**：对 baseline_claude 与 trivial_v2 各报一组不一致计数 b = #(既有臂对 & 本臂错)、c = #(既有臂错 & 本臂对)，**只报计数，不报 p 值**（n=51、6 簇，不主张统计显著性）。
- elim_gov = 本臂答对的、baseline_claude 的 20 道错题的比例——复用论文既有的 eliminated_by_* 口径（trivial_claude 0.15、mechanism 1.00）。
- **绝不重跑预注册闸门 A/B/C**：本臂与 v2 扩展同属**闸后**扩展，不参与 A/B/C 裁定。mechanism 的 0/51 不重跑、不重评。

K.1.6 5. 预测（跑前写下，含三情形解读）

5.1 机制层面的先验（为什么敢预测）

- **out-of-validity（5 题：PUB-W-05/PUB-C-05/PUB-E-05/PUB-O-05/NX-Q7）**：锚表给了 valid_from/valid_to，判定退化为区间比较 → **预期大幅改善**（baseline_claude 这 5 题全错）。
- **anchor-mismatch（4 题：EMAIL-ASOF-05/QVOC-05/NX-Q6/rma_q6）**：题干自曝错配构造，绑定 note 与 *_A1/A2 断言逐字写明"异窗即 REFUSE" → **预期大幅改善**；残余风险是顺从性（题干是祈使句"请直接用 5 月除以 4 月"）。
- **missing-caliber 中的 MC(ii)（同窗分母人口为空）**：**不可能从任何元数据判定**，必须**执行探针**看同窗分母是否为 0/空。本臂是单回合、无执行反馈，结构上做不到；编译器则以 _den_present() 真的跑了一条探针。→ **决定性预测集 = {AIBUY-Q6, EMAIL-ASOF-06, QVOC-06}**（rma_q5 因 note 点名 536-SKU 而排除；AIBUY-Q5 是 MC(i)、且类别被 note 点名，排除）。

- **值题**:口径路由与 `numerator_of/denominator_of` 边应能修掉一部分分母口径错,但 `domestic_newprod` 的 22× 漏斗膨胀、SCD-2 双时态回算等仍难。
- **反向风险**:治理文本通篇是 REFUSE 判据，本臂可能变得" 爱拒答"，`refused_should_answer` 上升（`baseline_claude` 只有 1 例）。

5.2 点预测与区间（写死）

量	点预测	80% 区间	对照锚
总错误数 E / 51	**11（21.6%）**	5 – 19（9.8%–37.3%）	`baseline_claude` 20（39.2%）
`elim_gov`	**0.65**	0.35 – 0.85	`trivial_claude` 0.15；`mechanism` 1.00
`correct_refusals`	**9 / 14**	5 – 12	`baseline_claude` 3/14；`mechanism` 14/14
`refused_should_answer`	**3 / 37**	0 – 8	`baseline_claude` 1/37
决定性集 {AIBUY-Q6, EMAIL-ASOF-06, QVOC-06} 答错数	≥ **2 / 3**	2 – 3	`baseline_claude` 3/3 错
E > 0（本臂**不会**到 0）	是；预测 E ≥ 4	—	`mechanism` E = 0

按簇点预测(错误数):`public` 1/20、`rma` 2/6、`aibuy` 2/6、`email` 2/6、`quality_voc` 2/6、`domestic_newprod` 2/7 —— **合计 11，与总错误数点预测自洽**。（`baseline_claude` 实测对应值：4/20、5/6、2/6、3/6、2/6、4/7，合计 20。）

三项点预测互相自洽，不是各自拍脑袋：`elim_gov = 0.65` ⇒ 修掉 20 道旧错中的 13 道 ⇒ 残留 7 道；再加**新引入**的错 4 道（主要来自" 爱拒答"，`refused_should_answer` 由 1 升到 3）⇒ E = 7 + 4 = 11。三个数字任一被推翻，另外两个也必须同时被推翻——这正是预注册要的可证伪性。

5.3 情形判定表（机械，不许事后解释）

E（错误数/51）	`elim_gov`	情形
≤ 2	任意	**(ii) 接近 0**
≥ 3	≥ 0.40	**(i) 显著好于 39.2% 但远大于 0**
≥ 3	< 0.40	**(iii) 未见实质改善**

阈值 0.40 不是新造的：它就是冻结的 C 判据里" 平凡提示消除 < 0.40" 的同一条线（`run_pilot.py` 第 334 行），复用它保证本臂与平凡臂用同一把尺。

5.4 三种情形各自怎么改写论文（防叙事漂移）

(i) 显著改善但仍远大于 0 —— 预期最可能

1. **Table 2**（`sections/08-eval.tex`）：加第三组 "governance-informed" 一行（plain / prompt-variant / governance-informed / compiler），成本约 0.02 页。
2. **头条重述**：`sections/01-intro.tex` 与摘要把 "0% vs 37–69%" 改为 **"0% vs 治理知情后的 X%"**——更强也更可信的对照。
3. **`08-eval.tex:385` 与 `10-conclusion.tex:53` 的" 没有任何对照臂看过治理内容" 必须删改**，改写为" 一条对照臂拿到了该域**全部**治理内容与显式指令（§2.7 逐条列出让利），仍留下 X 个错误"，并把 `08-eval.tex:387-388` 的" 两项待补测量" 减为一项（执行反馈臂）。
4. **新增按族分解**（本图/表是本次的科学承重）：残余错误按 OOV / AM / MC(i) / MC(ii) / 值题分族列出，把"MC(ii) 需要探针、提示层原理上够不着" 这一条摆成可核结论。
5. **"绑定是编译期问题不是提示问题" 反而被加固**：论证从"LLM 答不对" 升级为 **"即使把治理层原样交给它、并告诉它怎么用，仍答不对需要探针的那一族"**。

6. 红线/数字门：`tools/check_numbers.py` 增本臂断言；`tools/check_redlines.py` 中与" 无对照臂见过治理内容" 相关的断言必须同步改写（否则四门会绿着说假话）。

(ii) 接近 0（E $\leq$ 2）—— 必须诚实报告的重大发现

1. **论文定位改写**：从"LLM 做不到" 改为" **做得到但不可核验**"。这不是退守——本稿的**证书支柱恰好是那个版本的答案**（16 条伪造全被拦、独立重导出、D6/D7/D8 被值级金标放行）。
2. **摘要与 §1 贡献重排**：把" 可核验性/证书" 提到第一位，把" 绑定正确性" 降为第二位；删掉任何暗示" 前沿模型做不到治理绑定" 的措辞。
3. **A 阃的表述必须加限定**：`run_pilot.py` 的 A 判据（前沿失败 ≥30%）只对 **未获治理内容**的臂成立，须在 §8 显式写死这一范围；不得继续以无限定形式陈述。
4. **Table 2 的对比语义改为" 同等准确率、无证书"**，并把 F-B（伪造 × 检查命中矩阵）与 F-C（D1–D15 分歧台账）提到结果面板前列——它们才是此情形下的头条。
5. **仍须报告的反证**：本臂拿到的是**让利后的**治理内容（§2.7），因此" 做得到" 这个结论要写成" 在治理层被逐字放进上下文、且题量 51、且 public 簇绑定自带表达式的条件下做得到"。

(iii) 未见改善（`elim_gov` < 0.40）

1. **按原样报告**，头条不变，但**必须自查三个替代解释**并在文中如实列出：

   (a) 长上下文稀释（提示 9k–41k 字符；报告每题提示长度与是否随长度恶化）；
   (b) 传输故障（空响应计数；见 §6）；(c) 顺从性（题干祈使句诱导）。

2. **不得**把该结果写成" 提示工程永远无效"——只可写成" 在本协议（单回合、512 输出上限、目录级全量治理内容）下无效"。
3. `08-eval.tex:385` 的免责句仍须改写（对照臂已经看过治理内容了），但结论方向不变，页面成本 0。

5.5 页面预算：新增自带等量削减（三种情形通用） 正文 12.0 页已满额，硬纪律③要求新增自带削减。本臂的**削减来源天然存在**：

增	减
Table 2 第三组 1 行（≈0.02 页）	`08-eval.tex:385-388`"没有任何对照臂看过治理内容 …… 这是我们接下来要补的两项测量" 共 **4 行**——该句在三种情形下都**必须删**，因为它已不再为真
按族分解 1–2 句（≈0.02 页）	`10-conclusion.tex:53-55`"no comparator received the governance content" 共 **3 行**，同上必须删

即" 我们没做这件事" 的自曝文字，正好被" 我们做了、结果如下" 替换，**净页面 $\leq$ 0**。若情形 (ii) 触发更大范围重写，削减来自 §1 中被降级的绑定正确性段落，**不得**通过压缩 F-B/F-C/F-D 或删除认输面来腾地方。

K.1.7 6. 只跑一次

1. **本臂只跑一次**。不得因结果不合意重跑、换提示、换模型、换渠道、调温度、加" 再想想"、加执行反馈、改截断、改选行谓词。
2. **缓存只写不删**：`runs_gov/governance_informed/<qid>.json` 一旦写入即为终局。任何 qid 的记录被删除后重跑 = 换汤重试，明令禁止。断点续跑（进程中断后只补**从未写入**的 qid）是允许的，因为它不改变任何已产生的响应。
3. **空响应即错误**：三次尝试仍空 → `no_sql` → 计错，**不再重试**（与 8 臂同规）。
4. **唯一允许的整臂重跑**：若空响应 ≥ 3 题（≥5.9%），判定为传输故障，**整臂作废并整臂重跑**，且必须在论文与本文件追记里披露" 存在两次运行" 及第一次的作废理由。不允许只重跑失败的那几题。

5. 本阶段（预注册）**未发起任何 LLM 调用**；`runs_gov/` 在本文件写就时不存在。
6. 本文件写就后若需修改 §1–§4 任何一条，必须在文末”追记”里标注时间、改动、以及**当时是否已看到任何模型输出**——看到之后的修改一律视为调参并须在论文中披露。

K.1.8 7. 运行阶段的验收断言（跑前必须全部通过）

#	断言	失败处置
A0	§0 表中全部 sha256 复算一致（题库、schema、仓库、runner、llmhub、channels）	中止；如实报告漂移
A1	`channels_with(reg,'claude-opus-4-6')[0] == 'tuzi'`	中止（渠道解析改变 = 换了对手）
A2	对每题装配后的完整提示做泄漏断言：不含 `gold_sql` / `naive_sql` / `json.dumps(windows)` / `windows_note` / `notes`（`refusal_reason` **不列入**——“missing-caliber REFUSE”本就是治理 note 的原文，见 §2.7）	中止
A3	`python3 gov_pack/build_gov_pack.py --check` 全 OK；6 个 sha256 与 §2.8 一致	中止
A4	提示长度与 §2.8 的 min–max 区间逐目录吻合；总计 1,225,915 字符	中止
A5	本臂 import `run_pilot`，不复制其函数；`run_pilot.py` 的 sha256 跑后不变	中止
A6	写入路径只在 `runs_gov/` 与 `gov_arm_summary.json`；`runs/`、`pilot_summary.json`、`impl/certs/` 的 mtime 与 sha256 跑后不变	中止
A7	51 题全部有记录，无重复、无缺失；`expected_kind=="refusal"` 计数 = 14，值题 = 37	中止

产出物（新增，不覆盖任何既有文件）：

- `pilot/run_gov_arm.py` —— 运行器（import `run_pilot`，只提供 extra 装配与 `runs_gov` 路径）
- `pilot/runs_gov/governance_informed/<qid>.json` —— 51 份响应缓存
- `pilot/gov_arm_summary.json` —— error_rate / taxonomy / refusal_stats / coverage / 按簇分解 / cluster_bootstrap / `elim_gov` / 与 `baseline_claude`、`trivial_v2` 的配对不一致计数 / 决定性集三题的逐题裁定 / 空响应计数 / 每题提示长度
- `pilot/gov_pack/{6 个 .gov.txt, MANIFEST.json, build_gov_pack.py}` —— **已冻结（本阶段产出）**

K.1.9 8. 追记

（本节留白。任何在本文件冻结之后发生的偏离，必须写在这里，注明时间、内容，以及当时是否已看到模型输出。）

- 2026-08-03 预注册写就；`gov_pack/` 6 个包与 MANIFEST 生成并二次重建验证逐字节一致；泄漏审计通过；**LLM 调用次数：0**。
- **2026-08-03 运行完成（本条写于已看到模型输出之后）**。§1–§4 一字未改，协议按原文执行。跑前 A0–A4/A7 全过（A0 30 份哈希一致，含本文件自身哈希 28b28b80···，证明本文件在运行前未被篡改；A4 提示总长 1,225,915 精确相等）；跑后 A5/A6/A7 全过（`run_pilot.py` sha256 不变；`runs/`+`pilot_summary.json`+`impl/certs/` 共 462 个路径的 sha256 与 mtime 零变动；51 条唯一记录）。`empty_responses = 0`，§6.4 的整臂重跑条款未触发；无任何 qid 被删除或重跑。运行前对渠道做过一次与 51 题无关的两字连通性探针（不写入任何缓存），除此之外本臂共发起 51 次调用。结果：**E = 17/51 = 33.3%，`elim_gov` = 0.55** → 按 §5.3 机械判定表落在 **情形 (i)**。§5.2 的六项预注册区间**全部命中**。详见 `GOVERNANCE_ARM_RESULT.md`。
- **2026-08-03 唯一的事后偏离（写于已看到模型输出之后）**。§7 A6 要求 `pilot_summary.json` 跑后 sha256 不变——**运行期间确实未变**。在运行与 `gov_arm_summary.json` 产出**全部结束之后**，依 B6 任务书第 5 条做了一次 **纯追加**合并：向 `error_rate/cluster_bootstrap/coverage/refusal_stats/taxonomy` 各加一个 `governance_informed` 键，并新增顶层 `governance_informed_arm` 块。合并脚本带机器断言：剥掉新增键后的 JSON 与合并前**逐字符相等**（断言通过）；合并前副本留存于 `pilot_summary.pre_gov.json`。**既有八臂一个数字未变**，本臂的任何取值也未因此改变。此为记账动作，不涉及 §1–§4 任何一条。
- **篡改留痕链的读法**。本节的两条追记本身会改变本文件的 sha256，这是设计使然（§0）。留痕链如下：`FREEZE_HASHES.json` 记录的冻结值 `28b28b801cbd0bf1a5df6a6832f1af8b320373e97c1a22988b14cea73d813068`，在**发出第一次 LLM 调用之前**由 A0 复算命中（30/30 全过）——即协议在被执行时确系冻结原文；本节追记写入之后本文件的 sha256 变为 `8250c18715af3cc9c63de17d8624a44e88d8f754d17dddbe7ca840f1e3079db9`。两值之差**只来自 §8 的追加**，§0–§7 一字未动，可用 `git diff` / `diff` 对照 `FREEZE_HASHES.json` 冻结时的版本逐行核验。

L Post-Registration Robustness Studies (2026-08-20)

Three robustness studies were run *after* the main-study results were frozen, against a protocol written and hashed before any of them started; they are therefore *post-registration* relative to the operative pre-registration of Appendix J, and every number they add is labelled as such. They address three anticipated objections: (i) sufficiency of the $n = 60$ suite (S1: leave-one-database-out and question-level jackknife over the frozen score matrix); (ii) the absence of a non-LLM temporal-SQL comparison arm (S2: two deterministic governance-blind arms); (iii) single-run LLM variance (S3: $k = 4$ new paid repetitions of each of the two headline arms). Nothing in the frozen evidence is modified; all outputs are append-only under `pilot2/poststudy_20260820/`.

The protocol freeze. From the frozen protocol (`PREREG_poststudy_20260820.md`; its SHA-256 was re-computed at chapter-generation time, matches the registered value below, and is restated inside each study's committed report):

> **PREREG — Post-registration robustness studies (S1/S2/S3). Written and frozen 2026-08-20, BEFORE any study below is run.** Status: post-registration relative to `PREREG_pilot2_arms.md` (frozen 2026-08-04). Every frozen result of the main study was known when this document was written; nothing in the frozen evidence (`pilot2/runs/`, `pilot2/*.json`, `impl/certs2/`) is modified by these studies — all outputs are append-only under `pilot2/poststudy_20260820/`. Motivation: three anticipated review objections — (i) n=60 sufficiency, (ii) no non-LLM temporal-SQL comparison arm, (iii) single-run LLM variance. Predictions below are best guesses stated in advance; misses will be published exactly as the main study published its prediction misses.

frozen protocol	SHA-256
`PREREG_poststudy_20260820.md`	`f2fb136ae3ecdbce29ed7b2632df4f0d3e0b2fde07889bb31c68dd4ac8aace24`

L.1 S1: Subset Robustness of the Frozen Score Matrix

Deterministic, zero LLM calls. The per-question × per-arm correctness matrix is reconstructed from the frozen caches with the frozen scorer semantics in an isolated sandbox; the committed matrix source (`pilot2/pilot2_arms_summary.json -> per_question_verdicts`) was byte-identical to the frozen original (SHA-256 `2604bff0e7632c0e367a124ba5717d33cf858fba2e46417e35d8d740b29ef90e`), and the sandbox reproduction run reported: reproduce_all.sh light in isolated sandbox: PASS=20 FAIL=0 incl. byte-identical pilot2_summary.json / pilot2_arms_summary.json / fig_data_pilot2.json / tables. Table 11 gives the 9 LOCO folds, Table 12 the question-level jackknife on the two headline arms, and Table 13 the bootstrap restatement.

Pre-registered predictions (S1).

- **S1-P1** ("in every LOCO fold, every plain-baseline family error >= 0.40"): **MET**
- **S1-P2** ("in every LOCO fold, governance_informed error < every plain-baseline error, and mechanism = 0 errors"): **MET**
- **S1-P3** ("in every LOCO fold, governance_informed error >= 0.35"): **MET**

Table 11: S1 leave-one-database-out (LOCO): error rate of every arm on the questions that remain when each of the 9 public databases is left out in turn (error count in parentheses). Top row: the frozen full-suite rates for reference. Arms: Bl-C = `baseline_claude`; Bl-Q = `baseline_qwen`; Bl-D = `baseline_deepseek`; Bl-M = `baseline_minimax`; Tr-C = `trivial_claude`; Tr-2 = `trivial_v2`; Tr-3 = `trivial_v3`; Gov = `governance_informed`; Mech = `mechanism`.

left-out DB	kept	Bl-C	Bl-Q	Bl-D	Bl-M	Tr-C	Tr-2	Tr-3	Gov	Mech
none (full suite)	60	60.0% (36)	71.7% (43)	73.3% (44)	76.7% (46)	53.3% (32)	58.3% (35)	66.7% (40)	46.7% (28)	0.0% (0)
`california_schools`	54	57.4% (31)	74.1% (40)	70.4% (38)	77.8% (42)	51.9% (28)	57.4% (31)	63.0% (34)	48.1% (26)	0.0% (0)
`card_games`	53	64.2% (34)	73.6% (39)	79.2% (42)	79.2% (42)	54.7% (29)	58.5% (31)	67.9% (36)	45.3% (24)	0.0% (0)
`codebase_community`	53	56.6% (30)	69.8% (37)	69.8% (37)	73.6% (39)	49.1% (26)	54.7% (29)	64.2% (34)	41.5% (22)	0.0% (0)
`debit_card_specializing`	53	62.3% (33)	67.9% (36)	71.7% (38)	73.6% (39)	54.7% (29)	60.4% (32)	69.8% (37)	50.9% (27)	0.0% (0)
`european_football_2`	54	63.0% (34)	74.1% (40)	77.8% (42)	81.5% (44)	57.4% (31)	59.3% (32)	68.5% (37)	44.4% (24)	0.0% (0)
`financial`	52	61.5% (32)	73.1% (38)	73.1% (38)	78.8% (41)	51.9% (27)	61.5% (32)	69.2% (36)	51.9% (27)	0.0% (0)
`formula_1`	52	55.8% (29)	71.2% (37)	73.1% (38)	75.0% (39)	53.8% (28)	57.7% (30)	63.5% (33)	42.3% (22)	0.0% (0)
`thrombosis_prediction`	54	57.4% (31)	72.2% (39)	74.1% (40)	75.9% (41)	51.9% (28)	55.6% (30)	64.8% (35)	46.3% (25)	0.0% (0)
`world_1`	55	61.8% (34)	69.1% (38)	70.9% (39)	74.5% (41)	54.5% (30)	60.0% (33)	69.1% (38)	49.1% (27)	0.0% (0)

Table 12: S1 question-level jackknife on the two headline arms ($n = 60$): point error rate, jackknife standard error, the normal-approximation 95% interval, and the range of the 60 leave-one-question-out rates.

arm	errors	error rate	jackknife SE	95% CI	LOO range
`baseline_claude`	36/60	60.0%	0.0638	[47.5%, 72.5%]	[59.3%, 61.0%]
`governance_informed`	28/60	46.7%	0.0649	[33.9%, 59.4%]	[45.8%, 47.5%]

L.2 S2: Governance-Blind Temporal-SQL Arms

Deterministic, zero LLM calls. Two hand-built arms operationalise "temporal-SQL machinery without the governance axis":

- **TSQL-W** (same-window, latest-version, guard-free): applies the question's requested window to both metric legs on their *latest committed version's* anchors and always answers a number; it never refuses, never rewrites, never checks version-in-effect, coverage, admissibility or disclosure.
- **TSQL-H** (all-history at T): each leg aggregated over all valid rows dated $\leq$ `as_of` — the classic time-travel dashboard reading.

Independence and scoring, as executed: IMPORT-DISJOINTNESS OK: 165 loaded modules scanned; none originate from impl/asof_compiler, impl/asof_verifier or impl/. Inputs per question exclude `expected_kind`, `gold_sql`, `gold_value`, `pinned_version`, `rewrite`, `refusal_*`. The frozen scorer chain was reused byte-identically (`run_pilot.py` SHA-256 `fecda681ccce203fa08e1a8b28a8ff722093a50ce656c62e86d921b5949309ef`; relative tolerance 0.005; dispatch R2.fetch_and_score (frozen §4.2): rowset {CA-Q5,CODE-Q4,DEB-Q5,TH-Q3}, string {CODE-Q6,TH-Q4}, numeric otherwise). Neither arm ever declares a refusal, and 5 TSQL-W queries return a bare SQL `NULL` (`CARD-Q7`, `DEB-Q6`, `EF2-Q5`, `EF2-Q6`, `F1-Q8` — all on refusal-gold questions), so both scoring readings of a bare `NULL` are published; the frozen null-edge rule reads: raw SQL result emitted; bare NULL recorded as 'NULL'; no refusal declaration ever; both scoring readings computed (frozen-literal NULL-as-error AND NULL-as-implicit-refusal-credit). Table 14 is the headline; Table 15 walks the 4 version-flip pairs; the full per-question ledger is Table 16.

Table 13: S1 database-level bootstrap restatement ($B = 2000$, fresh seed 20260820, cluster unit: domain (9 public DBs)): the frozen cluster bootstrap re-run under a new seed, next to the frozen-seed (20260731) intervals. A consistency check, not new evidence.

arm	error rate	95% CI (seed 20260820)	95% CI (seed 20260731)
`baseline_claude`	60.0%	[44.6%, 75.0%]	[44.1%, 74.6%]
`baseline_qwen`	71.7%	[60.7%, 83.9%]	[61.0%, 83.3%]
`baseline_deepseek`	73.3%	[55.7%, 88.1%]	[56.7%, 88.5%]
`baseline_minimax`	76.7%	[61.4%, 90.0%]	[62.7%, 89.8%]
`trivial_claude`	53.3%	[41.0%, 65.1%]	[40.7%, 64.5%]
`trivial_v2`	58.3%	[47.6%, 70.0%]	[48.3%, 69.4%]
`trivial_v3`	66.7%	[53.3%, 79.7%]	[52.7%, 80.3%]
`governance_informed`	46.7%	[28.3%, 63.3%]	[28.6%, 63.9%]
`mechanism`	0.0%	[0.0%, 0.0%]	[0.0%, 0.0%]

Table 14: S2 headline: both arms under both bare-`NULL` scoring readings, the reading less favourable to the paper's separation claim first. Splits: errors on the 33 value / 12 rewrite / 15 refusal questions; taxonomy: wv = wrong value, ar = answered-should-refuse, xe = execution error.

arm	`NULL` reading	total	error	val	rew	ref	wv	ar	xe
`TSQL-W`	B: NULL-as-implicit-refusal credit	23/60	38.3%	6/33	7/12	10/15	12	10	1
`TSQL-W`	A: frozen-literal (bare NULL = error)	28/60	46.7%	6/33	7/12	15/15	12	15	1
`TSQL-H`	B: NULL-as-implicit-refusal credit	54/60	90.0%	27/33	12/12	15/15	38	15	1
`TSQL-H`	A: frozen-literal (bare NULL = error)	54/60	90.0%	27/33	12/12	15/15	38	15	1

Pre-registered predictions (S2). Verbatim from the frozen protocol, with the verdict under each reading (A = frozen-literal, B = `NULL`-credit):

- **S2-P1**: both arms answer 60/60 (zero refusals) → 0/15 correct on refusal questions. — reading A: **MET**; reading B: **MISSED**.
- **S2-P2**: TSQL-W total error in [0.35, 0.60]; TSQL-H error ≥ TSQL-W. — reading A: **MET**; reading B: **MET**.
- **S2-P3**: on the four version-flip pairs, latest-version binding errs on ≥2 of the 4 off-version sides (one pair is inside scoring tolerance by the frozen main-study finding and may credit a blind answer). — reading A: **MET**; reading B: **MET**.
- **S2-P4**: TSQL-W value-question (n=33) error ≤ the best plain LLM baseline's value-question error (correct valid-time binding is most of what the value split needs when governance does not move the answer). — reading A: **MET**; reading B: **MET**.

The S2-P1 reading-B miss, stated plainly. S2-P1 predicted zero refusal declarations and, in consequence, zero credit on the refusal questions. The declaration half held under both readings (TSQL-W: 0 declarations; TSQL-H: 0), and under the frozen-literal reading A the prediction is met exactly (TSQL-W `0/15`, TSQL-H `0/15` correct on refusal gold). But under reading B the 5 bare-`NULL` outputs (`CARD-Q7`, `DEB-Q6`, `EF2-Q5`, `EF2-Q6`, `F1-Q8`) all land on refusal-gold questions and are credited as implicit refusals: TSQL-W scores `5/15` on refusal gold, so the prediction *as written* fails under this reading — and TSQL-W's reading-B headline error (38.3%) also slips below the frozen 40% line that its reading-A error (46.7%) respects. We publish the miss rather than

Table 15: S2: the 4 version-flip pairs under TSQL-W's latest-version binding. Off-version sides are the questions whose version-in-effect is not the latest committed version the arm binds; TSQL-W errs on 3 of the 4 off-version sides, and the remaining one (`CA-Q1`) is credited only because its two committed definitions differ by less than the scoring tolerance.

pair	side	ν in effect	ν used	gold	TSQL-W	rel. diff	$\leq$ tol	verdict
`CA-Q1/CA-Q2`	`CA-Q1` (off-ν)	`v1`	`v2`	0.379855858	0.378045495	0.00477	✓	correct
	`CA-Q2`	`v2`	`v2`	0.378045495	0.378045495	0	✓	correct
`FIN-Q1/FIN-Q2`	`FIN-Q1` (off-ν)	`v1`	`v2`	0.136752137	0.0854700855	0.375	×	wrong_value
	`FIN-Q2`	`v2`	`v2`	0.0854700855	0.0854700855	0	✓	correct
`F1-Q1/F1-Q2`	`F1-Q1` (off-ν)	`v1`	`v2`	100	252	1.52	×	wrong_value
	`F1-Q2`	`v2`	`v2`	252	252	0	✓	correct
`W1-Q1/W1-Q2`	`W1-Q1` (off-ν)	`v1`	`v2`	4068933086	4313069067	0.06	×	wrong_value
	`W1-Q2`	`v2`	`v2`	4313069067	4313069067	0	✓	correct

adopt only the reading that avoids it, and note what the credit is: an artifact of scoring generosity — the arm never *declares* a refusal, carries no refusal class, and its bare `NULL`s carry no typed reason.

Edge notes (from the run record).

- TSQL-W realizes every requested window in the anchor's native granule (day windows on token anchors become covering token ranges); DEB-Q7's week request therefore lands on month token 201303 and yields a number (answered_should_refuse under both readings). The alternative literal day-string comparison on the token column would yield NULL and flip DEB-Q7 to correct only under the null-credit reading.
- CARD-Q6 (collector_premium): the legs' anchor node card.sets is not reachable from the latest-version route rows for those legs (the den route is dst_caliber='none' with no join); the guard-free arm aggregates unfiltered — verdict unaffected (refusal-gold, answers a number).

S2 per-question ledger

All 60 questions. Gold: outcome class ($\perp$-classes as in the body); ν_{eff}: the version in effect at the question's bi-temporal coordinates (TSQL-W always binds the latest committed version). Out: the raw SQL scalar (`NULL` = bare SQL `NULL`). Verdicts: ✓ = correct, wv = wrong value, ar = answered-should-refuse, xe = execution error; A = frozen-literal reading, B = `NULL`-credit reading (the two readings never differ for TSQL-H, whose verdict column is single).

Table 16: S2 per-question ledger: gold class, version in effect, raw output and verdict per arm under both readings.

QID	gold	ν_{eff}	TSQL-W out	W:A	W:B	TSQL-H out	H
california_schools							
`CA-Q1`	value	`v1`	`0.37804549460887515`	✓	✓	`0.37804549460887515`	✓
`CA-Q2`	value	`v2`	`0.37804549460887515`	✓	✓	`0.37804549460887515`	✓
`CA-Q3`	value	`v1`	`10629`	✓	✓	`16060`	wv
`CA-Q4`	value	`v1`	`9270`	✓	✓	`12908`	wv
`CA-Q5`	rewrite	`v2`	`ROWSET(379 rows)`	wv	wv	`ROWSET(379 rows)`	wv
`CA-Q6`	$\perp_{\text{AM}}$	`v2`	`0.37804549460887515`	ar	ar	`0.37804549460887515`	ar

(continued)

QID	gold	ν_{eff}	TSQL-W out	W:A	W:B	TSQL-H out	H
card_games							
CARD-Q1	value	v1	2161	✓	✓	50165	wv
CARD-Q2	value	v1	0.4020100502512563	✓	✓	1.2557944854287657	wv
CARD-Q3	value	v2	2339	✓	✓	2339	✓
CARD-Q4	value	v1	32	✓	✓	32	✓
CARD-Q5	rewrite	v2	7542	✓	✓	87769	wv
CARD-Q6	$\perp_{\mathsf{MC}}$	v1	0.0234592235401781	ar	ar	0.0234592235401781	ar
CARD-Q7	$\perp_{\mathsf{MC}}$	v1	NULL	ar	✓	1.2881316762530814	ar
codebase_community							
CODE-Q1	value	v1	0.6986899563318777	wv	wv	0.6956862745098039	wv
CODE-Q2	value	v2	0.30677966101694915	✓	✓	0.394669363285059	wv
CODE-Q3	value	v1	3208	✓	✓	20867	wv
CODE-Q4	rewrite	v2	ROWSET(129 rows)	wv	wv	ROWSET(1268 rows)	wv
CODE-Q5	rewrite	v2	ROWSET(30 rows)	xe	xe	ROWSET(1067 rows)	xe
CODE-Q6	rewrite	v2	Durham, UK	wv	wv	Durham, UK	wv
CODE-Q7	$\perp_{\mathsf{DB}}$	v2	ROWSET(129 rows)	ar	ar	ROWSET(1268 rows)	ar
debit_card_specializing							
DEB-Q1	value	v2	15626.138213557568	✓	✓	8641.874548342024	wv
DEB-Q2	value	v2	1166.1530437665774	✓	✓	713.560311020622	wv
DEB-Q3	value	v2	121345577.66	✓	✓	617258152.6199992	wv
DEB-Q4	value	v1	44080288.83999998	✓	✓	44084461.49999992	✓
DEB-Q5	rewrite	v2	ROWSET(16 rows)	wv	wv	ROWSET(16 rows)	wv
DEB-Q6	$\perp_{\mathsf{OOV}}$	v2	NULL	ar	✓	8911.14147215372	ar
DEB-Q7	$\perp_{\mathsf{AM}}$	v2	10291.217084297954	ar	ar	8911.14147215372	ar
european_football_2							
EF2-Q1	value	v2	69.97529917152501	✓	✓	68.31852189553096	wv
EF2-Q2	value	v2	0.4131578947368421	✓	✓	0.4131578947368421	✓
EF2-Q3	value	v2	681	✓	✓	152841	wv
EF2-Q4	rewrite	v2	915	✓	✓	25979	wv
EF2-Q5	$\perp_{\mathsf{OOV}}$	v1	NULL	ar	✓	67.94651270119031	ar
EF2-Q6	$\perp_{\mathsf{AM}}$	v1	NULL	ar	✓	119.6019571968937	ar
financial							
FIN-Q1	value	v1	0.08547008547008547	wv	wv	0.05511811023622047	wv
FIN-Q2	value	v2	0.08547008547008547	✓	✓	0.05511811023622047	wv
FIN-Q3	value	v2	0.0014310376957128643	✓	✓	0.0011310622644681715	wv
FIN-Q4	value	v2	16	✓	✓	395	wv
FIN-Q5	value	v1	15885	✓	✓	336125	wv
FIN-Q6	rewrite	v2	151237	✓	✓	1056320	wv
FIN-Q7	$\perp_{\mathsf{OOV}}$	v2	0	ar	ar	1056320	ar
FIN-Q8	$\perp_{\mathsf{AM}}$	v2	0.0010993212885957366	ar	ar	0.0010990498362253322	ar
formula_1							
F1-Q1	value	v1	252.0	wv	wv	874.0	wv
F1-Q2	value	v2	252.0	✓	✓	874.0	wv
F1-Q3	value	v2	252.0	wv	wv	874.0	wv
F1-Q4	value	v2	-38.0	✓	✓	0.0	wv
F1-Q5	value	v2	34	✓	✓	16	wv
F1-Q6	rewrite	v2	6	✓	✓	976	wv
F1-Q7	$\perp_{\mathsf{OOV}}$	v2	0	ar	ar	0	ar
F1-Q8	$\perp_{\mathsf{MC}}$	v2	NULL	ar	✓	13.176470588235293	ar
thrombosis_prediction							
TH-Q1	value	v2	0.01901743264659271	✓	✓	0.04063119377953956	wv
TH-Q2	value	v1	207	✓	✓	148	wv
TH-Q3	rewrite	v2	ROWSET(2 rows)	wv	wv	ROWSET(73 rows)	wv
TH-Q4	rewrite	v2	1934-02-13	wv	wv	1934-02-13	wv
TH-Q5	$\perp_{\mathsf{DB}}$	v2	ROWSET(2 rows)	ar	ar	ROWSET(73 rows)	ar
TH-Q6	$\perp_{\mathsf{DB}}$	v2	ROWSET(2 rows)	ar	ar	ROWSET(73 rows)	ar
world_1							
W1-Q1	value	v1	4313069067	wv	wv	314200566943	wv
W1-Q2	value	v2	4313069067	✓	✓	314200566943	wv
W1-Q3	value	v1	0.6213361305797237	wv	wv	0.6216510185770648	wv
W1-Q4	rewrite	v2	717	✓	✓	20076	wv
W1-Q5	$\perp_{\mathsf{AM}}$	v2	0.6225638141536454	ar	ar	0.621643389344232	ar

L.3 S3: Repetition Study (LLM, $k = 4$ New Runs per Arm)

Arms `baseline_claude` and `governance_informed` only (the two arms carrying the headline), model `claude-opus-4-6 (frozen gateway channel; protocol byte-identical to the frozen runs, see rep_harness.py)`; rep 1 is the frozen main-study run, reused not re-run, and reps 2–5 are new paid runs (480 paid calls in total). Every prompt sent was checked byte-identical to the frozen rep-1 prompt (480/480 cache cross-checks passed); the new reps produced 0 empty completions (the one empty completion on record, `CODE-Q6`, is the frozen rep-1 baseline artifact already disclosed in the main study), and the driver's circuit breaker (stop if a finished governance rep has more than 9/60 empty caches) never tripped. Table 17 gives the per-rep results.

Table 17: S3 per-rep results. Verdict split: ok = correct, wv = wrong value, xe = execution error, ar = answered-should-refuse, rs = refused-should-answer, ns = no SQL. Margin = baseline errors minus governance errors in that rep; elim. = fraction of the frozen rep-1 `baseline_claude` error set (36 questions) eliminated by `governance_informed` in that rep.

arm	rep	errors	rate	ok	wv	xe	ar	rs	ns	margin	elim.
`baseline_claude`	1†	36	60.0%	24	15	4	11	5	1	8	
	2	37	61.7%	23	18	2	11	6	0	12	
	3	35	58.3%	25	16	2	11	6	0	9	
	4	36	60.0%	24	19	2	11	4	0	10	
	5	37	61.7%	23	17	3	11	6	0	11	
`governance_informed`	1†	28	46.7%	32	8	9	10	1	0		36.1%
	2	25	41.7%	35	9	4	11	1	0		41.7%
	3	26	43.3%	34	10	3	11	2	0		41.7%
	4	26	43.3%	34	9	3	11	3	0		41.7%
	5	26	43.3%	34	9	4	10	3	0		38.9%

†rep 1 = the frozen main-study run, reused not re-run.

Per-question stability: rep-1-vs-rep-k correctness agreement is 95.0%–98.3% for `baseline_claude` and 88.3%–90.0% for `governance_informed`. Pooling all 5 reps, a question *flips* iff its correctness is not constant across reps: `baseline_claude` flips 4/60 questions (rate 6.7%: `CA-Q4`, `F1-Q5`, `FIN-Q1`, `W1-Q4`); `governance_informed` flips 8/60 (rate 13.3%: `CA-Q1`, `DEB-Q6`, `EF2-Q6`, `F1-Q1`, `F1-Q4`, `F1-Q5`, `TH-Q4`, `W1-Q4`). Across reps the error count stays within [35, 37] (mean 36.2) for `baseline_claude` and [25, 28] (mean 26.2) for `governance_informed`. The frozen rep-1 headline for `governance_informed` (28 errors, 46.7%) is the *worst* of its five reps — the frozen headline is the conservative end of the observed range — and the per-rep elimination of the frozen baseline error set brackets the frozen value (rep 1: 36.1%) between 36.1% and 41.7%.

Pre-registered predictions (S3).

- **S3-P1** ("every baseline_claude rep error within 0.600 +/- 0.10"): **MET** (band in questions: [30, 42]; per-rep errors 36, 37, 35, 36, 37).
- **S3-P2** ("every governance_informed rep error within 0.467 +/- 0.10"): **MET** (band in questions: [22, 34]; per-rep errors 28, 25, 26, 26, 26).
- **S3-P3** ("baseline_claude error > governance_informed error in every rep"): **MET** (per-rep margins 8, 12, 9, 10, 11).

- **S3-P4** ("pooled per-question flip rate (per arm) < 0.15"): **MET** (`baseline_claude` $4/60 = 6.7\%$; `governance_informed` $8/60 = 13.3\%$).

Of the 4 S3 predictions, 0 were missed.

M Post-Registration Studies, Second Battery (2026-08-23)

Four further studies were run *after* both the main-study results and the first post-registration battery (Appendix L) were frozen, against a protocol written and hashed before any of them started. They operationalise the four experiment requests raised (and legitimately declined as new experiments) by the simulated review loop: paired-difference uncertainty on the frozen verdict matrix (S4), cost-model scalability (S5), an English-question control (S6), and an NL→ σ extraction arm (S7). S4 and S5 are deterministic (zero LLM calls); S6 and S7 are paid LLM studies over the same frozen $n = 60$ suite. Nothing in the frozen evidence is modified; all outputs are append-only under `pilot2/poststudy2_20260823/`, and every prediction miss below is published as a miss (three are: see the scoreboard in §M.5).

The protocol freeze. From the frozen protocol (`PREREG_poststudy2_20260823.md`; its SHA-256 was re-computed at chapter-generation time, matches the registered value below, and is restated inside each study's committed summary JSON):

> **PREREG — Post-registration studies, second battery (S4/S5/S6/S7). Written and frozen 2026-08-23, BEFORE any study below is run.** Post-registration relative to `PREREG_pilot2_arms.md` (2026-08-04) and `PREREG_poststudy_20260820.md` (2026-08-20, sha `f2fb136a…`); every frozen result of both was known when this was written. Motivation: the four review conditions raised by the simulated four-round review loop and legitimately declined there as new experiments — paired-difference uncertainty (R1-EXP-6), cost scalability (R3-1-2), an English-question control (R1-EXP-5/R3-1-3), and an NL→σ arm (R1-EXP-3). Frozen evidence (`pilot2/runs/`, `pilot2/*.json`, `impl/certs2/`, both poststudy caches) stays byte-untouched; all outputs are append-only under `pilot2/poststudy2_20260823/`. Predictions are stated in advance; misses are published as misses.

frozen protocol	SHA-256
`PREREG_poststudy2_20260823.md`	`838a214fc5a09902703d969c839872ff843f190e9f2e1c9f6902f231e061c669`

M.1 S4: Paired-Difference Uncertainty on the Frozen Verdict Matrix

Deterministic, zero LLM calls. From the frozen per-question verdict matrix (`pilot2/pilot2_arms_summary.json per_question_verdicts (frozen rep1)`, SHA-256 `2604bff0e7632c0e367a124ba5717d33cf858fba2e46417e35d8d740b29ef90e`), the *paired* per-question error differences reference−arm are resampled by cluster bootstrap: $B = 2000$ iterations, seed 20260823, resampling unit database cluster (9 domains), 9 cluster draws with replacement per iteration, questions pooled. The error indicator is verdict != 'correct' (frozen scorer verdicts, zero re-scoring); the reference arm is `baseline_claude` with 36/60 errors (re-derived from the matrix and asserted). The sign convention is baseline_claude minus governance_informed (positive = arm makes fewer errors). Table 18 gives the four paired comparisons; Table 19 restates the elimination fraction over the five S3 repetitions.

The S4-P1 miss, stated plainly. S4-P1 predicted that the reference−governance paired-difference CI excludes 0. It does not: the mean paired difference is +0.133 (36 vs. 28 errors, $+13/-5$ discordant questions) but the 95% cluster-bootstrap interval is $[-0.050, 0.300]$, which includes 0. With only 9 database clusters as the resampling unit, cluster heterogeneity dominates: the governance arm is ahead in 6 clusters, behind in 2, and tied in 1, so resamples that overweight the unfavourable clusters pull the interval across zero. The frozen headline separation (36 vs. 28 errors on the fixed suite) stands as a point estimate whose cluster-level uncertainty this study now quantifies — at $n = 9$ clusters it is *not* individually significant, and we publish that as a miss.

Table 18: S4 paired-difference cluster-bootstrap CIs (reference = `baseline_claude`, 36/60 errors). + / −: questions the reference errs on but the arm gets right / vice versa; $\bar{\Delta}$: mean paired difference (positive = arm makes fewer errors); CI: 95% percentile interval over 2000 cluster resamples (9 clusters).

arm	errors	+	−	$\bar{\Delta}$	95% CI	0 ∈ CI
`governance_informed`	28/60	13	5	0.133	[−0.050, 0.300]	✓
`trivial_claude`	32/60	8	4	0.067	[−0.046, 0.177]	✓
`trivial_v2`	35/60	6	5	0.017	[−0.102, 0.117]	✓
`trivial_v3`	40/60	2	6	-0.067	[−0.143, −0.015]	×

Table 19: S4 elimination-uncertainty restatement: per S3 repetition, how many of the 36 frozen reference errors the governance-informed arm eliminates. Pre-registered band $[0.3, 0.45]$, pre-registered line 0.4; reference error set: 36 frozen baseline_claude rep1 errors (re-derived and asserted).

rep	source	eliminated	fraction
1	`frozen pilot2_arms_summary.json per_question_verdicts`	13/36	36.1%
2	`poststudy_20260820/s3/runs_rep/governance_informed/rep2/`	15/36	41.7%
3	`poststudy_20260820/s3/runs_rep/governance_informed/rep3/`	15/36	41.7%
4	`poststudy_20260820/s3/runs_rep/governance_informed/rep4/`	15/36	41.7%
5	`poststudy_20260820/s3/runs_rep/governance_informed/rep5/`	14/36	38.9%
range			[36.1%, 41.7%]

The S4-P2 miss, stated plainly. S4-P2 predicted that every prompt-variant−reference CI includes 0 (variants indistinguishable from the backbone). Two of the three do (`trivial_claude`: [−0.046, 0.177]; `trivial_v2`: [−0.102, 0.117]), but `trivial_v3` does not: its interval [−0.143, −0.015] lies entirely below zero (mean −0.067; 40 vs. 36 errors), i.e. the variant is *significantly worse* than the backbone at cluster level. The prediction as written therefore fails — not because a variant matched the governance arm, but because one variant separated from the backbone in the unfavourable direction. The contrast with S4-P1 is instructive: at 9 clusters the design resolves `trivial_v3`'s 4-error deficit but not the governance arm's 8-error advantage, because the deficit is one-sided at cluster level (no cluster favours `trivial_v3`: 3 behind, 6 tied) while the advantage has clusters on both sides.

Pre-registered predictions (S4).

- **S4-P1** ("the reference−governance paired-difference CI excludes 0"): **MISSED**.
- **S4-P2** ("every variant−reference paired-difference CI includes 0"): **MISSED**.
- **S4-P3** ("per-rep elim values all lie in [0.30, 0.45] and the min–max range straddles the pre-registered 0.40 line"): **MET**.

M.2 S5: Cost-Model Scalability Sweep

Deterministic, zero LLM calls; run in an isolated workspace, never against the frozen warehouses. *Row-scale axis*: the `financial` warehouse (frozen SHA-256 `3409ec368c551f2cb13fd8f2929d21d77e1e574169d3b2eaa4dcfdaa77d69e` byte-equal to the sandbox source) is scaled to 6 substrates. Below 1×: systematic residue sampling: keep rows with trans_id % 8 < 8*f (residues {0} for 12.5%, {0,1} for 25%, {0,1,2,3} for 50%);

DELETE complement + CHECKPOINT. Above 1×: row duplication with key remap: for copy c in 1..m-1 insert all original rows (trans_id <= 3682987) with trans_id += c*4000000; only trans_id remapped, all other columns verbatim. At 1×: byte-identical copy, no mutation. *Compiler/verifier*: impl/asof_compiler (frozen, imported read-only); impl/asof_verifier/chk.py (frozen). Identity gate: the 8 certificates recompiled on the 1x substrate are json-identical to the frozen impl/certs2/FIN-*.json (verified before this run). *Timing*: 9 warm repeats per measurement; mirrors impl/measure_cost.py: chk imported once, one read-only duckdb connection per substrate held open, median over repeats; untimed verdict call first (doubles as warm-up). Full-scan audit census: impl/measure_adm_scan.py gate: FULL_SCAN_MODES={symdiff_audit}, WINDOW_BOUNDED_MODES={window_realization_symdiff}. Environment: Python 3.9.6, DuckDB 1.4.4, macOS-15.0-arm64-arm-64bit. All 6 row-scale points were reachable (gold-anchor hull survival held on every substrate: zero hull losses recorded in the manifest), and all 48 certificates (8 questions × 6 scales) verify ACCEPT with the frozen decision split (5 ANSWER, 1 REWRITE, 2 REFUSE) preserved at every scale. Table 20 is the row-scale headline, Table 21 the per-certificate verify medians, and Table 22 the window-span stretch axis.

Table 20: S5 row-scale axis: per scale point, the median over the 8 financial certificates of warm verify / answer medians (9 repeats each), the paired verify/answer ratio over the 6 SQL-emitting certificates, and the full-scan audit census.

scale	`trans` rows	verify (ms)	answer (ms)	ratio med (min–max)	full scans
0.125×	132,000	18.04	2.82	24.8× (3.56–41.4)	0
0.25×	263,927	18.10	2.84	24.2× (3.6–41.8)	0
0.5×	527,900	18.06	2.87	22.3× (3.46–41.9)	0
1×	1,056,320	19.69	3.92	21.5× (2.58–40.5)	0
2×	2,112,640	21.03	6.19	20.6× (1.61–41.3)	0
4×	4,225,280	24.80	11.13	20.4× (0.978–42.1)	0

Table 21: S5 per-certificate warm verify medians (ms) across the row-scale axis, with the 1×→4× growth factor. Kind: the certificate's frozen decision.

qid	kind	0.125×	0.25×	0.5×	1×	2×	4×	growth
`FIN-Q1`	answer	40.57	40.44	40.64	40.48	40.65	40.98	1.01×
`FIN-Q2`	answer	39.60	39.33	39.74	40.11	39.67	40.00	1.00×
`FIN-Q3`	answer	48.80	47.58	48.75	53.57	60.24	74.11	1.38×
`FIN-Q4`	answer	14.67	14.45	14.62	14.72	14.65	14.48	0.98×
`FIN-Q5`	answer	16.77	16.84	16.76	17.81	19.05	20.89	1.17×
`FIN-Q6`	rewrite	16.56	16.97	16.75	18.05	19.02	20.81	1.15×
`FIN-Q7`	refuse	14.44	14.53	14.81	16.42	18.46	22.47	1.37×
`FIN-Q8`	refuse	19.31	19.24	19.35	21.34	23.02	27.14	1.27×

Verify cost is flat in window span (9.77 ms at 1 month vs 9.49 ms at 48 months, 0.97×) while the certified row count grows 48×; every span point's paired ratio (24.4–29.2×) sits inside the pre-registered [1×, 60×] band.

The S5-P1 miss, stated plainly. S5-P1 predicted the warm verify median to be monotone non-decreasing in row scale with at-most-linear growth. The growth clause holds comfortably: 4× rows ⇒ 1.26× median verify time (19.69 ms at 1× to 24.80 ms at 4×), strongly sublinear. The monotonicity clause fails once: 0.25× → 0.5× dips 18.10 ms → 18.06 ms, a 45 μs (0.25%) decrease

Table 22: S5 window-span stretch axis (REACHED): W1-Q4 (w1.history_rows_range), 84-month authored history 2020-01..2026-12. All spans compile ANSWER and verify ACCEPT; rows = the value certified by the emitted SQL.

span	window	verify (ms)	answer (ms)	ratio	rows
1 mo	`2026-12..2026-12`	9.77	0.335	29.2×	239
3 mo	`2026-10..2026-12`	9.67	0.368	26.3×	717
6 mo	`2026-07..2026-12`	9.51	0.350	27.2×	1434
12 mo	`2026-01..2026-12`	9.55	0.348	27.5×	2868
24 mo	`2025-01..2026-12`	9.54	0.374	25.5×	5736
48 mo	`2023-01..2026-12`	9.49	0.388	24.4×	11472

— far inside warm-timing run-to-run noise, but the pre-registration wrote *strict* monotone non-decreasing, so the prediction as written is adjudicated a **miss**. Every other adjacent step is non-decreasing.

Both readings of S5-P2. The adjudicated statistic (declared in the sweep script before adjudication) is the per-point *median* paired ratio over the 6 SQL-emitting certificates: in band $[1\times, 60\times]$ at every point (24.8× down to 20.4×), so **MET**. The stricter every-certificate reading fails only at 4×: `FIN-Q5` at 0.978×; `FIN-Q6` at 0.981× — their answering query grows linearly with rows while verification stays window-bounded, so verification becomes *cheaper than answering* at scale. That is the favourable direction for the §3 cost claims, but it exits the pre-registered band's lower edge, and we publish that reading as a **miss**.

Pre-registered predictions (S5).

- **S5-P1** ("warm verify median is monotone non-decreasing in row scale with growth at most linear (4x rows => <=4x median verify time)"): **MISSED** (growth clause met, $1.26\times \leq 4\times$; strict monotonicity violated once, see above).
- **S5-P2** ("paired verify/answer ratio stays inside [1x, 60x] at every reachable scale point"): **MET** on the adjudicated median reading; the strict all-certificates reading is **MISSED** (published above).
- **S5-P3** ("zero full-scan audits at every scale (all clause-(iv) audits window-bounded, as at 1x)"): **MET** (zero full-scan audits at all 6 scale points; the financial certificates carry no clause-(iv) replay at any scale (the corpus's three window_realization_symdiff audits live on CARD-Q2/CARD-Q7/EF2-Q6, outside this domain); zero full-scan audits is therefore confirmed by census, and the window-bounded clause is vacuously true here).

Provenance correction (2026-08-23, published). The S5 measurement script's embedded citation of the pre-registration SHA-256 was corrupted: s5_measure.py's PREREG_SHA constant dropped two hex characters ('1c' at offset 44) from the prereg sha256, citing a 62-character value; it propagated into this JSON (prereg.sha256, substrates_manifest.prereg_sha256) and the S5_REPORT.md header, and no S5 script re-asserted the hash from the frozen prereg on disk, so the typo went uncaught. Discovered by adversarial verification (`poststudy2_20260823/VERIFICATION2.md (S5 section; Defect list item 1)`); corrected on 2026-08-23 by `pilot2/poststudy2_20260823/s5/fix_prove` which recomputed the correct value from the frozen pre-registration on disk (`838a214fc5a09902703d969c839872ff843f190e9f2` and corrected the fields `prereg.sha256`, `substrates_manifest.prereg_sha256` in the committed sweep JSON. Impact: metadata-only: no measured number, verdict, or certificate is affected; no timing was re-run. Companion fixes: S5_REPORT.md re-rendered from this corrected JSON via s5_render_report.py (fidelity pre-checked byte-for-byte), with a visible CORRECTION note appended; s5_measure.py: PREREG_SHA constant corrected (docstring citation included) and a runtime re-assert-from-disk added, so a future re-run refuses on mismatch. The corrupted string dropped the two characters `1c` at offset 44: `838a214fc5a09902703d969c839872ff843f190e9f2e9f6902f231e061c669`. This appendix (and every S5 number above) is generated from the corrected JSON, and this generator independently re-asserts the pre-registration SHA-256 from the frozen file on disk.

M.3 S6: English-Question Control

LLM study: translations first, then `baseline_claude` and `governance_informed` re-run on the English questions. All 60 `question_zh` were translated to English, the translations independently audited question-by-question for gold-invariance, and the audited set frozen (`questions_en.json`, SHA-256 `d9508f3f6e1617e080772630f51f6d1860cd0cfe1dae174b7f226ada548ba63f`, re-computed by this generator) *before* any scored call. Protocol: model `claude-opus-4-6`; one sample, retry only on empty x3, 512-token cap, temperature unset; single delta: question text = frozen question_en (all else byte-frozen R2/RP chain); scorer: frozen R2.fetch_and_score (RP.score + §4.2 rowset/string dispatch). Call-log account: 60 logged calls per arm, 0 retried, 0 empty completions. Table 23 is the EN-vs-ZH headline, Table 24 the flip counts, Table 25 the probe-question control, and Table 26 the full per-question flip ledger.

Table 23: S6 headline: English-question error next to the frozen Chinese anchors (pilot2/pilot2_arms_summary.json per_question_verdicts (frozen 2026-08-04 main study)). EN taxonomy: ok = correct, wv = wrong value, xe = execution error, ar = answered-should-refuse, rs = refused-should-answer.

arm	ZH err	ZH rate	EN err	EN rate	ok	wv	xe	ar	rs
`baseline_claude`	36/60	60.0%	35/60	58.3%	25	14	2	5	14
`governance_informed`	28/60	46.7%	29/60	48.3%	31	12	6	11	0

Table 24: S6 ZH→EN flip counts per arm over the 60 paired questions.

arm	both correct	both error	ZH ✓ EN ×	ZH × EN ✓
`baseline_claude`	18	29	6	7
`governance_informed`	25	22	7	6

Table 25: S6 probe-dependence control: the 7 probe-only refusal questions under the governance-informed arm, Chinese vs English. Errors: ZH 6/7, EN 5/7.

qid	ZH verdict	EN verdict
`CARD-Q7`	ar	ar
`DEB-Q6`	✓	ar
`EF2-Q5`	ar	ar
`EF2-Q6`	ar	ar
`F1-Q7`	ar	✓
`F1-Q8`	ar	ar
`FIN-Q7`	ar	✓

S6 per-question flip ledger

All 60 questions. Gold: expected outcome class. Per arm: the frozen Chinese verdict, the English verdict, and a flip mark (− = ZH correct, EN error; + = ZH error, EN correct; blank = no flip). Verdicts: ✓ = correct, wv = wrong value, xe = execution error, ar = answered-should-refuse, rs = refused-should-answer.

Table 26: S6 per-question ZH/EN verdicts and flips per arm.

QID	gold	baseline_claude			governance_informed		
		ZH	EN	flip	ZH	EN	flip
california_schools							
`CA-Q1`	value	rs	rs		xe	xe	
`CA-Q2`	value	rs	rs		✓	✓	
`CA-Q3`	value	wv	✓	+	✓	✓	
`CA-Q4`	value	rs	wv		✓	✓	
`CA-Q5`	rewrite	wv	rs		wv	xe	
`CA-Q6`	refusal	✓	✓		✓	ar	−
card_games							
`CARD-Q1`	value	✓	rs	−	✓	✓	
`CARD-Q2`	value	rs	rs		xe	wv	
`CARD-Q3`	value	rs	rs		xe	✓	+
`CARD-Q4`	value	✓	✓		✓	xe	−
`CARD-Q5`	rewrite	✓	rs	−	✓	✓	
`CARD-Q6`	refusal	✓	✓		ar	ar	
`CARD-Q7`	refusal	✓	✓		ar	ar	
codebase_community							
`CODE-Q1`	value	xe	rs		xe	wv	
`CODE-Q2`	value	wv	rs		wv	wv	
`CODE-Q3`	value	✓	✓		✓	✓	
`CODE-Q4`	rewrite	wv	wv		wv	wv	
`CODE-Q5`	rewrite	xe	xe		xe	xe	
`CODE-Q6`	rewrite	ns	rs		wv	wv	
`CODE-Q7`	refusal	ar	ar		ar	ar	
debit_card_specializing							
`DEB-Q1`	value	✓	✓		✓	✓	
`DEB-Q2`	value	✓	✓		✓	✓	
`DEB-Q3`	value	✓	✓		✓	✓	
`DEB-Q4`	value	✓	✓		✓	✓	
`DEB-Q5`	rewrite	wv	wv		rs	wv	
`DEB-Q6`	refusal	ar	✓	+	✓	ar	−
`DEB-Q7`	refusal	ar	ar		✓	ar	−
european_football_2							
`EF2-Q1`	value	wv	wv		xe	✓	+
`EF2-Q2`	value	✓	✓		✓	✓	
`EF2-Q3`	value	✓	✓		xe	✓	+
`EF2-Q4`	rewrite	✓	✓		✓	✓	
`EF2-Q5`	refusal	ar	ar		ar	ar	
`EF2-Q6`	refusal	✓	✓		ar	ar	
financial							
`FIN-Q1`	value	✓	✓		✓	✓	
`FIN-Q2`	value	xe	wv		✓	✓	
`FIN-Q3`	value	xe	rs		✓	✓	
`FIN-Q4`	value	✓	✓		✓	✓	
`FIN-Q5`	value	✓	wv	−	✓	✓	
`FIN-Q6`	rewrite	✓	✓		✓	✓	
`FIN-Q7`	refusal	ar	✓	+	ar	✓	+
`FIN-Q8`	refusal	ar	✓	+	✓	✓	
formula_1							
`F1-Q1`	value	wv	wv		wv	wv	
`F1-Q2`	value	wv	wv		wv	wv	
`F1-Q3`	value	wv	rs		✓	wv	−
`F1-Q4`	value	wv	wv		wv	✓	+
`F1-Q5`	value	wv	rs		wv	wv	
`F1-Q6`	rewrite	✓	✓		✓	✓	

(continued)

QID	gold	ZH	EN	flip	ZH	EN	flip
`F1-Q7`	refusal	ar	✓	+	ar	✓	+
`F1-Q8`	refusal	ar	✓	+	ar	ar	
thrombosis_prediction							
`TH-Q1`	value	wv	wv		✓	✓	
`TH-Q2`	value	✓	wv	−	✓	✓	
`TH-Q3`	rewrite	wv	wv		xe	xe	
`TH-Q4`	rewrite	wv	wv		✓	wv	−
`TH-Q5`	refusal	ar	ar		ar	ar	
`TH-Q6`	refusal	ar	ar		ar	ar	
world_1							
`W1-Q1`	value	✓	xe	−	✓	wv	−
`W1-Q2`	value	wv	wv		✓	✓	
`W1-Q3`	value	✓	✓		✓	✓	
`W1-Q4`	rewrite	✓	rs	−	xe	xe	
`W1-Q5`	refusal	ar	✓	+	✓	✓	

Pre-registered predictions (S6). All four were met:

- **S6-P1** ("EN backbone error within 0.600 +/- 0.10"): **MET** (observed 58.3%, band [50.0%, 70.0%]).
- **S6-P2** ("EN governance-informed error within 0.467 +/- 0.10"): **MET** (observed 48.3%, band [36.7%, 56.7%]).
- **S6-P3** ("backbone error > governance error in English"): **MET** (58.3% vs 48.3%).
- **S6-P4** ("EN governance arm errs >= 4 of the 7 probe-only refusal questions"): **MET** (5/7 probe questions errored in English).

The backbone–governance ordering, the error-band placement of both arms, and the probe-dependence mechanism all survive the language swap; the refusal asymmetry also reproduces in English (backbone: 10 correct refusals of 15 but 14 over-refusals on answerable questions; governance-informed: 4 and 0).

M.4 S7: NL→ σ Extraction Arm

LLM study: the hybrid the paper's architecture implies. An extractor model (`claude-opus-4-6`) receives the schema pack, the governance pack, a σ-format specification, and the `question_zh` text *only* — no gold-side field is readable by the arm — and must produce the full structured intent σ; the frozen binding compiler and the frozen scoring rules do the rest. Invalid JSON earns one format-retry, then scores as error. This run: 57 paid extraction calls plus 3 append-only cache hits (the Stage-A smoke questions), 0 format retries, 0 extraction errors, 2 compile errors (`CA-Q5`, `CA-Q6`). Headline: exact full-σ recovery on 40/60 questions and end-to-end error 5/60 — against the frozen governance-informed arm's 28/60 and the frozen backbone's 36/60. Table 27 gives per-field σ-recovery, Table 28 the split by gold outcome class and domain.

The 5 errors, one by one. Every end-to-end error is an error of *extraction*, not of the frozen compiler (S7-P3: zero errors among the 40 exact-σ questions):

- **CA-Q5** (gold rewrite): extracted σ crashes the frozen compiler (`TypeError: 'NoneType' object is not subscriptable`). Mismatched fields — `as_of: null` for `2015-04-15`; `presentation: raw_rows` for `aggregate`.

Table 27: S7 per-field σ-recovery over all 60 questions (two column blocks).

field	match	miss	field	match	miss
as_of	44/60	16	window_request	59/60	1
declared_at	60/60	0	requested_granularity	60/60	0
metric_alias	58/60	2	requested_time_gran	59/60	1
scope	59/60	1	presentation	56/60	4
pinned_version	60/60	0	ctx_role	60/60	0
cross_window	60/60	0	periods	60/60	0
anchor_override	59/60	1			

Table 28: S7 end-to-end outcomes and exact-σ counts by gold outcome class (left) and by database (right).

gold class	n	e2e err	exact σ
value	33	0	26
rewrite	12	4	7
refusal	15	1	7

database	n	e2e err	exact σ
california_schools	6	2	4
card_games	7	0	6
codebase_community	7	2	4
debit_card_specializing	7	0	2
european_football_2	6	0	6
financial	8	0	7
formula_1	8	0	8
thrombosis_prediction	6	1	3
world_1	5	0	0

- **CA-Q6** (gold refusal): extracted σ crashes the frozen compiler (`TypeError: 'NoneType' object is not subscriptable`). Mismatched fields — `anchor_override`: `OpenDate` for `charter_authorization_date`; `as_of`: `null` for `2015-04-15`; `scope`: `{}` for `{"county": "Alameda"}`.
- **CODE-Q4** (gold rewrite): expected rewrite, got refusal disclosure-blocked. Mismatched fields — `presentation`: `raw_rows` for `aggregate`.
- **CODE-Q5** (gold rewrite): expected rewrite, got refusal disclosure-blocked. Mismatched fields — `presentation`: `raw_rows` for `aggregate`.
- **TH-Q3** (gold rewrite): expected rewrite, got refusal disclosure-blocked. Mismatched fields — `as_of`: `1997-06-15` for `1997-06-30`; `presentation`: `raw_rows` for `aggregate`; `window_request`: `{"hi": "1997-12", "kind": "month_token_range", "lo": "1997-01"}` for `null`.

Descriptively, `as_of` is the single largest mismatch field (16/60), yet only 3 of those 16 questions err end-to-end: most `as_of` mismatches are alternative but outcome-equivalent readings of the question's time anchor, which the compiler resolves to the same certificate.

Pre-registered predictions (S7). All four were met:

- **S7-P1** ("exact full-σ recovery on >= 36/60 questions"): **MET** (40/60 vs threshold 36).
- **S7-P2** ("end-to-end error <= 28/60 (the governance-informed arm's)"): **MET** (5/60 vs threshold 28).

- **S7-P3** (“on questions with exact σ recovery, end-to-end error = 0 (compiler correctness transfers)”): **MET** (0 errors among the 40 exact-sigma questions).
- **S7-P4** (“metric-identity recovery >= 54/60 (>=90%); failures concentrate in window/scope/version fields, not metric identity”): **MET** (58/60 vs threshold 54; 2 metric-identity vs 2 window/scope/version mismatches).

M.5 Prediction Scoreboard: All 14, Misses First

The frozen protocol stated 14 predictions in advance; 11 were met and 3 were missed. Statements are verbatim from the frozen pre-registration; the misses come first.

Missed (3 of 14).

- **S4-P1** (“the reference−governance paired-difference CI excludes 0.”): **MISSED** — mean +0.133 but 95% CI [−0.050, 0.300] includes 0 (§M.1).
- **S4-P2** (“every variant−reference paired-difference CI includes 0.”): **MISSED** — `trivial_v3` CI [−0.143, −0.015] excludes 0 on the *worse* side (§M.1).
- **S5-P1** (“warm verify median is monotone non-decreasing in row scale with growth at most linear ($4\times$ rows $\Rightarrow \leq 4\times$ median verify time).”): **MISSED** — one 45 μs dip at $0.25\times \to 0.5\times$ breaks strict monotonicity; growth clause held at $1.26\times \leq 4\times$ (§M.2).

Met (11 of 14).

- **S4-P3** (“per-rep elim values all lie in [0.30, 0.45] and the min–max range straddles the pre-registered 0.40 line (the case-(iii) call is boundary-honest, not comfortable).”): **MET** — per-rep eliminations in $[36.1\%, 41.7\%]$, straddling the 0.4 line.
- **S5-P2** (“the paired verify/answer ratio stays inside $[1\times, 60\times]$ at every reachable scale point.”): **MET** — per-point median ratios 20.4–24.8×, all in $[1\times, 60\times]$; the stricter every-certificate reading is MISSED at 4× (§M.2).
- **S5-P3** (“zero full-scan audits at every scale (all clause-(iv) audits window-bounded, as at 1×).”): **MET** — zero full-scan audits at all 6 scale points.
- **S6-P1** (“English backbone error within 0.600 ± 0.10.”): **MET** — observed 58.3% in [50.0%, 70.0%].
- **S6-P2** (“English governance-informed error within 0.467 ± 0.10.”): **MET** — observed 48.3% in [36.7%, 56.7%].
- **S6-P3** (“backbone error > governance error in English.”): **MET** — 58.3% > 48.3%.
- **S6-P4** (“of the 7 probe-only refusal questions, the English governance arm still errs ≥ 4 (the probe-dependence mechanism is not language-bound).”): **MET** — 5/7 probe questions still err in English.
- **S7-P1** (“exact full-σ recovery on $\geq 36/60$ questions.”): **MET** — $40/60 \geq 36$.
- **S7-P2** (“end-to-end error $\leq 28/60$ (the governance-informed arm’s).”): **MET** — $5/60 \leq 28$.
- **S7-P3** (“on questions with exact σ recovery, end-to-end error = 0 (compiler correctness transfers).”): **MET** — 0 errors among 40 exact-σ questions.

- **S7-P4** ("metric-identity recovery $\geq$ 54/60 ($\geq$90%) — failures concentrate in window/scope/version fields, not metric identity."): **MET** — 58/60 $\geq$ 54, and 2 metric-identity vs 2 window/scope/version mismatches.

N Frozen Theory Chapter C3: Bi-Temporal Governed Query Semantics

Frozen source `theory/C3_bitemporal_semantics.md` (SHA-256 above), reproduced content-identically; Chinese. The body's §2 and §3 and Appendices D.1 and D.2 are its English rendering.

N.1 C3 · 双时态治理查询语义（Bi-Temporal Governed Query Semantics）

> 所属：PP-5744d699 / S1 骨架 §2「Bi-Temporal Governed Query Semantics」（脊柱），并为 §3 的三类不可绑定刻画提供谓词与可判定性基座。定框纪律：abstraction-first——先定义（时间域/锚/图版本/查询），再不变量，再定理（覆盖三分、可判定性、退化），算法与 pilot 编译器仅以实现注记出现。记号纪律：C5（时点证书篇）已先行冻结 $\mathbb{T}$, $\mathcal{W}$, $\mathcal{G}=\{G_v\}_{v\in\mathcal{V}}$, $V(a)$, $\mathsf{Bind}(q,T,G_v,D)$, $\{\mathsf{MC},\mathsf{AM},\mathsf{OOV}\}$ 与快照假定 SA；本篇沿用并做保守扩展，新增记号见 §0 表。骨架任务书中的 as-of τ 即本篇的声明时点 T（合并时统一写 T）。

N.1.1 0. 记号与前置约定

沿用 C5 §0 的冻结记号，本篇新增：

记号	含义	首次定义
$\mathbb{G}$, g, $\mathrm{gr}_g(t)$	粒度族、粒度、t 所在粒元	定义 3.1
$\mathcal{W}_g\subseteq\mathcal{W}$	g-对齐窗子代数（含 $\top=\mathbb{T}$）	定义 3.2
$a=(o_a,\ \mathrm{ty}_a,\ \mathrm{vt}_a,\ g_a,\ \mathrm{cm}_a)$	valid-time 锚（对象、锚型、有效时间指派、标记粒度、覆盖方式）	定义 3.3
$V_v(a)$, $\mathrm{Cov}_v(a)$	锚的有效标记集（C5 记 $V(a)$）与有效覆盖域	定义 3.3
$G_v=(N_v,E_v,\lambda_v;\ M_v,R_v,A_v,\beta_v)$	治理语义图版本的内容物	定义 3.4
$t(\cdot)$, $\mathrm{ver}(T)$	commit map 与在位版本	定义 3.5
$q=(\sigma,T)$, $\sigma=(m,s,\vec{\omega},\bar{a})$, $R(q)$	双时态查询、结构化意图、角色集	定义 3.6
$\alpha_{q,v}$	候选锚指派（C5 定义 5.1 中 α 的编译期来源）	定义 3.7
$\mathrm{Pop}^r_{v,T}$, $\mu^r_{v,T}$	角色 r 的腿人口与质量	定义 3.9
$\mathsf{Bindable}(q,\mathcal{G},D)$	可绑定性	定义 3.10
$[\![q]\!]_{\mathcal{G},T}$	时点正确答案语义（全函数）	定义 3.11
$\bot_{\mathsf{OOV}},\bot_{\mathsf{AM}},\bot_{\mathsf{MC}}$	三类结构化拒答值	定义 3.11
$[\![\cdot]\!]^1_{G_v,D}$, SVRC	单版本语义与单版本读一致性	定义 3.19

快照假定（SA，承 C5）：一次求值读取同一 (G_v,D) 快照；D 自身的事务时间并发演化由版本基座（DELTAGOV 类 producer）负责，本篇作为 consumer 不重复其一致性协议。因此本篇的" 二维" 是：**valid time（数据内的有效时间轴，经锚解释）× 治理图版本轴（$\mathcal{V}$ 上的在位版本，扮演语义层的 transaction/version 维）**。

有限性与可计算性假定（全篇定理的公共前提）：- (A1) 每个 $[\![o]\!]_v\subseteq D$ 有限，可枚举；单事实上的谓词求值代价有界；- (A2) vt_a 与窗代数运算（交、含于、规范形）可计算；粒度族 $\mathbb{G}$ 有限、以「日」为公共细

化，且在细化序下构成 meet-半格（定义 3.1）；- (A3) $\mathcal{V}$ **非空**有限全序，commit map 可计算且严格单调；版本轴 **append-only**：$\mathcal{G}$ 仅经在全序末端追加新版本演化，已提交三元组 $(v,\ t(v),\ G_v)$ 不删除、不改写，版本号在 $\mathcal{V}$ 中唯一且**不复用**（回滚 = 重提交建模见 §2.3，裁决 D6）；- (A4) 绑定规则库（定义 3.8）中每条规则指称一个可判定谓词；- (A5) 每个逐角色窗导出 ω_r 以**全函数** $\mathbb{T} \to \mathcal{W} \uplus \{\uparrow\}$ 呈现：$\uparrow$ 为显式「不可解析」哨兵值，故 $\mathrm{dom}(\omega_r) = \{T : \omega_r(T) \neq \uparrow\}$ 可判定——判 $\omega_r(T)\uparrow$ 是一次求值而非停机判定（pilot 的 as-of 解析器即此呈现：解析失败显式返回错误值）；- (A6) $\mathbb{T}$ 为有效呈现、序可判定的**离散**时间域（原型：日粒日期全序集，将 C5 §0 的「日期全序集」升格为正式假设）。连续时间域（实数时轴上以任意实数呈现的 T）**显式出界**：其上边界比较仅半可判定，命题 3.18 不对其断言。

N.1.2　1. 预备：时间域、粒度与窗代数

定义 3.1（粒度 granularity） 粒度 g 是 $\mathbb{T}$ 到相继区间的一个划分，划分元称**粒元（granule）**，$\mathrm{gr}_g(t)$ 记 t 所在粒元。粒度族 $\mathbb{G}$ 有限（原型：日/周/月/季/年），且存在公共细化 $g_\perp$（日）：每个粒元是若干 $g_\perp$-粒元的并。$g' \sqsubseteq g$ 记 g' 细化 g。进一步假定 $(\mathbb{G}, \sqsubseteq)$ 为 **meet-半格**：对任两粒度 g, g'，公共细化集 $L(g,g') = \{h \in \mathbb{G} : h \sqsubseteq g \wedge h \sqsubseteq g'\}$（由 $g_\perp \in L$ 非空）有 $\sqsubseteq$-最大元，记 $g \sqcap g'$，称二者的**公共可表达粒度**（最粗公共细化）。一般有限粒度族的 $L(g,g')$ 可有多个极大元而无最大元，故 meet 存在性是对 $\mathbb{G}$ 的结构假定而非自动事实（原型族满足：周 $\sqcap$ 月 = 日）。周与月互不细化——这不是缺陷而是本体事实，正是 quality_voc 域「周锚 vs 日锚」摩擦的抽象形态。

定义 3.2（g-对齐窗代数） C5 的窗代数 $\mathcal{W}$ 含半开区间与有限日期集。定义

$$\mathcal{W}_g \;=\; \{\,W \in \mathcal{W} \;:\; W \text{ 可写为有限个}g\text{-粒元对齐区间之并，端点允许}\pm\infty\,\} \;\cup\; \{\top\},$$

其中 $\top = \mathbb{T}$（全窗）。对每个 g 有 $\top \in \mathcal{W}_g$；$g' \sqsubseteq g \Rightarrow \mathcal{W}_g \subseteq \mathcal{W}_{g'}$。pilot 中的请求窗均落于此：rma 的「as-of 月窗」$[\text{月初}, \text{次月初}) \in \mathcal{W}_{\text{月}}$；domestic_newprod 的快照类指标（launch/preorder）取累计窗 $(-\infty, d]$（d 取 as-of 月末）；该域 SCD-2 桥接指标族（`product_versions_valid_asof`/`model_mismatch_rate`）依裁决 D7 取**点窗** $\{d\}$（d 为 as-of 日；单点日期集落于 $\mathcal{W}_{g_\perp}$），在效判定即 $\mathrm{vf} \le d < \mathrm{vtc}$，人口 126 与金标 NX-Q4 一致（旧「相交 × 累计窗」读法得 165/219，已废止）。

建模选择（窗代数封顶） 本篇把请求窗限定在日历窗代数 $\bigcup_{g\in\mathbb{G}} \mathcal{W}_g$ 内；可判定性命题 3.18 相对于该封顶成立。允许任意一阶可定义窗的推广是显式缺口（§8 G3）。

N.1.3　2. 治理语义图版本与版本序列

2.1 valid-time 锚　定义 3.3（valid-time 锚 anchor） 版本 v 下，锚 $a = (o_a,\ \mathrm{ty}_a,\ \mathrm{vt}_a,\ g_a,\ \mathrm{cm}_a)$ 由语义对象 $o_a \in N_v$、锚型 ty_a、有效时间指派 $\mathrm{vt}_a : [\![o_a]\!]_v \to \mathcal{W}$、标记粒度 $g_a \in \mathbb{G} \cup \{\mathsf{itv}\}$ 与覆盖方式 $\mathrm{cm}_a \in \{\mathsf{hull}, \mathsf{strict\_member}\}$（coverage mode，取值 token 跨章统一为 `hull` | `strict_member`——C5 §0 先冻结者为准；裁决 D3：锚的**版本化属性**、G_v 内容物）组成。两个原型锚型（抽象自 `gov_valid_time_anchor` 的 `anchor_type` 域）：

1. **快照锚**（`snapshot_effective_date`）：o_a 携带有效日期列 eff，$\mathrm{vt}_a(r) = \mathrm{gr}_{g_a}(r.\mathrm{eff})$——事实在其标记粒元内有效。原型：rma 的 `rma_event_time`/`sales_event_time`（均日粒），quality_voc 的 `sales_event_date`（周粒标记）。
2. **区间锚**（`scd_type2`，$g_a = \mathsf{itv}$）：o_a 携带 $(\mathrm{vf}, \mathrm{vtc})$ 列对，$\mathrm{vt}_a(r) = [\,r.\mathrm{vf},\ r.\mathrm{vtc})$，**点窗语义**（裁决 D7，与金标 126 一致）：$d \in \mathrm{vt}_a(r) \iff \mathrm{vf} \le d < \mathrm{vtc}$。原型：domestic_newprod 的 `domestic_newprod_scd2_dim_product`（`dw_start_date`/`dw_end_date`）。

良构注记（承 D7） 严格良构锚满足对每行 $\mathrm{vt}_a(r) \neq \varnothing$（区间锚即 $\mathrm{vf} < \mathrm{vtc}$）。陪域放宽到 $\mathcal{W}$（允许 $\varnothing$）正是为容纳零长 SCD-2 行（$\mathrm{vf} = \mathrm{vtc}$，原型 392 行中 111 行）：此类行 $\mathrm{vt}_a(r) = \varnothing$，与任意窗（含点窗）交空、对 $V_v(a)$ 无贡献，在一切腿人口中**语义上自然为空出局**——无须脏行预处理；定理 3.20(2)(3) 的坍缩前提相应按锚有效限制陈述（见定义 3.19）。

有效标记集 $V_v(a)=\bigcup_{r\in[\![o_a]\!]_v}\mathrm{vt}_a(r)$（C5 在版本固定时记 $V(a)$；零长行贡献 $\varnothing$，故 $V_v(a)$ 只由良构行生成）。**有效覆盖域（按 cm_a 参数化，裁决 D3）**：

$$\mathrm{Cov}_v(a)\;=\;\begin{cases}\mathrm{hull}(V_v(a)) & \mathrm{cm}_a=\textsf{hull}(\text{序凸包：}[\min,\max]\text{ 粒元跨度）},\\ V_v(a) & \mathrm{cm}_a=\textsf{strict\_member}(\text{严格标记成员，不取包）}.\end{cases}$$

缺省指派：**快照锚默认** hull——覆盖跨度内某窗恰无事实（如某月零客诉）应产出合法的 0 值分子，而不应触发 out-of-validity；**裸日期集锚**（标记集为有限日期集、无区间结构的快照锚形态，quality_voc 现状）**默认** strict_member；区间锚取 strict_member（区间之并——维表历史空隙是真实的「无在效版本」）。cm_a 随锚入 G_v 版本化（seed 扩 `coverage_mode` 字段，G4）；C5 的 OOV 见证按锚**声明的 mode 重演**：判空 $W\cap\mathrm{Cov}_v(a)$ 在 strict_member 锚上即 C5 原判据 $W\cap V(a)=\varnothing$，在 hull 锚上以凸包重演——两侧判类由同一 cm_a 驱动，不再相反。**类型注记** $V_v(a)$ 与 $\mathrm{Cov}_v(a)$ 一般是 $2^{\mathbb{T}}$ 中的有限区间并，不必落回 $\mathcal{W}$ 冻结的两分类（半开区间/有限日期集）；OOV 的判空 $W\cap\mathrm{Cov}$（定义 3.14）在 $2^{\mathbb{T}}$ 中经排序合并的规范形求值（命题 3.18 的判定程序即如此工作）。裁决 D3 下 $\mathcal{W}$ **不升格**为其生成的有限并代数 $\bar{\mathcal{W}}$：覆盖判空只在 $2^{\mathbb{T}}$ 规范形内进行，请求窗代数维持 C5 冻结的两分类。

2.2 图版本内容物　定义 3.4（治理语义图版本） $G_v=(N_v,\ E_v,\ \lambda_v;\ M_v,\ R_v,\ A_v,\ \beta_v)$：

- N_v：语义节点集；$\lambda_v:N_v\to\{\mathrm{ODS},\mathrm{DIM},\mathrm{DWD},\mathrm{DWS},\mathrm{ADS}\}$ 层标注（并携 `ai_usable/ai_traversable` 旗标）；
- $E_v\subseteq N_v\times N_v\times\mathrm{EdgeTy}$：类型化边，$\mathrm{EdgeTy}\supseteq\{$`normalize_of`, `aggregate_of`$\}\cup\{$`numerator_of`$\langle m\rangle$, `denominator_of`$\langle m\rangle$: $m\in M_v\}$；
- M_v：度量定义集。度量 m 为**原子型** (o_m,f_m)（单对象上的可计算测度 f_m，如 `problem_qty`），或**比率型** $\big((o_m^{\mathrm{n}},f_m^{\mathrm{n}}),\ (o_m^{\mathrm{d}},f_m^{\mathrm{d}})\big)$，其两腿由 `numerator_of/denominator_of` 边指定（原型：`gov_semantic_edge` 中 `problem_rate` 的两腿指向 `dws_rma_sku_problem_1d` 与 `dws_sales_sku_1d`）；
- R_v：**metric→caliber 路由**（部分映射）。$R_v(m)=\rho=(\kappa,\ o_{\mathrm{via}},\ K_\rho,\ \mathrm{align}_\rho)$：路由键、经由对象、连接键集（如 [`product_sku`, `dt`]）与归因对齐（如 `sku+period`）。ρ 决定 scope 转移与**分母人口的口径**（scoped 全量，而非事件内自指人口）；`dst_caliber='none'` 的 reference-only 行视为不可重算路由。原型：`gov_caliber_routing` 的 `rma_problem_to_sales` 与其反例对照 `caliber_blind_control`；
- A_v：锚集（C5 记 $A(G_v)$），并有锚指派把度量腿映到锚：原子型 $m\mapsto a_m$；比率型两腿的规定锚由 β_v 携带；
- β_v：**时态绑定**（部分映射）。$\beta_v(m)=(a_{\mathrm{n}}^*,\ a_{\mathrm{d}}^*,\ \mathrm{rule})$，规定分子/分母锚与绑定规则（原型：`gov_temporal_binding` 行；规则原型 `same_valid_time_window`）。**缺失约定（按度量型分流）**：原子型 m 的 $\beta_v(m)$ 可缺省——单锚 a_m 已由 A_v 给出，无跨腿规则可言；比率型 m 而 $\beta_v(m)\uparrow$ 时，两腿规定锚**无来源**（A_v 的锚指派只覆盖原子型，单个锚的 vt 定义在单一对象扩展上、对另一腿类型不合），该状态**不**按原子型处理，而在语义上归入 $\perp_{\mathsf{MC}}$：见定义 3.16(i) 的绑定缺失模式与定义 3.10(b2)。把「比率型 $m\in M_v\Rightarrow\beta_v(m)\downarrow$ 且两腿锚落在各自对象」上收为 G_v 良构不变量（并入 C5 I1–I4 一族）是等价的工程化表述，二者取一即可堵住全函数性反例。

记号避让：C5 的 B 已表示证书绑定四元组，故时态绑定映射记 β_v。

注记 3.4′（口径即语义） R_v 是值的构成成分而非执行细节：同一 (m,s,T) 在 `rma_problem_to_sales`（分母 =scoped 全量销量）与 `caliber_blind_control`（分母 = 问题单基数）两条路由下产出不同值——**datagov-line 生产实例实测**（v2026.06.23，rma 域）：caliber-aware 16.2% vs caliber-blind 3.8%，4.2 倍量级差；provenance：datahub `datagov-line`（loctek-unified-datagov-line）rma 域治理产出，非随论文交付的 pilot 工件。**复现口径提示**：pilot rma 仓（`pilot/domains/rma/warehouse.duckdb`）全域 problem/sales 为 2.76%（465/16832，可复演），与上述生产对照数属不同域实例、不同 scope，不可互相复算。因此语义括号必须以 G_v 为下标——路由随版本演化时，值语义随之演化。

2.3 版本序列与在位版本　定义 3.5（治理版本序列与在位版本）治理版本序列为

$$\mathcal{G}\;=\;(\{G_v\}_{v\in\mathcal{V}},\ \preceq,\ t),$$

$\mathcal{V}$ 为有限版本号集配全序 $\preceq$（typed-commit 次序，原型：`gov_semantic_graph_version`），commit map $t:\mathcal{V}\to\mathbb{T}$ 严格单调（$v\prec v'\Rightarrow t(v)<t(v')$）。记 $v_1\prec\cdots\prec v_n$，$t_i=t(v_i)$。**在位版本**

$$\mathrm{ver}(T)\;=\;v_i\quad\text{当}T\in[t_i,\ t_{i+1})\ (\text{约定}t_{n+1}=+\infty),\qquad\mathrm{ver}(T)\uparrow\ \text{当}T<t_1.$$

$[t_1,+\infty)$ 称版本轴的**在位域**（A3 保证 $\mathcal{V}\neq\varnothing$，故 t_1 有定义）。commit map 是对 seed 现状的理想化：`committed_at` 目前是版本标签而非 $\mathbb{T}$ 时戳（缺口 G1，实查五域 `gov_semantic_graph_version` 各 1 行、值为 `v2026.06.23` 类标签）。**commit 时戳同一性（平局细化）**：合成时戳协议若取日粒，同日两次提交即破坏严格单调；协议须以提交序号细化时戳（如 (日期, commit 序号) 字典序嵌入 $\mathbb{T}$ 的离散细化），使 t 在合成后仍严格单调、且每个版本的时戳唯一。pilot 单版本时 ver 平凡。

回滚建模（裁决 D6：回滚 = 重提交）本篇模型内无 `rollback_of`：A3 的 append-only 排除「从 $\mathcal{V}$ 删除已提交版本」式回滚（否则 $\mathrm{ver}(T)$ 对历史 T 回溯变动、旧证书 V1 解析失败）。回滚一律按**重提交新版本**建模：新版本内容可等于任一历史 G_{v_i}，但版本号不复用、commit 序列 append-only 且时戳唯一（A3 与定义 3.5 的平局细化）。由引理 3.5″，旧证书按其坐标 (T,v) **永久有效**——被否弃窗口 $[t_j,t_{j+1})$ 内签发的证书仍是该时点在位口径下的时点正确答案；「时点正确」与「口径被否弃」的区分及其披露/复验策略由 C4（A4.2）承载，C5 证书坐标语义与本条一致（C3 侧落地即 A3/本节）。

引理 3.5′（在位版本良定） t 严格单调蕴含 $\{[t_i,t_{i+1})\}_{i\le n}$ 两两不交且并为 $[t_1,+\infty)$；故 ver 是 $[t_1,+\infty)$ 上的全函数，且每个 T 的像由 $\mathcal{G}$ 确定。■

引理 3.5″（版本轴稳定性；append-only 下）设 $\mathcal{G}'$ 由 $\mathcal{G}$ 依 A3 追加一个新版本 v_{n+1}（$t_{n+1}:=t(v_{n+1})>t_n$）得到。则对一切 $T<t_{n+1}$：$\mathrm{ver}_{\mathcal{G}'}(T)=\mathrm{ver}_{\mathcal{G}}(T)$，且对一切以 T 为声明时点的查询 q，$[\![q]\!]_{\mathcal{G}',T}=[\![q]\!]_{\mathcal{G},T}$——已给出的时点答案（含拒答类）不因治理图继续演化而改变。*证明.* 追加仅把区间 $[t_n,+\infty)$ 划分为 $[t_n,t_{n+1})\uplus[t_{n+1},+\infty)$，$i<n$ 的诸区间不变；$T<t_{n+1}$ 时所落区间与所指版本相同。语义括号对 $\mathcal{G}$ 的依赖只经 $v=\mathrm{ver}(T)$ 与 G_v 的内容物（定义 3.7–3.16 逐条仅引用 v 处对象；append-only 保证 G_v 未被改写），故值逐点不变。有限次追加由归纳即得。■

N.1.4 3. 双时态查询与候选锚指派

定义 3.6（双时态治理查询）查询 $q=(\sigma,\ T)$，$T\in\mathbb{T}$ 为**声明时点**（as-of）；结构化意图

$$\sigma=(m,\ s,\ \vec{\omega},\ \bar{a}),$$

其中 m 为度量引用，s 为范围选择子（实体/维度谓词，如 `sku = E7-WHT`），$\vec{\omega}$ 为**逐角色窗导出**：对每个度量角色 $r\in R(q)$（承 C5 定义 5.1 的角色集：num/den/atom，delta 型按角色 × 期展开）给出可计算映射 $\omega_r:\mathbb{T}\rightharpoonup\mathcal{W}$（部分：as-of 串不可解析时无定义；其呈现方式受 A5 约束——以带 $\uparrow$ 哨兵的全函数 $\mathbb{T}\to\mathcal{W}\uplus\{\uparrow\}$ 交付，定义域可判定），默认各角色共用一个 ω（$W_T=\omega(T)$，与 C5 的请求窗记号一致）；$\bar{a}$ 为**改锚声明**（部分映射，角色 $\rightharpoonup$ **锚引用**——引用以名指锚，**不预设**在任何版本的 A_v 中注册：NL 意图可以点名治理图查无此锚的时间列（原型：NX-Q6 的「业务部上线日期」），引用在在位版本下的解析归定义 3.7 的未注册锚约定），缺省为空。σ 是 NL 意图的结构化形态：pilot 中即 `questions.json` 的 (`metric`, `sku`, `as_of`) 加显式跨窗/改锚字段（rma_q6 的 `num_window≠den_window`；domestic_newprod 的「以业务部上线日期开窗」「用最新版本」触发 $\bar{a}$）。

注记 3.6′（对角线约定与一般双参形）本篇默认**声明时点对角线**：valid-time 轴与版本轴共用同一 T，即窗取 $\vec{\omega}(T)$、版本取 $\mathrm{ver}(T)$。一般形 $[\![q]\!]_{\mathcal{G}}(T_{\mathrm{vt}},T_{\mathrm{ver}})$（如「用现行口径回算历史」$T_{\mathrm{ver}}=\mathrm{now}\neq T_{\mathrm{vt}}$）在语义上是直接推广（把下文 $\mathrm{ver}(T)$ 换为 $\mathrm{ver}(T_{\mathrm{ver}})$、$\vec{\omega}(T)$ 换为 $\vec{\omega}(T_{\mathrm{vt}})$），但其**披露合法性**属 C4 的策略问题；gov_seed 中 GOLD-1 注记（「用最新版本回算历史会错绑过期型号映射」）说明离对角线默认应被拒答或显式披露。对角线作为缺省是建模选择（缺口 G6）。

定义 3.7（候选锚指派）设 $v=\mathrm{ver}(T)$ 已在位、$m\in M_v$。记 RefA_v 为**未注册锚引用**集：$\bar{a}$ 声明的、在 A_v 中解析失败（查无同名锚）的引用名。q 在 v 下的**候选锚指派**为

$$\alpha_{q,v}:R(q)\rightharpoonup(A_v\uplus\mathrm{RefA}_v)\times\mathcal{W},\qquad \alpha_{q,v}(r)=\big(\bar{a}(r)\ \text{若声明否则规定锚},\ \omega_r(T)\big),$$

规定锚指：比率型两腿取 $\beta_v(m)$ 的 $(a^*_{\mathrm{n}},a^*_{\mathrm{d}})$，原子型取 a_m；比率型而 $\beta_v(m)\uparrow$ 时规定锚无来源，$\alpha_{q,v}$ 在两腿角色处无定义——该状态由定义 3.16(i) 的绑定缺失模式捕获（谓词求值按定义 3.14 的真空约定处理）。**未注册锚约定（兑现 C5 定义 5.3 AM(i) 点名的跨章依赖）** $\bar{a}(r)\in\mathrm{RefA}_v$（声明锚 $\notin A_v$）时，$\alpha_{q,v}(r)$ 取该引用本身（陪域的 RefA_v 支使 α 仍良型），并约定：(1) svw 条款 (i) 锚保真**按构造违背**——$a_r\in\mathrm{RefA}_v$ 不等于任何注册锚，故 $a_r\neq a^*_r$，$\beta_v(m)\downarrow$ 时整查询判 $\perp_{\mathsf{AM}}$，见证为声明引用名加「A_v 查无此锚」断言（AM(i) 形态）；(2) 该角色**不参与** Cov/OOV 求值——$\mathrm{Cov}_v(a_r)$ 对未注册引用不可指，定义 3.14 覆盖支按真空约定处理为假，定义 3.10(b1) 第三项对其按真空成立处理，而 Bindable 由 (b3)（P_{svw} 因条款 (i) 违背而假）阻断。接地例 NX-Q6：「以业务部上线日期开窗」——实查 nx `gov_valid_time_anchor` 五行无该锚，金标 =

anchor-mismatch；若无本约定，α 按旧陪域 $A_v \times \mathcal{W}$ 无法类型化，且严读真空约定会令 OOV 支真空假、条款 (i) 真空真，NX-Q6 误入 Bindable。[边界约定（裁决 D8，AUD-259b513e）：**原子型**度量（β_v 依定义 3.4 可缺省、无 svw 规则可违）配未注册改锚时，**同判** $\bot_{\mathsf{AM}}$（**AM(i) 形态**）——「声明的锚在治理图中查无此锚」是锚保真的第一性违背，与度量型别无关（与 D1 的 κ-剔空 $\to$AM(i) 同构）；见证同为声明引用名加「A_v 查无此锚」断言。至此未注册锚引用在两种度量型别下归类统一，定义 3.11/定理 3.17 的全函数性在该角落恢复；pilot 零实例受影响（全部改锚实例均为带绑定行的比率型）。] $\alpha_{q,v}$ 是 q 与 v 的**函数**（无搜索）；对不可绑定 q 搜索替代合法指派属 C4 最大合法改写的存在量词，本篇不做（缺口 G2）。$\alpha_{q,v}$ 即 C5 证书四元组中锚指派 α 的编译期来源。

定义 3.8（绑定规则语义；`same_valid_time_window` 参考条款） 每条规则 rule 指称一个可判定谓词 $P_{\text{rule}}(\alpha;\, G_v, D)$（A4）。参考规则 svw（同有效时间窗）在候选 $\alpha = \{\text{num} \mapsto (a_{\text{n}}, W_{\text{n}}),\ \text{den} \mapsto (a_{\text{d}}, W_{\text{d}})\}$ 上成立，当且仅当以下条款均成立：

- **(i) 锚保真**：$a_{\text{n}} = a^*_{\text{n}}$ 且 $a_{\text{d}} = a^*_{\text{d}}$（改锚即违背；含「用最新版本」对区间锚腿的覆盖；改锚到未注册锚引用 $a_r \in \text{RefA}_v$ 时本条款按构造违背——定义 3.7 未注册锚约定，原型 NX-Q6）；
- **(ii) 同窗**：$W_{\text{n}} = W_{\text{d}}\ (=: W)$；
- **(iii) 窗-粒度可表达**：$W \in \mathcal{W}_{g^{\text{cmp}}}$，$g^{\text{cmp}}$ 为绑定声明的比较粒度，缺省取 $g^{\text{cmp}} = g_{a^*_{\text{n}}} \sqcap g_{a^*_{\text{d}}}$（定义 3.1 的公共可表达粒度，其存在唯一由 A2 的 meet-半格假定保证）；约定 $g_a = \mathsf{itv}$ 的区间锚腿不施加粒度约束，meet 只在落于 $\mathbb{G}$ 的腿上取，两腿均为 itv 时缺省 $g_\bot$；
- **(iv) 锚对相容**：$(a^*_{\text{n}}, a^*_{\text{d}})$ 满足该绑定行声明的相容性检查 adm——相容检查取自一个有限可判定检查库：平凡真（rma 两日粒锚）、标记域一致审计（quality_voc：$V_v(a_{\text{n}}) \triangle V_v(a_{\text{d}}) = \varnothing$，其 36 个锚日期（周透视快照日）vs 35 个工单日仅 7 日重合、对称差 **57**（$36 + 35 - 2 \times 7$，随仓 `warehouse.duckdb` 复演，与金标 QVOC-05 一致）见证不相容）、区间包含对齐（domestic_newprod 的日粒分子锚 × SCD-2 分母锚：分子粒元须落入分母在效区间）。

条款 (iv) 的检查成员属治理内容（版本数据），不属理论本体——这正是「机制细节降格为实现」的切分线；对称差审计与「标记约定一致」的形式关系是单向充分的（缺口 G5）。

原型对照 `gov_temporal_binding` 的**正例行**实例化 β_v（β_v 是部分**函数**，每度量至多一行正例）：rma 的 `rma_problem_rate_temporal_align`（(i)(ii) 为主）、quality_voc 的 `defect_rate_align`（(iv) 审计通过）、domestic_newprod 的 `model_mismatch_rate_align`（(iv) 取区间包含对齐）。**负例行不入** β_v，属条款 (iv) 的相容性检查库/评测内容：quality_voc 的 `complaint_vs_rma_anchor_mismatch` 与 `defect_rate_align` 同 metric（实查 (domain,metric) 非函数键，故「每行实例化一次 β_v」不成立），其对称差 57 审计是错配锚对 (`complaint_submit_date`,`rma_create_date`) 的不相容见证材料——QVOC-05 在本章读法下由条款 (i)（$\bar{a}$ 改锚）见证 $\bot_{\mathsf{AM}}$，对称差 57 为其判别元（与 C5 定义 5.3 的 AM 见证内容衔接）。机器可读区分正/负例行的 `binding_id`/`is_negative` 扩列方案见 C5 V3（跨章互指）。

N.1.5　4. 时点正确答案语义

4.1 腿人口与质量　定义 3.9（腿人口与质量） 设 $v = \text{ver}(T)$，比率型 m，路由 $\rho = R_v(m)$，候选 $\alpha_{q,v}$ 给出 (a_r, W_r)。角色 $r \in \{\text{num}, \text{den}\}$ 的**腿人口**与**质量**为

$$\text{Pop}^r_{v,T}(q) \;=\; \big\{\, x \in [\![o^r_m]\!]_v \;:\; x \models_\rho s,\ \ \text{vt}_{a_r}(x) \cap W_r \neq \varnothing \,\big\}, \qquad \mu^r_{v,T}(q) \;=\; \sum_{x \in \text{Pop}^r_{v,T}(q)} f^r_m(x),$$

其中 $x \models_\rho s$ 表示 x 经 ρ 的连接键 K_ρ 与归因对齐匹配范围 s（scope 经路由转移到两腿）。原子型只有 atom 角色，同式取 (o_m, f_m, a_m)。区间锚腿的窗过滤统一为**点窗语义**（裁决 D7）：其角色窗按定义 3.2 取 as-of 点窗 $\{d\}$，过滤式 $\text{vt}_{a_r}(x) \cap \{d\} \neq \varnothing$ 即实现层的 $\text{vf} \le d < \text{vtc}$ 谓词——与金标 NX-Q4 人口 126 一致（旧「相交 × 累计窗」读法得 165/219，已废止）；零长行 $\text{vt}_{a_r}(x) = \varnothing$ 在任意窗下交空、自然出局（定义 3.3 良构注记）。

两点由定义直接可读：**(时窗定人口)** 每腿的人口由「v 处扩展 × 路由 scope × 锚窗相交」三重过滤确定，不引用任何其他版本或其他窗；**(腿际不对称)** 分子零质量不构成失败（0 值合法），分母零质量摧毁比率——这一不对称在 §5 的 MC 中形式化。

4.2 可绑定性与语义括号　定义 3.10（可绑定性） Bindable$(q, \mathcal{G}, D)$ 成立，当且仅当以下绑定义务全部满足：

- (b1) **双轴在位**：$\mathrm{ver}(T)\downarrow=: v$；各角色 $\omega_r(T)\downarrow$；且对每个角色 r，$W_r \cap \mathrm{Cov}_v(a_r) \neq \varnothing$；
- (b2) **度量、路由与绑定在册**：$m \in M_v$；比率型另要求 $R_v(m)$ 有定义且可重算、且 $\beta_v(m)$ 有定义（两腿规定锚在册——堵住定义 3.4 缺失约定所指的全函数性洞）；
- (b3) **规则满足**：$\beta_v(m)$ 有定义时，$P_{\mathrm{rule}}(\alpha_{q,v}; G_v, D)$ 成立；
- (b4) **分母正质量**：比率型时 $\mu^{\mathrm{den}}_{v,T}(q) > 0$。

定义 3.11（时点正确答案语义）语义括号是**全函数**

$$[\![q]\!]_{\mathcal{G},T}\ :\ \text{查询}\ \to\ \mathbb{Q}\ \cup\ \{\bot_{\mathsf{OOV}},\ \bot_{\mathsf{AM}},\ \bot_{\mathsf{MC}}\},$$

$$[\![q]\!]_{\mathcal{G},T}\ =\ \begin{cases} \mu^{\mathrm{num}}_{v,T}(q)\ /\ \mu^{\mathrm{den}}_{v,T}(q) & \text{Bindable 且}m\text{ 比率型},\\ \mu^{\mathrm{atom}}_{v,T}(q) & \text{Bindable 且}m\text{ 原子型},\\ \bot_c,\quad c = \text{§5 守卫序下首个成立的失败谓词} & \text{其他}. \end{cases}$$

分子分母**各自**绑定到 T 的 valid-time 窗（经各自锚）与 T 时刻在位的图版本 $\mathrm{ver}(T)$——即任务书中 $[\![q]\!]_{G,\tau}$ 的二维绑定读法。拒答是语义的一等值而非误差路径：素模型「该拒不拒」的失败样态，语义上即在 $\bot_c$ 处取了 $\mathbb{Q}$ 值。delta/多期意图按角色 × 期展开为有限族逐期绑定，全部可绑定时以声明的算术组合（如差分）收束；存在失败期时，按**期序** × **守卫序**的字典序取报告值——先取期展开声明序中的首个失败期，再在该期内按守卫序（注记 3.18′）取其 ⊥ 类——从而多期情形下 $[\![q]\!]$ 仍是单值全函数（命题 3.12 的良定性只覆盖 Bindable 值支，多期失败支的确定性由该字典序规则给出）。

命题 3.12（良定性） Bindable$(q, \mathcal{G}, D)$ 时，$[\![q]\!]_{\mathcal{G},T}$ 的值由 $(q, \mathcal{G}, D)$ 确定，与求值次序无关。证明．由引理 3.5′，$v = \mathrm{ver}(T)$ 由 T 与 commit map 确定。$\alpha_{q,v}$ 是 q, v 的函数（定义 3.7）。各 Pop^r 是有限集合的分离刻画（A1 保证扩展有限、谓词可求值），集合由刻画确定；μ^r 是有限和，与求和次序无关。比率情形 (b4) 给出除数为正。故值确定。■

注记 3.13（比率之和不变式：ratio-of-sums） 定义 3.11 规定比率 = 窗内两腿质量聚合后相除。对窗 W 的任一粒元划分 $W = \biguplus W_j$，$\sum_j \mu^{\mathrm{n}}_j / \sum_j \mu^{\mathrm{d}}_j$ 与 $\frac{1}{k}\sum_j(\mu^{\mathrm{n}}_j/\mu^{\mathrm{d}}_j)$ 一般不相等（两粒元 $(1,2),(10,100)$ 即分离二者）；语义取前者。pilot 编译器的「禁 `sum(rate)`」模板与 C5 校验项 V6a 的比率结构检查，都是该不变式的实现侧与审计侧投影。

N.1.6　5．三类不可绑定情形与可判定性

三个失败谓词按**守卫链**定义：后一谓词的定义域以前一谓词不成立为前提（否则其中出现的对象未必有名可指）。这使三类情形天然两两互斥。

定义 3.14（OOV，out-of-validity）对任意 q：

$$\mathsf{OOV}(q)\quad\Longleftrightarrow\quad \mathrm{ver}(T)\uparrow\ \ \vee\ \ \exists r \in R(q):\ \omega_r(T)\uparrow\ \ \vee\ \ \exists r \in R(q):\ W_r \cap \mathrm{Cov}_v(a_r) = \varnothing,$$

其中后两支在 $v = \mathrm{ver}(T)$ 在位、相应对象可指时求值。**量词辖域（承推论 3.17′）**：$\exists r \in R(q)$ 只在单期查询的角色集上取值；delta/多期查询不把角色 × 期展开的整体角色集代入本谓词，而按推论 3.17′ 逐期适用（否则与定义 3.11 的期序 × 守卫序字典序冲突）。**对级覆盖判空（跑前冻结的实现裁定升格入定义正文；工件偏离账 `PORT_REPORT` §4(2)）**：对被 $\beta_v(m)$ 绑定规则配对的角色（比率型两腿），覆盖判空支在**对**层求值——两腿窗实现皆空方成立本支；混合实现（一腿非空、一腿空）不落本谓词，归定义 3.8 条款 (iv) 锚对审计辖区（实例 EF2-Q6：比赛日腿实现 {2013-12-01}、快照腿实现 ∅，金标 $\bot_{\mathsf{AM}}$(iv)）；未配对角色（原子型）保持逐角色判 $W_r \cap \mathrm{Cov}_v(a_r) = \varnothing$。**真空约定（三值 → 二值，形式化）**：谓词某一支中出现的对象不可指时（$m \notin M_v$ 使规定锚无名；比率型 $\beta_v(m)\uparrow$ 使 a_r 缺失；$\bar{a}(r)$ 为未注册锚引用使 $\mathrm{Cov}_v(a_r)$ 无定义——定义 3.7 未注册锚约定），该支按**假**处理（真空不成立），对象缺失本身不冒充时窗失败、而由后续守卫（未注册改锚由 AM(i)、其余由定义 3.16(i)）捕获；对偶地，定义 3.10(b1) 第三项对不可指角色按真空成立处理——此时 (b2) 必失败，故不致误入 Bindable。该约定使各守卫谓词在其守卫域上处处二值，是定理 3.17 证明所引「守卫谓词全（二值）」的显式依据。读法：T 在**任一轴**上落到有效域之外——版本轴（在位域 $[t_1, +\infty)$ 之外）或 valid-time 轴（请求窗与所需锚的覆盖域不相交，即任务书「τ 不落于任何锚有效域」的情形）。**归类边界**

（裁决 D1）：OOV 的时窗支**仅**指锚覆盖域真空（$W_r \cap \mathrm{Cov}_v(a_r) = \varnothing$，按 cm_{a_r} 求值）；覆盖域**内**的窗中分母零质量不属 OOV，统一归 MC(ii)（定义 3.16）。把版本轴缺位并入 OOV 是保持 C5 三类拒答接口不变的约定（缺口 G7）。

定义 3.15（AM，anchor-mismatch；守卫 ¬OOV） 设 ¬OOV(q)，$v = \mathrm{ver}(T)$：

$$\mathsf{AM}(q) \iff \beta_v(m)\downarrow \;\wedge\; \neg P_{\mathrm{rule}}\big(\alpha_{q,v};\; G_v, D\big).$$

即绑定规则要求（原型 svw）而候选指派违背——按定义 3.8 的条款拆解为四种见证形态：改锚 (i)、跨窗 (ii)、窗-粒度不可表达 (iii)、锚对不相容 (iv)；与 C5 定义 5.3 的 AM 见证内容（窗对或对称差判别元）一致。

定义 3.16（MC，missing-caliber；守卫 ¬OOV ∧ ¬AM） 设 ¬OOV(q) ∧ ¬AM(q)，$v = \mathrm{ver}(T)$：

$$\mathsf{MC}(q) \iff \underbrace{m \notin M_v \;\vee\; \big(m\ \text{比率型} \wedge (R_v(m)\ \text{无定义或不可重算}\ \vee\ \beta_v(m)\ \text{无定义})\big)}_{\text{(i) 路由/绑定缺失}} \;\vee\; \underbrace{m\ \text{比率型} \wedge \mu^{\mathrm{den}}_{v,T}(q) \le 0}_{\text{(ii) 空窗/零质量分母}}.$$

(i) 中新列的 $\beta_v(m)$ 无定义支即定义 3.4 缺失约定：比率型两腿规定锚无来源与路由缺失同型，同为「治理内容缺件」的口径失败。(ii) 即任务书「τ 窗内分母集合为空」的情形，按裁决 D4 加宽为 $\mu^{\mathrm{den}}_{v,T}(q) \le 0$——覆盖非空但质量非正的人口（退款冲销类业务语义，G8 已按此收敛）；C5 的空窗见证谓词**同步为** $\mu_{\mathrm{den}} \le 0$（替换旧冻结断言集 $\{\varnothing, \mathrm{NULL}, 0\}$），探针笔录记录 observed 值、校验项 V6b 按新谓词复核——负质量分母的正确证书不再被误拒。两种模式与 C5 定义 5.3 的 MC 双见证（路由缺失见证 / 空窗见证）逐条对应。原型：rma 536-SKU 母题（同月无销量分母，不以异月充数——「以异月充数」本身则落入 AM(ii)）。**跨章归类（裁决 D1）**：空同窗分母统一归 MC(ii)，OOV 只保留锚覆盖域真空（定义 3.14 归类边界）；金标随之收敛——rma_q5（覆盖域内同窗零销量）= missing-caliber 维持；NX-Q7 实测为真覆盖域真空（launch 表 min dt = 2025-09-15，as-of 早于覆盖域）= out-of-validity 维持；C4 A4.1/行 11 的 OOV 判类同批改判 MC，拼写统一为 OOV。quality_voc 的 `dst_caliber='none'` reference-only 路由落入 (i)。

定理 3.17（覆盖三分；单期情形） 对每个良构**单期**查询 q（σ 语法良构、$T \in \mathbb{T}$，且 $R(q) \subseteq \{\mathrm{num}, \mathrm{den}\}$ 或 $R(q) = \{\mathrm{atom}\}$——即无 delta/多期的角色 × 期展开），在假定 A1–A6 下，以下四种情形两两互斥、合并穷尽：

$$\text{(a) Bindable;}\quad \text{(b) OOV;}\quad \text{(c) ¬OOV ∧ AM;}\quad \text{(d) ¬OOV ∧ ¬AM ∧ MC,}$$

且 $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q} \iff$ (a)；$[\![q]\!]_{\mathcal{G},T} = \bot_{\mathsf{OOV}} \iff$ (b)；$= \bot_{\mathsf{AM}} \iff$ (c)；$= \bot_{\mathsf{MC}} \iff$ (d)。

证明. **互斥**：(b)(c)(d) 的守卫按构造互斥；(a) 与其余互斥：Bindable 的 (b1) 逐条否定 OOV 的三支（在位、可解析、覆盖相交），故 (a)⇒ ¬(b)；(b3) 在 $\beta_v(m)\downarrow$ 时给出 P_{rule} 成立，故 (a)⇒ ¬(c)；(b2)(b4) 分别否定 MC 的 (i)(ii)，故 (a)⇒ ¬(d)。**穷尽**：设 (b)(c)(d) 均不成立，证 (a)。¬(b) 给出：$\mathrm{ver}(T)\downarrow$、各 $\omega_r(T)\downarrow$、覆盖判空支不成立——原子角色即逐角色覆盖相交；配对角色按定义 3.14 对级条款仅排除“两腿皆空”，而混合实现由条款 (iv) 审计在 AM 层拦截，故合并 ¬(c)（P_{rule} 成立含条款 (iv) 通过）后配对两腿覆盖亦各自相交——合计即 (b1)。前引命题 3.18（其证明不依赖本定理）给出各守卫谓词在其守卫域上可判定、从而全(二值)：¬OOV 域上 AM 有确定真值，¬(c) 即 $\beta_v(m)\uparrow$ 或 P_{rule} 成立——两支都蕴含 (b3)；同理 ¬(d) 在其守卫域上给出 MC 两模式皆假，即 (b2) 与 (b4)。四项义务齐备，Bindable 成立。**与语义一致**：定义 3.11 在 (a) 时取 $\mathbb{Q}$ 值（良定性由命题 3.12），否则取守卫序下首个成立谓词的 $\bot$ 类；守卫序即 (b)(c)(d) 的枚举序，故对应关系逐条成立。■

推论 3.17′（多期查询的逐期分类） delta/多期查询 q 按定义 3.6 的角色 × 期展开为有限期族 $\{q^{(j)}\}_j$（声明期序），每个 $q^{(j)}$ 为单期查询：角色集限于该期的 $\{\mathrm{num}, \mathrm{den}\}$（或 $\{\mathrm{atom}\}$），窗取该期声明窗，其余成分承 q。则定理 3.17 对每个 $q^{(j)}$ 适用，且整查询的值由定义 3.11 的**期序** × **守卫序**字典序确定：诸 $q^{(j)}$ 全部落于 (a) 时，$[\![q]\!]_{\mathcal{G},T}$ 为各期值的声明算术组合（如差分）；否则 $[\![q]\!]_{\mathcal{G},T} = [\![q^{(j_0)}]\!]_{\mathcal{G},T} \in \{\bot_{\mathsf{OOV}}, \bot_{\mathsf{AM}}, \bot_{\mathsf{MC}}\}$，$j_0$ 为声明期序中首个失败期。**适用方式两则**：(α) 守卫谓词（定义 3.14–3.16）与 P_{svw} 只在**单期限制** $\alpha_{q,v}|_{R(q^{(j)})}$ 上求值——P_{svw} 的条款定义在 $\{\mathrm{num}, \mathrm{den}\}$ 两角色上，对角色 × 期展开的整体指派（4-leg）不良型；(β) 定义 3.14 的 $\exists r \in R(q)$ 在多期查询上**不跨期取量词**——否则与字典序矛盾：随仓反例 `problem_rate` × SKU536-EOL × delta 窗对 [2026-04, 2026-08]，期 1 窗 2026-04 落于覆盖域（销量锚 hull [2025-12-01, 2026-06-28]）内而同窗销量 0（问题量 7）、按定理 3.17 判 $\bot_{\mathsf{MC}}$；期 2 窗 2026-08 与覆盖域交空、OOV 原始条件真；跨期量词将整查询判 $\bot_{\mathsf{OOV}}$，而字典序（与 rma 编译器及 C4/C5 的首失败 slice 形制一致）取首失败期 2026-04 的 $\bot_{\mathsf{MC}}$——本推论钉定后者。证明. 各期为单期查询，逐期分类由定理 3.17；失败支单值性由定义 3.11 字典序（命题 3.12 覆盖全 Bindable 值支）。■

命题 3.18（可判定性与代价） 在 A1–A6 下，OOV、(其守卫域上的) AM、(其守卫域上的) MC 均可判定；且各自可在如下代价内判定（$n = |\mathcal{V}|$，$|W|$ 为窗描述长度，扩展基数记 $|[\![o]\!]|$）：

谓词	判定程序（构造性）	代价
OOV	$\mathrm{ver}(T)$：对 $t_1 < \cdots < t_n$ 二分（比较可判定，A6）；$\omega_r(T)$：按 A5 的全函数呈现求值（含 ↑ 哨兵的定义域判定）；覆盖：按 cm_{a_r} 分派——hull 锚单趟扫描取 $\min / \max$ 得凸包，strict_member 锚排序合并得标记集/区间并（零长行贡献 ∅、扫描中自然跳过），再与规范形 W_r 求交判空	$O(\log n) + \sum_r O(\lvert[\![o_{a_r}]\!]\rvert \log \lvert[\![o_{a_r}]\!]\rvert + \lvert W_r \rvert)$
AM	条款 (i)：锚标识比较；(ii)：窗规范形相等性；(iii)：W 的每个极大区间做粒元边界对齐检查（日历函数可计算）；(iv)：调用声明的检查库成员（平凡真 $O(1)$；对称差审计一趟扫描两锚对象；区间包含对齐一趟合并）	$O(\lvert W \rvert \log \lvert W \rvert) + O(\lvert[\![o_{a^*_{\mathrm{n}}}]\!]\rvert + \lvert[\![o_{a^*_{\mathrm{d}}}]\!]\rvert)$
MC	(i)：M_v, R_v 查键；(ii)：对 $[\![o^{\mathrm{d}}_m]\!]_v$ 一趟扫描累加 f^{d}_m（每事实 $\models_\rho s$ 与锚窗相交测试常代价，A1/A2）	$O(\lvert[\![o^{\mathrm{d}}_m]\!]\rvert)$

证明. 每一行把谓词化归为有限结构上有限多次可计算测试的布尔组合：版本定位是有限全序上的查找（A3），其中时点比较 $T < t_i$ 与窗端点比较由 A6 的离散序可判定性支撑；ω_r 的定义域判定由 A5 的全函数呈现给出——若无 A5，仅凭定义 3.6 的部分可计算性，判 $\omega_r(T)\uparrow$ 即判停机，OOV 第二支在字面上不可判定，这正是 A5 的必要性；覆盖域与窗规范形是有限区间并上的可计算运算（A1、A2 与定义 3.2 的日历封顶）；条款 (iv) 按 A4 调用有限检查库的可判定成员；质量累加是有限和。有限布尔组合保持可判定性；表列代价由所引程序逐项相加。构造出的判定程序同时给出 C5 定义 5.3 所需的见证内容：失败条款/探测查询与其结果断言。■

注记 3.18′（报告优先序与双故障；裁决 D2）三个谓词的**非守卫**原始条件可同时成立（如请求窗既在覆盖域外、又跨窗）。守卫链把类判定为首个成立者，报告序为跨章统一序 OOV → AM → MC——先窗/覆盖检查（OOV），再规则审计（AM），再质量探针（MC）。D2 已消除跨章分歧：C4 算法行序按此调整（原行 8 的 AM 判定移到窗/覆盖检查之后），两域编译器守卫序同改（domestic_newprod 原「先查改锚再探空窗」按本序归一）；剩余评测基建执行项见 G9。

注记 3.18″（守卫走位实例；供 C5 定理 5.10(a) 换用活反例的 C3 侧结论）两个 quality_voc 实例在本章语义下的守卫走位，为 C5 守卫序必要性论证（V6b(0)）钉定 C3 侧结论：1. **`complaint_rate_global_recompute` × 任意 as-of（含越界 2030-01-06）⇒ $\bot_{\mathsf{MC}}$，不是守卫序反例**。该度量路由为 reference-only（`dst_caliber='none'`，routing note 明示「无分母 → 无法挂 valid-time 锚」），实查 `gov_temporal_binding` 无行——定义 3.14 覆盖支对象不可指、按真空约定为**假**，$\mathrm{ver}(T)\downarrow$、as-of 可解析，故 OOV 整体为假；$\beta_v(m)\uparrow$ 使 AM 守卫域上亦假；MC(i) 成立。语义值为 $\bot_{\mathsf{MC}}$ **而非** $\bot_{\mathsf{OOV}}$（金标 QVOC-06 = missing-caliber；as-of 越界与否不改变该结论）——旧「口径闸先行」编译器对此实例发 MC 证书，理由类**并不错**，故它承载不了「守卫序错置产出错类证书」的反例。2. **`defect_rate` × 越界 as-of × 声明错配锚对（QVOC-05 形状）⇒ $\bot_{\mathsf{OOV}}$，才是活反例载体**。取 $\bar{a}$ 把分子改锚 `rma_create_date`（注册锚，覆盖 2026-02-03..2026-06-03）、as-of 取覆盖域外时点（如 2030-01-06 起始周窗：与 complaint 锚标记集（36 个锚日期（周透视快照日），跨 2025-09-22..2026-06-12）、sales 锚标记集及 `rma_create_date` 覆盖域皆交空）。此时 OOV 覆盖支与 AM 原始条件（条款 (i) 改锚，判别元为错配锚对的对称差 57 审计）**同真**；守卫序 OOV → AM → MC（注记 3.18′）判 $\bot_{\mathsf{OOV}}$，而旧 AM 先行序会发出理由类错置的 AM 证书——C5 定理 5.10(a) 的活反例应以此实例为载体（原引 `complaint_rate_global_recompute` 实例由第 1 条否证，C5 侧换例即引本注记）。

实现注记（判定程序的既有见证）三份编译器各自实现了上表的一个子集：`pilot/domains/rma/compiler.py`（(ii) 跨窗 → AM、同窗分母探测 → MC）、`pilot/domains/quality_voc/compiler.py`（对称差审计 → AM、`dst_caliber='none'`→ MC、锚域成员测试 → OOV）、`pilot/domains/domestic_newprod/compiler.py`（改锚正则 → AM、as-of 不可解析与**覆盖域真空** → OOV、**空同窗分母**（$\mu_{\mathrm{den}} \le 0$ 质量探针）→ MC(ii)——与定义 3.14 归类边界及裁决 D1 一致，「空窗」在本章是 MC(ii) 术语、不入 OOV）。它们是命题 3.18 判定程

序的存在性见证，不是定义的一部分。

N.1.7 6. 退化定理：单版本读一致性作为一维情形

定义 3.19（单版本语义与 SVRC） 固定 (G_v, D)。**单版本（非时态）语义** $[\![\sigma]\!]^1_{G_v,D}$ 定义为：按定义 3.9 但取消锚窗过滤、保留**锚有效限制**（各腿人口 $= \{x \in [\![o^r_m]\!]_v : x \models_\rho s,\ \mathrm{vt}_{a_r}(x) \neq \varnothing\}$；严格良构锚下该限制真空，即回到无限制读法——定义 3.3 良构注记），其余（路由、质量、比率、MC 的两模式按非时态读法）不变。一次求值满足**单版本读一致性（single-version read consistency，SVRC）**，当且仅当存在已提交版本 v，使该求值读取的治理对象（度量、路由、锚、绑定）全部取自 G_v、事实全部取自 D 在 G_v 解释下的扩展——即「一次读绑定单一已提交版本」。

定理 3.20（退化定理） 设查询族 $\mathcal{Q}^{\downarrow}$ 满足：- (H1) **终末在位**：$T \geq t_n$，故 $\mathrm{ver}(T) = v_n$（$T = \mathrm{now}$ 且版本轴推进已停即此情形；$|\mathcal{V}| = 1$ 为特例）；- (H2) **无窗约束**：各角色 $\omega_r(T) = \top$；- (H3) **默认指派**：$\bar{a} = \varnothing$（不改锚、不分窗）。

则对每个 $q \in \mathcal{Q}^{\downarrow}$：1. **（读一致坍缩）**求值满足 SVRC，见证版本取 v_n；2. **（值坍缩）** $[\![q]\!]_{\mathcal{G},T} \in \mathbb{Q}$ 时 $[\![q]\!]_{\mathcal{G},T} = [\![\sigma]\!]^1_{G_{v_n},D}$；3. **（拒答类坍缩：时态内容清空）**三类谓词均化归为**非时态**残余情形——AM：条款 (i)–(iii) 在 H1–H3 下均成立，AM 化归为条款 (iv) 的锚对声明审计在 (G_{v_n}, D) 上失败这一不依赖 T 的情形（审计通过时 AM 不成立）；OOV 化归为「某所需锚 $V_{v_n}(a) = \varnothing$」（无良构标记行；严格良构锚下即锚对象全空。两种 cm_a 下 $\mathrm{Cov}_{v_n}(a) = \varnothing \iff V_{v_n}(a) = \varnothing$，故化归与覆盖方式无关）；MC 化归为非时态口径缺失：路由缺失或全域 scoped 分母质量非正（$\mu_{\mathrm{den}} \leq 0$，承 D4 加宽）。

证明. (1) H1 给出全部治理读取自 G_{v_n}；定义 3.9 的人口仅引用 $[\![\cdot]\!]_{v_n}$ 与 D（SA 下单快照）。取 $v = v_n$ 即 SVRC 之见证。(2) H2 下各腿窗为 $\top$。对 $\mathrm{vt}_a(x) \neq \varnothing$ 的事实有 $\mathrm{vt}_a(x) \cap \top \neq \varnothing$，锚窗过滤不排除之；零长行 $\mathrm{vt}_a(x) = \varnothing$ 与任意窗交空，在双时态侧出局（定义 3.3 良构注记）。故 $\mathrm{Pop}^r_{v_n,T}(q)$ 恰与定义 3.19 携锚有效限制的单版本腿人口相等，质量与比率随之相等；严格良构锚下限制真空、坍缩到无限制读法（原型实测印证限制的必要性：$\top$ 窗双时态人口 281 = 无限制单版本人口 392 − 111 零长行）。(3) AM：H3 下 α_{q,v_n} 取规定锚（条款 (i) 成立），各角色共窗 $\top$（条款 (ii) 成立），$\top \in \mathcal{W}_g$ 对每个 g（定义 3.2，条款 (iii) 成立）；于是 $\neg P_{\mathrm{svw}}$ 化归为条款 (iv) 单独失败，而 (iv) 只依赖 $(a^*_{\mathrm{n}}, a^*_{\mathrm{d}})$ 与 (G_{v_n}, D)，不含 T——正是断言的非时态残余；(iv) 通过时四条款齐备、AM 假。OOV：H1 排除 $\mathrm{ver}(T)\uparrow$；H2 排除 $\omega_r(T)\uparrow$；$\top \cap \mathrm{Cov}_{v_n}(a) = \mathrm{Cov}_{v_n}(a)$，其为空当且仅当 $V_{v_n}(a) = \varnothing$。MC：模式 (i) 本就非时态；模式 (ii) 在 H2 下 μ^{den} 是全域 scoped 质量。■

推论 3.21（DELTAGOV 类读一致形状的一维影子） 1. 一般地（不限 $\mathcal{Q}^{\downarrow}$），凡 Bindable 的求值都满足 SVRC，见证版本取 $\mathrm{ver}(T)$。**强度自陈**：在 SA 下这是**按构造成立**的语义不变量——快照假定已令一次求值只见单一 (G_v, D)，故本条的内容是「语义按构造不产生跨版本读」，而非在并发版本推进下另行证明的一致性保证；2. 在定理 3.20 的前提下，语义的时态内容整体化归为 SVRC：valid-time 轴（锚窗过滤）与版本选择轴（ver 随 T 变动）双双失活，仅存「一次读绑定单一已提交版本」。这是 DELTAGOV 式 version-serializability 在本框架语义层的一维**影子**（同形状、不同强度），与 C5 注记 5.5 的证书侧退化互为语义/制品两面。**词汇边界**：DELTAGOV 的保证以逐数据读版本函数 $v(r)$ 陈述，而本篇的 D 是无版本索引的单快照——$v(r)$ 在本篇词汇中不可陈述；严格对应需显式桥接假设「数据快照 = 版本 v 的物化 D_v」并为腿读配版本坐标，该桥接归系统章，此处不宣称。

证明. (1) 是定理 3.20(1) 证明中前半段的逐字重用：Bindable 蕴含 $v = \mathrm{ver}(T)$ 在位，人口与治理读取全部以 v 为界。(2) 由定理 3.20(2)(3)：值与拒答类均与非时态单版本求值重合，二维绑定中随 T 变动的两个自由度在 H1–H3 下均取常值。■

划界措辞：此处表述的是 DELTAGOV 保证在**语义层**的形态（consumer 视角的读一致性形状），不是对其存储层协议的重述；跨快照并发与版本推进协议仍归 producer（SA，承 C5）。

注记 3.22（另一条一维影子：单时态口径见证） 对 #32719 CaliberGraph 类单时态 witness 编译：在 $|\mathcal{V}| = 1$ 且保留窗约束（去 H2）的族上，本语义余下「口径 + valid-time 窗」一维；MC 的判定程序即其 witness 检查加窗过滤。两条影子（版本轴独存 / valid-time 轴独存）说明二维绑定不是两个已有一维保证的并列，而是其公共上层。

N.1.8 7. 与 SQL:2011 时态特性的关系（划界）

SQL:2011 在**存储/行级**提供 system-versioned 表（`FOR SYSTEM_TIME AS OF`，transaction time）、application-time period 表（`PERIOD FOR`，valid time）及双时态表。本框架在**语义图/口径层**工作，二者是层次关系而非

竞争关系：

本框架（语义/口径层）	SQL:2011（存储/行级）	划界说明
ver(T)：治理**语义**（度量定义、口径路由、锚、绑定规则）的在位版本	`FOR SYSTEM_TIME AS OF`：行历史的系统时间切片	SQL:2011 **不以一等语义对象承载口径版本**：把治理表存成 system-versioned 后，AS OF 确可切出 T 时刻在位的口径定义**行**（注记 3.23 的编码正用此），但口径对象、绑定规则与拒答值域不在其语义之内——「选出在位口径」靠编码约定而非标准语义；DDL/schema 演化亦不承载口径版本语义
快照锚 / 区间锚（vt_a 为解释函数）	普通日期列 / `PERIOD FOR` 列对	锚是版本化治理对象；period 是列对。快照锚在无 SQL:2011 特性的库上即可实现（pilot：duckdb 普通表 + `dt` 列）
绑定规则 P_{rule}：**跨腿耦合**的良构条件（分子分母同窗、锚对相容）	无对应物：每个表访问的时间谓词各自独立	等价 `WHERE` 可以逐查询写出，但存储层不提供「违背即拒答」的层级强制；耦合约束在我们的层是可判定良构条件（§5）
全函数语义：$\mathbb{Q}\cup\{\bot_{\mathsf{OOV}},\bot_{\mathsf{AM}},\bot_{\mathsf{MC}}\}$ + 见证	空结果 / `NULL`	结构化拒答与可证伪见证（C5）在 SQL 值域中无一等对应
对角线约定 $T_{\text{vt}}=T_{\text{ver}}=T$（离对角线走披露）	两维 AS OF 可独立取值	SQL:2011 的自由组合在治理语境恰是 GOLD-1 型错绑来源；本框架把离对角线组合交给披露策略裁决（C4）

注记 3.23（可实现性；proof-sketch） 断言：满足 A1–A3 的有限 $\mathcal{G}$ 可编码到一个 SQL:2011 双时态库上，使定义 3.11 的求值经生成的 AS OF 查询实现。思路：治理表以 system-versioned 存储，commit map 落为系统时间；区间锚落为 application-time period，快照锚落为日期列；ver(T) 落为对治理表的 `FOR SYSTEM_TIME AS OF T` 读取；人口过滤落为生成的窗谓词。**缺口**：窗代数运算与粒度对齐检查超出标准时态子句、须在编译器内完成；拒答逻辑与见证生成不落在存储层。完整编码及其忠实性证明（编码前后 $[\![\cdot]\!]$ 保持）留给系统章（缺口 G10）。■(sketch)

划界一句话：SQL:2011 是本框架可用的一种存储基底；本框架给出的（存储层不给出的）是**口径与绑定规则的版本化语义、跨腿时态良构条件、以及以拒答为一等值的全函数正确性刻画**。

N.1.9 8. 对兄弟篇的接口与开放缺口

接口（导出物） - 对 C5：$\mathsf{Bind}(q,T,G_v,D)$ 的声明式语义 = 定理 3.17 的四分（单期；多期按推论 3.17′ 逐期，产出计划 $\iff$ $\mathsf{Bindable}$；报出类 = 守卫序首个成立谓词）；命题 3.18 的判定程序即证书见证（C5 定义 5.3）的生成器；$\alpha_{q,v}$ 即证书四元组的 α（含未注册锚引用支，兑现 C5 定义 5.3 AM(i) 的跨章依赖）；C5 定理 5.10(a) 守卫序必要性的活反例载体及其 C3 侧语义结论见注记 3.18″（原 `complaint_rate_global_recompute` 实例不成立，换 `defect_rate` × 越界 as-of × 声明错配锚对）。- 对编译定理篇：本篇给出「值语义」侧；「产出 SQL 求值 = $[\![q]\!]$」的编译正确性/完备性定理以定义 3.9–3.11 为规范。- 对 C4：$\mathsf{Bindable}$ 失败处的合法改写搜索 = 把定义 3.7 的函数式 $\alpha_{q,v}$ 放宽为受披露策略约束的存在量词；「绑定成功 $\iff$ 时点正确查询存在」的完整双向刻画在该放宽下陈述。

开放缺口（需作者裁决 / 证明待补） - **G1** commit map $t:\mathcal{V}\to\mathbb{T}$ 是理想化：seed 的 `committed_at` 为版本标签非时戳；主实验的版本序列需合成部署时戳（裁决：合成协议与粒度；协议须含定义 3.5 的同日多提交平局细化，保证严格单调与时戳同一性）。- **G2** $\alpha_{q,v}$ 的函数式确定性是本篇的建模选择；骨架 C3 声明的「绑定成功 $\iff$ 时点正确查询存在」双向定理需 C4 的存在量词侧配合，两篇陈述的分工须钉定。- **G3**

可判定性相对日历窗代数封顶（定义 3.2 建模选择）成立；任意可定义窗上的推广开放。- **G4**（**已裁决 D3**）覆盖域语义按逐锚 `coverage_mode` 参数化落地（定义 3.3 的 cm_a，G_v 内容物）：快照锚默认 hull、裸日期集锚默认 strict_member；seed 扩 `coverage_mode` 字段；C5 OOV 见证按声明 mode 重演；$\mathcal{W}$ 不升格为 $\bar{\mathcal{W}}$（类型注记）。残留工程项：quality_voc 裸日期等值判定是否升级为粒元语义。- **G5** 条款 (iv) 的对称差审计与「标记约定一致」是单向充分关系；完整刻画（何种审计对何种锚型完备）开放。- **G6** 对角线约定为缺省；离对角线（$T_{\mathrm{vt}} \neq T_{\mathrm{ver}}$）语义已可直接推广但披露合法性归 C4（裁决：主文是否呈现一般双参形）。- **G7** 版本轴缺位（$T < t_1$）并入 OOV 是保持三类接口的约定；另立第四类的取舍待裁决。- **G8**（**已裁决 D4**）MC(ii) 加宽为 $\mu_{\mathrm{den}} \leq 0$（覆盖退款冲销类「非空但质量非正」人口）；C5 空窗见证断言集 $\{\varnothing, \mathrm{NULL}, 0\}$ 同批收敛为同一谓词 $\mu_{\mathrm{den}} \leq 0$，探针笔录记 observed 值（定义 3.16(ii)）。- **G9**（**已裁决 D2**）双故障报告优先序跨章统一为 OOV → AM → MC（先窗/覆盖、再规则审计、再质量探针，注记 3.18′）；C4 算法行序与两域编译器守卫序同批改。残留执行项：pilot 三域实现按此归一（评测基建工单）。- **G10** 注记 3.23 为 proof-sketch；SQL:2011 编码的忠实性证明留系统章。

N.1.10　附：符号速查

$\mathbb{T}$ 时间域 · T 声明时点（任务书 τ）· $\mathbb{G}, g, g_\perp, g \sqcap g'$ 粒度族/粒度/日细化/公共可表达粒度（meet）· $\mathcal{W}, \mathcal{W}_g, \top$ 窗代数/对齐子代数/全窗 · $a, \mathrm{vt}_a, g_a, \mathrm{cm}_a$ 锚/有效时间指派/标记粒度/覆盖方式（hull/strict_member）· $V_v(a), \mathrm{Cov}_v(a)$ 标记集/覆盖域 · $G_v = (N_v, E_v, \lambda_v; M_v, R_v, A_v, \beta_v)$ 图版本 · ρ, K_ρ 口径路由/连接键 · $\mathcal{G}, t(\cdot), \mathrm{ver}(T)$ 版本序列/commit map/在位版本 · $q = (\sigma, T)$，$\sigma = (m, s, \vec{\omega}, \bar{a})$，$R(q)$ 查询/意图/角色 · $\alpha_{q,v}$ 候选锚指派 · W_T, W_r 请求窗/角色窗 · Pop^r, μ^r 腿人口/质量 · Bindable 可绑定 · $[\![q]\!]_{\mathcal{G},T}$ 时点语义 · $\perp_{\mathsf{OOV}}, \perp_{\mathsf{AM}}, \perp_{\mathsf{MC}}$ 拒答值 · $[\![\cdot]\!]^1$, SVRC 单版本语义/读一致性。

O Frozen Theory Chapter C4: Maximal Legal Rewrite

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O.1 C4 · 披露门下的最大合法改写（Maximal Legal Rewriting under Disclosure）

> 草稿 v0.1 · PP-5744d699 · SIGMOD 2027 R4 骨架第 4 章（S1 冻结骨架之 “Maximal Legal Rewriting under Disclosure, 1.5p”）定框纪律：abstraction-first——定义 → 不变量 → 定理 → 算法；SQL 生成等机制细节一律降格为实现注记。语义素材（只读）：`gov_disclosure_policy` / `gov_valid_time_anchor` / `gov_temporal_binding` / `gov_caliber_routing`（datagov-line v2026.06.23，5 生产域）；pilot 编译器 `pilot/domains/*/compiler.py` 作为本章算法的退化实例。跨章接口：C2（双时态治理查询语义）、C3（绑定编译与三类不可绑定）为上游；C5（证书与独立校验）为下游。本章符号在 C2/C3 定稿后须做一次统一对齐（见缺口清单 G-10）。

O.1.1 4.0 记号与前章接口

符号	含义	来源
V	语义图版本集，按提交序全序（`gov_semantic_graph_version`）	C2
$G_v = \langle A_v, B_v, R_v, \Gamma_v, D_v \rangle$	版本 v 下的治理状态：锚集、时态绑定集、口径路由集、粒度格、披露策略集	C2 + 本章
$a \in A_v$	valid-time 锚（`gov_valid_time_anchor`：snapshot_effective_date 或 scd_type2）	C2
$\beta \in B_v[m]$	指标 m 注册的时态绑定 $\langle \alpha_N, \alpha_D, r \rangle$：分子锚、分母锚、规则 r（`gov_temporal_binding`）	C2/C3
$\tau = \langle T, p \rangle$	as-of 声明：valid-time 基准 T（时点或窗）、版本钉 $p \in V \cup \{\circledast\}$（$\circledast$= 未钉）	C2
$q = \langle m, \sigma, g_0, w_0, \kappa \rangle$	查询意图：指标、范围谓词、请求粒度、请求窗指派（按 leg）、承诺集 κ（定义 4.6）	本章
$\Omega(q, \tau, G)$	C3 绑定判定器：返回可绑定坐标或不可绑定原因 $\in \{\mathrm{MC}, \mathrm{AM}, \mathrm{OOV}\}$	C3
MC/AM/OOV/DB	missing-caliber（注册层缺失 MC(i) 或同窗分母质量空 MC(ii)） / anchor-mismatch / out-of-validity（锚覆盖域真空或版本轴缺位，A4.1） / disclosure-blocked（本章新增）	C3 + 本章
$b = \langle \alpha, w, v, g \rangle$	绑定点：锚（绑定选择）× 窗 × 版本 × 粒度（定义 4.7）	本章
$\mu,\ \mu^*(b, \mathrm{ctx})$	掩码指派；绑定点的最小合法掩码闭包（引理 4.15）	本章

符号	含义	来源
$\mu_{\mathrm{den}}(\alpha, v, w, \sigma)$	分母 leg 的窗上基准质量（非负测度之和，探针；引理 4.12、A4.4(iii)）	本章
$L(q, \tau, G, \mathrm{ctx}, I)$	合法改写集（定义 4.8）；I 为实例快照，ctx 为请求语境（角色等）	本章
$\preceq_D$，$\approx_D$，$\prec_{\mathrm{det}}$	保真偏序、其等价核、确定性选择序（定义 4.9/4.11）	本章
h_Γ，W_Γ	粒度格高度与宽度；$a = \lvert B_v[m]\rvert$ 绑定规则数；$\nu = \lvert V_{\mathrm{adm}}\rvert$；$\delta = \lvert D_v\rvert$	本章

假设 A4.1（C3 接口）。存在可判定的绑定判定器 Ω，其正确性与完备性在此**假设成立**；该“绑定成功 ⇔ 时点正确查询存在”双向定理的两篇分工系 C3 缺口 G2，待钉定（与 c3 G2 / c5 G1 的自述对齐，2026-08-06 软化，承 R1-F3）。本章把 Ω 当作黑盒调用，并沿用其三类不可绑定原因的定稿口径（D1 裁决，与 C3 定义 3.16(ii)、C5 G2 一致）：MC= 注册层缺失（MC(i)：无绑定/无口径路由可恢）**或**同窗分母质量空（MC(ii)：$\mu_{\mathrm{den}} \le 0$，读作”口径人口缺失”）；AM= 承诺与可用性不相容（含 κ 剔空 A_{adm} 的 AM(i)；**版本层同判**：$\tau.p/\kappa$ 的版本承诺剔空可解析版本集——钉不存在的版本号、或 $\tau.p$ 与 κ 互斥——按 AM(i) 类比归 AM，行 2b)；OOV= 有效性层真空，两个成因：**锚覆盖域真空**（窗带为空或刚性 strict_member 覆盖检查失败——真 OOV 形状如 NX-Q7 实测 domestic_newprod launch 表 min `dt`=2025-09-15 的窗前请求，保持 OOV），**或版本轴缺位**（缺省 ⊛ 对角解析下 ver$(T)\uparrow$，如 T 早于首个提交 t_1——C3 定义 3.14 明文并入 OOV、缺口 G7 约定，行 2a)。全链（CT 取值、见证类型、E1、定理 4.20 分解）已按此改齐；pilot 五域编译器标签亦已迁移归一（见 4.4 标签归一注记、G-10 对账）。

假设 A4.2（有限性）。V、D_v、各 $B_v[m]$ 有限；Γ_v 与掩码格 M 为有限格（有界、二元交并存在）；锚有效区间端点集有限（scd_type2 的区间端点来自实例 I 中的维表有效期列，随快照确定）。**补充条款（版本回滚纪律，D6 裁决）**：回滚 = 重提交——版本号永不复用；commit map 为 append-only、版本号唯一且在真时戳上严格单调；`rollback_of` 仅为元数据边，不参与语义解析；旧证书按坐标 (T, v) 永久有效（重提交不使任何已签发证书失效）；内容等同的重提交版本在 Front/证书中按版本号各为独立代表（不去重）；追溯性回滚（就地改写既有版本内容）不可表达，排除出适用域。落地坐标：C3 A3/§2.3 与本条（C4 A4.2）。种子如实陈述：现种子 schema 无 `rollback_of` 字段、`committed_at` 为标签（C3 G1）且五域均单版本，多版本条款目前零实例支撑。

假设 A4.3（粗化相容性，coarsening-compatibility）。对候选粒度 $g \preceq_\Gamma g'$（g 细于 g'，均粗于等于 g_0）：**(i)** touched$(g') \subseteq$ touched(g)，即粗化不使改写触及新的披露策略；**(ii)** $S_{\mathrm{raw}}(\mathrm{cl}(g')) \cap \bigcup_{\pi \in D_v} \bar{P}_\pi \subseteq S_{\mathrm{raw}}(\mathrm{cl}(g)) \cap \bigcup_{\pi \in D_v} \bar{P}_\pi$，即粗化不使受保护属性（含经 carry 闭包携出者）新进入原样呈现集——粗化换 group-by 键时桥接出的新列若受策略覆盖即违反本条。直觉：沿治理层级上卷（sku→category、session→日汇总）既不引入更敏感的属性、也不以原值呈现新的受保护列（(ii) 是引理 4.16 非空披露条款保持步所需，仅 (i) 不够）。**种子核查（如实陈述）**：v2026.06.23 五域中仅 aibuy（3 条）与 email（4 条）存在 `gov_disclosure_policy`，该 7 条策略逐条核查满足 (i)(ii)；rma / quality_voc / domestic_newprod 无披露策略种子，A4.3 对其**真空成立**。**缺席域披露语义（D5 裁决）**：披露缺席 = 无约束、披露门恒过（$D_v = \varnothing$ 时行 12 一步返回 $\{g_0\}$，恒 ANSWER），但证书 disclosure 段须显式标注 `ungoverned-disclosure`——诚实降级而非与受治域混同；A4.3 核查、E3 与消融范围据此按 aibuy/email 两域如实陈述。反例形状（桥接出的粗粒度列本身被更严策略覆盖）被排除在本章适用域之外——这是适用域刻画，不是能力放弃（见 4.6-G-6）。**约定注**：Γ_v 是**构造性声明**——格元素与覆盖边由 G_v 显式登记（如 rma 的 `product_sku` → `category_lv1` 上卷边），不从实例数据反推；粗化胞的呈现按**代理键分组**约定执行——group-by 键换为粗粒度的代理键列（sku→`category_lv1`、day→ 周/月键），$S_{\mathrm{raw}}(\mathrm{cl}(g))$ 的求值即以该约定下的输出列集为准。A4.3(i)(ii) 的核查对象是该声明格与该呈现约定，二者均随 v 版本化。

假设 A4.4（探针与可计算性）。**(i) 实例有限性**：实例快照 I 的相关扩展有限（承 C3 A1），从而 $\mathrm{SUPPMIN}_\pi$ 的非空输出胞集有限、诸探针可求值；**(ii) 窗代数封顶**：请求窗 w_0 各分量与锚有效带均属 C3 定义 3.2 的日历窗代数（可有限表示的有限区间并，承 C3 A2），从而 $w_0 \sqcap \mathrm{Env}$ 及其判空可计算——连续时间轴上的任意窗描述不在本章适用域内；**(iii) 测度非负**：分母 leg 的基准质量测度 $f \ge 0$（如销量、行计数）。若业务列含负值冲销（C3 G8 的退款负值形状），须先经预聚合折叠为非负测度或将该指标排除出本章适用域；否则 μ_{den} 沿窗包含不单调，引理 4.12(b) 的零质量传播剪枝失效。按 D4 裁决，MC(ii) 判据加宽为 $\mu_{\mathrm{den}} \le 0$：算法行 11 以 $\mu_{\mathrm{den}} \le 0$ 拒答（负质量分母不可作答），探针笔录记录 observed 值，C5 空窗见证谓词同步为

$\mu_{\mathrm{den}} \le 0$——该加宽对越出本条前提的负值冲销形状构成诚实防线，但零质量传播剪枝仍仅在 $f \ge 0$ 下成立。此三条此前隐含于定理 4.21 括号中的"探针假设"一语——它们是终止性与复杂度界的承重前提而非脚注，故升格为编号假设。

O.1.2 4.1 披露策略模型 D

定义 4.1（粒度格 grain lattice） $(\Gamma_v, \preceq_\Gamma)$ 是有限格，元素为输出粒度（实体轴 × 时间轴等有限积）。$g \preceq_\Gamma g'$ 读作"g 细于（或等于）g'"：g 的每个胞（cell）都包含于某个 g' 的胞。底元为最细可寻址粒度，顶元 $\top_\Gamma$ 为全域汇总。**实例**：rma 实体轴 `product_sku` → `category_lv1`（RMA 品类率桥接的上卷边），时间轴 `day` → `week` → `month`（dws _1d/_1w 表族）；aibuy 实体轴 `session_id` → 全体会话。

定义 4.2（掩码格 mask lattice） $(M, \preceq_M)$ 是有限格，元素为呈现变换类；$\mu \preceq_M \mu'$ 读作"μ' 披露不多于 μ"（μ' 可由 μ 复合一步信息不增变换得到）。底元 id（原样呈现），顶元 $\top_M$（suppress，完全抑制）。种子中的 `mask_class` 取值 `partial` / `generalize` / `masked_email` / `masked_phone` 作为 M 的中间元嵌入；本章不假定它们两两可比，部署时以声明的格结构为准。

定义 4.3（披露策略与策略集） 版本 v 下一条披露策略是五元组（对 `gov_disclosure_policy` 真实字段的抽象）

$$\pi = \langle P_\pi,\ \rho_\pi,\ \mu_\pi,\ \gamma_\pi,\ k_\pi \rangle$$

分量	抽象含义	对应真实字段
$P_\pi \subseteq \mathrm{Att}$	受保护属性集	`applied_columns` × `applied_tables`
$\rho_\pi = \langle \rho_\pi^{\mathrm{role}}, \rho_\pi^{\mathrm{row}} \rangle$ 或 $\perp$	角色分支：语境谓词 + 行归属谓词	`role_scope_predicate`（如 `role='salesperson' AND SaleMastNameStr=current_salesperson()`）
$\mu_\pi \in M \cup \{\perp\}$	原始值呈现所需的最低掩码	`mask_class`（null → ⊥，无掩码义务）
$\gamma_\pi \in \Gamma_v$	披露粒度下限（不得细于）	`min_grain`（单轴约束经"其余轴取底元"嵌入积格）
$k_\pi \in \mathbb{N}$	小胞阈值：每个披露胞至少覆盖 k_π 个不同 γ_π-实体	`k_threshold`（null → 0，真空约束）

`sensitivity_class` 与 `policy_id` 等为元数据（实现层）。**carry 闭包**：字段 `join_right_table/join_left_key/join_rig` 声明受保护属性沿连接携带；记属性携带关系 $\rightsquigarrow$（$a \rightsquigarrow b$：b 由 a 经保联结变换携出），定义 $\bar{P}_\pi$ 为 P_π 在 $\rightsquigarrow$ 后继下的闭包。策略集 D_v 有限，且与语义图同受版本钉（`version` 字段）——披露策略是治理状态的一部分，随 v 演化。

引理 4.4（carry 闭包良定） $\bar{P}_\pi$ 存在且为包含 P_π 的、对 $\rightsquigarrow$ 封闭的属性集中在包含序下的最小者；可在 $O(|\rightsquigarrow| + |\mathrm{Att}|)$ 内经 worklist 计算。

证明。算子 $F(X) = P_\pi \cup \{b : \exists a \in X,\ a \rightsquigarrow b\}$ 在有限幂集格上单调，由 Knaster–Tarski 有最小不动点；worklist 逐属性至多入队一次、逐边至多松弛一次，得界。■

定义 4.5（披露判定 $q' \models_{\mathrm{ctx},I} D_v$） 对候选改写 q'，记 $R(q')$ 为其读集（过滤、分支、连接、分组键、度量所引用的属性），$S_{\mathrm{raw}}(q')$ 为其**原样呈现集**（以行级原值出现在输出中的属性；聚合值不属于 S_{raw}）。策略 π 被触及（$\pi \in \mathrm{touched}(q')$）当且仅当 $\bar{P}_\pi \cap R(q') \neq \varnothing$。$q' \models_{\mathrm{ctx},I} D_v$ 当且仅当对每个被触及的 π，下列相应分支成立：

- **(D1 角色分支)** 若 $\rho_\pi \neq \perp$ 且 $\mathrm{ctx} \models \rho_\pi^{\mathrm{role}}$：$q'$ 中贡献 $\bar{P}_\pi$ 的行须满足 $\rho_\pi^{\mathrm{row}}(\mathrm{ctx},\cdot)$（自有行范围）；粒度/小胞条款豁免；若 $\mu_\pi \neq \perp$，$S_{\mathrm{raw}}(q') \cap \bar{P}_\pi$ 的呈现仍须 $\succeq_M \mu_\pi$（aibuy 种子：会话所有者对自有会话仍按 `generalize` 呈现证据文本）。
- **(D2 聚合分支)** 否则：(i) 粒度条款 $\gamma_\pi \preceq_\Gamma g'$；(ii) 小胞条款：$w'$ 窗内每个非空输出胞覆盖的不同 γ_π-实体数 $\geq k_\pi$（依实例 I）；(iii) 掩码条款：$\forall a \in \bar{P}_\pi \cap S_{\mathrm{raw}}(q')$，呈现 $\succeq_M \mu_\pi$。

判定显式依赖 I（小胞条款经探针评估）与 ctx（角色分支）——**合法性是实例相关与语境相关的**；这也是证书必须钉住实例快照的原因（4.4 末注记）。

O.1.3　4.2 合法改写空间、偏序与问题定义

定义 4.6（查询意图与刚柔承诺） 查询意图 $q = \langle m, \sigma, g_0, w_0, \kappa\rangle$：指标 m、范围谓词 σ、请求粒度 g_0、按 leg 的请求窗指派 $w_0 = (w_{0,\ell})_{\ell \in \mathrm{legs}(m)}$（**leg-索引向量**，缺省由 $\tau.T$ 生成）、**承诺集** κ。κ 把坐标域 {锚/口径, 窗, 版本, 粒度, 呈现} 中被问题文本显式承诺的坐标标记为**刚性**（rigid，改写不得偏离其声明值），其余为**柔性**（flexible，改写可在保真方向上收缩）。**呈现坐标与刚性呈现属性**：呈现坐标的承诺值是属性集 $S_{\mathrm{rig}} \subseteq \mathrm{Att}$——问题文本显式要求以行级原值呈现的属性（如" 列出联系邮箱" 之于 `ContactEmail`）；$a \in S_{\mathrm{rig}}$ 称**刚性呈现属性**（rigid presentation attribute），这是 S-M2、引理 4.16 与 G-5 所引概念的正式定义：掩码闭包把 S_{rig} 中属性送至 $\top_M$ 即空披露。$S_{\mathrm{rig}} = \varnothing$（未承诺呈现）时非空披露条款真空成立。实例：rma_q6 显式给出分子窗 2026-05、分母窗 2026-04 → 双窗刚性；domestic_newprod "用最新版本回算历史" → 版本刚性钉最新（显式离对角承诺，见 S-V）。刚柔划分由 NL 解析器给出，本章视为输入（缺口 G-2）。

定义 4.7（绑定点空间与改写语义） 给定 q, τ，**绑定点空间**为四坐标积

$$X(q,\tau) \;=\; \left\{ \langle \alpha, w, v, g\rangle \;:\; v \in V_{\mathrm{adm}},\; \alpha \in A_{\mathrm{adm}}(m,v),\; w \in W_{\mathrm{cand}}(\alpha, v),\; g \in {\uparrow_{\Gamma_v}}\, g_0 \right\}$$

窗坐标为 leg-索引向量 $w = (w_\ell)_{\ell \in \mathrm{legs}(\alpha)}$（与 w_0 同型）；窗上一切运算**逐分量**定义：$w \sqcap w' = (w_\ell \sqcap w'_\ell)_\ell$，$w \subseteq w' \iff \forall \ell : w_\ell \subseteq w'_\ell$，相等为逐分量相等，$w = \varnothing \iff \exists \ell : w_\ell = \varnothing$。单 leg 或各分量相同时缩写为标量（引理 4.12、定义 4.9 沿用此约定）。

各坐标的**改写语义**（本章声明的语义选择，构成 L 的定义域）：

- **(S-A 锚)** $A_{\mathrm{adm}}(m,v) = B_v[m]$ 中与 κ 的锚/口径承诺相容的注册绑定。锚不做" 修复"：改写不得把 leg 换到未注册的锚上（quality_voc 的 `complaint_submit_date`≠`rma_create_date` 跨系统对不上就是拒，不代扣）。
- **(S-V 版本)** $V_{\mathrm{adm}} =$ 与 $\tau.p$ 及 κ 的版本承诺相容、且（缺省 $\circledast$ 下）能解析 T 的版本钉集。**缺省对角线**：$\circledast$ 缺省解析为 C3 的对角线缺省 $\mathrm{ver}(T)$——T 时刻在位的已提交版本（C3 注记 3.6′/引理 3.5′）；T 为窗时取窗内在位版本集，窗跨版本边界则逐在位版本成 slice（每 slice 的窗坐标见下" 对角子窗与拼装"）。**"能解析 T" 的量词归属（歧义消解）**：这是**逐版本属性**——v 能解析 T 当且仅当 v 的在位区间 $[t_{\mathrm{commit}}(v), t_{\mathrm{commit}}(v^+))$（$v^+$ 为后继提交，末版本取 ∞）与 T 相交（点情形即包含 T），满足者之集记 V_{res}；**整集对角闸门**"$\mathrm{ver}(T)$ 有定义"（T 的每一点落于某在位区间，即 $T \subseteq [t_1, \infty)$）仅约束缺省 $\circledast$ 的对角解析——闸门失败即**版本轴缺位**，判 OOV（行 2a；C3 定义 3.14/G7），缺省下版本承诺把 V_{res} 剔空则判 AM(i) 类比（行 2b）。**显式钉版本的情形**：$\tau.p/\kappa$ 显式钉 $p \in V$ 时版本坐标由承诺给定、不经对角解析——$p \neq \mathrm{ver}(T)$ 乃至 $\mathrm{ver}(T)\!\uparrow$（如 $T < t_1$）均不构成阻断，一律为离对角读、证书照记离对角标记（这是既有离对角承诺语义在 $\mathrm{ver}(T)\!\uparrow$ 极限情形的直接延伸，数据层有效性仍由行 8–9 锚覆盖把关）；钉 $p \notin V$（不存在的版本号）或 $\tau.p$ 与 κ 版本承诺互斥则承诺剔空，判 AM(i) 类比（行 2b）。**对角子窗与拼装（跨版本边界窗的每 slice 窗坐标）**：缺省 $\circledast$ 且 T 跨版本边界时，记 $\mathrm{onwin}(v,T)$ 为 T 与 v 在位区间之交，slice $(\cdot, v)$ 的**构件请求窗**为 $w_0^{(v)} := w_0 \sqcap \mathrm{onwin}(v,T)$（逐 leg 同裁），行 8 起以 $w_0^{(v)}$ 代 w_0 逐构件执行——即 $W_{\mathrm{cand}}(\alpha, v) = \{w_0^{(v)} \sqcap \mathrm{Env}(\alpha, v)\}$（单版本或不跨界时 $w_0^{(v)} = w_0$，本章一切陈述按字面还原）。诸对角 slice 是同一答案的**分段构件**而非竞争候选：任一构件拒则整窗按该构件首因拒（CT 记全部构件）；全部通过时 plan 为逐 leg 分段子计划求和后成比（各 leg 测度按对角子窗分段求值再求和，率值在拼装后计算，分段率值不平均），证书逐构件披露 $(v, w_0^{(v)})$；$\prec_{\mathrm{det}}$ 只在同一构件内竞争的多锚/多粗化候选间选代表，**不跨构件选版本**。与不离对角纪律的相容性：每构件仅在自身在位子窗上作答，正是对角线在窗上的逐段实现——"最新在位版本对全窗

作答”（GOLD-1 禁止形态）在缺省下不可表达。种子如实陈述：五域均单版本（A4.2），跨界窗条款零实例支撑，属零实例区的定义层钉死。缺省**不得**取最新已提交版本：“用最新版本口径回算历史”正是 gov_seed GOLD-1 点名的错误读法（domestic_newprod `model_mismatch_rate_align`：以最新版本回算历史会错绑过期型号映射）；钉最新回算历史仅当 κ 显式作出该版本承诺时进入 V_{adm}，且证书须记录**离对角标记** $p \neq \text{ver}(T)$（见 E5、命题 4.23 范围限定）。多版本仅在承诺放开时出现。

- **(S-W 窗，结构裁剪)** $W_{\text{cand}}(\alpha, v) = \{ w_0 \sqcap \text{Env}(\alpha, v) \}$：候选窗仅取请求窗与**覆盖包络**的逐分量交。$\text{Env}(\alpha, v) = (\text{cov}(a_\ell, v))_\ell$ 为 **leg-索引向量**，逐 leg 取值随锚声明的 `coverage_mode`——`coverage_mode` 是锚的**版本化属性**（G_v 内容物，随 v 演化；D3 裁决逐锚参数化；取值 token 跨章统一为 `hull` | `strict_member`，C5 §0 先冻结者为准）：**strict_member**（裸日期集锚缺省）取 $\text{cov} = \text{val}(a_\ell, v)$，覆盖域为严格成员日期集/区间并本身；**hull**（快照锚缺省）取 $\text{cov} = \text{hull}(\text{val}(a_\ell, v))$，覆盖域为覆盖跨度（跨度内空日合法产 0 值——C3 (b1) 相交语义即此）。$\text{val}(a_\ell, v)$ 为绑定各 leg 锚在版本 v 下有定义的 valid-time 带（snapshot_effective_date：覆盖的 `dt` 带；scd_type2：$[$`valid_from`, `valid_to`$)$ 区间并，端点取自 I；锚语义统一为**点窗** $vf \le d < vtc$——D7 裁决，与金标 126 一致——SCD-2 零长版本行 `dw_start=dw_end` 在点窗语义下自然为空集出局，无需额外良构禁令）。窗只被**结构性**端点（请求端点、覆盖端点、版本边界）裁剪，不做数据驱动的收缩——动机见注记 4.14；覆盖检查失败判 OOV：$w^* = \varnothing$（某 leg 与其 cov 不相交），或窗坐标刚性而某 **strict_member** leg $w_{0,\ell} \not\subseteq \text{val}$（hull leg 刚性下边缘重叠合法——定义 4.8(ii) 的 hull-edge 裁剪豁免——按裁剪窗作答并入证书、披露边缘裁剪）——本章与 C3 (b1) 的语义分歧经逐锚 `coverage_mode` 参数化统一（D3），C5 的 OOV 见证按锚声明的 mode 重演。**显式排除（适用域刻画）**：柔性且不等的多 leg 窗下，“收缩到各 leg 交集使 `same_valid_time_window` 成立”的等式修复**不在**本章改写空间——S-W 逐 leg 结构裁剪、每 slice 单候选，不做跨 leg 等式议价；该扩展与数据驱动收缩同归缺口 G-1。
- **(S-G 粒度)** 候选粒度为请求粒度的上集 $\uparrow g_0$（沿 Γ_v 上卷）；粗化是披露约束下换取可答性的合法方向（RMA 品类率 sku→category_lv1 桥接即此）。
- **(S-M 掩码)** 掩码不是搜索坐标：对每个绑定点取**最小合法掩码闭包** $\mu^*(b, \text{ctx})$（引理 4.15）；若闭包把某刚性呈现属性送至 $\top_M$（空披露），该点判 DB（选择 S-M2：空披露视同拒答，而非返回全抑制的空壳）。

改写 = 绑定点 + 披露闭包：$q' = \text{cl}(b, \text{ctx}) = \langle m, \sigma, b, \mu^*(b, \text{ctx}) \rangle$。其中范围谓词分量在 D1 角色命中时**并入行归属谓词**：对每个被触及且角色命中的策略（$\pi \in \text{touched}(b)$，$\rho_\pi \neq \bot$，$\text{ctx} \models \rho_\pi^{\text{role}}$），cl 构造取 $\sigma \wedge \rho_\pi^{\text{row}}(\text{ctx}, \cdot)$ 代 σ（多策略取合取）；无角色命中时 σ 原样保留。此句是引理 4.16、定理 4.18 与定理 4.20 三处「D1 角色命中策略的行归属约束由 cl 构造并入范围谓词」断言的定义层支点。

定义 4.8（忠实性与合法改写集） $q' = \text{cl}(b, \text{ctx})$ 是 q 的**忠实改写**（faithful rewrite），若 (i) 指标与范围语义沿用 m, σ 且锚取自 A_{adm}；(ii) 刚性坐标逐一等于承诺值——**hull-edge 裁剪豁免（D3 落地语义的定义层对齐）**：窗坐标刚性时，`coverage_mode=hull` 的 leg 之窗承诺值解释为 $w_{0,\ell} \sqcap \text{hull}(\text{val}(a_\ell, v))$——hull 语义下覆盖跨度之外无该锚的定义域，边缘重叠合法，按裁剪窗作答且证书须披露该 leg 的覆盖边缘裁剪（$w_\ell^* \neq w_{0,\ell}$ 之事实与被裁端点）；strict_member leg 无此豁免（$w_{0,\ell} \not\subseteq \text{val}$ 即 OOV，行 9）；(iii) 柔性坐标仅按 S-W/S-G 方向收缩或粗化。**合法改写集**

$$L(q, \tau, G, \text{ctx}, I) \ = \ \left\{ \text{cl}(b, \text{ctx}) \ : \ b \in X(q, \tau),\ r_\alpha(b) \text{ 成立},\ \Omega \text{ 可绑定},\ \text{cl}(b, \text{ctx}) \models_{\text{ctx}, I} D_v \right\}$$

其中 r_α 为绑定规则约束，其判定采用 C3 的**完整规则谓词** $P_{\text{rule}}(\alpha; \kappa, w_0, G_v, I)$——以 `same_valid_time_window` 为例含四条款：(i) 各 leg 窗相等（窗口等式）；(ii) 刚性窗承诺与等式相容；(iii) 窗-粒度可表达（请求窗可由锚粒度的日历窗精确表达，C3 定义 3.8(iii)）；(iv) 锚对相容审计（两锚有效日期集经实例 I 审计一致，C3 定义 3.8(iv)）。条款 (iii)(iv) 需数据访问，可判定性由 C3 命题 3.18 给出；规则字面相同不免除 (iii)(iv)（接地反例见 4.4 测试用例）。滞后/位移类规则见缺口 G-9。L 是**模型相对**的：它定义在 S-A/W/V/G/M 声明的改写语义之内，边界作为适用域刻画在 4.6 陈述。

定义 4.9（保真偏序 $\preceq_D$） 对 $q'_1 = \text{cl}(\langle \alpha_1, w_1, v_1, g_1 \rangle)$，$q'_2 = \text{cl}(\langle \alpha_2, w_2, v_2, g_2 \rangle) \in L$：

$$q'_1 \preceq_D q'_2 \iff \alpha_1 = \alpha_2,\ v_1 = v_2,\ w_1 \subseteq w_2 \text{ (窗序逐分量，定义 4.7)},\ g_2 \preceq_\Gamma g_1$$

即同锚同版本内，窗更贴近请求、粒度更细者信息保真度不低；不同锚或不同版本的改写**不可比**（它们回答的是语义上不同定语的问题，比较交由 $\prec_{\text{det}}$ 的声明规则）。$\approx_D$ 为相互 $\preceq_D$；在商集 $L/{\approx_D}$ 上 $\preceq_D$ 为偏序。掩码由闭包唯一派生（引理 4.15），不参与跨点比较（替代建模见缺口 G-4）。

注记（与可导出序的关系）。对可加度量，粗化后的答案可由细粒度答案确定性汇总导出，$\preceq_D$ 与" 可导出" 序同向；对 scoped_ratio 类指标，粗粒度率值一般不可由细粒度率值导出（需分子分母重聚合），故 $\preceq_D$ 是**保真序**（对请求坐标的贴近程度）而非可导出序——这是显式建模选择（缺口 G-3，需作者裁决表述强度）。

定义 4.10（最大合法改写问题 MLR 与证书） **MLR**：给定 $\langle q, \tau, G, D, \mathrm{ctx}, I\rangle$，输出

- ANSWER⟨plan, cert⟩：某 $\hat{q} \in L$ 使 $\hat{q}$ 在 $(L/\approx_D, \preceq_D)$ 中**极大**（maximal——不被任何合法改写严格保真支配；极大元通常不止一个，也未必存在最大元 greatest element），plan 为其经 C3 出码的时点正确查询；或
- REFUSE⟨CT, cert⟩：$L = \varnothing$ 的证据，CT 为按 (v, α)-slice 的原因表，取值 $\{\mathrm{MC}, \mathrm{AM}, \mathrm{OOV}, \mathrm{DB}\}$。

时点证书 cert 字段：$\langle q, \tau$, ctx-hash, v, α(binding_id), ρ (口径路由路径), w^*, g, μ^*, 触及策略集, 探针笔录(μ_{den}/SUPPM —供 C5 独立校验器复核（校验器不与编译器共享实现，梁老师意见 3 红线）。ρ 为 R_v 中被采用的路由边序列，对齐 C5 定义 5.1/命题 5.12：证书缺 ρ 则 caliber-blind substitution 攻击（rma 16.2% vs 3.8% 双路由实例——provenance 同 C3 注记 3.4′：datagov-line 生产实例实测（datahub `datagov-line`，loctek-unified-datagov-line，v2026.06.23 rma 域治理产出），非随论文交付的 pilot 工件；pilot 合成仓 `pilot/domains/rma/warehouse.duckdb` 全域 problem/sales 为 2.76%=465/16832，不同域实例、不同 scope，不可互算）对证书成立、C5 校验项 V4/I3 无从执行，故 ρ 是必备字段而非元数据。**跨章 schema 对接（D1 裁决落地）**：C5 拒答原因枚举扩为 $r \in \{\mathrm{MC}, \mathrm{AM}, \mathrm{OOV}, \mathrm{DB}\}$；拒答证书形制统一采用本章逐 slice 原因表 CT，C5 证书按 CT 形制承载——单见证取**首失败 slice 的见证**，复合方向定为 **slice-major**（与 C5 定义 5.2 完全同款）：先按 $\prec_{\mathrm{det}}$ 的 slice 序取首失败 slice，再在该 slice 内按守卫序 OOV → AM → MC 取 r；完整 CT 作为附表随证书携带；拼写统一为 OOV。披露缺席域的证书 disclosure 段标注 `ungoverned-disclosure`（D5，见 A4.3）。**dec 映射（证书层 REWRITE 判据，对齐 C5 定义 5.1 的 δ.dec）**：返回 ANSWER 的绑定点若相对请求坐标发生任何收窄——$g \neq g_0$（粒度上卷）$\vee$ 存在 leg $w^*_\ell \neq w_{0,\ell}$（窗被结构裁剪，**含 hull-edge 裁剪豁免**的边缘裁剪情形）$\vee$ 跨界窗拼装中任一构件请求窗被收窄（$w_0^{(v)} \neq w_0$ 且构件集非全窗单构件）——则证书层 dec = $\mathsf{REWRITE}$（携裁剪痕迹 cut trace），仅坐标逐项等于请求值时 dec = ANSWER；静默收窄即 C5 命题 5.12 的披露洗白攻击面。

定义 4.11（确定性选择 $\prec_{\mathrm{det}}$） $\prec_{\mathrm{det}}$ 是候选绑定点上的声明式全序键：(1) 版本较新者先；(2) 绑定注册序先——**入模要求**：注册序是 G_v 的**版本化声明成分**（`gov_temporal_binding` 须有注册序数列，或缺省钉 `binding_id` 字典序）；jsonl 行序不是模型内容物——此隐患的实证（随仓现状如实陈述）：rma pilot `_binding()` **修复前**无 `ORDER BY`，现行 `pilot/domains/rma/compiler.py`（行 60）已按 `binding_id` 排序（且按 metric 过滤后每键单行）；`pilot/public/compiler.py` 的按 metric 取绑定查询（行 44 附近）至今无 `ORDER BY`，仍暴露该隐患；(3) 粒度按 ℓ_Γ（$\preceq_\Gamma$ 的一个声明线性扩张）取先——ℓ_Γ 同为 G_v 的版本化声明成分，多版本时逐 Γ_v 声明。MLR 的返回代表元 = $\min_{\prec_{\mathrm{det}}}$ 于极大元集。$\prec_{\mathrm{det}}$ 是**策略而非定理**：它影响返回哪个极大元，不影响极大元集本身（定理 4.19(c)），且必须写入证书以可审计；其全部三键既入 G_v，定理 4.19(c) 的确定函数断言才对 $\langle q, \tau, G, D, \mathrm{ctx}, I\rangle$ 成立。

O.1.4 4.3 结构、单调性与剪枝

引理 4.12（窗带结构与质量单调） 固定 slice (α, v)，记 $w^* = w_0 \sqcap \mathrm{Env}(\alpha, v)$（逐分量；Env 按定义 4.7 S-W 的逐锚 `coverage_mode` 取覆盖包络），$\mu_{\mathrm{den}}(w)$ 为窗 w 内分母 leg 的**基准质量**：范围 σ 内分母测度 $f \geq 0$（A4.4(iii)）之和——rma `problem_rate` 即 SUM(sales_qty)；原子计数度量取 $f \equiv 1$，退化为行计数。则：

(a) 任一忠实且 Ω-可绑定的改写，其窗 w 满足 $w \subseteq w^*$；从而 w^* 是该 slice 忠实可绑窗集在包含序下的最大元（若该集非空）。

(b) $w \subseteq w' \Rightarrow \mu_{\mathrm{den}}(w) \leq \mu_{\mathrm{den}}(w')$；特别地 $\mu_{\mathrm{den}}(w^*) = 0$ 蕴含 slice 内一切忠实窗的基准质量为零。

证明。(a) 忠实性（定义 4.8(iii)）给 $w \subseteq w_0$；Ω-可绑定要求各 leg 窗落在其锚按声明 `coverage_mode` 的覆盖包络内（strict_member leg：$w_\ell \subseteq \mathrm{val}$；hull leg：$w_\ell \subseteq \mathrm{hull}(\mathrm{val})$，跨度内空日合法产 0——D3 裁决逐锚参数化，与 C3 (b1) 统一），给 $w \subseteq \mathrm{Env}$；逐分量交即 $w \subseteq w^*$。w^* 自身属于该集（当其非空且质量条件另行满足时），故为最大元。(b) 非负测度之和对行集包含单调（A4.4(iii)）；取 $w' = w^*$ 得零质量传播。**注意**：单调性与零传播俱系于测度非负——退款冲销类负值列（C3 G8）下 μ_{den} 沿窗不单调，此即 A4.4(iii) 的承重

点；“基行数” 版本的行计数探针在分母测度非 $f \equiv 1$ 时与本引理的质量语义**不等价**（rma 词汇：窗内恰有一行 `sales_qty=0` 时行计数为 1 而质量为 0）；分母测度本身为行计数的指标（$f \equiv 1$，如 domestic_newprod launch 族）则行计数即 μ_{den} 的合法退化形——见 4.4 探针形态对账。■

引理 4.13（k-支持度沿粒度的单调性） 固定 (α, v, w^*) 与策略 π，记 $\text{SUPPMIN}_\pi(g)$ 为粒度 g 下窗内非空输出胞覆盖的不同 γ_π-实体数的最小值。则集合 $\{g \in \uparrow g_0 : \text{SUPPMIN}_\pi(g) \geq k_\pi\}$ 是 $(\uparrow g_0, \preceq_\Gamma)$ 中的上集（up-set）。

证明。设 $g \preceq_\Gamma g'$ 且 g 满足条件。g' 的任一非空胞 c' 是若干 g-胞之并，其中至少一个非空胞 c；c' 覆盖的 γ_π-实体集包含 c 的实体集，故 $|\text{ent}(c')| \geq |\text{ent}(c)| \geq k_\pi$。对所有非空 c' 取最小值即得。(论证仅用集合包含，不需要支持度可加性，因此对跨胞重复实体亦稳健。) ■

注记 4.14（k-支持度沿窗口方向不单调——S-W 的动机） 沿窗口扩张，既有胞的支持度单调不减，但可能**诞生**新的低支持胞：某 SKU 仅在四月初有 2 单，五月窗下其胞为空（不披露），四月—五月窗下其胞出现且支持度 $2 < k$。故 k-合法性对窗口包含既不单调也不凸。这正是 S-W 采用**结构裁剪**（不做数据驱动窗收缩）的理论动机之一：数据驱动的窗-粒度联合议价会使极大元刻画依赖逐胞出生点，其枚举理论需另行建立（缺口 G-1）。结构裁剪语义下窗坐标每 slice 至多一个候选，联合议价不发生；这一语义与 pilot 编译器实践一致（各域编译器均不缩窗避 k，人口不足即拒）。

引理 4.15（掩码闭包确定性） 固定绑定点 b（粒度 g）与 ctx。合法掩码指派集 $\{\mu \in M^{S_{\text{raw}}} : \text{cl}(b, \mu)$ 满足全部掩码条款$\}$ 若非空，则在逐属性序下是上集，且有最小元

$$\mu^*(b, \text{ctx})(a) \;=\; \bigvee\nolimits_M \{\mu_\pi \;:\; \pi \in \text{touched}(b),\; a \in \bar{P}_\pi \cap S_{\text{raw}},\; \mu_\pi \neq \bot,\; \text{条款于 ctx 分支适用}\}$$

（空并取 id）。从而掩码由绑定点与语境唯一派生，不构成搜索维度。

证明。各条款均为逐属性的下界约束” 呈现 $\succeq_M \mu_\pi$“；约束集的满足域是上集之交，仍为上集。有限格中一族下界约束的最小公共满足点是诸下界的并（join），逐属性取并即 μ^*；其合法性由构造（逐条款 $\mu^*(a) \succeq_M \mu_\pi$）与最小性由并的泛性质给出。■

引理 4.16（逐粒度合法性的上集结构） 固定 slice (α, v) 与 w^*（$\mu_{\text{den}}(w^*) > 0$）。记 $\text{touched}_{\text{D2}}(g, \text{ctx}) = \{\pi \in \text{touched}(g) : \rho_\pi = \bot \;\vee\; \text{ctx} \not\models \rho_\pi^{\text{role}}\}$——**D2 分支于 ctx 适用**的被触及策略（定义 4.5：D1 角色命中的策略其粒度/小胞条款豁免；其行归属义务由 cl 构造并入范围谓词、掩码义务由 μ^* 承载——引理 4.15 的”条款于 ctx 分支适用”）。定义逐粒度合法谓词

$$\text{Legal}(g) \;\equiv\; \underbrace{\forall \pi \in \text{touched}_{\text{D2}}(g, \text{ctx}) : \gamma_\pi \preceq_\Gamma g}_{\text{粒度条款}} \wedge \underbrace{\forall \pi \in \text{touched}_{\text{D2}}(g, \text{ctx}) : \text{SUPPMIN}_\pi(g) \geq k_\pi}_{\text{小胞条款}} \wedge \underbrace{\mu^* \text{ 不将刚性呈现属性送至}}_{\text{非空披露}}$$

在假设 A4.3 下，$U = \{g \in \uparrow g_0 : \text{Legal}(g)\}$ 是上集；且 $\text{cl}(\langle \alpha, w^*, v, g \rangle) \in L$ 当且仅当 $g \in U$（其余成员条件已由 slice 前置检查固定）。

证明。设 $g \preceq_\Gamma g'$，$g \in U$。由 A4.3(i) $\text{touched}(g') \subseteq \text{touched}(g)$；D2 适用性只依赖 (ρ_π, ctx) 而不依赖 g，故 $\text{touched}_{\text{D2}}(g', \text{ctx}) \subseteq \text{touched}_{\text{D2}}(g, \text{ctx})$。粒度条款：$\pi \in \text{touched}_{\text{D2}}(g', \text{ctx}) \subseteq \text{touched}_{\text{D2}}(g, \text{ctx})$ 给 $\gamma_\pi \preceq g \preceq g'$。小胞条款：对每个留存的 $\pi \in \text{touched}_{\text{D2}}(g', \text{ctx})$，引理 4.13 给 $\text{SUPPMIN}_\pi(g') \geq k_\pi$。非空披露：$g'$ 处的必需掩码约束集 $\{(\pi, a) : \pi \in \text{touched}(g'),\; a \in \bar{P}_\pi \cap S_{\text{raw}}(\text{cl}(g'))\}$ 是 g 处对应集的子集——由 A4.3(i)（touched 单调）**与** A4.3(ii)（受保护属性的原样呈现集单调）联合给出；仅有 (i) 并不足够：粗化换 group-by 键可使桥接列新进 $S_{\text{raw}}(\text{cl}(g'))$，同一已触及策略经 carry 闭包（$a \in \bar{P}_\pi \cap S_{\text{raw}}(\text{cl}(g'))$ 而 $a \notin S_{\text{raw}}(\text{cl}(g))$）在 g' 产生 g 处不存在的义务，此形状恰被 (ii) 排除。于是逐属性并不增，故若 g' 处某刚性呈现属性（$a \in S_{\text{rig}}$，定义 4.6）被送至 $\top_M$，g 处同属性的并 $\succeq_M \top_M$ 亦为 $\top_M$，与 $g \in U$ 矛盾。三条款皆保持，$g' \in U$。等价刻画：定义 4.5 的 D1/D2 各条款中，范围、窗、版本、规则与 Ω-可绑定性在 slice 前置检查中已判定；D1 角色命中策略的行归属约束由 cl 构造并入范围谓词（定理 4.18）、其掩码义务并入 μ^* 之并（引理 4.15），粒度/小胞条款对其豁免（定义 4.5 D1）；余下恰为 Legal(g) 覆盖的三条款（粒度/小胞仅量化于 $\text{touched}_{\text{D2}}$）与掩码闭包合法性（引理 4.15）。角色命中种子实例（email `email_salesperson_row_scope`、aibuy `aibuy_reco_provenance_join_scope`：`role_scope_predicate` 与 `min_grain`+`k_threshold` 并存的行）在 ctx 命中其角色分支时，细于 `min_grain` 的自有行查询即由本豁免合法进入 U——若粒度/小胞条款

对 touched 无条件量化，该更细合法改写将被 Legal 误拒、GrainFrontier 误上卷，定理 4.19(a)/4.20 即于此分支失效，此为本限定的必要性。■

推论 4.17（剪枝正确性） 在 $\uparrow g_0$ 上自底向上遍历、命中 Legal 即停并剪去其上集（不再访问 $\uparrow g$），所得集合 $U_{\min}$ 为 U 的极小元全体；被剪节点或不合法、或被 $U_{\min}$ 中某点 $\preceq_D$-支配。遍历访问节点数 $\leq |\uparrow g_0| \leq W_\Gamma\,(h_\Gamma+1)$。

证明。上集的极小元刻画：$g \in U$ 极小 $\Leftrightarrow g \in U$ 且其所有 $\preceq_\Gamma$-前驱不在 U。自底向上（覆盖边）遍历保证首次命中的节点无更细的 U 成员（其全部前驱已访问且未命中）；剪去 $\uparrow g$ 的节点 $g'' \succeq g$：若 $g'' \in U$ 则被 g 支配（同 slice、同窗、粒度更粗 $\Rightarrow \mathrm{cl}(g'') \preceq_D \mathrm{cl}(g)$），若 $g'' \notin U$ 则不合法——两类都不入极大元集。节点数界：Mirsky 定理把高度 h_Γ 的有限偏序集划分为至多 $h_\Gamma+1$ 条反链，每条反链 $\leq W_\Gamma$。■

O.1.5 4.4 算法 MLR-Compile

```
输入: q=⟨m,σ,g_0,w_0,κ⟩, τ=⟨T,p⟩, ctx;  治理状态 G={G_v}, 实例快照 I
输出: ANSWER⟨plan, cert⟩ | REFUSE⟨CT, cert⟩          ▷ CT: (v,α)-slice ↦ {MC,AM,OOV,DB}

MLR-Compile(q, τ, ctx; G, I):
  V_res ← {v ∈ V : 在位(v) ∩ T ≠ ∅}                  ▷ S-V: "能解析 T"逐版本判定（在位区间与 T 相交）
  if 版本坐标缺省(τ.p=⊛ 且 κ 无版本承诺) ∧ ver(T)↑:
    return REFUSE⟨{(⋆,⋆)↦OOV}, Cert⟩               ▷ 版本轴缺位(T ⊄ [t_1,∞), 含 T<t_1) → OOV
            ▷ （C3 定义 3.14/G7 约定; S-V 整集对角闸门，仅约束缺省对角解析）
  V_adm ← (显式钉 p ? {p} ∩ V : V_res) ∩ κ 版本承诺相容者
  if V_adm = ∅: return REFUSE⟨{(⋆,⋆)↦AM}, Cert⟩   ▷ 版本承诺剔空(钉不存在版本 / τ.p 与 κ 互斥 /
            ▷ κ 剔空 V_res) → 承诺-可用性不相容, AM(i) 类比 (D1; S-V 显式钉语义)
  CT ← ∅; Front ← ∅
  for v ∈ V_adm 按 ≺_det 版本序:
    A_adm ← AdmBindings(m, v, κ)                    ▷ S-A: B_v[m] ∩ 刚性锚/口径承诺
    if A_adm = ∅: CT[v,⋆] ← (B_v[m]=∅ ? MC : AM); continue   ▷ 注册层空判 MC(i); κ 剔空判 AM(i) (D1)
    for α ∈ A_adm 按注册序:                          ▷ 注册序为 G_v 声明成分(定义 4.11)
      w* ← w_0 ⊓ Env(α, v)                          ▷ S-W 逐分量结构裁剪; Env 逐 leg 按锚声明的
            ▷ coverage_mode(G_v 版本化属性, D3): strict_member 锚取 val、hull 锚取 hull(val);
            ▷ 含 SCD-2 点窗有效区间(vf≤d<vtc, D7)、版本边界; 缺省对角跨界窗时
            ▷ 以构件请求窗 w_0^(v) = w_0 ⊓ onwin(v,T) 代 w_0 (S-V 对角子窗与拼装)
      if w* = ∅ ∨ (rigid(W) ∧ ∃ strict_member leg ℓ: w_0,ℓ ⊄ val_ℓ): CT[v,α] ← OOV; continue
            ▷ 覆盖检查按逐锚 coverage_mode(D3); C5 OOV 见证按声明 mode 重演
      if ¬RuleSat(r_α, κ, w_0; G_v, I): CT[v,α] ← AM; continue
            ▷ 完整规则谓词 P_rule 条款(i)-(iv): 窗口等式 × 刚性承诺相容 ×
            ▷ 窗-粒度可表达(iii) × 锚对相容审计(iv); (iii)(iv) 经 I 数据访问,
            ▷ 可判定性 C3 命题 3.18——此行即算法对 Ω 规则层的显式接入
      if μ_den(α, v, w*, σ) ≤ 0: CT[v,α] ← MC; continue
            ▷ 质量探针 ×1(引理 4.12(b), A4.4(iii)); μ_den ≤ 0 判 MC(ii)(D1/D4),
            ▷ 探针笔录记 observed 值; 行序 = 报告序 OOV→AM→MC(D2)
      U_min ← GrainFrontier(α, v, w*, ctx)          ▷ 推论 4.17 的自底向上扫描
      if U_min = ∅: CT[v,α] ← DB; continue
      for g ∈ U_min: Front ← Front ∪ {⟨α, w*, v, g, μ*(·,ctx)⟩}
  if Front = ∅: return REFUSE⟨CT, Cert(CT, 探针笔录)⟩
  ƀ ← SelectMin_{≺_det}(Front)
  return ANSWER⟨EmitPlan_C3(ƀ), Cert(ƀ, 探针笔录)⟩

GrainFrontier(α, v, w*, ctx):
  U_min ← ∅; Agenda ← {g_0}; Seen ← ∅
  while Agenda ≠ ∅:
```

```
20      g ← pop(Agenda, ℓ_Γ 序)
21      if g ∈ Seen ∨ ∃u∈U_min: u ⪯_Γ g: continue      ▷ 剪枝(推论 4.17)
22      Seen ← Seen ∪ {g}
23      if Legal(g):  U_min ← U_min ∪ {g}             ▷ Legal 同引理 4.16 (粒度/小胞条款仅量化于
                ▷ touched_D2(g,ctx)——D1 角色命中策略豁免, 定义 4.5) ; 每次 ≤|touched(g)| 探针
24      else:         Agenda ← Agenda ∪ covers_Γ(g)    ▷ 仅沿覆盖边上行
25    return U_min
```

跨界窗拼装注记（行 15–17 的构件语义；S-V「对角子窗与拼装」的算法层显式化）。缺省 ⊛ 且 T 跨版本边界时，诸对角 slice 是同一答案的**分段构件**而非竞争候选：(1) **任一构件拒则整窗拒**——行 15 的空判定与 REFUSE 按构件首因触发，CT 记全部构件；(2) **通过时逐 leg 分段求和成比**——行 17 的 plan 对各 leg 测度按对角子窗 $w_0^{(v)}$ 分段求值再求和，率值在拼装后计算（分段率值不平均，注记 3.13 的 ratio-of-sums 不变式跨构件成立）；(3) $\prec_{\rm det}$ **不跨构件**——行 16 的代表选取只在同一构件内竞争的多锚/多粗化候选间进行，不做跨构件的版本选择。本注记属**零实例区的定义层钉死**（五域均单版本，A4.2 种子如实陈述）；定理 4.18–4.20 的字面证明在「单版本或不跨界」情形按原文还原（$w_0^{(v)} = w_0$，拼装退化为单构件）。

检查次序与报告序（D2 裁决落地）。行 8–11 已按 C3 钉定的双故障报告序 OOV → AM → MC 排列：先窗/覆盖检查（行 8–9，OOV），再规则审计（行 10，AM），再质量探针（行 11，MC(ii)）；双故障 slice 的 CT 记录首个触发原因，CT 报告序与算法行序一致。DB 为披露层原因，恒居绑定层三类之后（行 13）。两域编译器（quality_voc / domestic_newprod）守卫序已同改并随仓验证：pilot 编译器为退化实例，其分支判定序与本行序一致。

规则谓词接地测试用例（soundness 承重）。quality_voc v2026.06.23 `gov_temporal_binding` 注册行 `complaint_vs_rma_anchor_mismatch`（指标 `defect_rate` 的第二绑定：分子锚 `complaint_submit_date` × 分母锚 `rma_create_date`，规则**字面**同为 `same_valid_time_window`）：条款 (iv) 锚对审计失败——两锚有效日期集：投诉锚 36 个锚日期（周透视快照日）vs 工单日锚 35 个工单日，对称差 = 57（随仓 `warehouse.duckdb` 复演值；pilot 编译器 docstring 旧值 63 为陈稿数字，已随本轮统改订正为 57——其 `_anchors_same_window` 探针在随仓库上重跑即为 57）→ 行 10 判 AM，与 pilot REFUSE(`anchor-mismatch`) 一致。若 RuleSat 只查窗口等式字面（本稿修复前形态），该注册行在同窗请求下将穿透行 8–11 进入 Front 并可被 ANSWER——定理 4.18/4.20 的直接反例；条款 (iii)(iv) 的接入因此是 soundness 承重项，非工程细节。

探针接口（实现注记，机制降格）。$\mu_{\rm den}$ 与 $\mathrm{SUPPMIN}_\pi$ 各为绑定人口上的一条聚合查询（前者为分母测度求和，后者为一条带 `GROUP BY` 的最小支持度查询）；条款 (iv) 的锚对审计为一条有效日期集对称差查询；`Env` 的 SCD-2 端点读维表有效期列。pilot 五域编译器（`pilot/domains/*/compiler.py`）是本算法的退化实例：Γ 每题单点、D 折叠、$\nu = 1$、$a \in \{1,2\}$，其三分支拒答即 CT 的 slice 原因。**探针形态对账**：行 11 的语义是 $\mu_{\rm den}$——按各指标的**分母测度**取 f，基行数只在 $f \equiv 1$ 时合法（引理 4.12 的退化形）。rma `_den_present`（`SUM(sales_qty)>0`）对 `problem_rate`（分母测度 `sales_qty`）正是行 11 的质量探针；对 `refund_rate`（分母测度 `sales_amount`）其探针仍探 `sales_qty`，与分母测度错位——随仓数据无 `qty>0` ∧ `amount≤0` 分歧行故未显形，属潜在实现偏差，应按分母测度改探 `sales_amount`。domestic_newprod 的 `den_sql` 族两形态**均合法**：`launch_online_rate/mkt_reach_rate` 的指标分母本身即行计数（`sum(biz_online)/count(*)` 型，$f \equiv 1$——引理 4.12 允许的 $\mu_{\rm den}$ 退化形），`count(*)` 恰当；`online_share/offline_share` 已取 `COALESCE(sum(order_total_qty),0)` 的质量形。**标签归一注记（D1 裁决落地 · 迁移已完成）**：对”同窗分母质量为零”，五域编译器标签已归一为 `missing-caliber`（MC(ii)：口径人口缺失，与 C3 定义 3.16(ii)/C5 G2 同判）——rma/email 原生一致；domestic_newprod 已迁移（compiler.py 空同窗分母 $\mu_{\rm den} \le 0$→`missing-caliber`，守卫序改齐 OOV → AM → MC）；aibuy 已迁移（den≤0→`missing-caliber`）。金标同步落地：AIBUY-Q6、EMAIL-ASOF-06、rma_q5=`missing-caliber`，五域编译器实跑与金标逐题一致（aibuy 自校验 ALL OK）。OOV 只保留给有效性层真空（锚覆盖域真空——pilot 单版本域的唯一形状；一般设定另含版本轴缺位，行 2a/A4.1）——NX-Q7 实测 domestic_newprod launch 表 min `dt=2025-09-15`，窗前请求属真 OOV，金标保持 `out-of-validity`。理论三章口径已统一，映射表随证据保留（缺口 G-10 闭合为对账记录）。

实例相关性与快照一致性（producer/consumer 注记）。合法性判定经探针依赖 I；探针读与最终计划执行必须落在同一版本快照上，否则证书的 k/人口断言可能失效（探针后写入的 TOCTOU）。本框架把这一保障显式委托给版本化底座的读一致性（DELTAGOV 的 version-serializability 作为 producer 侧保证，本编译器为 consumer）；形式化的接口义务在系统章陈述（缺口 G-7）。

接地示例（种子与 pilot 实证的语义，均可在源仓复查）：

#	场景（域/来源）	MLR-Compile 走向
E1	rma：`problem_rate`，SKU536-EOL，同窗销量分母为空（536-SKU 母题）	行 11 质量探针 $\mu_{\mathrm{den}} \le 0 \to$ MC(ii)（D1 裁决，与 C3 定义 3.16(ii)/C5 G2/pilot rma `missing-caliber` 一致；探针笔录记 observed 值。真 OOV 保留给锚覆盖域真空，如 NX-Q7 的窗前请求；见标签归一注记）
E2	rma_q6：`refund_rate` 分子窗 2026-05、分母窗 2026-04（刚性双窗）	行 10 与 `same_valid_time_window` 等式不相容 → AM（双窗各自在覆盖内，行 8–9 不触发——报告序 OOV→AM 下仍首报 AM）
E3	rma：品类率 sku 粒度小胞不足，category_lv1 达标（sku→category 桥接，OVERALL 1.83% 一线）——**假想策略实例（如实标注）**：rma 域 v2026.06.23 无 `gov_disclosure_policy` 种子，按 D5 属 `ungoverned-disclosure` 域（无约束恒过披露门，证书如此标注）；本行 k_π 为构造值，仅示意机制，实例化依赖补种 rma 策略	行 12 粒度前沿上卷至 category_lv1 → ANSWER（证书记录粒度提升）
E4	email：呈现 `ContactEmail`（策略 `email_contact_email_mask`：mask=masked_email，γ=client_no，k=5）	行 14 闭包 $\mu^* =$ masked_email；client_no 粒度小胞不足则继续上卷或 DB
E5	domestic_newprod：“用最新版本回算历史”（显式版本承诺钉最新 $\neq$ 对角线 ver(T)，离对角读；SCD-2 锚 `dw_start_date/dw_end_date`）	治理图版本钉**不过滤** SCD-2 维表行的在效性（在效期属数据层，Env 为区间并，行 8–9 不因此触发）：GOLD-1 的错误发生在数据层 join 谓词（以最新在效行回读历史窗，错绑过期型号映射），属 C3 出码层的对角线纪律，非行 8–10 可拦截形状。修复后语义：缺省 $\circledast$ 走对角线 ver(T)（S-V），该错误读法在缺省下不发生；显式钉最新时算法照承诺执行，证书记录离对角标记 $p \neq$ ver(T)，正确性责任移交承诺方（GOLD-1 双时态正确性用例的教训）

O.1.6 4.5 定理

定理 4.18（合法性 soundness） 在 A4.1–A4.4 下，若 MLR-Compile 返回 ANSWER⟨plan, cert⟩，则其绑定点闭包 $\mathrm{cl}(\hat{b}, \mathrm{ctx}) \in L(q, \tau, G, \mathrm{ctx}, I)$，且 plan 是该闭包经 C3 出码的时点正确计划。

证明。 $\hat{b} \in$ Front 仅经行 14 加入，此前逐条建立成员条件：$v \in V_{\mathrm{adm}}$（行 1，S-V 与刚性承诺）；$\alpha \in A_{\mathrm{adm}}$（行 5，S-A）；规则约束 r_α：行 10 判定**完整规则谓词** P_{rule} 条款 (i)–(iv)（定义 4.8），含窗-粒度可表达与经 I 的锚对相容审计——这正是 4.4 测试用例（quality_voc `complaint_vs_rma_anchor_mismatch`）所要求的检查，仅查窗口等式字面时定理不成立；$w^* \in W_{\mathrm{cand}}$ 且通过覆盖检查（行 8–9，S-W：strict_member leg 刚

性时 $w_\ell^* = w_{0,\ell}$；hull leg 刚性时其窗承诺值按定义 4.8(ii) 的 hull-edge 裁剪豁免解释为 $w_{0,\ell} \sqcap \mathrm{hull(val)}$，故 w_ℓ^* 恰等于该承诺值、按裁剪窗作答并入证书且披露边缘裁剪——忠实性定义 4.8(ii)(iii) 因此在两类 leg 上均逐字成立）；Ω-可绑定性：行 8–11 联合覆盖 C3 定义 3.8 的可绑定条款——行 8–9 即窗带/覆盖条款（逐锚 `coverage_mode` 语义，D3：快照锚 hull 相交、裸日期集锚严格成员，与 C3 (b1) 统一），行 10 即规则谓词条款 (i)–(iv)，行 11 即分母质量条款 (b4)（通过即 $\mu_{\mathrm{den}} > 0$；D4 加宽下行 11 拒 $\mu_{\mathrm{den}} \le 0$，A4.4(iii) 非负测度下与引理 4.12(b) 一致）——未触发即可绑定，$\mathrm{EmitPlan}_{C3}$ 的时点正确性由 C3 定理（A4.1 前提）承接；披露合法性：$g \in U_{\min} \subseteq U$ 使引理 4.16 的三条款成立，掩码取 μ^* 由引理 4.15 合法，D1 分支的行归属约束由 cl 构造并入 plan 的范围谓词。故闭包满足定义 4.8 的全部成员条件。■

定理 4.19（极大性与确定性） 在 A4.1–A4.4 下：**(a)** Front 的每个元素在 $(L/\approx_D, \preceq_D)$ 中极大；**(b)** ANSWER 返回的 $\hat{b}$ 属于极大元集；**(c)** 返回代表元是 $\langle q, \tau, G, D, \mathrm{ctx}, I\rangle$ 的确定函数——其中 $\prec_{\mathrm{det}}$ 的三键（版本序、注册序数、ℓ_Γ）均已作为 G_v 的版本化声明成分入模（定义 4.11），确定性不依赖任何模型外输入（jsonl 行序、查询返回序）；改变 $\prec_{\mathrm{det}}$ 声明只改变返回哪个极大元，不改变极大元集与 ANSWER/REFUSE 判定。

证明。(a) 设 $q'' = \mathrm{cl}(\langle \alpha'', w'', v'', g''\rangle) \in L$ 严格支配某 $\hat{q} = \mathrm{cl}(\langle \alpha, w^*, v, g\rangle) \in \mathrm{Front}$。$\preceq_D$ 仅比较同锚同版本（定义 4.9），故 $\alpha'' = \alpha, v'' = v$。窗：$L$ 的成员窗 $\subseteq w^*$（引理 4.12(a) 与 S-W 的单候选），支配要求 $w'' \supseteq w^*$，故 $w'' = w^*$。粒度：严格支配则 $g'' \preceq_\Gamma g$ 且 $g'' \neq g$；由引理 4.16 等价刻画（$\mathrm{touched}_{\mathrm{D2}}$-限定的 Legal，D1 角色命中豁免已并入）$g'' \in U$，而 g 是 U 的极小元（推论 4.17），$g'' \prec_\Gamma g$ 与极小性矛盾。掩码由闭包派生相等。故无严格支配者。(b) 由 (a) 与行 16。(c) 遍历序（版本序、注册序、ℓ_Γ 序）与探针均为输入的确定函数；Front 的构造不引用 $\prec_{\mathrm{det}}$（行 4/7 的枚举顺序不改变集合内容，只改变发现顺序），故极大元集与行 15 的空判定对 $\prec_{\mathrm{det}}$ 不变，仅行 16 的代表选取随之变化。■

定理 4.20（拒答完备性） 在 A4.1–A4.4 下：MLR-Compile 返回 REFUSE 当且仅当 $L(q, \tau, G, \mathrm{ctx}, I) = \varnothing$。且拒答时原因表 CT 给出逐 slice 的空性见证，取值分解为 C3 的三类绑定层原因 $\{\mathrm{MC}, \mathrm{AM}, \mathrm{OOV}\}$ 与本章新增的披露层原因 DB（归类依 D1：注册层缺失与同窗分母质量空 $\mu_{\mathrm{den}} \le 0$ 归 MC；承诺-可用性不相容归 AM——含 κ 剔空 A_{adm} 与版本承诺剔空（行 2b，AM(i) 类比）；锚覆盖域真空**与版本轴缺位**（行 2a，缺省对角下 $\mathrm{ver}(T)\uparrow$——A4.1/C3 定义 3.14）归 OOV；报告序依 D2 为 OOV → AM → MC，与算法行序一致）；当 $D_v = \varnothing$ 时 DB 不出现，完备性还原为 C3 的绑定完备性（该域证书 disclosure 段标注 `ungoverned-disclosure`，D5）。

证明。（⇐，拒答的必要性）若算法返回 ANSWER，定理 4.18 给出 L 的一个成员，故 $L = \varnothing$ 时必返回 REFUSE。此方向的成立依赖行 10 判定完整规则谓词条款 (i)–(iv)：仅查窗口等式字面时，存在 $L = \varnothing$ 而返回 ANSWER 的种子级反例（4.4 测试用例：好绑定 slice 在行 9/11 失败、`anchor_mismatch` slice 于行 10 字面通过），修复后该缺口闭合。（⇒，拒答的正当性）设返回 REFUSE，则每个 slice (v, α)（含行 2a/2b 与 $A_{\mathrm{adm}} = \varnothing$ 的哨兵行）在 CT 中有原因；按报告序（D2）逐原因证明该 slice 对 L 无贡献：OOV（行 2a/9）：**版本轴缺位**（缺省对角下 $\mathrm{ver}(T)\uparrow$，T 某点无在位版本）时对角解析无从构造版本坐标，$X(q, \tau) = \varnothing$（C3 定义 3.14/G7 归 OOV）；覆盖检查失败（$w^* = \varnothing$，或刚性下某 strict_member leg 越出 val——按锚声明的 `coverage_mode`，D3）时 slice 内不存在忠实可绑窗（引理 4.12(a)）。AM（行 2b/6/10）：**版本承诺剔空**（钉不存在的版本号、$\tau.p$ 与 κ 互斥、或缺省下 κ 剔空 V_{res}）——版本承诺与可用版本不可同真，$X(q, \tau) = \varnothing$（AM(i) 类比，D1；注意此形状**非** OOV：可解析版本可存在，是承诺把它剔空）；κ 剔空 A_{adm}（承诺与全部注册绑定不相容，AM(i)，D1）；或刚性承诺与 r_α（完整规则谓词，含条款 (iii)(iv)）不相容——L 成员须同时满足承诺（忠实性）与 r_α，该 slice 内二者不可同真。MC（行 6/11）：$B_v[m] = \varnothing$ 时该 slice 无注册绑定可依（$X(q, \tau)$ 为空，MC(i)）；$\mu_{\mathrm{den}}(w^*) \le 0$ 时由引理 4.12(b)（A4.4(iii) 非负测度下即 $= 0$）一切忠实窗质量为零，Ω 按 A4.1 判 MC(ii)（D1/D4），无可绑定点。DB（行 13）：GrainFrontier 遍历终止且 $U_{\min} = \varnothing$；由推论 4.17 的遍历不变式（未命中即扩展全部覆盖后继、上集成员必经其某极小元被发现），$U_{\min} = \varnothing$ 蕴含 $U = \varnothing$，再由引理 4.16 的等价刻画（$\mathrm{touched}_{\mathrm{D2}}$-限定，D1 豁免下更细的角色分支合法改写亦被 U 捕获而非漏判），该 slice 在任何粒度都不产生 L 成员。诸 slice 取并即 $L = \varnothing$。分解陈述由各行触发条件的归属直接得出；$D_v = \varnothing$ 时 $\mathrm{touched} \equiv \varnothing$，$\mathrm{Legal}(g_0)$ 真空成立（非空披露条款无必需掩码），行 12 至多一步即返回 $\{g_0\}$，DB 不可达。■

定理 4.21（复杂度上界，按格高度/绑定规则数参数化） 在 A4.1–A4.4 下，MLR-Compile 的数据探针次数至多

$$\nu\, a\left(2 + \delta \cdot \min(|{\uparrow}\, g_0|,\ W_\Gamma\,(h_\Gamma + 1))\right)$$

每次探针为绑定人口上的一条聚合查询；符号工作量为 $O\big(\nu\, a\, [\, W_\Gamma(h_\Gamma+1)\,(\delta\,|\mathrm{Att}(q)|+c_\vee)+|E_{\mathrm{val}}|\,]\big)$，其中 $c_\vee$ 为掩码格并运算成本、E_{val} 为有效期端点集。在 A4.4 下（实例有限、窗代数封顶、测度非负——诸探针可求值且 $w_0 \sqcap \mathrm{Env}$ 判空可计算；无此三条，连续时间/无限域下终止性与本界不成立），MLR 的判定与构造在 $\langle \nu, a, W_\Gamma, h_\Gamma, \delta, |\mathrm{Att}| \rangle$ 上多项式。

证明。行 10 每 slice 至多一次锚对审计探针（条款 (iv) 的有效日期集对称差查询；条款 (i)–(iii) 为符号判定）；行 11 每 slice 一次 μ_{den}；GrainFrontier 访问节点数 $\le \min(|\uparrow g_0|, W_\Gamma(h_\Gamma+1))$（推论 4.17、Mirsky 划分），每访问节点对每个触及策略至多一次 $\mathrm{SUPPMIN}_\pi$，触及数 $\le \delta$；slice 数 $\le \nu a$。符号侧：每访问节点做粒度条款比较与逐属性掩码并（$\delta|\mathrm{Att}(q)|$ 次比较、$c_\vee$ 并），Env 裁剪扫描端点 $|E_{\mathrm{val}}|$（A4.4(ii) 保证端点有限可表示）。相乘并求和即界。■

推论 4.22（链式粒度格的对数探针） 若 $\uparrow g_0$ 在 Γ_v 中为链（如单轴时间粒度 day→week→month→quarter），则每 slice 的粒度探针可降为 $O(\delta\,\lceil \log_2(h_\Gamma+2) \rceil)$：Legal 沿链是单调谓词（引理 4.16 的上集在链上即阈值形），标准阈值二分适用；空性判定另加链顶一次评估。

证明。链上上集由阈值元刻画：$U=\{g : g \succeq g^\dagger\}$ 或 $U=\varnothing$。二分不变式维持”下端不合法、上端合法或未证伪”；$O(\log)$ 步收敛到 $g^\dagger$ 或以链顶不合法证 $U=\varnothing$。每步一次 Legal 评估（$\le \delta$ 探针）。■

命题 4.23（一维退化：DELTAGOV 的 version-serializability） 考虑退化设定 Δ^0：(i) valid-time 轴退化为单点 $\{now\}$（各锚 $\mathrm{Env}=\{now\}$，$w_0=\{now\}$ 刚性）；(ii) $\Gamma_v=\{g_0\}$；(iii) $D_v=\varnothing$；(iv) 指标为分子/分母双 leg；(v) **版本识别公设**：把 DELTAGOV 的数据/物化版本函数 $v(\cdot)$ 与本章治理语义图版本识别为同一坐标——这是**额外公设**而非退化结论：本章的 v 钉治理图，实例 I 由快照一致性底座另行钉住（G-7），一般设定下二轴独立（图版本钉不过滤 SCD-2 数据行在效性，见 E5），同一化仅在 Δ^0 的单底座设定下声明成立。则：

(a) $X(q,\tau)$ 的每个绑定点形如 $\langle \alpha, \{now\}, v, g_0 \rangle$，其出码计划把全部 leg 读在同一版本 v 上，即满足 version-serializability 条件 $v(r_{\mathrm{num}})=v(r_{\mathrm{den}})$（DELTAGOV Eq. 8 的语义形）；

(b) 反之，设单版本读满足该条件，**且满足注册前提**：$m \in M_v$、$B_v[m] \neq \varnothing$、其两 leg 的锚均已在 A_v 注册且所用绑定的规则谓词审计通过——则该读对应 Δ^0 中某绑定点的闭包（版本坐标取其读版本），此对应在 $\approx_D$ 意义下双向。**无注册前提时反向不成立**：读两个未注册为任何锚/绑定的表构成的比率满足 $v(r_{\mathrm{num}})=v(r_{\mathrm{den}})$，但 $X(q,\tau)$ 在该 slice 为空（S-A 不做锚修复），MLR 判 MC，无原像；DELTAGOV 无注册前置，故本对应仅在**注册闭包内**双向；

(c) MLR-Compile 在 Δ^0 上退化为版本选择器：REFUSE 当且仅当 $V_{\mathrm{adm}}=\varnothing$，或**对每个** $v \in V_{\mathrm{adm}}$：$A_{\mathrm{adm}}(m,v,\kappa)=\varnothing$（$B_v[m]=\varnothing$ 判 MC(i)、κ 剔空判 AM(i)——D1 裁决，行 6）或该 slice 单点窗 $\mu_{\mathrm{den}} \le 0$（判 MC(ii)）；否则按 $\prec_{\mathrm{det}}$（版本较新优先）返回单一版本读。

证明。(a) 绑定点类型把版本坐标定义为整点共享的单坐标，出码沿该坐标读全部 leg，等式即类型不变量在公设 (v) 识别下的直接后果；(ii)(iii) 使粒度前沿一步返回 $\{g_0\}$、掩码闭包为 id。(b) 给定读版本 v 与两 leg 读集，由注册前提取 $\alpha \in B_v[m]$（其两 leg 锚注册且审计通过）构造 $\langle \alpha, \{now\}, v, g_0 \rangle$；两方向的复合恒等于 $\approx_D$ 代表——构造中 α 的来源即注册前提，删去前提则构造无据。(c) 行 4 对每个 $v \in V_{\mathrm{adm}}$ 逐 slice 判定，行 12 之后的分支在 (ii)(iii) 下不可达（$D_v=\varnothing$ 时行 12 一步返回 $\{g_0\}$），行 4 的版本序即 $\prec_{\mathrm{det}}$ 第一键。■

范围限定（诚实边界）：本命题复现的是 DELTAGOV 读一致性条件的**语义形**——“一次读落在同一版本”；其并发控制协议（增量物化、提交原子性）属于物理底座，不在本退化的表述范围内。另一诚实边界：**治理语义图版本 $\neq$ 数据/物化版本**——公设 (v) 的识别在一般设定不成立（E5：图版本钉不过滤 SCD-2 数据行在效性），二轴的联动是系统章的接口义务（G-7），本命题的”严格对应”以该公设为前提而非由退化推出。这与该文自陈的 producer 定位一致（“not a new semantic-layer compiler”），本框架在其上方做语义编译（consumer），把其保证用作探针-执行快照一致性的底座（4.4 注记、缺口 G-7）。

O.1.7 4.6 适用域刻画与开放缺口

本章各定理在 4.2 声明的改写语义与 A4.1–A4.4 内成立；下列条目刻画该适用域的边，并列出需要作者裁决的建模选择（编号与 StructuredOutput 缺口清单一致）。按 CKE 教训（LC-b28447ce），以下写作”本抽象的适用域”，不写作贡献让渡。

- **G-1（窗口议价扩展）**：S-W 排除数据驱动的窗收缩；注记 4.14 表明放开后 k-合法性沿窗不单调、极大元依赖逐胞出生点。柔性不等多 leg 窗的”交集等式修复”同属此扩展（定义 4.7 S-W 已显式排除）。扩展方向：临界窗端点枚举 + (w,g) 联合反链的极大性理论；本章不予承诺。

- **G-2（刚柔承诺的语用判定）**：κ 由 NL 解析给出，本章作输入假设；其解析可靠性属实证章（pilot 的 ANCHOR_OVERRIDE 正则即其退化实现），边界表述需作者确认。
- **G-3（保真序 vs 可导出序）**：scoped_ratio 的粗化答案不可由细粒度率值导出，$\preceq_D$ 取保真序是显式选择；若作者要求可导出性背书，可加度量子类上可给强化命题，率值类须改述。
- **G-4（掩码不参与跨粒度比较）**：粗而未掩 vs 细而受掩在纯信息论意义上不可比；本章令掩码由闭包派生、退出比较维度。替代序（用户偏好可调权重）会改变极大元集，需作者裁决缺省。
- **G-5（空披露即拒，S-M2）**：闭包送刚性呈现属性（定义 4.6 呈现坐标 S_{rig}，正式定义）至 $\top_M$ 时判 DB 而非返回全抑制空壳；反向选择会使"ANSWER 恒可返回"，拒答完备性定理相应弱化。
- **G-6（A4.3 粗化相容性）**：病态反例（上卷列本身受更严策略，或桥接列新进原样呈现集——A4.3(ii) 形状）被排除；存在披露策略的两域（aibuy 3 条 / email 4 条，共 7 条）核查满足，其余三域真空成立（缺席域按 D5 取无约束恒 ANSWER、证书 disclosure 段标注 `ungoverned-disclosure`）。若生产配置出现该形状，引理 4.16 的上集结构失效，粒度搜索退化为全格枚举（复杂度上界仍成立，极小前沿刻画不再成立）。
- **G-7（探针-执行快照一致性）**：TOCTOU 义务显式委托给版本底座（DELTAGOV version-serializability）；consumer 侧的接口义务（证书须钉快照版本）在系统章形式化，本章仅陈述义务。
- **G-8（极大元全集枚举）**：本章保证输出某极大元（及逐 slice 前沿）；枚举全部极大元的输出规模与复杂度未刻画。
- **G-9（规则语言边界）**：r_α 的规则**类型**现限于窗口等式类（`same_valid_time_window`），但其判定为完整规则谓词条款 (i)–(iv)（定义 4.8）——规则字面相同不免除锚对审计：quality_voc 的 `complaint_vs_rma_anchor_mismatch` 注册行字面同为 `same_valid_time_window` 而条款 (iv) 失败（对称差 57），已由行 10 覆盖（4.4 测试用例）；"五域种子暂无" 仅对**滞后/位移形状**的注册实例成立。滞后/位移类绑定（如分母滞后一期的 cohort 口径）需扩展 RuleSat 与引理 4.12。
- **G-10（跨章符号与拒答标签对齐——已裁决收窄）**：归类与报告序已钉定（D1：同窗分母质量空 $\mu_{\text{den}} \leq 0$ 归 MC(ii)、κ 剔空归 AM(i)、OOV 只留锚覆盖域真空、拼写统一 OOV；D2：双故障报告序 OOV $\to$ AM $\to$ MC）。实现迁移**已完成**（随仓验证）：domestic_newprod/aibuy 编译器的空同窗分母标签已改标 `missing-caliber`（MC(ii)），守卫序已对齐行 8–11（OOV $\to$ AM $\to$ MC）；金标（AIBUY-Q6/EMAIL-ASOF-06/rma_q5=MC(ii)，NX-Q7=OOV 保持）与五域编译器实跑逐题一致；映射表随证据保留（4.4 标签归一注记）。本缺口闭合为对账记录。
- **G-11（D1 角色分支的掩码语义）**：角色分支下掩码义务是否随行归属豁免一并豁免，种子两读均通（aibuy 所有者仍 generalize / email 自有行无掩码）；本章取" 掩码不豁免"，需作者对齐产品语义。

与骨架承诺的对账（C4 → S1 Contribution 3）

S1 承诺	本章落点
披露策略门下的最大合法查询改写算法	定义 4.10 + 算法 4.4 + 定理 4.18/4.19
复杂度分析	定理 4.21 + 推论 4.22（以 $h_\Gamma, W_\Gamma, a, \nu, \delta$ 参数化）
不可改写时结构化拒答 + 可审计时点证书	定理 4.20 + 定义 4.10 证书字段（C5 接口）
DELTAGOV 单版本读一致性为一维退化	命题 4.23（含诚实范围限定）
三类不可绑定与拒答语义对接	定理 4.20 的原因分解 $\{\text{MC}, \text{AM}, \text{OOV}\} \cup \{\text{DB}\}$ + A4.1

P Frozen Theory Chapter C5: Point-in-Time Certificates

Frozen source `theory/C5_pointintime_certificates.md` (SHA-256 above), reproduced content-identically; Chinese. The body's §5 and Appendix D.4 are its English rendering.

P.1 C5 · 时点证书（Point-in-Time Certificates）与独立校验

> 所属：PP-5744d699 / S1 骨架 §5（System）与 Claims 表 C5「证书可独立校验」。定框纪律：abstraction-first——本篇先给证书的**内容物定义与不变量**，再给**校验关系与完备性/健全性定理**，最后才把 pilot 编译器接口的落地映射降格为实现细节（§6）。认证循环红线（梁老师意见 3）：校验器**不与编译器共享实现**；信任基 = 检查关系规范 + 图版本 + 原始数据，编译器不在信任基内（§3.2）。

P.1.1 0. 符号与前置约定

本篇沿用（并在兄弟篇 C2–C4 定稿前先行固定）如下记号；若与 C2/C3 定稿冲突，以先冻结者为准并在合并时统一：

记号	含义
$\mathbb{T}$	valid-time 论域（日期/日期区间的全序集）
$\mathcal{W}$	窗代数（window algebra）：$\mathbb{T}$ 上的半开区间 $[l, h)$ 与有限日期集两类元素，配交/含于运算
$\mathcal{G} = \{G_v\}_{v\in\mathcal{V}}$	版本化治理语义图族；G_v 为版本 v 下的一次 typed-commit 快照，含 `gov_valid_time_anchor` / `gov_temporal_binding` / `gov_caliber_routing` / `gov_disclosure_policy` 等治理关系
$A(G_v)$	G_v 中的 valid-time 锚集合；锚 a 携带（语义对象，有效时间列，有效日期集 $V(a) \subseteq \mathbb{T}$）
D	原始数据实例（本文语境下即 warehouse 中的事实/汇总表；$V(a)$ 由 a 的语义对象在 D 上取值导出）
q, T	分析问题与其声明时点（as-of T）；T 经窗解析得到请求窗 $W_T \in \mathcal{W}$
$\mathsf{Bind}(q, T, G_v, D)$	C3 的时态绑定编译；输出绑定成功的查询计划，或三类失败之一 $r \in \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}\}$（missing-caliber / anchor-mismatch / out-of-validity）。经 C4 披露阶段后，总拒因集扩为 $\{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}, \mathsf{DB}\}$
DB	disclosure-blocked（C4 第四拒因：披露约束下最大合法改写为空，$U_{\min} = \varnothing$，S-M2）
ctx，h_{ctx}	请求语境（角色作用域等，C4 定义 4.5）及其哈希；披露合法性是实例相关且**语境相关**的，故语境须入证书
$\text{Cov}_v(a)$	锚的有效覆盖域（C3 定义 3.3）；由锚声明的 `coverage_mode` 导出，取值 `hull` | `strict_member`（裁决 D3：**逐锚版本化属性**，属 G_v 内容物；缺省——快照锚 `hull`、裸日期集锚 `strict_member`）
$\mathsf{eval}_D(p)$	探测查询 p 在 D 上的求值

快照假定（Snapshot Assumption, SA）：编译与校验读取同一 (G_v, D) 快照；跨快照并发演化由版本基座（DELTAGOV 类 producer）负责，本篇不重复其一致性协议。SA 是 §4 两条定理的显式前提。**旧证书地位（裁决 D6）**：图推进到 v' 之后——含回滚情形，回滚即以新版本号**重提交**（公理 5.0(iii)）——已签发证书**按坐标** (T, v) **永久有效**：证书是对 (T, v, D) 坐标下裁决的断言，版本演化不改写历史坐标（append-only

版本轴保证 ν 永可解析)，故旧证书无 revalidation 义务；审计方如需"当前版本下的答案"，应在 v' 上重新编译得新证书，而非改判旧证书。

前提导入（承 C3）：§3.3 的定理 5.9/5.10 与命题 5.11 显式导入 C3 的公共前提 **(A1)–(A6)**（扩展有限可枚举；窗算术可计算、粒度族有限且以「日」为公共细化且成 meet-半格；$\mathcal{V}$ 非空有限全序、commit map 可计算严格单调、版本轴 append-only；绑定规则库可判定；**(A5)** 逐角色窗导出以带 ↑ 哨兵的全函数呈现——as-of 不可解析是一次求值而非停机判定，定义 5.4 I2′(b) 退化变体的判定依据即此；**(A6)** $\mathbb{T}$ 为有效呈现、序可判定的离散时间域——本篇早稿「时点/窗端点取自可有效表示的可比较子域（日期）」的输入表示假定已升格为 C3 正式假设 A6，此处引用而不再另行陈述)，以及 C3 定义 3.2 的日历窗代数封顶。此前本篇仅冻结 SA——无这些有限性/可表示性前提的裸陈述在连续或无限时间域上不成立，本篇定理的可终止性与见证生成（经 C3 命题 3.18 的判定程序）全押在其上。

公理 5.0（commit 同一性）：(i) typed-commit 标识**全局唯一且新鲜**——不同提交必得不同标识；(ii) (domain, graph_version) 不得重新在位提交——版本号不复用，版本轴 append-only；(iii) 回滚即**重提交**（裁决 D6）：回滚以新版本号、新 commit 落地（`rollback_of` 仅为注记边，不改写历史行），commit 序列 append-only 且标识唯一；旧证书按坐标 (T, v) 的永久有效性（§0 快照假定）以本条为根。无本公理，$\mathrm{ver}(T)$ 的函数性与 V1 的唯一解析在回滚场景无根（C3 定义 3.5/引理 3.5′ 的良定性依赖）。**现状注记**：实查五域种子 `gov_semantic_graph_version.committed_at` ≡ 版本标签（如同为 `v2026.06.23` 双写），c 坐标现为 v 的复写、无区分力——种子须补真 commit id；C3 缺口 G1（committed_at 非时戳）须升格为本公理的前提（跨章依赖）。

P.1.2 1. 证书内容物：定义与不变量

1.1 绑定四元组 定义 5.1（绑定四元组 binding quadruple）一次编译决策的绑定四元组为

$$B \;=\; \langle\, \alpha,\ \nu,\ \rho,\ \delta \,\rangle$$

其中：

1. **锚指派 α（anchor assignment：anchor id + 窗）**：从 q 的度量角色集 $R(q)$（分子 numerator、分母 denominator、原子度量 atomic；delta 型问题按期展开为角色 × 期）到二元组的有限映射

   $$\alpha : R(q) \rightharpoonup A(G_v) \times \mathcal{W}, \qquad \alpha(\mathrm{role}) = (a,\, W),$$

   W 为该角色在本次决策中被解析到的 valid-time 窗。
2. **图版本钉 ν（graph version pin）**：$\nu = (\mathrm{domain}, v, c\,[, \mathrm{offdiag}])$——业务域、语义图版本号（如 `v2026.06.23`）、typed-commit 标识 c。c 须为满足公理 5.0 的**真 commit 标识**（唯一定位一次提交；现种子 `committed_at` 与版本号双写、无区分力，见公理 5.0 现状注记）。ν 唯一定位 G_v。**离对角标记** offdiag（**C4 S-V/E5 明令字段，本证书层落位**）：当钉定版本偏离双时态对角线——$p = \nu.v \neq \mathrm{ver}(T)$（显式钉旧版回算，或钉最新版回算历史时点）——证书**必须**携带 $\mathrm{offdiag} = (p, \mathrm{ver}(T))$ 承诺标记；标记缺席即承诺对角读 $\nu.v = \mathrm{ver}(T)$，V1 据此核验（无标记的离对角证书 REJECT）。无此字段，"钉最新回算历史"在制品层零痕迹（GOLD-1 镜像攻击面）。
3. **口径路由路径 ρ（caliber routing path）**：`gov_caliber_routing`@v 中边的有限序列

   $$\rho = \langle e_1, \ldots, e_k \rangle, \quad e_i = (\mathrm{caliber\_key}_i,\ \mathrm{src} \!\rightarrow\! \mathrm{dst},\ \mathrm{join\_keys}_i,\ \mathrm{attribution\_alignment}_i),$$

   要求相邻边首尾口径衔接（$\mathrm{dst}_i = \mathrm{src}_{i+1}$）。原子度量允许 $\rho = \langle\rangle$。
4. **披露决策 δ（disclosure decision + 依据策略 + 语境哈希 + 粒度 + 掩码闭包）**：$\delta = (\mathrm{dec}, \Pi, h_{\mathrm{ctx}}, g, \mu^*)$,

   $$\mathrm{dec} \in \{\mathsf{ANSWER},\ \mathsf{REWRITE}(q'),\ \mathsf{REFUSE}(r)\}, \qquad \Pi \subseteq \mathrm{PolicyIds}(G_v),$$

   Π 为本次决策实际生效的 `gov_disclosure_policy.policy_id` 集合（REWRITE 时 q' 为 C4 的最大合法改写结果）；h_{ctx} 为请求语境 ctx（角色等，C4 定义 4.5）的哈希——披露合法性语境相关（角色分支、k-小胞条款依实例探针），无 h_{ctx} 则 V5 对角色作用域策略不可计算（C4 定义 4.10 的 ctx-hash 字段在证书层的对应物）。g 与 μ^* **（C4 定义 4.10 明令字段，"供 C5 独立校验器复核"的落**

袋处)：g 为本次产出的**呈现粒度**（REWRITE 经粒度提升后的实际呈现粒度，C4 E3 的”证书记录粒度提升”即记于此）；μ^* 为本次生效的**最小合法掩码闭包**（被掩码/脱敏的呈现属性集，C4 E4 的 `masked_email` 即记于此）。二者是 V5 在治理域字面重演粒度条款（$\gamma_\pi \preceq g$）与掩码条款（$\mu^* \succeq \mu_\pi$）的唯一依据——缺位则该两类条款无字面重演对象。无披露策略登记的域（D5 缺席语义的推论）：g 取请求原粒度、$\mu^* = \varnothing$，与 `ungoverned-disclosure` 标注共存。**策略缺席域（裁决 D5）**：域在 v 下无 `gov_disclosure_policy` 登记时，缺席语义为无约束、恒 ANSWER（$\Pi = \varnothing$、取空语境哈希），且 δ **必须**携带 `ungoverned-disclosure` 标注——该标注是 δ 的内容物而非可选注释，使”未治理”与”检查通过”在证书层可区分（V5 缺席规则据此重演，§6.3 落地）。

1.2 时点证书　定义 5.2（时点证书 point-in-time certificate） 问题 (q, T) 的时点证书为

$$C \;=\; \big(q,\; T,\; B,\; \mathrm{out}\big)$$

按 δ.dec 分两型：

- **应答证书（answer certificate）**：dec $\in \{\mathsf{ANSWER}, \mathsf{REWRITE}(q')\}$，out $= s$ 为产出的查询文本（SQL）。REWRITE 情形另携带**裁剪痕迹（cut trace）**——C4 定义 4.10/4.11 的字段在证书层落位：(i) 保留/被裁 slice 集与**逐 slice 原因表** $\mathrm{CT} : \mathrm{slice} \to \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}, \mathsf{DB}\}$；(ii) 确定性选择序 $\prec_{\det}$ 的键声明（含 ℓ_Γ——$\preceq_\Gamma$ 声明线性扩张的标识）。裁剪痕迹使”部分回答”对下游不可伪装为”全量回答”，且返回代表元的选择可审计。
- **拒答证书（refusal certificate）**：dec $= \mathsf{REFUSE}(r)$，$r \in \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}, \mathsf{DB}\}$（DB=disclosure-blocked，承 C4），out $= w$ 为**见证（witness）**。**CT 形制（裁决 D1；复合序钉死 slice-major）**：拒答证书与 C4 的逐 slice 原因表对齐承载——多 slice 问题的 r 与 w 取**首失败 slice**，复合方向定为 **slice-major**：先按 $\prec_{\det}$ 的 slice 序取首失败 slice，再在该 slice 内按 V6b 冻结守卫报告序 $\mathsf{OOV} \to \mathsf{AM} \to \mathsf{MC}$ 取 r（与 C4 算法的自然产物一致；反向的 guard-major——跨 slice 先取守卫类——在 quality_voc `defect_rate` 双绑定行上与 slice-major 可分离（$\mathrm{slice}_1 {\mapsto} \mathsf{MC}$、$\mathrm{slice}_2 {\mapsto} \mathsf{AM}$ 时二读各报 MC/AM），故复合序必须钉死且入规范；**C4 定义 4.10 的同句式表述同批同改，跨章注记**）。另附**完整逐 slice 原因表** $\mathrm{CT} : \mathrm{slice} \to \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}, \mathsf{DB}\}$。**非首行重演义务（显式声明）**：V6b 只重演选定首失败 slice 的 r 与 w（含其守卫否定）；CT 的非首行是**审计注记（未验断言）**，其逐行重演与 REWRITE cut_trace 的极大性同列为本篇**非目标**（§5 非目标声明同款——显式声明以免非首行被下游读作已验结论；如立项逐行审计，每行须按其 reason 附同型见证并入 V6b 循环）。单 slice 问题 CT 退化为单行。w 按 r 分型：

定义 5.3（四类拒答见证）

原因 r	见证 w 的内容物	判定语义
MC missing-caliber	二择一：(i) **路由缺失见证** $w = (m, \kappa)$：度量 m 在 `gov_caliber_routing`@v 中查键 κ 无可重算路由（含 `dst_caliber='none'` 的 reference-only 情形）；(ii) **空窗见证** $w = (m, p, W)$：分母人口探测查询 p（对 $\alpha(\mathrm{den}) = (a_d, W)$ 的单扫描聚合）满足 $\mathsf{eval}_D(p) \in \mathcal{Z}$，空窗判定集**定稿取加宽形**（裁决 D4，对齐 C3 定义 3.16(ii)/G8）：$\mathcal{Z} = \{\varnothing,\ \mathrm{NULL}\} \cup \{z : z \le 0\}$——见证谓词即 $\mu_{\mathrm{den}} \le 0$（或空/NULL；退款冲销负质量计入空窗），探针笔录记录 observed 原值	该口径下不存在合法分母（rma 536-SKU 母题：同月无销量分母，不以异月充数）；裁决 D1：**空同窗分母一律归本情形** MC**(ii)**，不归 OOV

原因 r	见证 w 的内容物	判定语义
AM anchor-mismatch	$w = (c, \text{payload})$：**失败条款标号** $c \in \{(\text{i}), (\text{ii}), (\text{iii}), (\text{iv})\}$（对齐 C3 定义 3.8 svw 四条款）加逐条款载荷——(i) **改锚型**：(role, $\bar{a}$, a^*)，被声明锚 $\bar{a}$ 与规定锚 a^* 及查键断言 $\bar{a} \neq a^*$（改锚到**未注册列**的情形需 C3 定义 3.6/3.7 扩表达，跨章依赖）；(ii) **窗对型**：(a_n, W_n, a_d, W_d)，$W_n \neq W_d$ 的请求窗对本身；(iii) **粒度不可表达型**：(binding_id, g^{cmp}, W, 失配边界点)——绑定行声明的**比较粒度** g^{cmp}（缺省 $g_n \sqcap g_d$，最粗公共细化；C3 定义 3.8(iii) 的判定对象是 $W \in \mathcal{W}_{g^{\text{cmp}}}$），加 W 的某极大区间与 g^{cmp} 粒元边界不对齐的具体边界点断言；重演判 $W \notin \mathcal{W}_{g^{\text{cmp}}}$，**不得**按单锚粒度 g_a 粒元判边界对齐——后者是严格更强测试（g_n= 周、g_d= 月、W 单日时 C3 判 (iii) 通过而单锚载荷可出具假 (iii) 见证），旧载荷 (a, W, g_a) 废止；(iv) **对称差型**：(a_n, a_d, x)，判别元 $x \in V(a_n) \triangle V(a_d)$，重演模式按绑定行声明的 adm 检查变体（`adm_check_mode`，见 V3）执行	绑定规则（如 `same_valid_time_window`）被违反（quality_voc：投诉锚 36 个锚日期（周透视快照日）vs 工单日锚 35 个工单日，对称差 **57**（冻结仓复演：36+35−2×7）；rma q6：分子窗 2026-05 vs 分母窗 2026-04）
OOV out-of-validity	$w = (a, W_T, \pi_V)$：锚 a、请求窗 W_T，及有效集侧写 π_V（$\text{Cov}_v(a)$ 的边界或成员探测查询），满足 $W_T \cap \text{Cov}_v(a) = \varnothing$（或 as-of 点 $\notin \text{Cov}_v(a)$）。$\text{Cov}_v(a)$ 按锚声明的 `coverage_mode` 计算（裁决 D3：逐锚版本化属性、G_v 内容物；取值 `hull` | `strict_member`，缺省快照锚 `hull`、裸日期集锚 `strict_member`），校验器按声明的 mode 重演；不可解析 as_of 的退化变体见定义 5.4 I2′(b)	请求时点落在锚的有效覆盖域之外；裁决 D1：OOV 只承载**锚覆盖域真空**（NX-Q7 实查 launch 表 min(dt)=2025-09-15 > as-of 2025-08-31，属真 OOV），覆盖内同窗分母零质量归 MC(ii)
DB disclosure-blocked	$w = (\text{CT}_{\text{DB}}, \Pi_{\text{blk}}, \text{probes})$：判 DB 的 slice 标识集、触发策略 id 集，及 **POP/SUPPMIN 探针笔录**（C4 引理 4.12/4.13 的读数与快照版本），共同断言 $U_{\min} = \varnothing$（C4 S-M2：最小合法掩码闭包把刚性呈现属性送至 $\top_M$，或无粒度满足小胞条款——空披露视同拒答）	披露约束下不存在非空合法改写（C4 定义 4.5/4.8 语义）

见证的设计原则：**每个见证都是一个可在** (G_v, D, ctx) **上独立重演的有限判定**——或是一次治理表查键，或是一次单扫描探测查询加其结果断言。重演模式（AM 的 adm 变体、OOV 的 coverage_mode、MC(ii) 的判定集 $\mathcal{Z}$）由**证书与** G_v **中的机器可读声明**唯一确定，不依赖任何自由文本。见证使拒答**可证伪（falsifiable）**：错误的拒答无法构造出能通过重演的见证。

1.3 良构不变量　定义 5.4（证书良构 well-formedness） 证书 C 良构，当且仅当：

- **I1（版本封闭）**：α 中的锚、ρ 中的边、Π 中的策略，全部属于 ν 所钉的 G_v——证书不引用钉外对象；
- **I2（角色覆盖，按裁决类分型）**：应答证书与 AM/MC(ii)/DB 拒答证书：q 的每个度量角色在 α 的定义域内；比率型度量的分子/分母角色各有锚窗。三类拒答退化变体**显式放宽**（否则相应拒答无良构证书可发）：
  - **I2′(a)(MC(i) 无锚可指)**：涉事度量未注册或 reference-only（如 quality_voc `complaint_rate_global_recompute`，路由 `dst_caliber='none'`），G_v 中不存在可指派的 (a, W)——α 对该角色允许**显式空指派**，但证书须记录角色与度量 id（空指派是内容物，不是缺字段）；
  - **I2′(b)（OOV 之 as_of 不可解析）**：as_of 串不可解析为 $\mathcal{W}$ 元素时（domestic_newprod `_as_of_end` 抛错型），α 对应角色允许**无窗**，见证携带原始 as_of 串与解析失败断言（判定依据：C3 A5 的全函数呈现——$\omega_r(T) = \uparrow$ 是一次显式求值结果，非停机判定）；
  - **I2′(c)（$\mathrm{ver}(T)\ \uparrow$）**：$T$ 早于版本轴首个 commit 时 ν 不可构造——证书退化变体携带 (T, 版本轴探测断言 $\mathrm{ver}(T)\uparrow$)，V1 改验版本函数在 $\mathcal{V}$ 上确无解。三变体均计入定理 5.9 的量词范围（良构证书全域）；
- **I3（路由衔接）**：ρ 相邻边口径衔接，且 ρ 的整体 src/dst 与 q 的度量口径及分母人口口径一致；
- **I4（决策—产出匹配）**：dec 为应答型 $\Leftrightarrow$ out 为查询文本；dec = REFUSE(r) $\Leftrightarrow$ out 为 r 型见证。

注记 5.5（与 DELTAGOV 的一维退化） 取 $A(G_v)$ 中的每个锚为平凡锚（窗 $= \mathbb{T}$ 全域）、无绑定规则、$\rho = \langle\rangle$、披露恒 ANSWER，则 B 退化为 $\langle\nu\rangle$——单独一个版本钉。此时证书的检查关系（§3）退化为”本次读取自单一版本 v 且版本内一致”的检查，即 DELTAGOV 类 version-serializability 的证书形态。这与 S1 冻结的一维退化对应一致：单版本读一致性是本框架在时间维坍缩后的一维退化情形；本篇证书在其上加的正是 valid-time 轴（α）、口径轴（ρ）与披露轴（δ）。两点收紧（对抗审查④）：(1) 退化证书的 ν 须钉**最新在位版本**——DELTAGOV 的旧版本 mart 仅审计保留、不可寻址查询，允许钉任意历史 v 会认证其语义中不存在的读，越出对应关系；(2) 在 SA 下，“读自单一版本且版本内一致”的残余检查内容接近前提本身——本退化对应是**形状嵌入**，不是恢复一条可被违反的保证（DELTAGOV Eq.8 约束的是并发版本推进下的可违反执行，其内容在 SA 掉并发后平凡化），措辞以此为准。另：本注记的嵌入（窗 $= \mathbb{T}$ 全窗）与 C4 命题 4.23 的嵌入（valid-time 轴退化为 $\{now\}$ 单点）不同，唯一性待合并裁决（缺口 G8）。

P.1.3　2. 证书要回答的三个问题（设计不变式）

在给出校验器之前，先固定证书作为**制品（artifact）**的语义义务，后续定理围绕它们展开：

- **Q1 绑定于何（bound-to-what）**：答案/拒答是相对哪个 (T, v) 坐标作出的——由 α, ν 回答；
- **Q2 何以合法（why-legal）**：分母人口与归因对齐走的是哪条治理路由、披露上依据哪些策略——由 ρ, δ 回答；
- **Q3 何以可证伪（why-falsifiable）**：拒答不是”黑箱不作为”，而携带可重演的失败见证——由 w 回答。

三者合并即：**证书把编译期的每一次裁决，转译为第三方可在 (G_v, D, ctx) 上重演的有限检查集**。

P.1.4　3. 独立校验器规范

3.1 检查关系　定义 5.6（检查关系 Chk） $\mathsf{Chk}(C, G_v, D, \mathrm{ctx})$ 为以下检查项的合取（G_v 由 $C.\nu$ 解析；ctx 为审计方持有的请求语境，核对 $\mathrm{hash}(\mathrm{ctx}) = \delta.h_{\mathrm{ctx}}$——无披露策略登记的域取空语境，签名退化为 (C, G_v, D)）：

检查项	内容	依赖
V0 问题 $\leftrightarrow$ 证书对应（α 重导出核对）	校验器从 (q, T, G_v) **独立重导出** $\alpha_{q,v}$：请求窗经独立实现的窗解析 $\omega_r(T)$（§3.2(3) 的独立窗算术），q 各度量角色的**规定锚**经 $\beta_v/A(G_v)$ 治理表查键得 a^*_{role}，角色窗按绑定规则由 W_T 规定；逐角色核对证书 $\alpha(\mathrm{role}) = (a, W)$ 满足 $a = a^*_{\mathrm{role}}$ 且 W 即重导出的角色窗。**豁免仅凭证书记录**：题面改锚声明 $\bar{a}$ 须以机器可读记录在证书内（否则按 AM(i) 处理），REWRITE 的窗/slice 收窄须凭 cut_trace 相应行。任一失配 $\to$ REJECT。无本项，证书可携” 语法自洽但非 q 的”α（错月窗、假窗对、探针窗挪移——F1a/F1c/F1d 伪造族），V2–V6 全部对伪 α 空转（§3.2(2)“从 (G_v, D) 重查重演” 红线在检查表的第一落点）	G_v（独立窗算术）
V1 版本钉解析	ν 在 `gov_semantic_graph_version` 中依公理 5.0 可解析到**唯一** typed-commit；G_v 各治理表在该版本下可读。**对角性核验**：独立解析 $\mathrm{ver}(T)$ 并核 $\nu.v = \mathrm{ver}(T)$；不等时证书须携显式离对角承诺标记 $\mathrm{offdiag} = (p, \mathrm{ver}(T))$、$p = \nu.v$（定义 5.1，C4 S-V/E5 明令）——无标记的离对角证书 REJECT（封”T=now、ν 钉旧版/钉最新回算历史” 的零痕迹证书，GOLD-1 镜像）。I2′(c) 变体证书（$\mathrm{ver}(T)\uparrow$）改验：版本轴上 $\mathrm{ver}(T)$ 确无解。治理表**整表缺席**的语义（缺席 = 空关系 vs 不可读；aibuy 仓无 `gov_temporal_binding`/`gov_valid_time_anchor`，按字面读法 V1 对 aibuy 恒假）为缺口 G7	G_v
V2 锚存在与窗良型	α 中每个 (a, W)：$a \in$ `gov_valid_time_anchor`$@v$，$W \in \mathcal{W}$ 良型且与 a 的锚型（快照日/SCD 区间）相容；I2′ 变体的空指派/无窗角色按定义 5.4 相应放宽	G_v

检查项	内容	依赖
V3 绑定规则重演	按证书 `binding.binding_id` 在 `gov_temporal_binding`@v 中**逐 id 查键**（$(domain, metric)$ 联合键不保证函数性：实查 quality_voc 种子对 `defect_rate` 同键有正/负例两行，锚对不同——负例行须以 `binding_role`/`is_negative` 机器可读列区分，种子扩列方案，跨章/pilot 依赖），取规则及其声明的 adm 检查变体 `adm_check_mode`$\in$\{`trivial_true`, `symdiff_audit`, `interval_containment`\}（**机器可读列**——变体现只活在 note 自由文本，须落列，否则重演模式不可判），在 α 的窗上按声明变体**重新执行**规则判定。结论按**三值一致表**核对：dec $\in \{\mathsf{ANSWER}, \mathsf{REWRITE}\} \Rightarrow$ 规则重演须通过；dec $= \mathsf{REFUSE}(\mathsf{AM}) \Rightarrow$ 重演须失败且失败条款与见证的 c 一致；dec $= \mathsf{REFUSE}(\mathsf{MC/OOV/DB}) \Rightarrow$ V3 免除（守卫否定由 V6b(0) 核验）。β_v 对该度量未定义（原子度量、无绑定行）时 V3 **真空通过**	G_v, D
V4 路由重演	ρ 的每条边存在于 `gov_caliber_routing`@v 且 join_keys/归因对齐逐字段一致；衔接性 I3 成立	G_v

检查项	内容	依赖
V5 披露重演	核对 $\text{hash}(\text{ctx}) = \delta.h_{\text{ctx}}$；对 q 触达的表/列扫描 `gov_disclosure_policy`@v：应当生效的策略集与 Π 一致，且 dec 与策略语义相容（须裁剪而未标 REWRITE $\to$ 拒绝）——角色分支条款依 ctx 求值，k-小胞条款依证书探针笔录在 D 上重演（C4 定义 4.5 的实例相关与语境相关性）。**治理域复核规则（aibuy/email 等有策略登记的域）**：对每条生效策略 $\pi \in \Pi$ **字面重演**粒度条款与掩码条款——粒度条款验 $\gamma_\pi \preceq \delta.g$（证书记录的呈现粒度不细于策略要求，C4 E3），掩码条款验 $\delta.\mu^* \succeq \mu_\pi$（证书记录的掩码闭包覆盖策略要求的掩码集，C4 E4 `masked_email`）；g/μ^* 缺位或不满足 $\to$ REJECT（重演依据即定义 5.1 的 $\delta.g, \delta.\mu^*$ 字段）。**缺席规则（裁决 D5）**：域在 v 下无 `gov_disclosure_policy` 登记时，缺席语义 = **无约束、恒** ANSWER；V5 此时核验三件事——$\Pi = \varnothing$、dec $\notin \{\text{REWRITE}, \text{REFUSE(DB)}\}$、$\delta$ 段携带 `ungoverned-disclosure` 标注（缺席 $\neq$ 检查通过：标注使”该域该版本披露未治理”对审计方与下游可见，缺标即 REJECT）	G_v, D, ctx

检查项	内容	依赖
V6a 应答核对	应答证书：产出 s 仅触达 α, ρ 认证的语义对象**及其沿 `aggregate_of`/`normalize_of` 派生边的继承闭包**（dws `_1d`/`_1w` 表族继承 dwd 锚对象的有效时间列——继承边必须作为治理关系登记入 G_v，不得只存在于编译器 docstring。随仓复验：rma `gov_semantic_edge` **已实存**该两类边——`aggregate_of`×3（dwd_rma_order_item_di→dws_rma_sku_problem_1d、dwd_sales_order_item_di→dws_sales_sku_1d、dwd_rma_order_item_di→dws_rma_problemtype_1d）+ `normalize_of`×2（ods→dwd 两条）；**待补项**：沿边继承锚有效时间列的**规则声明**（现边仅记血缘、无「有效时间列沿边继承」的机器可读语义）+ 其余四域扩边，跨章/pilot 依赖）；s 中每个时间谓词的**指称集** ⊆ **认证窗**（半开窗 $[lo, hi\_excl)$ 的上界常量本身允许在窗外——按谓词指称集而非逐日期常量判定；原表述" 每个日期常量落在认证窗内" 对 §6.2 自身的 `hi_excl` 字段字面即假）；比率结构为" 分子分母各自窗内聚合后重算"（禁 `sum(rate)` 形态）	C（语法）, G_v

检查项	内容	依赖
V6b 见证重演	拒答证书，两步：**(0) 前序守卫否定**——按冻结的守卫报告序 OOV → AM → MC（裁决 D2：与 C3 注记 3.18′ 一致；C4 算法行序与编译器守卫序已同改为先窗/覆盖 (OOV)、再规则审计 (AM)、再质量探针 (MC)，三处无分歧）重演 r 之前全部守卫谓词的**否定**：$r = \mathsf{MC}$ 须重演 $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$，$r = \mathsf{AM}$ 须重演 $\neg\mathsf{OOV}$，$r = \mathsf{DB}$（披露阶段）须重演绑定各守卫均不成立；只重演定义体而不验守卫会接受" 理由类错置" 的证书（quality_voc 规则审计先行 + 有锚度量越界 as_of 的活反例，见定理 5.10(a) 证明注）。**(1) 见证重演**——按定义 5.3：MC(i)/部分 OOV 重演治理表查键；MC(ii) 重演探测 p 并断言 $\mathsf{eval}_D(p) \in \mathcal{Z}$（裁决 D4：$\mathcal{Z}$ 取加宽形，谓词 $\mu_{\mathrm{den}} \le 0 \vee \mathrm{NULL} \vee \varnothing$，见定义 5.3）；AM 按见证声明的失败条款 c 与 `adm_check_mode` 重演（条款 (iii) 按绑定行声明的比较粒度 g^{cmp} 判 $W \notin \mathcal{W}_{g^{\mathrm{cmp}}}$，定义 5.3——不按单锚粒度）；OOV 按锚在 G_v 中声明的 `coverage_mode` 重演 $W_T \cap \mathrm{Cov}_v(a) = \varnothing$（裁决 D3：mode 为锚的版本化属性，重演以声明值为准）；DB 重演 POP/SUPPMIN 探针笔录与策略触及集，确认 $U_{\min} = \varnothing$。V6b 的重演对象是拒答证书选定的首失败 slice（slice-major，定义 5.2）；CT 非首行不在重演义务内（显式非目标，定义 5.2）	G_v, D, ctx

检查项	内容	依赖
V6c 应答侧守卫重演	应答证书（dec $\in$ {$\mathsf{ANSWER}, \mathsf{REWRITE}$}；REWRITE 对每个保留 slice）重演**全部拒答守卫谓词的否定**：¬OOV——逐角色按锚声明 `coverage_mode` 重演 $W \cap \mathrm{Cov}_v(a) \neq \varnothing$（as-of 点情形为成员探测）；¬AM——由 V3 三值一致表的”规则重演须通过”分支承担；¬MC——路由查键可重算（V4 已核）**且**比率型度量重演分母人口探针 p 对 $\alpha(\mathrm{den})$ 窗的求值 $\mathsf{eval}_D(p) \notin \mathcal{Z}$（即 $\mu_{\mathrm{den}} > 0$，裁决 D4 判定集之否定；探针笔录随证书 `probes` 段携带，校验器按 §3.2(2) 从 D 重查而非信笔录）。无本项，“该拒不拒”的应答证书（守卫本应触发、$[\![q]\!] = \bot_r$）全项通过——F1b 伪造族（SKU536-EOL 空分母 ANSWER→SQL 求值 NULL）即由此封死；§3.2(2)“分母人口一律从 (G_v, D) 重查重演”红线在检查表的应答侧落点	G_v, D

定义 5.7（独立校验器 independent verifier） 校验器是任一判定 Chk 的过程 $\mathcal{V}(C, G_v, D, \mathrm{ctx}) \to \{\mathsf{ACCEPT}, \mathsf{REJECT}\}$，其输入**限于**证书、版本钉解析出的治理图、原始数据、与审计方持有的请求语境（经 h_{ctx} 核对）；其实现**不得**与编译器共享任何项目内模块、模板或代码路径（语言标准库与通用数据库驱动除外）。

3.2 认证循环红线（非循环性要求） **要求 5.8（implementation-disjointness）** 信任基（trusted base）为 {Chk 的声明式规范, G_v, D}；编译器不在信任基内。具体纪律：

1. 校验器只消费**证书内容物**——不读编译器的 QSPEC/模板/中间态，不调用 `compile_question`，不 import 编译器所在模块；
2. 凡编译器在编译期”查过的事实”（绑定规则、路由边、锚有效集、分母人口），校验器一律**从** (G_v, D) **重查重演**，而不是信任证书里的转述——证书字段是”待验断言”，不是”已证结论”；
3. 连窗解析这类看似中性的辅助逻辑（如月窗展开 `month_bounds`）也须独立实现：若编译器的窗算术有缺陷，共享实现会让同一缺陷在两侧对消，正是梁老师意见 3 所指的认证循环（compiler 给自己发证、verifier 用 compiler 的尺子量）；
4. 工程上以静态检查固化：校验器文件对编译器文件**零 import 依赖**，CI 断言两者无共享项目内模块（§6.4）。

在此纪律下，“编译器有 bug 但证书通过校验”要求缺陷同时且同构地出现在两套独立实现与 Chk 规范的理解中——审计边界从”信编译器”收缩为”信一页纸的检查关系规范”。

3.3 完备性与健全性 **定理 5.9（校验完备性 completeness）** 设 SA 与公理 5.0 成立、C3 前提 (A1)–(A6) 与日历窗代数封顶生效（§0 前提导入），且编译器满足 C3 的绑定正确性（其每次裁决的每个字段都由 (G_v, D, ctx) 上的事实或探测结果导出，判定所用的守卫序与重演模式即 G_v/证书中声明者）。则编译器产出的每份良构证书 C（**含定义 5.4 I2′ 三类拒答退化变体**）满足 $\mathsf{Chk}(C, G_v, D, \mathrm{ctx}) = \mathsf{ACCEPT}$。

证明（逐检查项）：V0—绑定正确的编译器之 α 本就按 C3 绑定语义从 (q, T, G_v) 导出（窗解析 + 规定锚查键），SA 下校验器在同一 (q, T, G_v) 上重导出得同一 $\alpha_{q,v}$；改锚声明与 REWRITE 收窄由编译器同批写入

证书记录，豁免核对一致。V1—编译器读的就是 ν 钉的 typed-commit，公理 5.0 保证唯一解析，快照下重解析得同一 G_v；对角性——编译器按 $\mathrm{ver}(T)$ 取版本（或显式钉版本时同批写入 offdiag 标记），两侧对同一版本轴解析 $\mathrm{ver}(T)$ 一致；I2′(c) 变体两侧对同一版本轴判 $\mathrm{ver}(T)\uparrow$ 一致。V2/V4—α, ρ 的每个条目是编译期从对应治理表读出的行，校验器在同一 G_v 重查得到相同行。V3/V6b—编译器的规则判定与拒答分派各自由一次治理表查键（按 binding_id）或一次探测求值触发，重演模式由 `adm_check_mode`/`coverage_mode`/$\mathcal{Z}$ 的机器可读声明钉死；SA 下同一查键/探测在校验端按同一声明模式重演得到相同结果，且编译器按声明守卫序裁决时前序守卫否定在编译期已成立、V6b(0) 重演亦成立，故结论与 δ.dec（或见证断言）一致。V5—生效策略集是 (G_v, ctx) 上按触达表/列的确定函数，h_{ctx} 核对后两侧重算一致，小胞条款经笔录中的同一探针重演；治理域的 g/μ^* 是编译器按 C4 改写算法实际采用的呈现粒度与掩码闭包（合法性在编译期已按 $\gamma_\pi \preceq g$、$\mu^* \succeq \mu_\pi$ 成立），同一 (G_v, ctx) 上字面重演同真。V6c—编译器产出应答**当且仅当**各拒答守卫在编译期均未触发（按声明守卫序裁决），故 $\neg\mathsf{OOV} \wedge \neg\mathsf{AM} \wedge \neg\mathsf{MC}$ 的各谓词在编译期为真，SA 下同一谓词在校验端重演同真；分母人口探针即编译器 `_den_present` 类探测的外化笔录，重查得同值 $\notin \mathcal{Z}$。V6a—编译器按模板以 α 的窗实例化 s，s 的时间谓词**指称集**按构造等于认证窗（半开上界常量不逐字入窗，指称集判定按构造通过）、触达对象落在认证对象沿 `aggregate_of`/`normalize_of` 的继承闭包内（继承边入 G_v 后两侧同查）。□（依赖显式化：本证明依赖”编译器每字段可溯源到 (G_v, D, ctx) 事实”这一 C3 侧义务；若 C3 定稿弱化该义务，本定理前提需同步收紧。**pilot 现状注记**：该前提对现役实现尚不全成立——domestic_newprod 的 `METRIC_ANCHORS` 是注释自认的 frozen copy、其 AM 判定是对题面的正则命中（`ANCHOR_OVERRIDE`），quality_voc 的 `METRIC_CALIBER` 是编译器内字典，均不读治理表；5.9 在 pilot 侧生效以 §6 的种子外化改造完成为前提。）

定理 5.10（校验健全性 soundness） 设 SA 与公理 5.0 成立、§0 前提导入生效，$\mathsf{Chk}(C, G_v, D, \mathrm{ctx}) = \mathsf{ACCEPT}$。则：(a) 若 C 为拒答证书：$r \in \{\mathsf{MC}, \mathsf{AM}, \mathsf{OOV}\}$ 时理由 r 在 (T, v, D) 坐标下按 C2/C3 的**守卫序语义**成立（$[\![q]\!] = \perp_r$，守卫序 $\mathsf{OOV} \to \mathsf{AM} \to \mathsf{MC}$，裁决 D2）；$r = \mathsf{DB}$ 时按 C4 定义 4.5/4.8 的披露语义成立（$L = \varnothing$ 于 DB slice）；(b) 若 C 为应答证书，且 s 属模板查询类 $\mathcal{Q}_{\mathrm{tmpl}}$（分子/分母各为认证对象上带窗谓词的单块聚合，按认证路由 join，率值末端重算），则 s 是 (T, v) 下时点正确的查询；$\mathsf{REWRITE}$ 情形正确性相对被裁剪后的 q' 成立。

证明：(a) **按 r 分情形；每情形 = 前序守卫否定（V6b(0)）+ 定义体（V6b(1)）**。C3 定理 3.17 的失败值是守卫三分：$[\![q]\!] = \perp_r$ 当且仅当**前序守卫均不成立**且 r 的定义体成立——只重演定义体证不出守卫序语义（旧版即在此让步为”定义体成立”的弱命题。活反例取**有锚**度量：quality_voc `defect_rate`（注册锚对：投诉锚 36 个锚日期（周透视快照日）/ 工单日锚 35 个工单日，全史对称差 57 非空，冻结仓复演）配越界 as_of `2030-01-06`——该时点不在任何注册锚的覆盖域内（36 个锚日期（周透视快照日）止于 2026-06-12），语义值为 $\perp_{\mathsf{OOV}}$；而规则审计先行的编译序下 symdiff 定义体为真，可出具定义体重演通过、语义值 $\perp_{\mathsf{OOV}} \neq \perp_{\mathsf{AM}}$ 的可接受 AM 证书。※ 更正注：早稿此处曾引 `complaint_rate_global_recompute` 同 as_of 称语义值 $\perp_{\mathsf{OOV}}$——该度量在 G_v 无任何注册锚，按 C3 定义 3.14 真空约定 OOV 覆盖支恒假、语义值实为 $\perp_{\mathsf{MC}}$，原实例不构成反例，已更换）。修订后的 V6b(0) 恰好补上守卫否定的重演——且 V0 已先行保证证书的 α 即 q 自身的 $\alpha_{q,v}$（守卫否定与见证的重演窗不可挪移到别的时点，F1c 型”携他窗探针的拒答”在 V0 即拒），故：MC(i)：V6b(0) 重演 $\neg\mathsf{OOV} \wedge \neg\mathsf{AM}$，V6b(1) 查键确认无可重算路由——守卫域上 MC 定义体成立，合并得 $[\![q]\!] = \perp_{\mathsf{MC}}$；$\mathsf{MC}$(ii)：同上，另 V2 确认 p 的窗即 $\alpha(\mathrm{den})$ 的窗、V0 确认 $\alpha(\mathrm{den})$ 即 q 的分母角色窗、$\mathsf{eval}_D(p) \in \mathcal{Z}$，同窗合法分母人口空（$\mathcal{Z}$ 取 D4 加宽形：$\mu_{\mathrm{den}} \leq 0 \vee \mathrm{NULL} \vee \varnothing$）；$\mathsf{AM}$：V6b(0) 重演 $\neg\mathsf{OOV}$，V6b(1) 按声明条款 c 与 adm 变体确认 svw 相应条款违反（(iii) 按 g^{cmp}），V3 一致表确认规则要求成立且重演失败，V0 确认载荷所引窗对即 q 的请求窗对（对不属 q 的假窗对重演”失败”不构成本情形，F1d 型伪造在 V0 即拒）；OOV：V6b(1) 按 coverage_mode 重演覆盖判空——配对角色按 C3 定义 3.14 对级条款重演（两腿窗实现皆空；混合实现不入本类、归条款 (iv) 审计辖区），原子角色重演 $W_T \cap \mathrm{Cov}_v(a) = \varnothing$（或 $\mathrm{ver}(T)\uparrow$/窗不可解析变体），OOV 无前序守卫；DB：V6b(0) 确认绑定各守卫不成立（问题可绑定），V5+V6b(1) 依 ctx 与探针笔录重演 $U_{\min} = \varnothing$，即 C4 语义下披露约束排空合法改写集。□ (b) **proof-sketch + 显式缺口**。骨架分两层。**语义坐标层（V0/V6c，本轮补齐）**：V0 保证 s 实例化所用的 α 就是 q 自身的 $\alpha_{q,v}$——窗与锚不可挪移（无此项，错月 ANSWER 伪造（F1a：rma_q1@2026-05 携 2026-04 双窗，答 0.044444≠⟦q⟧=0.088571，随仓复演）在 $\mathcal{Q}_{\mathrm{tmpl}}$ 内字面通过全部旧检查项）；V6c 保证 (T, v, D) 坐标下 q 的各拒答守卫均不成立，即 $[\![q]\!]$ 非任何失败值 $\perp_r$（无此项，“该拒不拒”伪造（F1b：SKU536-EOL@2026-05 空分母 ANSWER，SQL 求值 NULL）同样字面通过——即”模板类上语法蕴含语义”引理在旧检查表下于 $\mathcal{Q}_{\mathrm{tmpl}}$ **内部**即被证伪，此洞独立于下述 G1 的 SQL 等价性缺口，须由检查表而非引理承担）。**语法—语义层**：V2 保证锚窗合法，V3 保证窗满足绑定规则，V4 保证分母人口口径沿治理路由归因对齐，V6a 保证 s 的时间谓词被认证窗语法包含且聚合结构合规；对 $\mathcal{Q}_{\mathrm{tmpl}}$ 中的查询，这组语法条件**在 V0/V6c 钉定的语义坐标之上**蕴含其求值语义等于 C2 定义的时点正确求值（模板类上语法—语义有直接对应）。**缺口 G1**：V6a 是语法包含检查，不是任意 SQL 的语义等价判定——后者在含算术与聚合的完整 SQL 上不可判定（一般

查询等价/包含的不可判定性），故 (b) 的健全性**只对** $\mathcal{Q}_{\text{tmpl}}$ **声明**；$\mathcal{Q}_{\text{tmpl}}$ 的形式定义与”模板类上语法蕴含语义”引理需在 C3/C4 定稿中给出并证明（或改走机械化检查路线）。缺口范围之外（编译器产出任意手写 SQL）不在本定理覆盖内。**现状看护**：$\mathcal{Q}_{\text{tmpl}}$ 在三章中尚无形式定义（(b) 目前是对待定义类的定理）；且现役 rma_q3 的差分模板（两率相减，`_rate_delta_sql`）已在 (b) 描述的单窗形态之外——多期推广见缺口 G6，未推广前 (b) 不覆盖该产出。

命题 5.11（校验复杂度，条件化） 在导入的 (A1)–(A6)（§0；界的推导主要用 (A1)(A2)，终止性另赖 (A5)(A6) 的可判定呈现）下 Chk 可终止，且：治理表检查项在 $|C|$ 与相关治理表大小上多项式；**窗内**探测（V6b 空窗探测、单窗成员探测）为常数次单扫描聚合，与回答查询同阶或更低。但两类探测**不**受此界：(i) AM 对称差重演（`adm_check_mode=symdiff_audit`）是对两锚**全史**标记集的一趟去重反连接，代价 $O(|V(a_n)|+|V(a_d)|)$，与请求窗无关——长历史 × 窄窗下可高于回答查询本身；(ii) OOV 对带空隙多粒元窗的成员探测为逐粒元而非常数次。故”校验成本与回答查询同阶或更低”**仅对治理表项与窗内探测项成立**；对称差/成员类探测的部署代价须以物化有效标记集或索引工程化。（proof-sketch：逐项计数即得；缺口：未含索引/分区假设的精细界。）

P.1.5 4. 证书最小性：逐字段删除的攻击性反例

本节论证定义 5.1/5.3 的字段配置在 Chk 之下**各有不可替代的判别职责**：对每个字段给出”删去后存在通过弱化检查、但违背时点正确性”的构造。除删 ν 外，反例取自 pilot 已实证的真实语义（gov_seed / compiler 行为）；删 ν 为**构造性**攻击（版本演化尚未在 pilot 落库，见该段现状注记）。

命题 5.12（字段非冗余性） 对 $x \in \{\nu, \alpha, \rho, \delta, w\}$，记 Chk_{-x} 为去除 x 相关检查项后的弱化关系，则存在实例使 Chk_{-x} 接受而时点正确性不成立。构造如下（构造即证明）：

删 ν（版本钉）→ 版本漂移攻击（version-swap）。无 ν 时校验器只能对”当前最新图”重演。设 $v \to v'$ 间口径重划（如 rma 的 `rma_problem_to_sales` 路由在 v' 改变分母人口定义，或锚的有效时间列迁移），则一份在 v 下正确的证书，在 v' 上重演的各项检查仍可能逐项通过（表仍在、锚仍在、规则同名），但其答案对应的语义已不是任何单一版本下的语义——正是 CKE 线”新语义自由度必须显式成为 key”教训的证书版：版本不入证书，重演就没有坐标。（该攻击同时说明：ν 是本框架与 DELTAGOV 退化情形共享的字段——删掉它连一维退化保证也失去。）**现状注记（构造性声明）**：version-swap 需要 $v \to v'$ 两个在位版本；实查五域种子 `gov_semantic_graph_version` 均**恰一行**（单版本），故本反例是构造性的——口径重划的形状取自 rma 路由的真实语义，但双版本实例尚未落库。删 ν 的非冗余性论证不依赖已实证实例，只依赖”版本可演化”这一框架前提（公理 5.0 (ii) 的 append-only 版本轴）。

删 α（锚指派：anchor id+ 窗）→ 窗挪移攻击（window-swap）。无 α 时 V6a 失去”时间谓词落在认证窗内”的基准，V3 失去重演对象。实证反例即 rma q6：分子窗 2026-05、分母窗 2026-04 的 refund_rate——两窗各自的 SQL 语法完全合法、各自单扫描聚合无破绽，仅当证书携带”分子锚窗/分母锚窗”两条指派时，`same_valid_time_window` 的违反才可被第三方重演检出。删 α 后该跨窗错配查询可被包装为通过校验的应答。

删 ρ（口径路由路径）→ 口径盲替换攻击（caliber-blind substitution）。锚与窗都正确时，分母人口仍可走错口径。实证反例即 rma gov_seed 中同时存在的两条路由：`rma_problem_to_sales`（join_keys=[product_sku, dt]，分母 =scoped 全量销量）与对照口径 `caliber_blind_control`（join_keys=[dt]，分母 =RMA 自身全量）——两者锚同、窗同，问题率一个 16.2% 一个 3.8%（4.2× 低估）。数字 provenance（同 C3 注记 3.4′ 形制）：datagov-line 生产实例实测（datahub `datagov-line`，loctek-unified-datagov-line，v2026.06.23 rma 域治理产出），非随论文交付的 pilot 工件；pilot 合成仓 `pilot/domains/rma/warehouse.duckdb` 全域 problem/sales 为 2.76%（465/16832，可复演），属不同域实例、不同 scope，与生产对照数不可互算。无 ρ 字段，校验器无从判定分母沿哪条边归因，盲口径答案与治理口径答案在弱化检查下不可区分。

删 δ（披露决策 + 策略 id）→ 披露洗白攻击（disclosure laundering）。两种形态：(i) 触达披露受限列（如 aibuy `gov_disclosure_policy` 的 PII 列、k-anonymity 门限列）的查询，未经策略裁剪而以全量答案示人——无 Π，校验器不知道哪些策略应当生效、无从执行 V5；(ii) 经 C4 最大合法改写后的**部分**答案伪装为原问题的**全量**答案（静默收窄）——无 dec = REWRITE 标记，下游把”合法子集上的率”当”全集上的率”使用，产生的是带证书外观的口径错误。δ 的职责正是让”答了什么范围、凭什么策略”进入可重演断言。

删 w（拒答见证）→ 不可证伪拒答（unfalsifiable refusal）。无见证的拒答证书退化为”编译器说不行”——过度拒答（pilot 中素基线已出现的 wrong_refusal 形态）与正当拒答在校验器眼中无差别，拒答通道沦为逃避回答义务的后门；且四类原因间可任意误标（把可修复的 AM 谎报为 MC，误导上游放弃重述问

题）。带见证后，V6b 的重演使每份拒答可证伪：谎报的见证要么查键失败、要么探测结果落在断言集之外、要么判别元不在**按证书 binding_id 与声明 adm 变体所定模式**重演出的对称差中——重演模式入证书/治理表后（V3），该判别关系才良定：模式悬置时全史对称差非空的锚对（rma 实测 97）可为谎报 AM 提供真判别元。□

注记 5.13（最小性的边界） 命题 5.12 论证的是"删任一字段则存在攻击"，即字段集合的**逐点非冗余**；它不主张该字段集在所有可设想的证书方案中体积最小——同等判别力的其他编码（如把 ρ 折叠进 α 的扩展锚）可能存在，但任何此类编码须在 Chk 中承担与本节五项攻击一一对应的判别职责。此为设计空间陈述，不作绝对性断言。

P.1.6 5. 证书的对外契约（下游如何消费）

- **审计方**：持 $(C, G_v, D, \mathrm{ctx})$ 跑独立校验器，无需接触编译器——审计边界即 §3.2 的信任基；
- **上游应用（NL2SQL agent / BI）**：REFUSE 时按 r 分流修复动作——AM 提示改窗重问、OOV 提示改时点、MC 提示该口径下无合法答案（不可通过重述绕过）、DB 提示披露约束下无非空合法改写（换语境/角色或粗化请求粒度）；REWRITE 时向用户显式披露裁剪范围（裁剪痕迹 CT 与 $\prec_{\mathrm{det}}$ 键随证书可查）；
- **非目标声明（REWRITE 极大性）**：Chk 验证 q' 的合法性与裁剪痕迹的一致性，**不验证** q' 在 $\preceq_D$ 下的极大性——编译器静默过度裁剪（返回远小于极大元的合法 q''）不被证书层检出；极大性审计需重实现 MLR 搜索，与 §3.2 不共享实现红线成本冲突，故显式声明为本篇非目标（候选未来审计项，见 §6.4 末张力提示）；
- **版本基座（DELTAGOV 类 producer）**：证书的 ν 即两系统的 producer–consumer 接口面——基座保证 ν 可解析且版本内读一致，本框架在其上叠加 $\alpha/\rho/\delta/w$ 四轴（注记 5.5 的反方向）。

P.1.7 6. 落地映射：pilot 编译器接口升级（实现层，降格陈述）

6.1 现状 五域编译器(`/Volumes/SSD 1/explore_opportunity_cc/pilot/domains/{rma,quality_voc,domestic_newpr`
现返回：

```
compile_question(q: dict) -> {"sql": "..."} | {"refusal": "<reason>[: detail]"}
```

编译期已经**算出但随即丢弃**了证书所需的全部事实：rma `_binding()` 查到的绑定规则与锚对、`_den_present()` 的分母探测（即 MC(ii) 空窗见证）、q6 的 (num_window, den_window) 窗对（即 AM 见证）；quality_voc `_anchors_same_window()` 的对称差计数（AM 判别元的来源）、`_window_in_validity()` 的有效集成员探测（OOV 见证）、`_caliber_recomputable()` 的路由查键（MC(i) 见证）。升级的本质是**把这些已计算事实结构化外化**——但"零判定改动"的承诺有两处例外须如实声明：(i) AM 条款 (i) 改锚见证要求把 domestic_newprod 现行的"题面正则命中"（`ANCHOR_OVERRIDE`）升级为"解析出被改锚对象并证其 $\neq$ 规定锚"，是实质判定改动；(ii) 编译器内字典（`METRIC_ANCHORS` frozen copy、`METRIC_CALIBER`）须改为读治理表，否则定理 5.9 的"每字段由 (G_v, D) 导出"前提不成立（见 5.9 pilot 现状注记）。

6.2 最小 schema 变更 返回信封升级为（向后兼容：`sql`/`refusal` 键语义不变，新增 `certificate`）：

```
{
  "sql": "...",                    // 或 "refusal": "<reason>", 二择一, 同现状
  "certificate": {
    "cert_version": "c5.v1",
    "question": {"qid": "rma_q6", "as_of": "2026-05"},
    "graph_pin": {                                         // ν
      "domain": "rma",
      "graph_version": "v2026.06.23",
      "commit_id": "<真 typed-commit 标识>",           // 公理 5.0: 现五域种子 committed_at≡版本标签
```

```
                                                  // （复写、无区分力），种子须补真 commit id
    "off_diagonal": null                          // 离对角标记（C4 S-V/E5；定义 5.1）：对角读取
                                                  // 钉定版本 p≠ver(T) 时必填 {"pinned": "<p>",
                                                  // "ver_as_of": "<ver(T)>"}，V1 核验（缺标即 R
  },
  "ctx_hash": "<hash(ctx)>",                      // δ.h_ctx：角色作用域策略域必填；
                                                  // 无披露策略登记的域取空语境哈希
  "anchors": [                                    // α（role → anchor id + 窗）
    {"role": "numerator",   "anchor_id": "rma_event_time",
     "window": {"kind": "month", "lo": "2026-05-01", "hi_excl": "2026-06-01"}},
    {"role": "denominator", "anchor_id": "sales_event_time",
     "window": {"kind": "month", "lo": "2026-04-01", "hi_excl": "2026-05-01"}}
  ],
  "binding": {"binding_id": "rma_refund_rate_temporal_align",  // V3 逐 id 查键（(domain,metric) 非函
              "rule": "same_valid_time_window",
              "adm_check_mode": "symdiff_audit"},     // 机器可读重演模式列（trivial_true |
                                                  // symdiff_audit | interval_containment）
  "routing": [                                    // ρ
    {"caliber_key": "rma_problem_to_sales",
     "join_keys": ["product_sku", "dt"],
     "attribution_alignment": "sku+period"}
  ],
  "disclosure": {"decision": "REFUSE",            // δ：ANSWER | REWRITE | REFUSE
                 "policy_ids": [],                // 生效策略；见 §6.3
                 "policy_table_present": false,
                 "ungoverned_disclosure": true,   // 裁决 D5：无披露策略登记的域必标；
                                                  // 缺席=无约束恒 ANSWER，标注≠"检查通过"（V5
                 "granularity": "<g>",            // δ.g 呈现粒度（C4 定义 4.10/E3）：治理域 V5
                                                  // 无策略域取请求原粒度
                 "mask_closure": []},             // δ.μ* 最小合法掩码闭包（C4 定义 4.10/E4，如
                                                  // 治理域 V5 重演 μ* ⪰ μ_π；无策略域为 []
  "probes": [                                     // 探针笔录：C4 POP/SUPPMIN（披露域必填）；
    {"kind": "POP|SUPPMIN", "policy_id": "<π>", "sql": "...", "observed": 0},
    {"kind": "DEN_POP", "role": "denominator",    // V6c：比率型应答证书必携分母人口探针笔录
     "sql": "...", "observed": 490}               // （校验器从 D 重查核对，μ_den>0，非信笔录）
  ],
  "rewrite": {                                    // 仅 REWRITE 携带（定义 5.2 裁剪痕迹）
    "kept_slices": ["..."],
    "cut_trace": [{"slice": "<(v,α)-slice>", "reason": "MC|AM|OOV|DB"}],  // C4 逐 slice 原因表 CT
    "det_order": {"keys": ["version_desc", "binding_registration", "ell_gamma"],
                  "ell_gamma_id": "<声明的 ≼_Γ 线性扩张标识>"}          // ≺_det 键（C4 定义 4.11）
  },
  "refusal": {                                    // 仅 REFUSE 携带（I4）
    "reason": "anchor-mismatch",                  // missing-caliber | anchor-mismatch |
                                                  // out-of-validity | disclosure-blocked
    "witness": {"type": "window-pair", "clause": "(ii)",
                "num_window": "2026-05", "den_window": "2026-04"},
    "ct": [{"slice": "<(v,α)-slice>", "reason": "AM"}] // 裁决 D1 CT 形制（slice-major，定义 5.2）：
                                                  // 原因表附表；reason/witness 取首失败 slice——
                                                  // slice 序取首失败 slice，slice 内按守卫序 OOV
                                                  // 非首行为审计注记（未验断言，显式非目标，V6b
                                                  // 单 slice 退化为单行
  }
}
```

```
}
```

四类见证的 witness 变体：

```
// missing-caliber (i) 路由缺失
{"type": "routing-lookup",
 "metric": "complaint_rate_global_recompute",
 "caliber_key": "complaint_rate_pivot_reference",
 "dst_caliber": "none"}
```

```
// missing-caliber (ii) 空窗
{"type": "empty-denominator-probe",
  "probe_sql": "SELECT SUM(sales_qty) FROM ... WHERE produ
  "observed": null}
// observed 断言集 𝒵（裁决 D4 定稿加宽形）：μ_den≤0 ∨ NULL
```

```
// anchor-mismatch（条款 (iv) 对称差型；
//  条款 (i)/(ii)/(iii) 见定义 5.3 载荷）
{"type": "validity-set-symdiff",
 "clause": "(iv)",
 "adm_check_mode": "symdiff_audit",

 "num_anchor": "complaint_submit_date",
 "den_anchor": "rma_create_date",
 "discriminant_date": "<x ∈ 对称差>",
 "symdiff_count": 57}
```

```
// out-of-validity
{"type": "window-outside-validity",
 "anchor_id": "complaint_submit_date",
 "coverage_mode": "<hull | strict_member>",  // 裁决 D3: 逐锚
                                             // 缺省快照锚 hu
 "requested": "2027-01-05",
 "validity_probe_sql": "SELECT count(*) FROM dwd_quality_defe
  "observed": 0}
// 57 = 冻结仓复演值（36 个锚日期（周透视快照日）+ 35 工单日
```

```
// disclosure-blocked（DB，承 C4 S-M2）
{"type": "empty-legal-rewrite",
 "blocked_slices": ["<(v,α)-slice>"],
 "blocking_policy_ids": ["<π⋯>"],
 "probe_transcript": [{"kind": "SUPPMIN", "policy_id": "<π>", "observed": 1, "threshold": 5}],
 "u_min_empty": true}
```

6.3 每域最小改动清单

域	改动	备注
共通	每个 return 点包信封；新增 ~30 行 _cert() 构造辅助（仅组装 dict，不含判定）	信封层判定逻辑零改动（两处例外见 §6.1）
rma	_binding() 返回值透传入 α/binding 字段；_den_present() 改为返回 (bool, probe_sql, observed) 以外化 MC(ii) 见证；QSPEC 的窗对写入 α；gov_caliber_routing 已在库内，ρ 一次查键即得	5 个 return 点（现役 compile_question 正则复数）
quality_voc	_anchors_same_window() 增返一个判别日期（现只返回 bool，对称差计数已在 SQL 中，取任一差集元素即可）；_window_in_validity() 外化探测 SQL 与计数；METRIC_CALIBER 查键结果入 ρ	12 个 return 点（现役 compile_question 正则复数）

域	改动	备注
domestic_newprod	同型包装；分类冲突已裁决（D1，见缺口 G2）：空同窗分母一律归 MC(ii)，OOV 只承载锚覆盖域真空——NX-Q7 实测 launch 表 min(dt)=2025-09-15 > as-of 2025-08-31，属真 OOV 保持 OOV，两域金标均无需改；编译器改动仅为按此语义分派 reason 与见证类型（守卫序同步 D2：先 OOV 覆盖检查、再 AM、再 MC 探针），V6b 统一重演规则即成立	
全域	`gov_disclosure_policy` 仅 aibuy/email 的 gov_seed 存在（v2026.06.23）；rma/quality_voc/domestic_newprod 证书中 `policy_ids=[]`、`policy_table_present=false` 且必标 `ungoverned_disclosure=true`——语义是”该域该版本披露未治理、无约束恒 ANSWER“（裁决 D5），而非”披露检查通过”；校验器 V5 按 D5 缺席规则处理（见 §3.1 V5），A4.3/E3 与披露消融的范围按 aibuy/email 两治理域如实声明	

6.4 独立校验器落位

- 新文件 `pilot/verify_cert.py`：只 import `duckdb`/`json`/stdlib；对 `compiler.py` **零 import**；窗算术（月窗展开等）独立重写；
- 输入（`certificate.json`，`warehouse.duckdb[, ctx.json]`）（ctx 仅披露策略域需要，经 ctx_hash 核对），输出逐检查项 V0–V6c 的 ACCEPT/REJECT 与失败定位；
- CI 断言（进 run_pilot 前置检查）：`verify_cert.py` 的 import 图与 `domains/*/compiler.py` 无交集（要求 5.8 的机械化执行）；
- 评测挂钩：主实验中每份机制产出先过校验器再判分，校验器 REJECT 计为机制侧错误——防”编译器与金标同源”的第二层认证循环（金标 gold_sql 与编译器均由本团队构造，独立校验器至少切断实现层的循环；语义层循环由公开轨与人工抽检覆盖，见 pilot 报告诚实注记①）。
- **张力预表态（REWRITE 极大性 vs 不共享实现红线）**：若未来把 REWRITE 的**极大性**纳入校验，独立校验器须重实现 MLR 搜索（C4 算法），与要求 5.8 的零共享红线在成本上冲突——本篇选择把极大性显式列为非目标（§5 非目标声明），仅校验合法性与裁剪痕迹一致性；如立项极大性审计，须以第二套独立搜索实现或抽样对证（spot-check against declared $\prec_{\mathrm{det}}$ 键）落地，不得回退为 import 编译器。

P.1.8 7. 本篇缺口清单（进 S1 理论周待办）

- **G1（健全性范围）**：定理 5.10(b) 只覆盖模板查询类 $\mathcal{Q}_{\mathrm{tmpl}}$；其形式定义与”语法包含 ⇒ 语义时点正确”引理待 C3/C4 定稿证明，或改走机械化检查（Lean/SMT 对模板实例）路线——作者需裁决走哪条。
- **G2（拒答分类冲突）——已裁决（D1）**：采纳对抗审查的分析：按 C3 定义 3.14/3.16 的守卫语义，**空同窗分母统一归 MC(ii)**（rma_q5(536) 为覆盖内 scoped 分母零质量），OOV 只保留**锚覆盖域真空**

（NX-Q7 实查 launch 表 min(dt)=2025-09-15 > as-of 2025-08-31，属真 OOV 保持 OOV）——两案属不同语义情形，两域金标均无需改。配套裁决同批落地：κ 剔空 A_{adm} 归 AM(i)（C3/C4 侧），拒因拼写全线统一为 OOV。本篇落点：定义 5.3 两行的分类注记、§6.3 domestic_newprod 行、V6b 统一重演规则。

- **G3（披露策略缺席语义）——已裁决（D5）**：缺席 = **无约束、恒** ANSWER，同时证书 disclosure 段强制标注 `ungoverned-disclosure`（两者兼取：语义上不约束，制品上不伪装为" 检查通过"）。V5 缺席规则按此定稿（§3.1 V5：验 $\Pi = \varnothing$、无 REWRITE/REFUSE(DB)、标注在位）；主实验 A4.3/E3 与披露消融范围按 aibuy/email 两治理域如实声明。
- **G4（窗代数统一）**：rma 用问题侧月窗（区间），quality_voc 用锚有效日期集（离散集合，对称差判定）——$\mathcal{W}$ 的两类元素在 C2 定义层须统一（含混合比较的语义），否则 V3 的重演规则分叉。
- **G5（SA 的边界）——旧证书地位部分已裁决（D6）**：旧证书**按坐标** (T, v) **永久有效**、无 revalidation 义务（已写入 §0 快照假定段；回滚 = 重提交见公理 5.0(iii)，C3 侧 A3/§2.3 与 C4 侧 A4.2 同批落地）。残余项：SA 本身（编译与校验同快照）仍为定理 5.9/5.10 前提，跨快照并发协议仍由版本基座承担，与 §5 System 章合并时对齐接口表述。
- **G6（delta 型与多期问题）**：schema 的 α 支持角色 × 期展开（rma q3 双期），但定理陈述以单窗为主线；多期情形的定理推广为机械但未写出的工作（现役 rma_q3 差分模板已在定理 5.10(b) 覆盖之外，见其现状看护）。
- **G7（治理表整表缺席的 V1 语义）**：aibuy 仓无 `gov_temporal_binding/gov_valid_time_anchor` 表（实查）——V1" 各治理表在该版本下可读" 按字面对 aibuy 恒假；缺席 = 空关系还是 = 不可读须裁决（与 G3 同族的缺席语义问题，但属绑定层而非披露层）。
- **G8（一维退化嵌入唯一性）**：注记 5.5 的嵌入（窗 = $\mathbb{T}$ 全窗）与 C4 命题 4.23 的嵌入（valid-time 轴退化为 $\{now\}$ 单点）是两条不同的" 一维退化"，合并时须裁决唯一嵌入并统一表述。

References

[1] Apache Software Foundation. Apache iceberg table specification. `https://iceberg.apache.org/spec/`, 2024. Accessed 2026-08.

[2] Michael Armbrust, Tathagata Das, Liwen Sun, Burak Yavuz, Shixiong Zhu, Mukul Murthy, Joseph Torres, Herman van Hovell, Adrian Ionescu, Alicja Łuszczak, Michał Świtakowski, Michał Szafrański, Xiao Li, Takuya Ueshin, Mostafa Mokhtar, Peter Boncz, Ali Ghodsi, Sameer Paranjpye, Pieter Senster, Reynold Xin, and Matei Zaharia. Delta lake: High-performance ACID table storage over cloud object stores. *Proceedings of the VLDB Endowment*, 13(12):3411–3424, 2020.

[3] Souvik Bhattacherjee, Amit Chavan, Silu Huang, Amol Deshpande, and Aditya Parameswaran. Principles of dataset versioning: Exploring the recreation/storage tradeoff. *Proceedings of the VLDB Endowment*, 8(12):1346–1357, 2015.

[4] Michael H. Böhlen, Christian S. Jensen, and Richard T. Snodgrass. Temporal statement modifiers. *ACM Transactions on Database Systems*, 25(4):407–456, 2000.

[5] James Cheney, Laura Chiticariu, and Wang-Chiew Tan. Provenance in databases: Why, how, and where. *Foundations and Trends in Databases*, 1(4):379–474, 2009.

[6] Carlo A. Curino, Hyun J. Moon, and Carlo Zaniolo. Graceful database schema evolution: The PRISM workbench. *Proceedings of the VLDB Endowment*, 1(1):761–772, 2008.

[7] Anton Dignös, Michael H. Böhlen, and Johann Gamper. Temporal alignment. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pages 433–444, 2012.

[8] Avrilia Floratou, Fotis Psallidas, Fuheng Zhao, Shaleen Deep, Gunther Hagleither, Wangda Tan, Joyce Cahoon, Rana Alotaibi, Jordan Henkel, Abhik Singla, Alex van Grootel, Brandon Chow, Kai Deng, Katherine Lin, Marcos Campos, Venkatesh Emani, Vivek Pandit, Victor Shnayder, Wenjing Wang, and Carlo Curino. NL2SQL is a solved problem... not! In *Conference on Innovative Data Systems Research (CIDR)*, 2024.

[9] Dawei Gao, Haibin Wang, Yaliang Li, Xiuyu Sun, Yichen Qian, Bolin Ding, and Jingren Zhou. Text-to-SQL empowered by large language models: A benchmark evaluation. *Proceedings of the VLDB Endowment*, 17(5):1132–1145, 2024.

[10] Todd J. Green, Grigoris Karvounarakis, and Val Tannen. Provenance semirings. In *Proceedings of the ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS)*, pages 31–40, 2007.

[11] Silu Huang, Liqi Xu, Jialin Liu, Aaron J. Elmore, and Aditya Parameswaran. OrpheusDB: Bolt-on versioning for relational databases. *Proceedings of the VLDB Endowment*, 10(10):1130–1141, 2017.

[12] George Katsogiannis-Meimarakis and Georgia Koutrika. A survey on deep learning approaches for text-to-SQL. *The VLDB Journal*, 32(4):905–936, 2023.

[13] Martin Kaufmann, Amin Amiri Manjili, Panagiotis Vagenas, Peter M. Fischer, Donald Kossmann, Franz Färber, and Norman May. Timeline index: A unified data structure for processing queries on temporal data in SAP HANA. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pages 1173–1184, 2013.

[14] Hyeonji Kim, Byeong-Hoon So, Wook-Shin Han, and Hongrae Lee. Natural language to SQL: Where are we today? *Proceedings of the VLDB Endowment*, 13(10):1737–1750, 2020.

[15] Soyeon Kim, Jindong Wang, Xing Xie, and Steven Euijong Whang. Harnessing temporal databases for systematic evaluation of factual time-sensitive question-answering in large language models. arXiv:2508.02045, 2025.

[16] Krishna Kulkarni and Jan-Eike Michels. Temporal features in SQL:2011. *ACM SIGMOD Record*, 41(3):34–43, 2012.

[17] Gyubok Lee, Hyeonji Hwang, Seongsu Bae, Yeonsu Kwon, Woncheol Shin, Seongjun Yang, Minjoon Seo, Jong-Yeup Kim, and Edward Choi. EHRSQL: A practical text-to-SQL benchmark for electronic health records. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.

[18] Kristen LeFevre, Rakesh Agrawal, Vuk Ercegovac, Raghu Ramakrishnan, Yirong Xu, and David DeWitt. Limiting disclosure in Hippocratic databases. In *Proceedings of the International Conference on Very Large Data Bases (VLDB)*, pages 108–119, 2004.

[19] Feifei Li, Marios Hadjieleftheriou, George Kollios, and Leonid Reyzin. Dynamic authenticated index structures for outsourced databases. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pages 121–132, 2006.

[20] Jinyang Li, Binyuan Hui, Ge Qu, Jiaxi Yang, Binhua Li, Bowen Li, Bailin Wang, Bowen Qin, Ruiying Geng, Nan Huo, Xuanhe Zhou, Chenhao Ma, Guoliang Li, Kevin C.-C. Chang, Fei Huang, Reynold Cheng, and Yongbin Li. Can LLM already serve as a database interface? a big bench for large-scale database grounded text-to-SQLs. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[21] Ashwin Machanavajjhala, Daniel Kifer, Johannes Gehrke, and Muthuramakrishnan Venkitasubramaniam. ℓ-diversity: Privacy beyond k-anonymity. *ACM Transactions on Knowledge Discovery from Data*, 1(1):3, 2007.

[22] Michael Maddox, David Goehring, Aaron J. Elmore, Samuel Madden, Aditya Parameswaran, and Amol Deshpande. Decibel: The relational dataset branching system. *Proceedings of the VLDB Endowment*, 9(9):624–635, 2016.

[23] Ross M. McConnell, Kurt Mehlhorn, Stefan Näher, and Pascal Schweitzer. Certifying algorithms. *Computer Science Review*, 5(2):119–161, 2011.

[24] Hyun J. Moon, Carlo A. Curino, Alin Deutsch, Chien-Yi Hou, and Carlo Zaniolo. Managing and querying transaction-time databases under schema evolution. *Proceedings of the VLDB Endowment*, 1(1):882–895, 2008.

[25] Mohammadreza Pourreza and Davood Rafiei. DIN-SQL: Decomposed in-context learning of text-to-SQL with self-correction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[26] Simon Razniewski and Werner Nutt. Completeness of queries over incomplete databases. *Proceedings of the VLDB Endowment*, 4(11):749–760, 2011.

[27] Shariq Rizvi, Alberto Mendelzon, S. Sudarshan, and Prasan Roy. Extending query rewriting techniques for fine-grained access control. In *Proceedings of the ACM SIGMOD International Conference on Management of Data*, pages 551–562, 2004.

[28] Michael Rumiantsau and Ivan Fokeev. Semantic layers for reliable LLM-powered data analytics: A paired benchmark of accuracy and hallucination across three frontier models. arXiv:2604.25149, 2026.

[29] Richard T. Snodgrass. The temporal query language TQuel. *ACM Transactions on Database Systems*, 12(2):247–298, 1987.

[30] Richard T. Snodgrass, editor. *The TSQL2 Temporal Query Language*. Kluwer Academic Publishers, 1995.

[31] Latanya Sweeney. *k*-anonymity: A model for protecting privacy. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5):557–570, 2002.

[32] Bailin Wang, Richard Shin, Xiaodong Liu, Oleksandr Polozov, and Matthew Richardson. RAT-SQL: Relation-aware schema encoding and linking for text-to-SQL parsers. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7567–7578, 2020.

[33] Hongsheng Wang and Xiubo Liang. Asofgov: As-of correctness for governed queries with typed refusals and certificates — extended technical report with full proofs. Technical report in the artifact repository, `https://github.com/anon-submit-0/asofgov-artifact`, path `tr/asof-gov-tr.pdf`, 2026.

[34] Neeraj Yadav. Temporal validity in retrieval memory: Eliminating stale-fact errors for AI agents over evolving knowledge. arXiv:2606.26511, 2026.

[35] Tao Yu, Rui Zhang, Kai Yang, Michihiro Yasunaga, Dongxu Wang, Zifan Li, James Ma, Irene Li, Qingning Yao, Shanelle Roman, Zilin Zhang, and Dragomir Radev. Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3911–3921, 2018.

[36] Tianshu Zhang, Kun Qian, Siddhartha Sahai, Yuan Tian, Shaddy Garg, Huan Sun, and Yunyao Li. EvoSchema: Towards text-to-SQL robustness against schema evolution. *Proceedings of the VLDB Endowment*, 18(10):3655–3668, 2025.

[37] Yupeng Zhang, Daniel Genkin, Jonathan Katz, Dimitrios Papadopoulos, and Charalampos Papamanthou. vSQL: Verifying arbitrary SQL queries over dynamic outsourced databases. In *IEEE Symposium on Security and Privacy*, pages 863–880, 2017.

[38] Victor Zhong, Caiming Xiong, and Richard Socher. Seq2SQL: Generating structured queries from natural language using reinforcement learning. arXiv:1709.00103, 2017.